Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
3310 ROUGHER AND SCAVENGER FLOTATION
DESIGN OF MISCELLANEOUS FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The purpose of this document is to analysis and design of **ROUGHER AND SCAVENGER FLOTATION** foundation. In this document design & analysis of **MISCELLANEOUS** foundation have been covered. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week

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3. SCOPE

The analysis has been carried out in Staad pro & designed by Excel Sheet. For member strength calculation force values are directly taken from STAAD Pro & designed in excel sheet. Superstructure support reaction taken from respective superstructure Staad pro.

4. DESIGN BASIS

4.1 Material

A. Concrete Strength of 28 days

Foundation and Structural Members:

35 MPa Blinding Concrete below foundation bases: 15 MPa

B. Reinforcement Grade

Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)

Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

C. Grouting

Grout shall be 50 mm cementitious of Strength 40 MPa.

4.2 Loading

A. Primary Load

LOAD 1 = DEAD LOAD (DL)

LOAD 2 = NORMAL OPERATING FLUID LOAD (FL1)

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LOAD 3 = MAX. FLUID LOAD (FL2)

LOAD 4 = LIVE LOAD (LL)

LOAD 5 = SEISMIC LOAD + X (SLX1)

LOAD 6 = SEISMIC LOAD - X (SLX2)

LOAD 7 = SEISMIC LOAD +Z (SLZ1)

LOAD 8 = SEISMIC LOAD -Z (SLZ2)

LOAD 9 = WIND LOAD + X (WLX1)

LOAD 10 = WIND LOAD - X (WLX2)

LOAD 11 = WIND LOAD + Z (WLZ1)

LOAD 12 = WIND LOAD - Z (WLZ2)

LOAD 13 = TEMPERATURE LOAD (TL)

LOAD 14 = LATERAL PIPE LOAD (PL)

4.3 Load combination

A. The following load combination are considered for member strength calculation.

101 1.4DL+1.4FL1

102 1.4DL+1.4FL2

103 1.2DL+1.2FL1+1.6LL

104 1.2DL+1.2FL2+1.6LL

105 1.2DL+1.2FL1+1.2TL+1.6LL

106 1.2DL+1.2FL2+1.2TL+1.6LL

107 1.4DL+1.4FL1+1.4PL

108 1.4DL+1.4FL2+1.4PL

109 1.2DL+1.2FL1+1.6LL+1.2PL

110 1.2DL+1.2FL2+1.6LL+1.2PL

111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL

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112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL

113 1.2DL+0.8WLX1

114 1.2DL+0.8WLX2

115 1.2DL+0.8WLZ1

116 1.2DL+0.8WLZ2

117 1.2DL+1.6WLX1+0.5LL

118 1.2DL+1.6WLX2+0.5LL

119 1.2DL+1.6WLZ1+0.5LL

120 1.2DL+1.6WLZ2+0.5LL

121 1.2DL+1.0SLX1+0.5LL+0.5FL1

122 1.2DL+1.0SLX2+0.5LL+0.5FL1

123 1.2DL+1.0SLZ1+0.5LL+0.5FL1

124 1.2DL+1.0SLZ2+0.5LL+0.5FL1

125 0.9DL+1.6WLX1

126 0.9DL+1.6WLX2

127 0.9DL+1.6WLZ1

128 0.9DL+1.6WLZ2

129 0.9DL+1.0SLX1

130 0.9DL+1.0SLX2

131 0.9DL+1.0SLZ1

132 0.9DL+1.0SLZ2

B) The following load combination are considered for base pressure, sliding & overturning check.

201 1.0DL+1.0FL1

202 1.0DL+1.0FL2

203 1.0DL+1.0FL1+1.0LL+1.0TL

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204 1.0DL+1.0FL2+1.0LL+1.0TL

205 1.0DL+1.0FL1+0.75LL+0.75TL

206 1.0DL+1.0FL2+0.75LL+0.75TL

207 1.0DL+1.0FL1+1.0PL

208 1.0DL+1.0FL2+1.0PL

209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL

210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL

211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL

212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL

213 1.0DL+1.0FL1+1.0WLX1

214 1.0DL+1.0FL1+1.0WLX2

215 1.0DL+1.0FL1+1.0WLZ1

216 1.0DL+1.0FL1+1.0WLZ2

217 1.0DL+1.0FL2+1.0WLX1

218 1.0DL+1.0FL2+1.0WLX2

219 1.0DL+1.0FL2+1.0WLZ1

220 1.0DL+1.0FL2+1.0WLZ2

221 1.0DL+1.0FL1+0.7SLX1

222 1.0DL+1.0FL1+0.7SLX2

223 1.0DL+1.0FL1+0.7SLZ1

224 1.0DL+1.0FL1+0.7SLZ2

225 1.0DL+1.0FL2+0.7SLX1

226 1.0DL+1.0FL2+0.7SLX2

227 1.0DL+1.0FL2+0.7SLZ1

228 1.0DL+1.0FL2+0.7SLZ2

229 1.0DL+1.0FL1+0.75WLX1+0.75LL

230 1.0DL+1.0FL1+0.75WLX2+0.75LL

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231 1.0DL+1.0FL1+0.75WLZ1+0.75LL

232 1.0DL+1.0FL1+0.75WLZ2+0.75LL

233 1.0DL+1.0FL2+0.75WLX1+0.75LL

234 1.0DL+1.0FL2+0.75WLX2+0.75LL

235 1.0DL+1.0FL2+0.75WLZ1+0.75LL

236 1.0DL+1.0FL2+0.75WLZ2+0.75LL

237 1.0DL+1.0FL1+0.525SLX1+0.75LL

238 1.0DL+1.0FL1+0.525SLX2+0.75LL

239 1.0DL+1.0FL1+0.525SLZ1+0.75LL

240 1.0DL+1.0FL1+0.525SLZ2+0.75LL

241 1.0DL+1.0FL2+0.525SLX1+0.75LL

242 1.0DL+1.0FL2+0.525SLX2+0.75LL

243 1.0DL+1.0FL2+0.525SLZ1+0.75LL

244 1.0DL+1.0FL2+0.525SLZ2+0.75LL

245 0.6DL+1.0WLX1

246 0.6DL+1.0WLX2

247 0.6DL+1.0WLZ1

248 0.6DL+1.0WLZ2

249 0.6DL+0.6FL1+0.7SLX1

250 0.6DL+0.6FL1+0.7SLX2

251 0.6DL+0.6FL1+0.7SLZ1

252 0.6DL+0.6FL1+0.7SLZ2

4.4 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm except where welded wire fabric is used for ties and spirals the minimum rebar size shall be 10 mm.

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4.5 Minimum concrete cover

	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation & Columns	50

4.6 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m2 at 1.5m depth below EGL (RL+ 913.000) refer Table 5 - BH-10 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.7 Vertical Modulus of Subgrade Reaction

The Vertical Modulus of Subgrade Reaction has been considered as 40500 kN/m^2/m at 1.5m depth below EGL (RL+ 913.000) refer Table 5 - BH-10 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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5.1 DESIGN CALCULATION OF FOOTING F12

Size of Columns at node	Along-X	Along-Z
1	0.55m	0.55m
2	0.55m	0.55m
Size of fdn. raft:	3.60m	2.20m
Thickness of raft:	0.50	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.35	m
U/S level of fdn:	-1.50	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

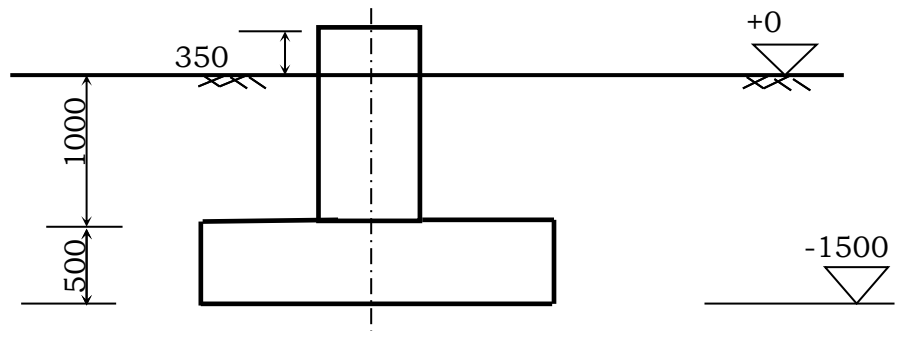
Calculation of Weight of Foundation :

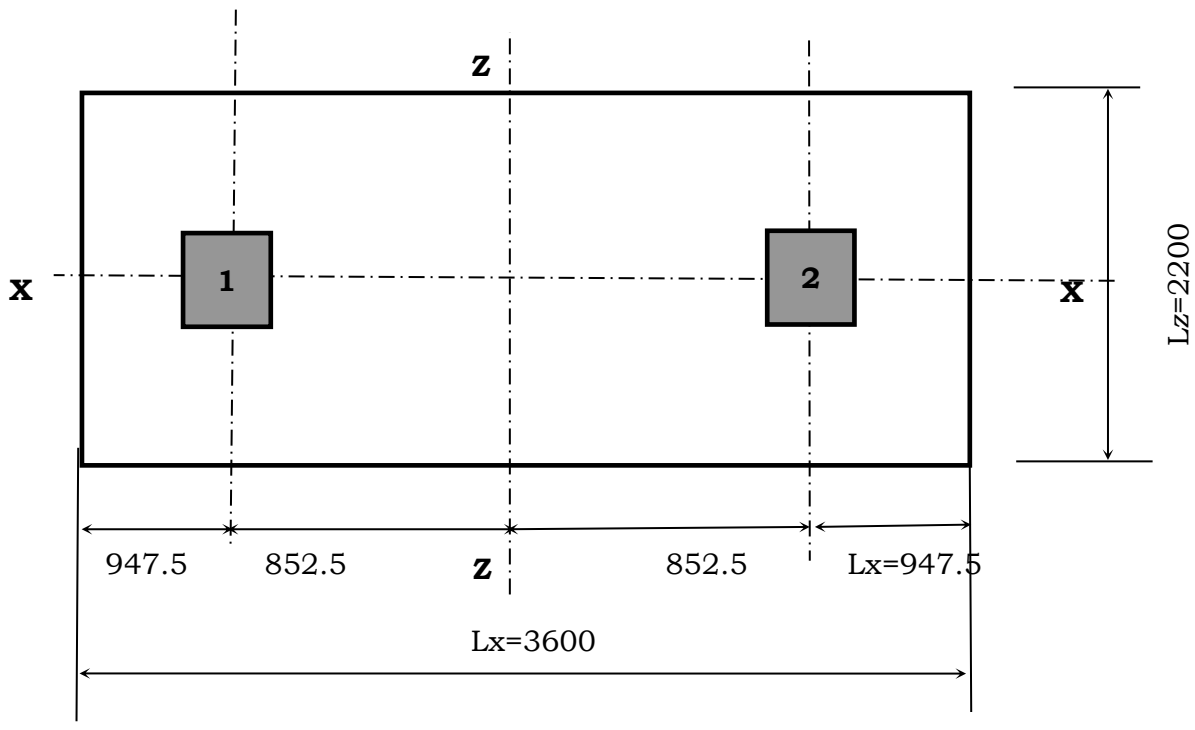
Weight of raft = (3.6*2.2*0.5*25) = 99.00KN

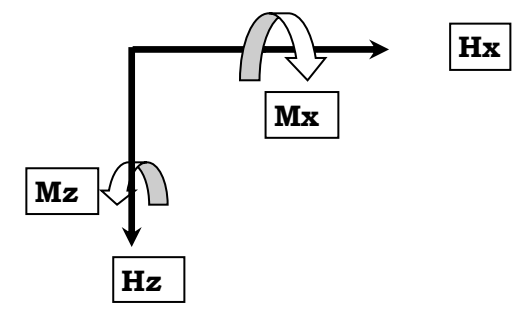
Weight of soil = ((3.6*2.2)-2*(0.55*0.55))*1*18 = 131.67KN

Total = 230.67KN

Submerged wt. = 230.67-3.6*2.2*1.5*10 = 111.87KN







H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F12

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0FL1	465.2	-75.4	120.6	0.0	0.0	56.4	-0.6	1.0	0.0	0.0
202 1.0DL+1.0FL2	509.9	-83.3	133.3	0.0	0.0	58.5	-0.7	1.1	0.0	0.0
203 1.0DL+1.0FL1+1.0LL+1.0TL	503.2	-127.2	203.5	0.0	0.0	82.3	-1.9	3.0	0.0	0.0
204 1.0DL+1.0FL2+1.0LL+1.0TL	547.9	-135.1	216.2	0.0	0.0	84.4	-2.0	3.2	0.0	0.0
205 1.0DL+1.0FL1+0.75LL+0.75TL	493.7	-114.3	182.8	0.0	0.0	75.8	-1.6	2.5	0.0	0.0
206 1.0DL+1.0FL2+0.75LL+0.75TL	538.4	-122.2	195.4	0.0	0.0	77.9	-1.7	2.7	0.0	0.0
207 1.0DL+1.0FL1+1.0PL	546.5	-84.8	135.7	0.4	0.6	91.0	-0.7	1.1	0.0	0.0
208 1.0DL+1.0FL2+1.0PL	591.2	-92.7	148.3	0.4	0.6	93.1	-0.8	1.3	0.0	0.0
209 1.0DL+1.0FL1+1.0LL+1.0TL+1.(	584.5	-136.6	218.6	0.4	0.6	116.9	-2.0	3.2	0.0	0.0
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.(	629.2	-144.5	231.2	0.4	0.6	119.0	-2.1	3.4	0.0	0.0
211 1.0DL+1.0FL1+0.75LL+0.75TL+	575.0	-123.7	197.8	0.4	0.6	110.4	-1.7	2.7	0.0	0.0
212 1.0DL+1.0FL2+0.75LL+0.75TL+	619.7	-131.6	210.5	0.4	0.6	112.5	-1.8	2.8	0.0	0.0
213 1.0DL+1.0FL1+1.0WLX1	519.6	-96.2	153.9	0.0	0.0	79.4	-1.2	1.9	0.0	0.0
214 1.0DL+1.0FL1+1.0WLX2	410.9	-54.6	87.4	0.0	0.0	33.4	0.0	0.0	0.0	0.0
215 1.0DL+1.0FL1+1.0WLZ1	437.0	-68.0	108.8	-4.9	-7.8	53.8	-0.6	1.0	-2.8	-4.5
216 1.0DL+1.0FL1+1.0WLZ2	493.4	-82.8	132.5	4.9	7.8	59.0	-0.6	1.0	2.8	4.5
217 1.0DL+1.0FL2+1.0WLX1	564.3	-104.1	166.6	0.0	0.0	81.5	-1.3	2.1	0.0	0.0
218 1.0DL+1.0FL2+1.0WLX2	455.6	-62.5	100.0	0.0	0.0	35.5	-0.1	0.2	0.0	0.0
219 1.0DL+1.0FL2+1.0WLZ1	481.7	-75.9	121.4	-4.9	-7.8	55.9	-0.7	1.1	-2.8	-4.5
220 1.0DL+1.0FL2+1.0WLZ2	538.1	-90.7	145.1	4.9	7.8	61.1	-0.7	1.1	2.8	4.5
221 1.0DL+1.0FL1+0.7SLX1	553.5	-104.4	167.0	0.0	0.0	70.8	-0.7	1.1	0.0	0.0
222 1.0DL+1.0FL1+0.7SLX2	376.9	-46.4	74.3	0.0	0.0	42.0	-0.5	0.8	0.0	0.0
223 1.0DL+1.0FL1+0.7SLZ1	414.5	-59.2	94.8	-0.1	-0.2	50.3	-0.5	0.8	-0.1	-0.2
224 1.0DL+1.0FL1+0.7SLZ2	515.9	-91.6	146.5	0.1	0.2	62.5	-0.7	1.1	0.1	0.2
225 1.0DL+1.0FL2+0.7SLX1	598.2	-112.3	179.6	0.0	0.0	72.9	-0.8	1.2	0.0	0.0
226 1.0DL+1.0FL2+0.7SLX2	421.6	-54.3	86.9	0.0	0.0	44.1	-0.6	1.0	0.0	0.0
227 1.0DL+1.0FL2+0.7SLZ1	459.2	-67.1	107.4	-0.1	-0.2	52.4	-0.6	1.0	-0.1	-0.2
228 1.0DL+1.0FL2+0.7SLZ2	560.6	-99.5	159.2	0.1	0.2	64.6	-0.8	1.2	0.1	0.2
229 1.0DL+1.0FL1+0.75WLX1+0.75	525.7	-92.4	147.9	0.0	0.0	81.5	-1.1	1.7	0.0	0.0
230 1.0DL+1.0FL1+0.75WLX2+0.75	444.2	-61.2	98.0	0.0	0.0	47.0	-0.2	0.2	0.0	0.0
231 1.0DL+1.0FL1+0.75WLZ1+0.75	463.8	-71.3	114.0	-3.7	-5.9	62.3	-0.6	1.0	-2.1	-3.4
232 1.0DL+1.0FL1+0.75WLZ2+0.75	506.1	-82.4	131.8	3.7	5.9	66.2	-0.6	1.0	2.1	3.4
233 1.0DL+1.0FL2+0.75WLX1+0.75	570.4	-100.3	160.5	0.0	0.0	83.6	-1.2	1.8	0.0	0.0
234 1.0DL+1.0FL2+0.75WLX2+0.75	488.9	-69.1	110.6	0.0	0.0	49.1	-0.3	0.4	0.0	0.0
235 1.0DL+1.0FL2+0.75WLZ1+0.75	508.5	-79.2	126.7	-3.7	-5.9	64.4	-0.7	1.1	-2.1	-3.4
236 1.0DL+1.0FL2+0.75WLZ2+0.75	550.8	-90.3	144.4	3.7	5.9	68.3	-0.7	1.1	2.1	3.4
237 1.0DL+1.0FL1+0.525SLX1+0.75	551.1	-98.6	157.7	0.0	0.0	75.0	-0.7	1.0	0.0	0.0
238 1.0DL+1.0FL1+0.525SLX2+0.75	418.7	-55.1	88.1	0.0	0.0	53.4	-0.5	0.9	0.0	0.0
239 1.0DL+1.0FL1+0.525SLZ1+0.75	446.9	-64.7	103.5	-0.1	-0.2	59.6	-0.5	0.9	-0.1	-0.2
240 1.0DL+1.0FL1+0.525SLZ2+0.75	522.9	-89.0	142.3	0.1	0.2	68.8	-0.7	1.0	0.1	0.2
241 1.0DL+1.0FL2+0.525SLX1+0.75	595.8	-106.5	170.3	0.0	0.0	77.1	-0.8	1.2	0.0	0.0
242 1.0DL+1.0FL2+0.525SLX2+0.75	463.4	-63.0	100.8	0.0	0.0	55.5	-0.6	1.0	0.0	0.0
243 1.0DL+1.0FL2+0.525SLZ1+0.75	491.6	-72.6	116.2	-0.1	-0.2	61.7	-0.6	1.0	-0.1	-0.2
244 1.0DL+1.0FL2+0.525SLZ2+0.75	567.6	-96.9	155.0	0.1	0.2	70.9	-0.8	1.2	0.1	0.2
245 0.6DL+1.0WLX1	131.7	-29.6	47.4	0.0	0.0	47.0	-0.7	1.1	0.0	0.0
246 0.6DL+1.0WLX2	23.0	12.0	-19.2	0.0	0.0	1.0	0.5	-0.9	0.0	0.0
247 0.6DL+1.0WLZ1	49.1	-1.4	2.3	-4.9	-7.8	21.4	-0.1	0.1	-2.8	-4.5
248 0.6DL+1.0WLZ2	105.5	-16.2	26.0	4.9	7.8	26.6	-0.1	0.1	2.8	4.5
249 0.6DL+0.6FL1+0.7SLX1	367.4	-74.2	118.8	0.0	0.0	48.3	-0.4	0.7	0.0	0.0
250 0.6DL+0.6FL1+0.7SLX2	190.9	-16.3	26.0	0.0	0.0	19.4	-0.3	0.5	0.0	0.0
251 0.6DL+0.6FL1+0.7SLZ1	228.4	-29.1	46.5	-0.1	-0.2	27.8	-0.3	0.5	-0.1	-0.2
252 0.6DL+0.6FL1+0.7SLZ2	329.8	-61.4	98.3	0.1	0.2	39.9	-0.4	0.7	0.1	0.2

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL+1.0FL1	752.3	-76.00	-226.9	0.0	0.0	6.0	NA	4.0	NA
202 1.0DL+1.0FL2	799.1	-84.00	-250.4	0.0	0.0	5.7	NA	3.8	NA
203 1.0DL+1.0FL1+1.0LL+1.0TL	816.2	-129.10	-152.3	0.0	0.0	9.6	NA	2.5	NA
204 1.0DL+1.0FL2+1.0LL+1.0TL	863.0	-137.10	-175.7	0.0	0.0	8.8	NA	2.5	NA
205 1.0DL+1.0FL1+0.75LL+0.75TL	800.2	-115.90	-171.0	0.0	0.0	8.4	NA	2.8	NA
206 1.0DL+1.0FL2+0.75LL+0.75TL	847.0	-123.90	-194.5	0.0	0.0	7.8	NA	2.7	NA
207 1.0DL+1.0FL1+1.0PL	868.2	-85.50	-251.5	0.4	0.6	6.2	1591.6	4.1	868.2
208 1.0DL+1.0FL2+1.0PL	915.0	-93.50	-275.0	0.4	0.6	6.0	1677.4	3.9	915.0
209 1.0DL+1.0FL1+1.0LL+1.0TL+1.(	932.1	-138.60	-176.8	0.4	0.6	9.5	1708.8	2.7	932.1
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.(	978.9	-146.60	-200.3	0.4	0.6	8.8	1794.6	2.7	978.9
211 1.0DL+1.0FL1+0.75LL+0.75TL+	916.1	-125.40	-195.6	0.4	0.6	8.4	1679.5	2.9	916.1
212 1.0DL+1.0FL2+0.75LL+0.75TL+	962.9	-133.40	-219.1	0.4	0.6	7.9	1765.3	2.9	962.9
213 1.0DL+1.0FL1+1.0WLX1	829.7	-97.40	-219.5	0.0	0.0	6.8	NA	3.4	NA
214 1.0DL+1.0FL1+1.0WLX2	675.0	-54.60	-234.4	0.0	0.0	5.2	NA	4.9	NA
215 1.0DL+1.0FL1+1.0WLZ1	721.5	-68.60	-216.9	-7.7	-12.3	6.0	64.5	4.2	37.5
216 1.0DL+1.0FL1+1.0WLZ2	783.1	-83.40	-236.8	7.7	12.3	6.0	70.0	3.8	40.7
217 1.0DL+1.0FL2+1.0WLX1	876.5	-105.40	-242.9	0.0	0.0	6.5	NA	3.3	NA
218 1.0DL+1.0FL2+1.0WLX2	721.8	-62.60	-257.9	0.0	0.0	5.0	NA	4.6	NA
219 1.0DL+1.0FL2+1.0WLZ1	768.3	-76.60	-240.5	-7.7	-12.3	5.8	68.7	4.0	39.9
220 1.0DL+1.0FL2+1.0WLZ2	829.9	-91.40	-260.4	7.7	12.3	5.7	74.2	3.6	43.1
221 1.0DL+1.0FL1+0.7SLX1	855.0	-105.10	-243.4	0.0	0.0	6.3	NA	3.3	NA
222 1.0DL+1.0FL1+0.7SLX2	649.6	-46.90	-210.4	0.0	0.0	5.6	NA	5.5	NA
223 1.0DL+1.0FL1+0.7SLZ1	695.5	-59.70	-214.9	-0.2	-0.4	5.8	1912.5	4.7	1390.9
224 1.0DL+1.0FL1+0.7SLZ2	809.1	-92.30	-238.9	0.2	0.4	6.1	2224.9	3.5	1618.1
225 1.0DL+1.0FL2+0.7SLX1	901.8	-113.10	-267.0	0.0	0.0	6.1	NA	3.2	NA
226 1.0DL+1.0FL2+0.7SLX2	696.4	-54.90	-233.9	0.0	0.0	5.4	NA	5.1	NA
227 1.0DL+1.0FL2+0.7SLZ1	742.3	-67.70	-238.4	-0.2	-0.4	5.6	2041.2	4.4	1484.5
228 1.0DL+1.0FL2+0.7SLZ2	855.9	-100.30	-262.4	0.2	0.4	5.9	2353.6	3.4	1711.7
229 1.0DL+1.0FL1+0.75WLX1+0.75	837.9	-93.50	-229.1	0.0	0.0	6.6	NA	3.6	NA
230 1.0DL+1.0FL1+0.75WLX2+0.75	721.9	-61.40	-240.4	0.0	0.0	5.4	NA	4.7	NA
231 1.0DL+1.0FL1+0.75WLZ1+0.75	756.8	-71.90	-227.3	-5.8	-9.3	6.0	89.5	4.2	52.2
232 1.0DL+1.0FL1+0.75WLZ2+0.75	803.0	-83.00	-242.2	5.8	9.3	6.0	95.0	3.9	55.4
233 1.0DL+1.0FL2+0.75WLX1+0.75	884.7	-101.50	-252.7	0.0	0.0	6.3	NA	3.5	NA
234 1.0DL+1.0FL2+0.75WLX2+0.75	768.7	-69.40	-263.9	0.0	0.0	5.2	NA	4.4	NA
235 1.0DL+1.0FL2+0.75WLZ1+0.75	803.6	-79.90	-250.8	-5.8	-9.3	5.8	95.0	4.0	55.4
236 1.0DL+1.0FL2+0.75WLZ2+0.75	849.8	-91.00	-265.8	5.8	9.3	5.8	100.5	3.7	58.6
237 1.0DL+1.0FL1+0.525SLX1+0.75	856.8	-99.30	-247.2	0.0	0.0	6.2	NA	3.5	NA
238 1.0DL+1.0FL1+0.525SLX2+0.75	702.8	-55.60	-222.4	0.0	0.0	5.7	NA	5.1	NA
239 1.0DL+1.0FL1+0.525SLZ1+0.75	737.2	-65.20	-225.8	-0.2	-0.4	5.9	2027.2	4.5	1474.3
240 1.0DL+1.0FL1+0.525SLZ2+0.75	822.4	-89.70	-243.8	0.2	0.4	6.1	2261.5	3.7	1644.7
241 1.0DL+1.0FL2+0.525SLX1+0.75	903.6	-107.30	-270.7	0.0	0.0	6.0	NA	3.4	NA
242 1.0DL+1.0FL2+0.525SLX2+0.75	749.6	-63.60	-245.9	0.0	0.0	5.5	NA	4.7	NA
243 1.0DL+1.0FL2+0.525SLZ1+0.75	784.0	-73.20	-249.3	-0.2	-0.4	5.7	2155.9	4.3	1567.9
244 1.0DL+1.0FL2+0.525SLZ2+0.75	869.2	-97.70	-267.2	0.2	0.4	5.9	2390.2	3.6	1738.3
245 0.6DL+1.0WLX1	409.4	-30.30	-23.7	0.0	0.0	31.1	NA	5.4	NA
246 0.6DL+1.0WLX2	254.7	12.50	-38.9	0.0	0.0	11.8	NA	8.1	NA
247 0.6DL+1.0WLZ1	301.2	-1.50	-21.2	-7.7	-12.3	25.6	26.9	80.3	15.6
248 0.6DL+1.0WLZ2	362.8	-16.30	-41.2	7.7	12.3	15.9	32.4	8.9	18.8
249 0.6DL+0.6FL1+0.7SLX1	646.4	-74.60	-152.5	0.0	0.0	7.6	NA	3.5	NA
250 0.6DL+0.6FL1+0.7SLX2	441.0	-16.60	-119.7	0.0	0.0	6.6	NA	10.6	NA
251 0.6DL+0.6FL1+0.7SLZ1	486.9	-29.40	-124.0	-0.2	-0.4	7.1	1338.9	6.6	973.7
252 0.6DL+0.6FL1+0.7SLZ2	600.4	-61.80	-148.1	0.2	0.4	7.3	1651.0	3.9	1200.7
Min.=	5.0	26.9	2.5	15.6					

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Base pressure check of footing F12

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	1	201	0	132.90	0
Min Px	1	201	0	132.90	0
Max Py	1003	208	0	177.71	0
Min Py	114	246	0	14.16	0
Max Pz	1	201	0	132.90	0
Min Pz	1	201	0	132.90	0

Gross Maximum Base pressure, p_{max} = 177.7 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 1.1 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 469.8 KN/m^2
> 177.7 KN/m^2
Hence OK

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Design of foundation raft F12

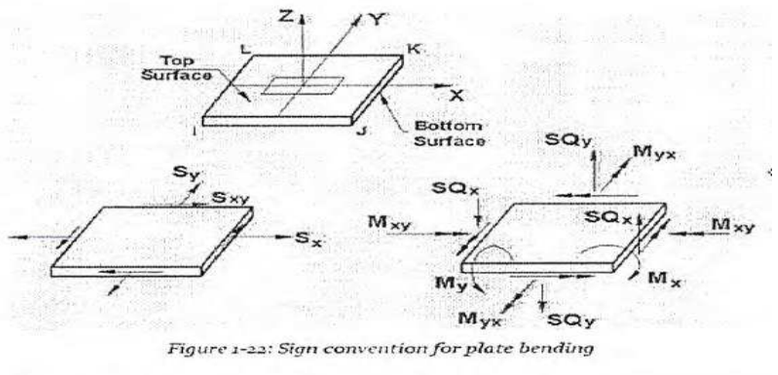
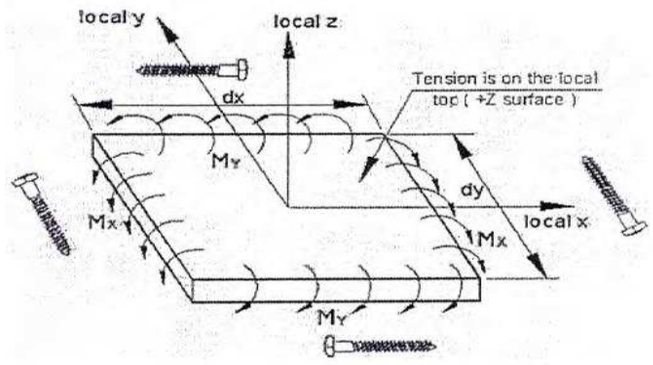


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F12_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1324	0.758	0.247	0	0	0	-153.2	-97.672	4.894	-158.127	-102.566	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Qx	1270	-0.353	0.298	0	0	0	-99.37	-24.251	0	-99.373	-24.251	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max Qy	1298	0.098	0.53	0	0	0	-132.5	-40.163	9.268	-141.725	-49.431	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Qy	1313	0.097	-0.529	0	0	0	-132.3	-39.914	-9.453	-141.781	-49.367	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max Sx	1343	0.204	-0.181	0	0	0	-63.63	-30.503	-15.782	-79.408	-46.285	101 1.4DL+1.4FL1
Min Sx	1343	0.204	-0.181	0	0	0	-63.63	-30.503	-15.782	-79.408	-46.285	101 1.4DL+1.4FL1
Max Sy	1343	0.204	-0.181	0	0	0	-63.63	-30.503	-15.782	-79.408	-46.285	101 1.4DL+1.4FL1
Min Sy	1343	0.204	-0.181	0	0	0	-63.63	-30.503	-15.782	-79.408	-46.285	101 1.4DL+1.4FL1
Max Sxy	1343	0.204	-0.181	0	0	0	-63.63	-30.503	-15.782	-79.408	-46.285	101 1.4DL+1.4FL1
Min Sxy	1343	0.204	-0.181	0	0	0	-63.63	-30.503	-15.782	-79.408	-46.285	101 1.4DL+1.4FL1
Max Mx	1334	-0.055	0.022	0	0	0	20.309	2.368	-1.01	21.319	3.378	130 0.9DL+1.0SLX2
Min Mx	1324	0.758	0.247	0	0	0	-153.2	-97.672	4.894	-158.127	-102.566	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max My	1334	-0.055	0.022	0	0	0	20.309	2.368	-1.01	21.319	3.378	130 0.9DL+1.0SLX2
Min My	1324	0.758	0.247	0	0	0	-153.2	-97.672	4.894	-158.127	-102.566	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max Mxy	1285	-0.307	-0.084	0	0	0	-54.71	-19.173	30.545	-85.256	-49.718	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Mxy	1268	-0.308	0.086	0	0	0	-54.84	-19.439	-30.577	-85.42	-50.016	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL

PLATE CENTER STRESS : FDN_F12_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1199	0.189	0.04	0	0	0	-50.34	-47.75	3.178	-53.52	-50.928	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Qx	1269	-0.231	0.092	0	0	0	-52.17	-9.498	0	-52.17	-9.498	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max Qy	1299	-0.007	0.253	0	0	0	-113.4	-10.498	6.328	-119.695	-16.826	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Qy	1312	-0.008	-0.252	0	0	0	-113.2	-10.39	-6.446	-119.694	-16.836	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max Sx	1341	0.111	-0.128	0	0	0	-60.97	-11.968	-14.356	-75.327	-26.324	101 1.4DL+1.4FL1
Min Sx	1341	0.111	-0.128	0	0	0	-60.97	-11.968	-14.356	-75.327	-26.324	101 1.4DL+1.4FL1
Max Sy	1341	0.111	-0.128	0	0	0	-60.97	-11.968	-14.356	-75.327	-26.324	101 1.4DL+1.4FL1
Min Sy	1341	0.111	-0.128	0	0	0	-60.97	-11.968	-14.356	-75.327	-26.324	101 1.4DL+1.4FL1
Max Sxy	1341	0.111	-0.128	0	0	0	-60.97	-11.968	-14.356	-75.327	-26.324	101 1.4DL+1.4FL1
Min Sxy	1341	0.111	-0.128	0	0	0	-60.97	-11.968	-14.356	-75.327	-26.324	101 1.4DL+1.4FL1
Max Mx	1340	0.004	0.013	0	0	0	12.225	-0.163	-0.828	13.053	-0.991	130 0.9DL+1.0SLX2
Min Mx	1299	-0.007	0.253	0	0	0	-113.4	-10.498	6.328	-119.695	-16.826	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max My	1218	-0.007	0.004	0	0	0	-4.185	0.036	0.626	-4.811	0.662	127 0.9DL+1.6WLZ1
Min My	1199	0.189	0.04	0	0	0	-50.34	-47.75	3.178	-53.52	-50.928	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max Mxy	1284	-0.231	-0.091	0	0	0	-52.05	-9.377	18.791	-70.839	-28.168	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Mxy	1269	-0.231	0.092	0	0	0	-52.17	-9.498	-18.87	-71.04	-28.368	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

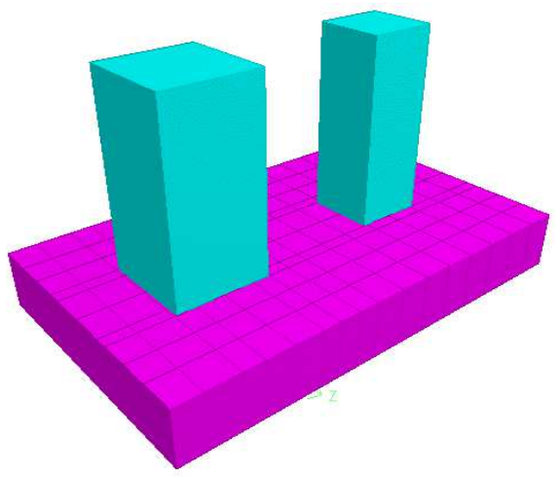
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
130 0.9D	21.32	top face	50	442	21.319	0.15	0.0003	153.5	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
112 1.2D	-158.13	bottom face	50	442	158.13	1.08	0.0026	1157.1		16	173.83	150	1341	18.91	23.64	0.053	tension control
112 1.2D	-85.26	bottom face	50	442	85.256	0.58	0.0014	618.4		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
112 1.2D	-85.42	bottom face	50	442	85.42	0.58	0.0014	619.6		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.189	1	94.5	53.52	0.002	0.442	431.73	435.484	758.323	431.73	1618	> 94.5	ok	323.8	φVc>Vu	not required
	Min Qx	0.231	1	115.5	52.17			435.117			435.12	1621	> 115.5	ok	326.3	φVc>Vu	not required

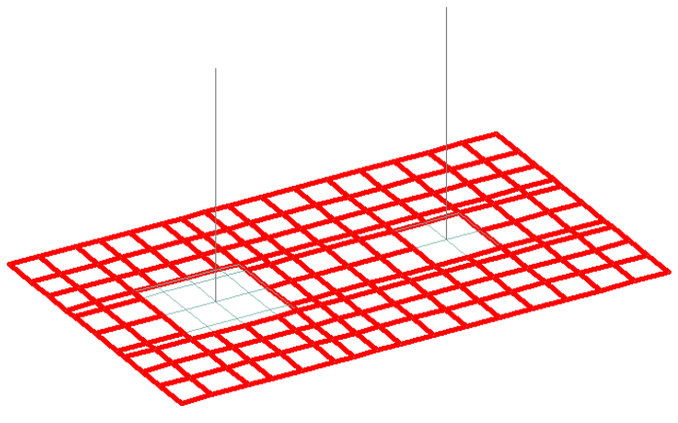
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
130 0.9D	3.38	top face	50	442	3.378	0.02	0.0001	24.3	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
112 1.2D	-102.57	bottom face	50	442	102.57	0.70	0.0017	745.5		16	252.82	150	1341	18.91	23.64	0.053	tension control
112 1.2D	-49.72	bottom face	50	442	49.718	0.34	0.0008	359.2		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
112 1.2D	-50.02	bottom face	50	442	50.016	0.34	0.0008	361.3		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

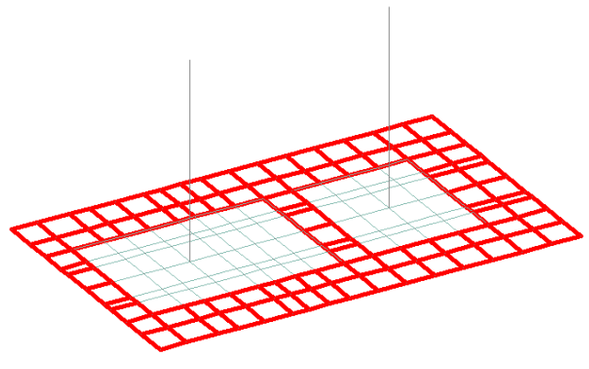
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	112 1.2D	0.253	1	126.5	16.826	0.002	0.442	475.204	435.484	fc'758.3	435.48	456	> 126.5	ok	326.6	φVc>Vu	not required
	112 1.2D	0.252	1	126	16.836			474.946			435.48	456	> 126	ok	326.6	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	16ϕT	**150**	c/c
At Top	16ϕT	**200**	c/c
Along Z Direction :			
At Bottom	16ϕT	**150**	c/c
At Top	16ϕT	**200**	c/c

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5.2 DESIGN CALCULATION OF FOOTING F12-A

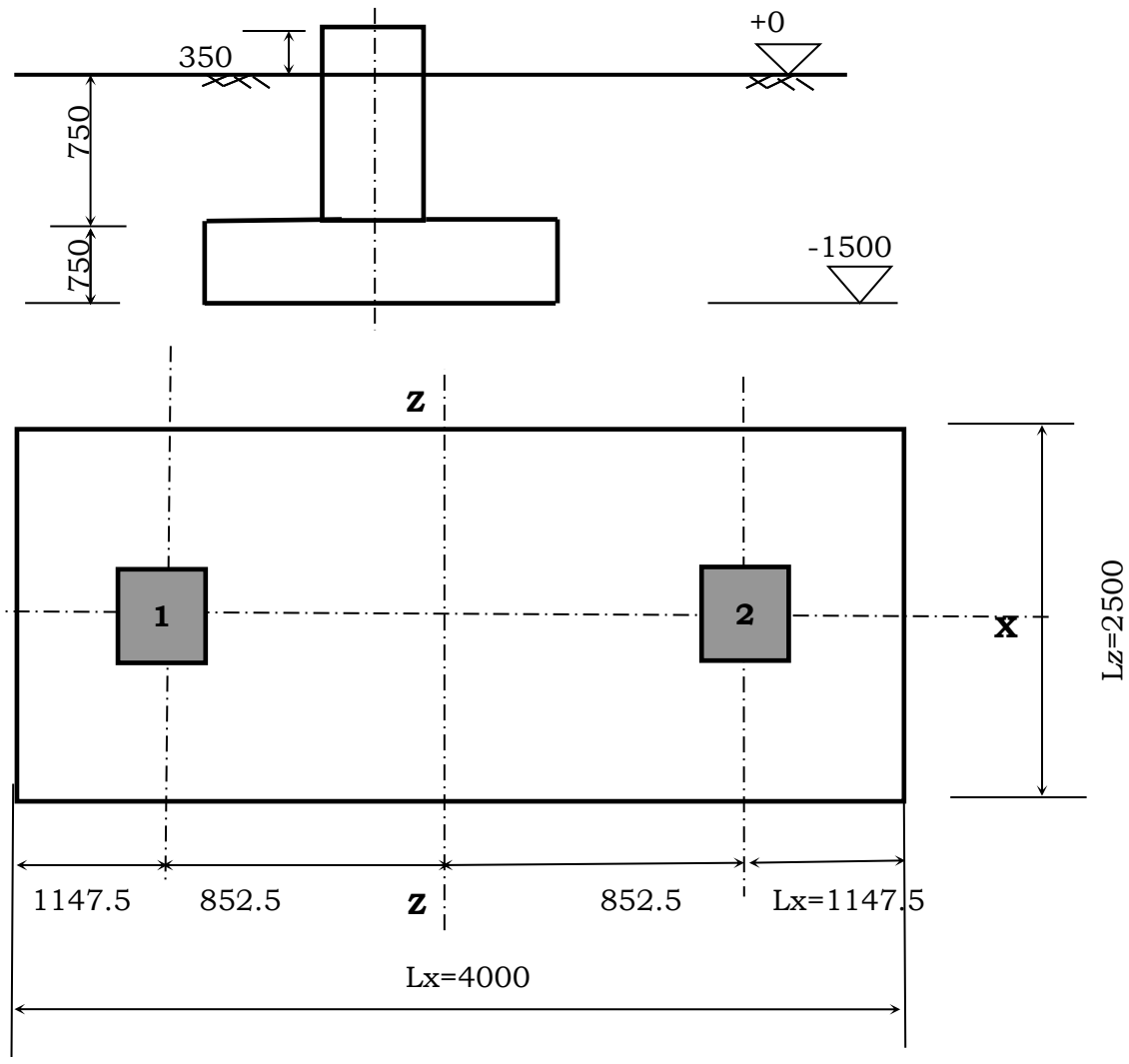
Size of Columns at node	Along-X	Along-Z
1	0.55m	0.55m
2	0.55m	0.55m
Size of fdn. raft:	4.00m	2.50m
Thickness of raft:	0.75	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.35	m
U/S level of fdn:	-1.50	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

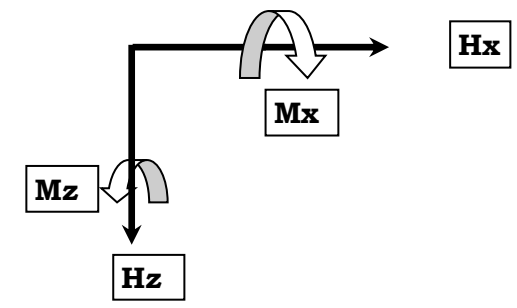
Calculation of Weight of Foundation :

Weight of raft =	(4*2.5*0.75*25) =	187.50KN
Weight of soil =	((4*2.5)-2*(0.55*0.55))*0.75*18 =	126.83KN
	Total =	314.33KN
Submerged wt. =	314.33-4*2.5*1.5*10 =	164.33KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F12-A

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0FL1	747.9	-147.9	236.6	-98.6	-157.8	97.5	-0.9	1.4	0.0	0.0
202 1.0DL+1.0FL2	824.8	-172.2	275.5	-108.6	-173.8	101.3	-0.9	1.4	0.0	0.0
203 1.0DL+1.0FL1+1.0LL+1.0TL	997.4	-322.5	516.0	-154.4	-247.0	181.1	-2.3	3.7	0.0	0.0
204 1.0DL+1.0FL2+1.0LL+1.0TL	1074.3	-346.8	554.9	-164.4	-263.0	184.9	-2.3	3.7	0.0	0.0
205 1.0DL+1.0FL1+0.75LL+0.75TL	935.0	-278.9	446.2	-140.4	-224.7	160.2	-2.0	3.1	0.0	0.0
206 1.0DL+1.0FL2+0.75LL+0.75TL	1011.9	-303.2	485.0	-150.4	-240.7	164.0	-2.0	3.1	0.0	0.0
207 1.0DL+1.0FL1+1.0PL	831.1	-151.2	241.9	-118.5	-189.6	114.6	-1.0	1.6	0.0	0.0
208 1.0DL+1.0FL2+1.0PL	908.0	-175.5	280.8	-128.5	-205.6	118.4	-1.0	1.6	0.0	0.0
209 1.0DL+1.0FL1+1.0LL+1.0TL+1.(	1080.6	-325.8	521.3	-174.3	-278.9	198.2	-2.4	3.8	0.0	0.0
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.(	1157.5	-350.1	560.2	-184.3	-294.9	202.0	-2.4	3.8	0.0	0.0
211 1.0DL+1.0FL1+0.75LL+0.75TL+	1018.2	-282.2	451.4	-160.4	-256.6	177.3	-2.1	3.3	0.0	0.0
212 1.0DL+1.0FL2+0.75LL+0.75TL+	1095.1	-306.5	490.3	-170.3	-272.6	181.1	-2.1	3.3	0.0	0.0
213 1.0DL+1.0FL1+1.0WLX1	785.0	-179.6	287.4	-97.9	-156.6	121.3	-2.3	3.7	0.0	0.0
214 1.0DL+1.0FL1+1.0WLX2	710.8	-115.2	184.3	-99.3	-158.9	73.2	0.7	-1.1	0.0	0.0
215 1.0DL+1.0FL1+1.0WLZ1	846.9	-162.0	259.2	-144.5	-231.2	102.5	-1.0	1.6	-2.8	-4.5
216 1.0DL+1.0FL1+1.0WLZ2	648.7	-133.8	214.1	-52.5	-84.0	92.5	-0.8	1.3	2.8	4.5
217 1.0DL+1.0FL2+1.0WLX1	861.9	-203.9	326.2	-107.9	-172.6	125.1	-2.3	3.7	0.0	0.0
218 1.0DL+1.0FL2+1.0WLX2	787.7	-139.5	223.2	-109.3	-174.9	77.0	0.7	-1.1	0.0	0.0
219 1.0DL+1.0FL2+1.0WLZ1	923.8	-186.3	298.1	-154.5	-247.2	106.3	-1.0	1.6	-2.8	-4.5
220 1.0DL+1.0FL2+1.0WLZ2	725.6	-158.1	253.0	-62.5	-100.0	96.3	-0.8	1.3	2.8	4.5
221 1.0DL+1.0FL1+0.7SLX1	809.4	-183.9	294.2	-100.4	-160.7	114.9	-1.0	1.6	0.0	0.0
222 1.0DL+1.0FL1+0.7SLX2	686.4	-111.9	179.1	-96.8	-154.8	80.1	-0.8	1.3	0.0	0.0
223 1.0DL+1.0FL1+0.7SLZ1	959.2	-177.7	284.2	-185.8	-297.3	114.7	-1.0	1.7	-0.1	-0.2
224 1.0DL+1.0FL1+0.7SLZ2	536.6	-118.2	189.0	-11.4	-18.2	80.4	-0.8	1.2	0.1	0.2
225 1.0DL+1.0FL2+0.7SLX1	886.3	-208.2	333.1	-110.4	-176.7	118.7	-1.0	1.6	0.0	0.0
226 1.0DL+1.0FL2+0.7SLX2	763.3	-136.2	218.0	-106.8	-170.8	83.9	-0.8	1.3	0.0	0.0
227 1.0DL+1.0FL2+0.7SLZ1	1036.1	-201.9	323.1	-195.8	-313.3	118.5	-1.0	1.7	-0.1	-0.2
228 1.0DL+1.0FL2+0.7SLZ2	613.5	-142.5	227.9	-21.4	-34.2	84.2	-0.8	1.2	0.1	0.2
229 1.0DL+1.0FL1+0.75WLX1+0.75	864.7	-187.5	300.0	-104.8	-167.7	131.3	-2.1	3.4	0.0	0.0
230 1.0DL+1.0FL1+0.75WLX2+0.75	809.0	-139.2	222.7	-105.9	-169.4	95.2	0.2	-0.2	0.0	0.0
231 1.0DL+1.0FL1+0.75WLZ1+0.75	911.1	-174.3	278.9	-139.8	-223.6	117.2	-1.1	1.8	-2.1	-3.4
232 1.0DL+1.0FL1+0.75WLZ2+0.75	762.5	-153.2	245.0	-70.8	-113.2	109.7	-1.0	1.6	2.1	3.4
233 1.0DL+1.0FL2+0.75WLX1+0.75	941.6	-211.8	338.9	-114.8	-183.7	135.1	-2.1	3.4	0.0	0.0
234 1.0DL+1.0FL2+0.75WLX2+0.75	885.9	-163.5	261.6	-115.9	-185.4	99.0	0.2	-0.2	0.0	0.0
235 1.0DL+1.0FL2+0.75WLZ1+0.75	988.0	-198.6	317.8	-149.8	-239.6	121.0	-1.1	1.8	-2.1	-3.4
236 1.0DL+1.0FL2+0.75WLZ2+0.75	839.4	-177.5	283.9	-80.8	-129.2	113.5	-1.0	1.6	2.1	3.4
237 1.0DL+1.0FL1+0.525SLX1+0.75	882.9	-190.7	305.1	-106.7	-170.7	126.4	-1.1	1.8	0.0	0.0
238 1.0DL+1.0FL1+0.525SLX2+0.75	790.8	-136.7	218.8	-104.0	-166.4	100.4	-1.0	1.6	0.0	0.0
239 1.0DL+1.0FL1+0.525SLZ1+0.75	995.3	-186.0	297.7	-170.8	-273.2	126.3	-1.2	1.8	-0.1	-0.2
240 1.0DL+1.0FL1+0.525SLZ2+0.75	678.4	-141.4	226.3	-39.9	-63.9	100.5	-0.9	1.5	0.1	0.2
241 1.0DL+1.0FL2+0.525SLX1+0.75	959.8	-215.0	344.0	-116.7	-186.7	130.2	-1.1	1.8	0.0	0.0
242 1.0DL+1.0FL2+0.525SLX2+0.75	867.7	-161.0	257.7	-114.0	-182.4	104.2	-1.0	1.6	0.0	0.0
243 1.0DL+1.0FL2+0.525SLZ1+0.75	1072.2	-210.3	336.5	-180.8	-289.2	130.1	-1.2	1.8	-0.1	-0.2
244 1.0DL+1.0FL2+0.525SLZ2+0.75	755.3	-165.7	265.1	-49.9	-79.9	104.3	-0.9	1.5	0.1	0.2
245 0.6DL+1.0WLX1	205.2	-57.1	91.3	-14.9	-23.8	58.7	-1.6	2.5	0.0	0.0
246 0.6DL+1.0WLX2	131.0	7.3	-11.7	-16.3	-26.1	10.6	1.4	-2.3	0.0	0.0
247 0.6DL+1.0WLZ1	267.1	-39.5	63.2	-61.5	-98.4	39.9	-0.3	0.4	-2.8	-4.5
248 0.6DL+1.0WLZ2	68.9	-11.3	18.0	30.5	48.8	29.9	-0.1	0.1	2.8	4.5
249 0.6DL+0.6FL1+0.7SLX1	510.2	-124.7	199.6	-61.0	-97.6	75.9	-0.6	1.0	0.0	0.0
250 0.6DL+0.6FL1+0.7SLX2	387.3	-52.8	84.4	-57.3	-91.7	41.1	-0.5	0.8	0.0	0.0
251 0.6DL+0.6FL1+0.7SLZ1	660.1	-118.5	189.6	-146.4	-234.2	75.7	-0.7	1.1	-0.1	-0.2
252 0.6DL+0.6FL1+0.7SLZ2	237.4	-59.0	94.4	28.1	44.9	41.4	-0.4	0.6	0.1	0.2

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL+1.0FL1	1159.7	-148.80	-316.5	-98.6	-157.8	7.3	9.2	3.1	4.7
202 1.0DL+1.0FL2	1240.4	-173.10	-339.9	-108.6	-173.8	7.3	8.9	2.9	4.6
203 1.0DL+1.0FL1+1.0LL+1.0TL	1492.8	-324.80	-176.2	-154.4	-247.0	16.9	7.6	1.8	3.9
204 1.0DL+1.0FL2+1.0LL+1.0TL	1573.5	-349.10	-199.6	-164.4	-263.0	15.8	7.5	1.8	3.8
205 1.0DL+1.0FL1+0.75LL+0.75TL	1409.5	-280.90	-211.2	-140.4	-224.7	13.3	7.8	2.0	4.0
206 1.0DL+1.0FL2+0.75LL+0.75TL	1490.2	-305.20	-234.7	-150.4	-240.7	12.7	7.7	2.0	4.0
207 1.0DL+1.0FL1+1.0PL	1260.0	-152.20	-367.3	-118.5	-189.6	6.9	8.3	3.3	4.3
208 1.0DL+1.0FL2+1.0PL	1340.7	-176.50	-390.7	-128.5	-205.6	6.9	8.2	3.0	4.2
209 1.0DL+1.0FL1+1.0LL+1.0TL+1.(	1593.1	-328.20	-227.1	-174.3	-278.9	14.0	7.1	1.9	3.7
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.(	1673.8	-352.50	-250.6	-184.3	-294.9	13.4	7.1	1.9	3.6
211 1.0DL+1.0FL1+0.75LL+0.75TL+	1509.8	-284.30	-262.2	-160.4	-256.6	11.5	7.4	2.1	3.8
212 1.0DL+1.0FL2+0.75LL+0.75TL+	1590.5	-308.60	-285.6	-170.3	-272.6	11.1	7.3	2.1	3.7
213 1.0DL+1.0FL1+1.0WLX1	1220.6	-181.90	-274.7	-97.9	-156.6	8.9	9.7	2.7	5.0
214 1.0DL+1.0FL1+1.0WLX2	1098.3	-114.50	-360.4	-99.3	-158.9	6.1	8.6	3.8	4.4
215 1.0DL+1.0FL1+1.0WLZ1	1263.7	-163.00	-373.8	-147.3	-235.7	6.8	6.7	3.1	3.4
216 1.0DL+1.0FL1+1.0WLZ2	1055.5	-134.60	-258.8	-49.7	-79.5	8.2	16.6	3.1	8.5
217 1.0DL+1.0FL2+1.0WLX1	1301.3	-206.20	-298.2	-107.9	-172.6	8.7	9.4	2.5	4.8
218 1.0DL+1.0FL2+1.0WLX2	1179.0	-138.80	-383.8	-109.3	-174.9	6.1	8.4	3.4	4.3
219 1.0DL+1.0FL2+1.0WLZ1	1344.4	-187.30	-397.2	-157.3	-251.7	6.8	6.7	2.9	3.4
220 1.0DL+1.0FL2+1.0WLZ2	1136.2	-158.90	-282.2	-59.7	-95.5	8.1	14.9	2.9	7.6
221 1.0DL+1.0FL1+0.7SLX1	1238.6	-184.90	-296.3	-100.4	-160.7	8.4	9.6	2.7	4.9
222 1.0DL+1.0FL1+0.7SLX2	1080.8	-112.70	-336.5	-96.8	-154.8	6.4	8.7	3.8	4.5
223 1.0DL+1.0FL1+0.7SLZ1	1388.2	-178.70	-434.0	-185.9	-297.5	6.4	5.8	3.1	3.0
224 1.0DL+1.0FL1+0.7SLZ2	931.3	-119.00	-198.7	-11.3	-18.0	9.4	64.7	3.1	33.0
225 1.0DL+1.0FL2+0.7SLX1	1319.3	-209.20	-319.7	-110.4	-176.7	8.3	9.3	2.5	4.8
226 1.0DL+1.0FL2+0.7SLX2	1161.5	-137.00	-359.9	-106.8	-170.8	6.5	8.5	3.4	4.4
227 1.0DL+1.0FL2+0.7SLZ1	1468.9	-202.90	-457.5	-195.9	-313.5	6.4	5.9	2.9	3.0
228 1.0DL+1.0FL2+0.7SLZ2	1012.0	-143.30	-222.1	-21.3	-34.0	9.1	37.2	2.8	19.0
229 1.0DL+1.0FL1+0.75WLX1+0.75	1310.3	-189.60	-321.8	-104.8	-167.7	8.1	9.8	2.8	5.0
230 1.0DL+1.0FL1+0.75WLX2+0.75	1218.5	-139.00	-386.0	-105.9	-169.4	6.3	9.0	3.5	4.6
231 1.0DL+1.0FL1+0.75WLZ1+0.75	1342.6	-175.40	-396.1	-141.9	-227.0	6.8	7.4	3.1	3.8
232 1.0DL+1.0FL1+0.75WLZ2+0.75	1186.5	-154.20	-309.9	-68.7	-109.8	7.7	13.5	3.1	6.9
233 1.0DL+1.0FL2+0.75WLX1+0.75	1391.0	-213.90	-345.2	-114.8	-183.7	8.1	9.5	2.6	4.8
234 1.0DL+1.0FL2+0.75WLX2+0.75	1299.2	-163.30	-409.4	-115.9	-185.4	6.3	8.8	3.2	4.5
235 1.0DL+1.0FL2+0.75WLZ1+0.75	1423.3	-199.70	-419.5	-151.9	-243.0	6.8	7.3	2.9	3.7
236 1.0DL+1.0FL2+0.75WLZ2+0.75	1267.2	-178.50	-333.3	-78.7	-125.8	7.6	12.6	2.8	6.4
237 1.0DL+1.0FL1+0.525SLX1+0.75	1323.6	-191.80	-338.0	-106.7	-170.7	7.8	9.7	2.8	5.0
238 1.0DL+1.0FL1+0.525SLX2+0.75	1205.5	-137.70	-368.2	-104.0	-166.4	6.5	9.1	3.5	4.6
239 1.0DL+1.0FL1+0.525SLZ1+0.75	1435.9	-187.20	-441.3	-170.9	-273.4	6.5	6.6	3.1	3.4
240 1.0DL+1.0FL1+0.525SLZ2+0.75	1093.2	-142.30	-264.9	-39.8	-63.7	8.3	21.5	3.1	11.0
241 1.0DL+1.0FL2+0.525SLX1+0.75	1404.3	-216.10	-361.4	-116.7	-186.7	7.8	9.4	2.6	4.8
242 1.0DL+1.0FL2+0.525SLX2+0.75	1286.2	-162.00	-391.6	-114.0	-182.4	6.6	8.8	3.2	4.5
243 1.0DL+1.0FL2+0.525SLZ1+0.75	1516.6	-211.50	-464.8	-180.9	-289.4	6.5	6.6	2.9	3.4
244 1.0DL+1.0FL2+0.525SLZ2+0.75	1173.9	-166.60	-288.4	-49.8	-79.7	8.1	18.4	2.8	9.4
245 0.6DL+1.0WLX1	578.2	-58.70	-31.1	-14.9	-23.8	37.2	30.4	3.9	15.5
246 0.6DL+1.0WLX2	455.9	8.70	-116.6	-16.3	-26.1	7.8	21.8	21.0	11.2
247 0.6DL+1.0WLZ1	621.3	-39.80	-130.1	-64.3	-102.9	9.6	7.5	6.2	3.9
248 0.6DL+1.0WLZ2	413.1	-11.40	-15.1	33.3	53.3	54.5	9.7	14.5	5.0
249 0.6DL+0.6FL1+0.7SLX1	900.4	-125.30	-169.6	-61.0	-97.6	10.6	11.5	2.9	5.9
250 0.6DL+0.6FL1+0.7SLX2	742.7	-53.30	-209.9	-57.3	-91.7	7.1	10.1	5.6	5.2
251 0.6DL+0.6FL1+0.7SLZ1	1050.1	-119.20	-307.5	-146.5	-234.4	6.8	5.6	3.5	2.9
252 0.6DL+0.6FL1+0.7SLZ2	593.1	-59.40	-72.1	28.2	45.1	16.5	16.4	4.0	8.4
Min.=						6.1	5.6	1.8	2.9

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Base pressure check of footing F12-A

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	5	201	0	143.85	0
Min Px	5	201	0	143.85	0
Max Py	525	227	0	294.18	0
Min Py	518	251	0	0.00	0
Max Pz	5	201	0	143.85	0
Min Pz	5	201	0	143.85	0

Gross Maximum Base pressure, p_{max} = 294.2 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2

Depth of soil above footing = 0.75 m

Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 463.5 KN/m^2

> 294.2 KN/m^2

Hence OK

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Design of foundation raft F12-A

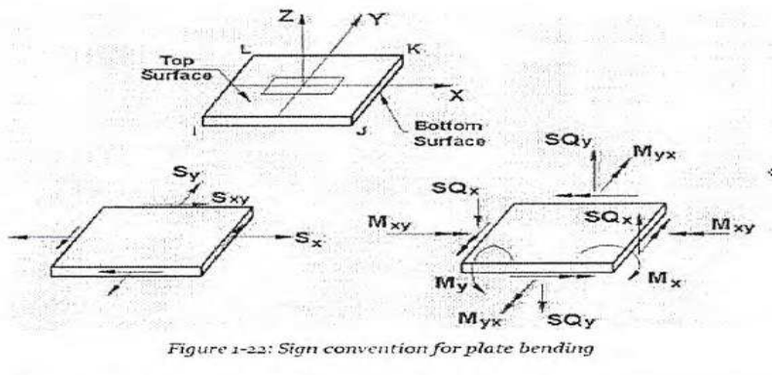
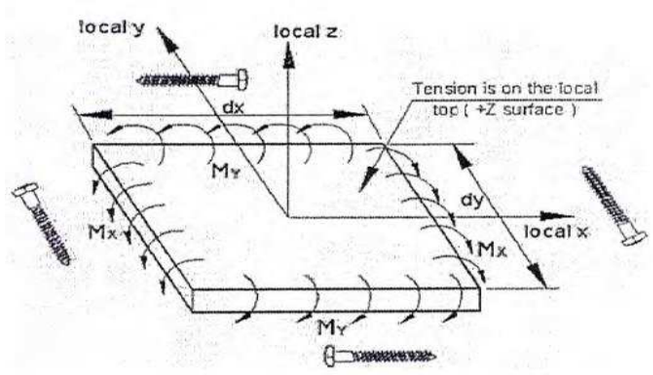


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F12-A_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	592	1.108	-1.351	0	0	0	-468.5	-322.21	150.203	-618.691	-472.413	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Qx	568	-0.582	-0.247	0	0	0	-113.4	-97.956	0	-113.431	-97.956	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max Qy	1009	0.024	0.496	0	0	0	-268.9	-49.396	47.221	-316.073	-96.617	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Qy	591	-0.072	-1.354	0	0	0	-228.2	-349.036	141.395	-369.613	-490.431	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max Sx	1140	0.005	-0.027	0	0	0	-0.293	-2.362	0.973	-1.266	-3.335	101 1.4DL+1.4FL1
Min Sx	1140	0.005	-0.027	0	0	0	-0.293	-2.362	0.973	-1.266	-3.335	101 1.4DL+1.4FL1
Max Sy	1140	0.005	-0.027	0	0	0	-0.293	-2.362	0.973	-1.266	-3.335	101 1.4DL+1.4FL1
Min Sy	1140	0.005	-0.027	0	0	0	-0.293	-2.362	0.973	-1.266	-3.335	101 1.4DL+1.4FL1
Max Sxy	1140	0.005	-0.027	0	0	0	-0.293	-2.362	0.973	-1.266	-3.335	101 1.4DL+1.4FL1
Min Sxy	1140	0.005	-0.027	0	0	0	-0.293	-2.362	0.973	-1.266	-3.335	101 1.4DL+1.4FL1
Max Mx	463	0.027	-0.013	0	0	0	66.432	-78.411	-49.688	116.12	-128.099	106 1.2DL+1.2FL2+1.2TL+1.6LL
Min Mx	592	1.108	-1.351	0	0	0	-468.5	-322.21	150.203	-618.691	-472.413	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max My	569	0.048	0.198	0	0	0	7.062	78.135	6.763	13.825	84.898	132 0.9DL+1.0SLZ2
Min My	591	-0.072	-1.354	0	0	0	-228.2	-349.036	141.395	-369.613	-490.431	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max Mxy	592	1.108	-1.351	0	0	0	-468.5	-322.21	150.203	-618.691	-472.413	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Mxy	1003	-0.172	-0.077	0	0	0	-123.9	46.682	-97.606	-221.46	144.288	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL

PLATE CENTER STRESS : FDN_F12-A_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	1125	0.259	-0.04	0	0	0	-153.9	-3.691	3.715	-157.605	-7.406	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Qx	1114	-0.253	-0.161	0	0	0	-158	-19.619	0	-158.045	-19.619	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max Qy	1087	-0.044	0.121	0	0	0	-212.3	-6.468	38.195	-250.446	-44.663	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Qy	1120	0.045	-0.279	0	0	0	-256	-25.398	15.78	-271.763	-41.178	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max Sx	1140	0.005	-0.027	0	0	0	-0.293	-2.362	0.973	-1.266	-3.335	101 1.4DL+1.4FL1
Min Sx	1140	0.005	-0.027	0	0	0	-0.293	-2.362	0.973	-1.266	-3.335	101 1.4DL+1.4FL1
Max Sy	1140	0.005	-0.027	0	0	0	-0.293	-2.362	0.973	-1.266	-3.335	101 1.4DL+1.4FL1
Min Sy	1140	0.005	-0.027	0	0	0	-0.293	-2.362	0.973	-1.266	-3.335	101 1.4DL+1.4FL1
Max Sxy	1140	0.005	-0.027	0	0	0	-0.293	-2.362	0.973	-1.266	-3.335	101 1.4DL+1.4FL1
Min Sxy	1140	0.005	-0.027	0	0	0	-0.293	-2.362	0.973	-1.266	-3.335	101 1.4DL+1.4FL1
Max Mx	1118	-0.023	0.047	0	0	0	15.141	7.352	-7.54	22.681	14.892	132 0.9DL+1.0SLZ2
Min Mx	1120	0.045	-0.279	0	0	0	-256	-25.398	15.78	-271.763	-41.178	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Max My	1081	-0.218	0.023	0	0	0	-150.6	10.937	-7.159	-157.768	18.096	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min My	572	-0.026	0.047	0	0	0	0.643	-63.366	4.663	5.306	-68.029	110 1.2DL+1.2FL2+1.6LL+1.2PL
Max Mxy	1089	0.039	0.102	0	0	0	-191.5	-12.57	43.47	-234.987	-56.04	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Mxy	1077	-0.155	-0.025	0	0	0	-65.15	3.538	-20.808	-85.959	24.346	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		750	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

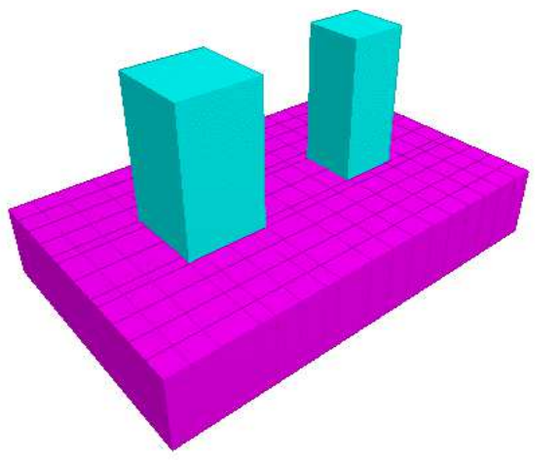
Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
106 1.2D	116.12	top face	50	692	116.12	0.32	0.0008	535.6	1245.6	16	161.48	150	1341	18.91	23.64	0.085	tension control
112 1.2D	-618.69	bottom face	50	688	618.69	1.75	0.0043	2946.0		25	166.69	150	3273.9	46.16	57.7	0.033	tension control
112 1.2D	-618.69	bottom face	50	688	618.69	1.75	0.0043	2946.0		25	166.69	150	3273.9	46.16	57.7	0.033	tension control
112 1.2D	-221.46	bottom face	50	692	221.46	0.62	0.0015	1026.7		16	161.48	150	1341	18.91	23.64	0.085	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.259	1	194.25	157.61	0.002	0.692	674.472	677.825	1187.24	674.47	2532	> 194.25	ok	505.9	φVc>Vu	not required
	Min Qx	0.253	1	189.75	158.05			673.969			673.97	2532	> 189.75	ok	505.5	φVc>Vu	not required

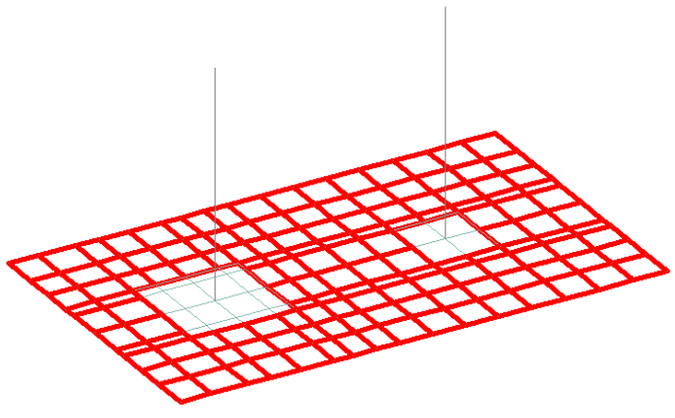
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
132 0.9D	84.90	top face	50	692	84.898	0.24	0.0006	391.0	1245.6	16	161.48	150	1341	18.91	23.64	0.085	tension control
112 1.2D	-490.43	bottom face	50	688	490.43	1.38	0.0034	2319.9		25	211.68	150	3273.9	46.16	57.7	0.033	tension control
112 1.2D	-472.41	bottom face	50	688	472.41	1.33	0.0032	2232.6		25	219.96	150	3273.9	46.16	57.7	0.033	tension control
112 1.2D	144.29	top face	50	692	144.29	0.40	0.0010	666.5		16	161.48	150	1341	18.91	23.64	0.085	tension control

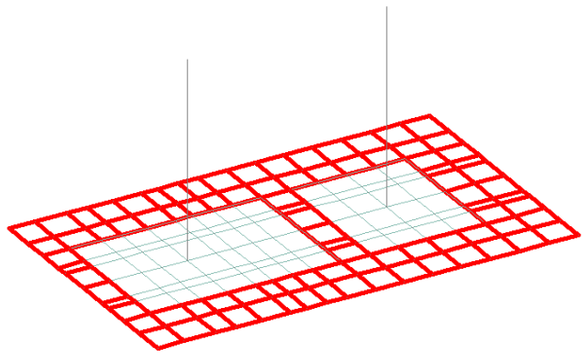
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	112 1.2D	0.121	1	90.75	44.663	0.002	0.692	687.082	677.825	fc'1187	677.83	711	> 90.75	ok	508.4	φVc>Vu	not required
	112 1.2D	0.279	1	209.25	41.178			735.193			677.83	711	> 209.25	ok	508.4	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom 25ϕT **150** c/c

At Top 16ϕT **150** c/c

Along Z Direction :

At Bottom 25ϕT **150** c/c

At Top 16ϕT **150** c/c

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5.3 DESIGN CALCULATION OF FOOTING F12-B

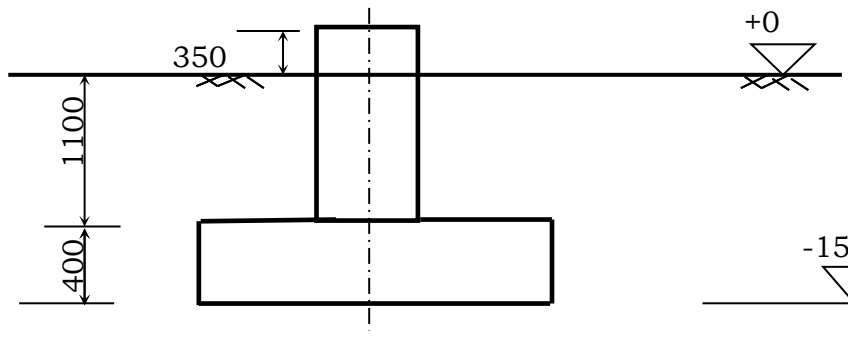
Size of Columns at node	Along-X	Along-Z
1	0.55m	0.55m
2	0.55m	0.55m
Size of fdn. raft:	4.00m	2.50m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.35	m
U/S level of fdn:	-1.50	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

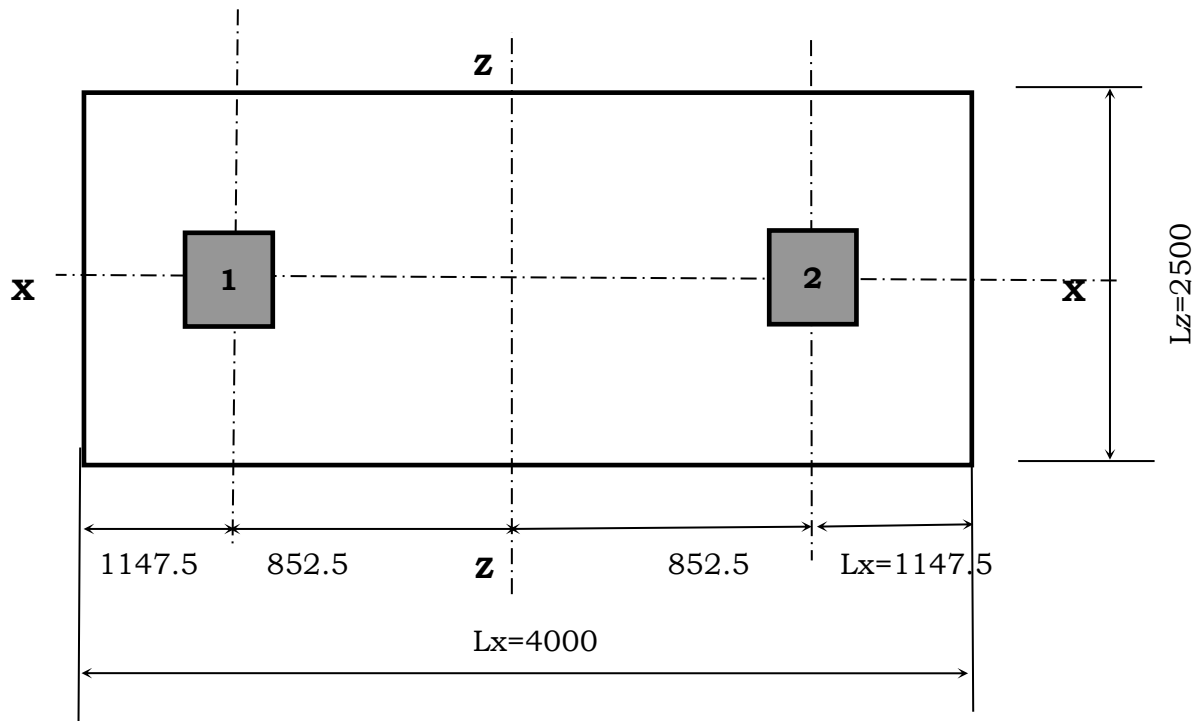
Calculation of Weight of Foundation :

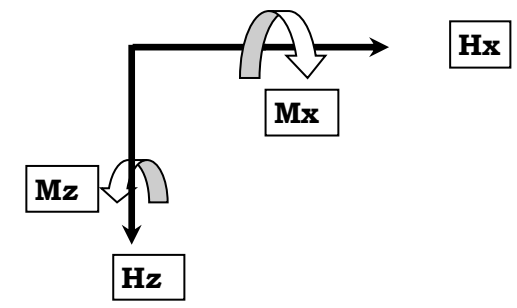
Weight of raft =	(4*2.5*0.4*25) =	100.00KN
Weight of soil =	((4*2.5)-2*(0.55*0.55))*1.1*18 =	186.02KN
	Total =	286.02KN
Submerged wt. =	286.02-4*2.5*1.5*10 =	136.02KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F12-B

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0FL1	94.8	-1.6	2.6	0.0	0.0	95.0	-0.9	1.4	0.0	0.0
202 1.0DL+1.0FL2	94.6	-1.8	2.9	0.0	0.0	96.2	-0.9	1.4	0.0	0.0
203 1.0DL+1.0FL1+1.0LL+1.0TL	-39.3	-33.5	53.6	-0.2	-0.3	309.0	-1.3	2.1	0.0	0.0
204 1.0DL+1.0FL2+1.0LL+1.0TL	-39.5	-33.7	53.9	-0.2	-0.3	310.2	-1.3	2.1	0.0	0.0
205 1.0DL+1.0FL1+0.75LL+0.75TL	-5.8	-25.5	40.8	-0.2	-0.2	255.5	-1.2	1.9	0.0	0.0
206 1.0DL+1.0FL2+0.75LL+0.75TL	-6.0	-25.7	41.2	-0.2	-0.2	256.7	-1.2	1.9	0.0	0.0
207 1.0DL+1.0FL1+1.0PL	208.7	13.1	-21.0	0.6	1.0	55.5	-1.9	3.0	0.0	0.0
208 1.0DL+1.0FL2+1.0PL	208.5	12.9	-20.6	0.6	1.0	56.7	-1.9	3.0	0.0	0.0
209 1.0DL+1.0FL1+1.0LL+1.0TL+1.(	74.6	-18.8	30.1	0.4	0.6	269.5	-2.3	3.7	0.0	0.0
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.(	74.4	-19.0	30.4	0.4	0.6	270.7	-2.3	3.7	0.0	0.0
211 1.0DL+1.0FL1+0.75LL+0.75TL+	108.1	-10.8	17.3	0.5	0.7	216.0	-2.2	3.5	0.0	0.0
212 1.0DL+1.0FL2+0.75LL+0.75TL+	107.9	-11.0	17.6	0.5	0.7	217.2	-2.2	3.5	0.0	0.0
213 1.0DL+1.0FL1+1.0WLX1	-18.2	-36.8	58.9	0.0	0.0	220.0	-1.6	2.6	0.0	0.0
214 1.0DL+1.0FL1+1.0WLX2	211.6	36.8	-58.9	0.0	0.0	-33.6	0.1	-0.2	0.0	0.0
215 1.0DL+1.0FL1+1.0WLZ1	77.3	-4.9	7.8	-0.9	-1.4	118.7	-0.8	1.3	-2.8	-4.5
216 1.0DL+1.0FL1+1.0WLZ2	112.3	1.7	-2.7	0.9	1.4	71.3	-1.0	1.6	2.8	4.5
217 1.0DL+1.0FL2+1.0WLX1	-18.4	-37.0	59.2	0.0	0.0	221.2	-1.6	2.6	0.0	0.0
218 1.0DL+1.0FL2+1.0WLX2	211.4	36.6	-58.6	0.0	0.0	-32.4	0.1	-0.2	0.0	0.0
219 1.0DL+1.0FL2+1.0WLZ1	77.1	-5.1	8.2	-0.9	-1.4	119.9	-0.8	1.3	-2.8	-4.5
220 1.0DL+1.0FL2+1.0WLZ2	112.1	1.5	-2.4	0.9	1.4	72.5	-1.0	1.6	2.8	4.5
221 1.0DL+1.0FL1+0.7SLX1	61.1	-8.2	13.1	0.0	0.0	134.4	-0.6	1.0	0.0	0.0
222 1.0DL+1.0FL1+0.7SLX2	128.5	5.0	-8.0	0.0	0.0	55.6	-1.2	1.9	0.0	0.0
223 1.0DL+1.0FL1+0.7SLZ1	92.9	-2.2	3.6	0.1	0.1	100.5	-0.9	1.4	-0.1	-0.2
224 1.0DL+1.0FL1+0.7SLZ2	96.7	-1.0	1.6	-0.1	-0.1	89.5	-0.9	1.4	0.1	0.2
225 1.0DL+1.0FL2+0.7SLX1	60.9	-8.4	13.4	0.0	0.0	135.6	-0.6	1.0	0.0	0.0
226 1.0DL+1.0FL2+0.7SLX2	128.3	4.8	-7.6	0.0	0.0	56.8	-1.2	1.9	0.0	0.0
227 1.0DL+1.0FL2+0.7SLZ1	92.7	-2.4	3.9	0.1	0.1	101.7	-0.9	1.4	-0.1	-0.2
228 1.0DL+1.0FL2+0.7SLZ2	96.5	-1.2	1.9	-0.1	-0.1	90.7	-0.9	1.4	0.1	0.2
229 1.0DL+1.0FL1+0.75WLX1+0.75	33.1	-30.6	48.9	0.0	0.0	212.5	-1.7	2.8	0.0	0.0
230 1.0DL+1.0FL1+0.75WLX2+0.75	205.4	24.7	-39.4	0.0	0.0	22.3	-0.5	0.7	0.0	0.0
231 1.0DL+1.0FL1+0.75WLZ1+0.75	104.7	-6.6	10.6	-0.7	-1.1	136.6	-1.1	1.8	-2.1	-3.4
232 1.0DL+1.0FL1+0.75WLZ2+0.75	131.0	-1.7	2.7	0.7	1.1	101.0	-1.3	2.0	2.1	3.4
233 1.0DL+1.0FL2+0.75WLX1+0.75	32.9	-30.8	49.2	0.0	0.0	213.7	-1.7	2.8	0.0	0.0
234 1.0DL+1.0FL2+0.75WLX2+0.75	205.2	24.5	-39.1	0.0	0.0	23.5	-0.5	0.7	0.0	0.0
235 1.0DL+1.0FL2+0.75WLZ1+0.75	104.5	-6.8	10.9	-0.7	-1.1	137.8	-1.1	1.8	-2.1	-3.4
236 1.0DL+1.0FL2+0.75WLZ2+0.75	130.8	-1.9	3.0	0.7	1.1	102.2	-1.3	2.0	2.1	3.4
237 1.0DL+1.0FL1+0.525SLX1+0.75	92.6	-9.1	14.5	0.0	0.0	148.3	-1.0	1.6	0.0	0.0
238 1.0DL+1.0FL1+0.525SLX2+0.75	143.1	0.8	-1.3	0.0	0.0	89.2	-1.4	2.3	0.0	0.0
239 1.0DL+1.0FL1+0.525SLZ1+0.75	116.4	-4.6	7.4	0.1	0.1	122.9	-1.2	1.9	-0.1	-0.2
240 1.0DL+1.0FL1+0.525SLZ2+0.75	119.2	-3.7	5.9	-0.1	-0.1	114.6	-1.2	1.9	0.1	0.2
241 1.0DL+1.0FL2+0.525SLX1+0.75	92.4	-9.3	14.9	0.0	0.0	149.5	-1.0	1.6	0.0	0.0
242 1.0DL+1.0FL2+0.525SLX2+0.75	142.9	0.6	-0.9	0.0	0.0	90.4	-1.4	2.3	0.0	0.0
243 1.0DL+1.0FL2+0.525SLZ1+0.75	116.2	-4.8	7.7	0.1	0.1	124.1	-1.2	1.9	-0.1	-0.2
244 1.0DL+1.0FL2+0.525SLZ2+0.75	119.0	-3.9	6.2	-0.1	-0.1	115.8	-1.2	1.9	0.1	0.2
245 0.6DL+1.0WLX1	-57.1	-35.7	57.1	0.0	0.0	177.7	-1.2	2.0	0.0	0.0
246 0.6DL+1.0WLX2	172.7	37.9	-60.7	0.0	0.0	-75.9	0.5	-0.7	0.0	0.0
247 0.6DL+1.0WLZ1	38.4	-3.8	6.0	-0.9	-1.4	76.4	-0.4	0.7	-2.8	-4.5
248 0.6DL+1.0WLZ2	73.4	2.8	-4.5	0.9	1.4	29.0	-0.6	1.0	2.8	4.5
249 0.6DL+0.6FL1+0.7SLX1	23.2	-7.5	12.1	0.0	0.0	96.4	-0.3	0.4	0.0	0.0
250 0.6DL+0.6FL1+0.7SLX2	90.6	5.6	-9.0	0.0	0.0	17.6	-0.8	1.3	0.0	0.0
251 0.6DL+0.6FL1+0.7SLZ1	55.0	-1.6	2.5	0.1	0.1	62.5	-0.5	0.9	-0.1	-0.2
252 0.6DL+0.6FL1+0.7SLZ2	58.8	-0.3	0.5	-0.1	-0.1	51.5	-0.5	0.9	0.1	0.2

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL+1.0FL1	475.8	-2.50	4.2	0.0	0.0	228.2	NA	76.1	NA
202 1.0DL+1.0FL2	476.8	-2.70	5.7	0.0	0.0	168.4	NA	70.6	NA
203 1.0DL+1.0FL1+1.0LL+1.0TL	555.7	-34.80	352.6	-0.2	-0.3	3.2	2315.5	6.4	1111.4
204 1.0DL+1.0FL2+1.0LL+1.0TL	556.7	-35.00	354.1	-0.2	-0.3	3.1	2319.7	6.4	1113.4
205 1.0DL+1.0FL1+0.75LL+0.75TL	535.7	-26.70	265.5	-0.2	-0.2	4.0	3348.3	8.0	1071.4
206 1.0DL+1.0FL2+0.75LL+0.75TL	536.7	-26.90	267.1	-0.2	-0.2	4.0	3354.5	8.0	1073.4
207 1.0DL+1.0FL1+1.0PL	550.2	11.20	-148.6	0.6	1.0	7.4	687.8	19.7	366.8
208 1.0DL+1.0FL2+1.0PL	551.2	11.00	-147.0	0.6	1.0	7.5	689.0	20.0	367.5
209 1.0DL+1.0FL1+1.0LL+1.0TL+1.(	630.1	-21.10	200.0	0.4	0.6	6.3	1312.8	11.9	630.1
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.(	631.1	-21.30	201.4	0.4	0.6	6.3	1314.8	11.9	631.1
211 1.0DL+1.0FL1+0.75LL+0.75TL+	610.1	-13.00	112.8	0.5	0.7	10.8	1089.5	18.8	488.1
212 1.0DL+1.0FL2+0.75LL+0.75TL+	611.1	-13.20	114.3	0.5	0.7	10.7	1091.3	18.5	488.9
213 1.0DL+1.0FL1+1.0WLX1	487.8	-38.40	264.6	0.0	0.0	3.7	NA	5.1	NA
214 1.0DL+1.0FL1+1.0WLX2	464.0	36.90	-268.1	0.0	0.0	3.5	NA	5.0	NA
215 1.0DL+1.0FL1+1.0WLZ1	482.0	-5.70	44.4	-3.7	-5.9	21.7	102.1	33.8	52.1
216 1.0DL+1.0FL1+1.0WLZ2	469.6	0.70	-36.1	3.7	5.9	26.1	99.5	268.4	50.8
217 1.0DL+1.0FL2+1.0WLX1	488.8	-38.60	266.1	0.0	0.0	3.7	NA	5.1	NA
218 1.0DL+1.0FL2+1.0WLX2	465.0	36.70	-266.6	0.0	0.0	3.5	NA	5.1	NA
219 1.0DL+1.0FL2+1.0WLZ1	483.0	-5.90	46.0	-3.7	-5.9	21.0	102.3	32.7	52.2
220 1.0DL+1.0FL2+1.0WLZ2	470.6	0.50	-34.6	3.7	5.9	27.2	99.7	376.5	50.9
221 1.0DL+1.0FL1+0.7SLX1	481.5	-8.80	76.6	0.0	0.0	12.6	NA	21.9	NA
222 1.0DL+1.0FL1+0.7SLX2	470.1	3.80	-68.2	0.0	0.0	13.8	NA	49.5	NA
223 1.0DL+1.0FL1+0.7SLZ1	479.4	-3.10	11.5	0.0	-0.1	83.5	5992.8	61.9	NA
224 1.0DL+1.0FL1+0.7SLZ2	472.2	-1.90	-3.1	0.0	0.1	301.0	5902.8	99.4	NA
225 1.0DL+1.0FL2+0.7SLX1	482.5	-9.00	78.1	0.0	0.0	12.4	NA	21.4	NA
226 1.0DL+1.0FL2+0.7SLX2	471.1	3.60	-66.7	0.0	0.0	14.1	NA	52.3	NA
227 1.0DL+1.0FL2+0.7SLZ1	480.4	-3.30	13.0	0.0	-0.1	74.1	6005.3	58.2	NA
228 1.0DL+1.0FL2+0.7SLZ2	473.2	-2.10	-1.6	0.0	0.1	575.5	5915.3	90.1	NA
229 1.0DL+1.0FL1+0.75WLX1+0.75	531.6	-32.30	204.6	0.0	0.0	5.2	NA	6.6	NA
230 1.0DL+1.0FL1+0.75WLX2+0.75	513.7	24.20	-194.8	0.0	0.0	5.3	NA	8.5	NA
231 1.0DL+1.0FL1+0.75WLZ1+0.75	527.3	-7.70	39.6	-2.8	-4.5	26.6	146.5	27.4	75.3
232 1.0DL+1.0FL1+0.75WLZ2+0.75	518.0	-3.00	-20.9	2.8	4.5	49.6	143.9	69.1	74.0
233 1.0DL+1.0FL2+0.75WLX1+0.75	532.6	-32.50	206.1	0.0	0.0	5.2	NA	6.6	NA
234 1.0DL+1.0FL2+0.75WLX2+0.75	514.7	24.00	-193.3	0.0	0.0	5.3	NA	8.6	NA
235 1.0DL+1.0FL2+0.75WLZ1+0.75	528.3	-7.90	41.1	-2.8	-4.5	25.7	146.8	26.8	75.5
236 1.0DL+1.0FL2+0.75WLZ2+0.75	519.0	-3.20	-19.4	2.8	4.5	53.6	144.2	64.9	74.1
237 1.0DL+1.0FL1+0.525SLX1+0.75	526.9	-10.10	63.6	0.0	0.0	16.6	NA	20.9	NA
238 1.0DL+1.0FL1+0.525SLX2+0.75	518.3	-0.60	-44.9	0.0	0.0	23.1	NA	345.5	NA
239 1.0DL+1.0FL1+0.525SLZ1+0.75	525.3	-5.80	14.8	0.0	-0.1	70.8	6566.5	36.2	NA
240 1.0DL+1.0FL1+0.525SLZ2+0.75	519.8	-4.90	3.9	0.0	0.1	268.1	6497.8	42.4	NA
241 1.0DL+1.0FL2+0.525SLX1+0.75	527.9	-10.30	65.2	0.0	0.0	16.2	NA	20.5	NA
242 1.0DL+1.0FL2+0.525SLX2+0.75	519.3	-0.80	-43.4	0.0	0.0	24.0	NA	259.7	NA
243 1.0DL+1.0FL2+0.525SLZ1+0.75	526.3	-6.00	16.3	0.0	-0.1	64.4	6579.0	35.1	NA
244 1.0DL+1.0FL2+0.525SLZ2+0.75	520.8	-5.10	5.4	0.0	0.1	193.9	6510.3	40.8	NA
245 0.6DL+1.0WLX1	406.6	-36.90	259.3	0.0	0.0	3.1	NA	4.4	NA
246 0.6DL+1.0WLX2	382.8	38.40	-273.3	0.0	0.0	2.8	NA	4.0	NA
247 0.6DL+1.0WLZ1	400.8	-4.20	39.1	-3.7	-5.9	20.5	84.9	38.2	43.3
248 0.6DL+1.0WLZ2	388.4	2.20	-41.4	3.7	5.9	18.8	82.3	70.6	42.0
249 0.6DL+0.6FL1+0.7SLX1	405.6	-7.80	74.9	0.0	0.0	10.8	NA	20.8	NA
250 0.6DL+0.6FL1+0.7SLX2	394.2	4.80	-69.9	0.0	0.0	11.3	NA	32.9	NA
251 0.6DL+0.6FL1+0.7SLZ1	403.5	-2.10	9.8	0.0	-0.1	82.4	5044.0	76.9	NA
252 0.6DL+0.6FL1+0.7SLZ2	396.3	-0.80	-4.8	0.0	0.1	164.3	4954.0	198.2	NA
Min.=						2.8	82.3	4.0	42.0

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Base pressure check of footing F12-B

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	7	201	0	50.98	0
Min Px	7	201	0	50.98	0
Max Py	690	204	0	104.41	0
Min Py	613	245	0	0.00	0
Max Pz	7	201	0	50.98	0
Min Pz	7	201	0	50.98	0

Gross Maximum Base pressure, p_{max} = 104.4 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 1.1 m
Unit weight of soil = 18 KN/m^3

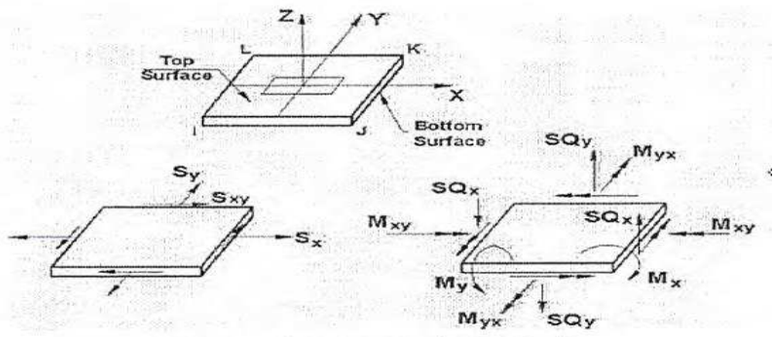
Gross Safe Bearing Capacity = 469.8 KN/m^2
> 104.4 KN/m^2
Hence OK

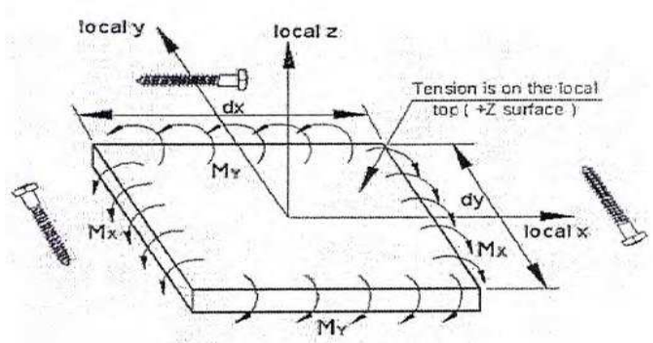
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Design of foundation raft F12-B





Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F12-B_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	696	0.368	0.09	0	0	0	-32.65	-47.225	6.241	-38.886	-53.466	106 1.2DL+1.2FL2+1.2TL+1.6LL
Min Qx	638	-0.491	0.199	0	0	0	-48.96	-49.532	0	-48.959	-49.532	118 1.2DL+1.6WLX2+0.5LL
Max Qy	645	-0.163	0.326	0	0	0	-32.2	-24.238	-2.268	-34.463	-26.506	118 1.2DL+1.6WLX2+0.5LL
Min Qy	742	-0.163	-0.326	0	0	0	-32.2	-24.238	2.268	-34.463	-26.506	118 1.2DL+1.6WLX2+0.5LL
Max Sx	1062	0.006	-0.008	0	0	0	-0.412	-1.927	-0.717	-1.129	-2.644	101 1.4DL+1.4FL1
Min Sx	1062	0.006	-0.008	0	0	0	-0.412	-1.927	-0.717	-1.129	-2.644	101 1.4DL+1.4FL1
Max Sy	1062	0.006	-0.008	0	0	0	-0.412	-1.927	-0.717	-1.129	-2.644	101 1.4DL+1.4FL1
Min Sy	1062	0.006	-0.008	0	0	0	-0.412	-1.927	-0.717	-1.129	-2.644	101 1.4DL+1.4FL1
Max Sxy	1062	0.006	-0.008	0	0	0	-0.412	-1.927	-0.717	-1.129	-2.644	101 1.4DL+1.4FL1
Min Sxy	1062	0.006	-0.008	0	0	0	-0.412	-1.927	-0.717	-1.129	-2.644	101 1.4DL+1.4FL1
Max Mx	638	0.287	-0.145	0	0	0	33.091	20.059	-2.987	36.078	23.046	125 0.9DL+1.6WLX1
Min Mx	680	-0.017	0.297	0	0	0	-53.75	-22.708	-6.215	-59.963	-28.923	106 1.2DL+1.2FL2+1.2TL+1.6LL
Max My	749	0.287	0.145	0	0	0	33.091	20.059	2.987	36.078	23.046	125 0.9DL+1.6WLX1
Min My	749	-0.491	-0.199	0	0	0	-48.96	-49.532	0.75	-49.709	-50.282	118 1.2DL+1.6WLX2+0.5LL
Max Mxy	675	0.112	-0.014	0	0	0	4.288	-9.922	16.09	20.378	-26.012	118 1.2DL+1.6WLX2+0.5LL
Min Mxy	766	0.112	0.014	0	0	0	4.288	-9.922	-16.09	20.378	-26.012	118 1.2DL+1.6WLX2+0.5LL

PLATE CENTER STRESS : FDN_F12-B_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	698	0.105	0.024	0	0	0	-3.803	-26.922	2.217	-6.02	-29.139	106 1.2DL+1.2FL2+1.2TL+1.6LL
Min Qx	636	-0.126	0.038	0	0	0	-5.376	-30.999	0	-5.376	-30.999	118 1.2DL+1.6WLX2+0.5LL
Max Qy	643	-0.089	0.085	0	0	0	-5.633	-21.777	-4.481	-10.114	-26.258	118 1.2DL+1.6WLX2+0.5LL
Min Qy	740	-0.089	-0.085	0	0	0	-5.633	-21.777	4.481	-10.114	-26.258	118 1.2DL+1.6WLX2+0.5LL
Max Sx	1062	0.006	-0.008	0	0	0	-0.412	-1.927	-0.717	-1.129	-2.644	101 1.4DL+1.4FL1
Min Sx	1062	0.006	-0.008	0	0	0	-0.412	-1.927	-0.717	-1.129	-2.644	101 1.4DL+1.4FL1
Max Sy	1062	0.006	-0.008	0	0	0	-0.412	-1.927	-0.717	-1.129	-2.644	101 1.4DL+1.4FL1
Min Sy	1062	0.006	-0.008	0	0	0	-0.412	-1.927	-0.717	-1.129	-2.644	101 1.4DL+1.4FL1
Max Sxy	1062	0.006	-0.008	0	0	0	-0.412	-1.927	-0.717	-1.129	-2.644	101 1.4DL+1.4FL1
Min Sxy	1062	0.006	-0.008	0	0	0	-0.412	-1.927	-0.717	-1.129	-2.644	101 1.4DL+1.4FL1
Max Mx	1015	-0.003	-0.014	0	0	0	18.257	-0.711	5.442	23.699	-6.153	126 0.9DL+1.6WLX2
Min Mx	1017	0.014	0.069	0	0	0	-32.93	-1.038	-4.157	-37.086	-5.195	106 1.2DL+1.2FL2+1.2TL+1.6LL
Max My	747	0.074	0.023	0	0	0	4.004	13.457	-0.7	4.704	14.157	125 0.9DL+1.6WLX1
Min My	636	-0.126	0.038	0	0	0	-5.376	-30.999	-2.081	-7.457	-33.08	118 1.2DL+1.6WLX2+0.5LL
Max Mxy	1007	0.05	0.025	0	0	0	-0.998	-1.345	9.816	-10.814	-11.161	118 1.2DL+1.6WLX2+0.5LL
Min Mxy	1042	0.05	-0.025	0	0	0	-0.998	-1.345	-9.816	-10.814	-11.161	118 1.2DL+1.6WLX2+0.5LL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

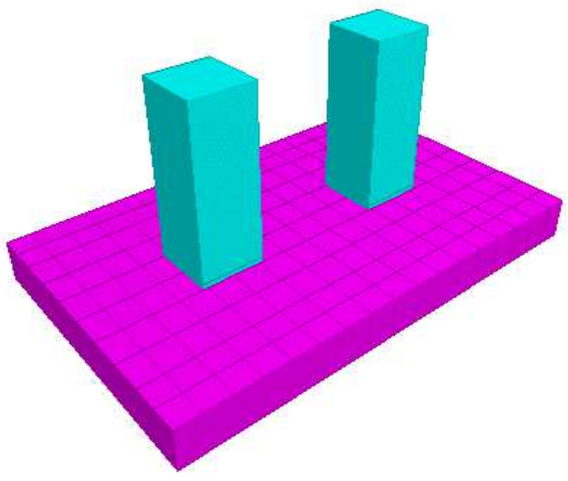
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
125 0.9D	36.08	top face	50	344	36.078	0.41	0.0010	335.3	619.2	12	182.72	150	754.3	10.64	13.29	0.075	tension control
106 1.2D	-59.96	bottom face	50	344	59.963	0.68	0.0016	559.8		12	182.72	150	754.3	10.64	13.29	0.075	tension control
118 1.2D	20.38	top face	50	344	20.378	0.23	0.0005	188.8		12	182.72	150	754.3	10.64	13.29	0.075	tension control
118 1.2D	20.38	top face	50	344	20.378	0.23	0.0005	188.8		12	182.72	150	754.3	10.64	13.29	0.075	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.105	1	42	6.02	0.002	0.344	356.396	338.444	590.188	338.44	1261	> 42	ok	253.8	φVc>Vu	not required
	Min Qx	0.126	1	50.4	5.376			366.976			338.44	1261	> 50.4	ok	253.8	φVc>Vu	not required

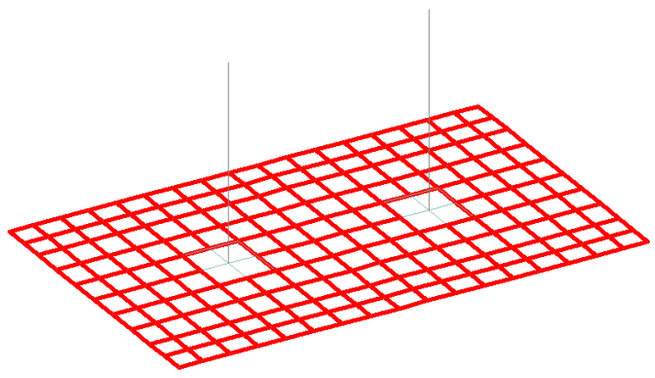
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
125 0.9D	23.05	top face	50	344	23.046	0.26	0.0006	213.6	619.2	12	182.72	150	754.3	10.64	13.29	0.075	tension control
118 1.2D	-50.28	bottom face	50	344	50.282	0.57	0.0014	468.5		12	182.72	150	754.3	10.64	13.29	0.075	tension control
118 1.2D	-26.01	bottom face	50	344	26.012	0.29	0.0007	241.2		12	182.72	150	754.3	10.64	13.29	0.075	tension control
118 1.2D	-26.01	bottom face	50	344	26.012	0.29	0.0007	241.2		12	182.72	150	754.3	10.64	13.29	0.075	tension control

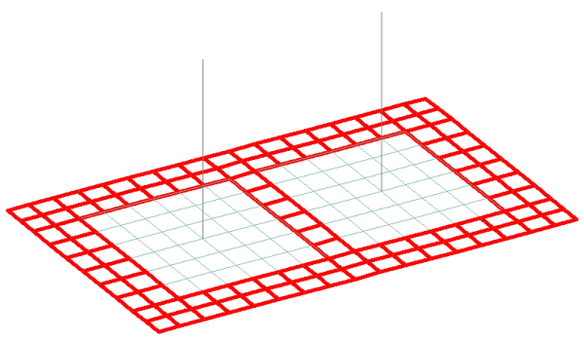
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	118 1.2D	0.085	1	34	26.258	0.002	0.344	331.333	338.444	fc'590.2	331.33	349	> 34	ok	248.5	φVc>Vu	not required
	118 1.2D	0.085	1	34	26.258			331.333			331.33	349	> 34	ok	248.5	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	12ϕT	**150**	c/c
At Top	12ϕT	**150**	c/c
Along Z Direction :			
At Bottom	12ϕT	**150**	c/c
At Top	12ϕT	**150**	c/c

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5.4 DESIGN CALCULATION OF FOOTING F13

Size of Columns at node	Along-Z	Along-X
1	0.50m	0.50m
2	0.50m	0.50m
Size of fdn. raft:	4.00m	1.50m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.35	m
U/S level of fdn:	-0.85	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

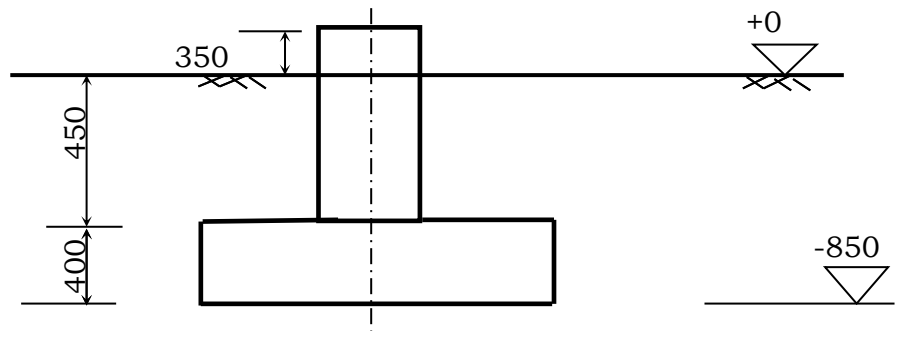
Calculation of Weight of Foundation :

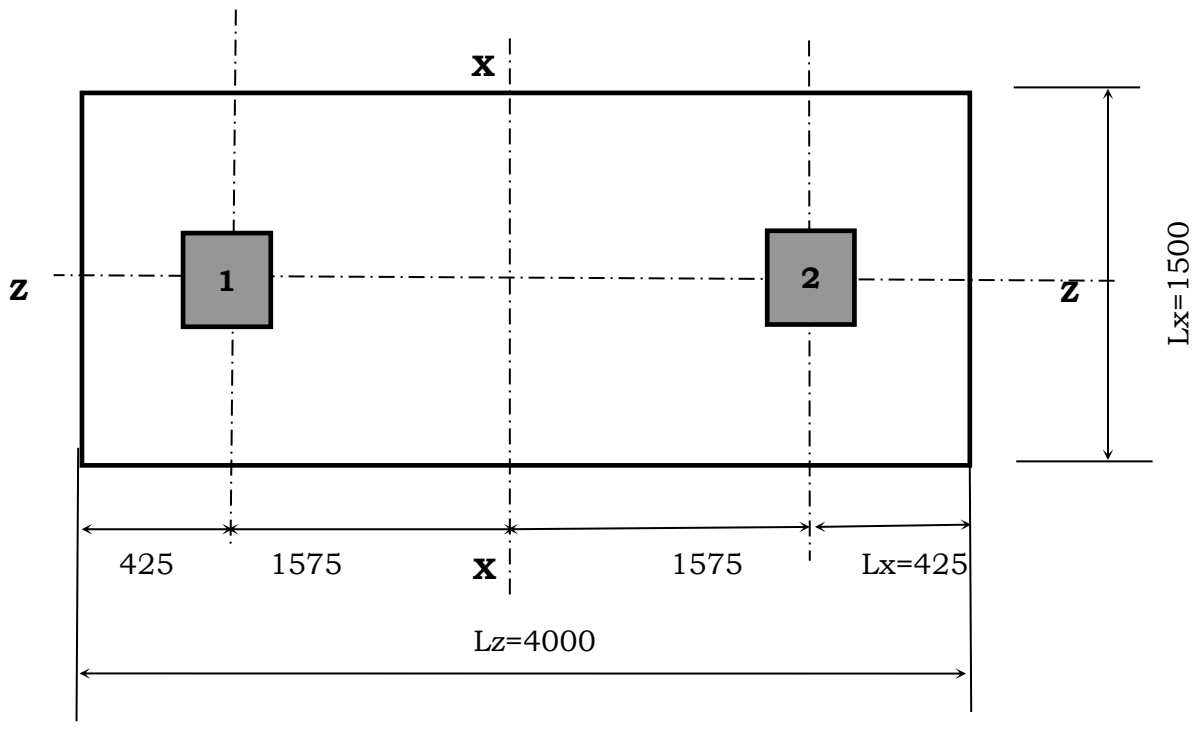
Weight of raft = (4*1.5*0.4*25) = 60.00KN

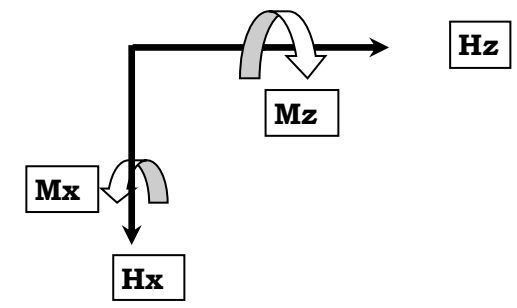
Weight of soil = ((4*1.5)-2*(0.5*0.5))*0.45*18 = 44.55KN

Total = 104.55KN

Submerged wt. = 104.55-4*1.5*0.85*10 = 53.55KN







H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F13

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0LL	48.0	4.6	-5.5	20.9	25.1	116.5	7.7	-9.2	-20.5	-24.6
202 1.0DL+0.75LL	41.5	3.9	-4.7	17.1	20.5	96.8	6.4	-7.7	-16.7	-20.1
203 1.0DL+1.0WLX1	9.8	-4.9	5.9	5.4	6.5	25.4	-4.2	5.0	-5.3	-6.4
204 1.0DL+1.0WLX2	34.0	8.7	-10.4	5.6	6.7	49.6	9.4	-11.3	-5.5	-6.6
205 1.0DL+1.0WLZ1	3.7	1.2	-1.4	0.2	0.2	55.7	4.1	-4.9	-10.7	-12.8
206 1.0DL+1.0WLZ2	40.1	2.6	-3.1	10.8	13.0	19.3	1.1	-1.3	-0.1	-0.1
207 1.0DL+0.75LL+0.75WLX1	32.4	-1.2	1.4	17.0	20.4	87.7	1.3	-1.6	-16.7	-20.0
208 1.0DL+0.75LL+0.75WLX2	50.6	9.0	-10.8	17.1	20.6	105.8	11.5	-13.8	-16.8	-20.2
209 1.0DL+0.75LL+0.75WLZ1	27.8	3.4	-4.1	13.1	15.7	110.4	7.6	-9.1	-20.7	-24.8
210 1.0DL+0.75LL+0.75WLZ2	55.1	4.5	-5.3	21.0	25.2	83.1	5.3	-6.4	-12.8	-15.3
211 0.6DL+1.0WLX1	1.0	-5.7	6.8	3.2	3.8	10.4	-5.2	6.3	-3.1	-3.8
212 0.6DL+1.0WLX2	25.2	7.9	-9.5	3.4	4.1	34.6	8.4	-10.0	-3.3	-4.0
213 0.6DL+1.0WLZ1	-5.1	0.4	-0.5	-2.0	-2.4	40.7	3.1	-3.7	-8.5	-10.2
214 0.6DL+1.0WLZ2	31.3	1.8	-2.2	8.6	10.3	4.3	0.1	-0.1	2.1	2.5

Check for overturning and sliding:

	Resultant Load at bottom of footing								
						Overturning		Sliding	
L/C No.	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL+1.0LL	269.1	0.40	108.4	12.3	-14.7	13.7	5.0	8.7	269.1
202 1.0DL+0.75LL	242.9	0.40	87.5	10.3	-12.4	14.7	5.6	9.4	242.8
203 1.0DL+1.0WLX1	139.8	0.10	24.7	-9.1	10.9	9.6	11.3	6.1	559.0
204 1.0DL+1.0WLX2	188.2	0.10	24.7	18.1	-21.7	6.5	15.3	4.2	752.6
205 1.0DL+1.0WLZ1	164.0	-10.50	69.3	5.3	-6.3	19.5	4.7	12.4	6.2
206 1.0DL+1.0WLZ2	164.0	10.70	-19.9	3.7	-4.4	27.9	16.5	17.7	6.1
207 1.0DL+0.75LL+0.75WLX1	224.7	0.30	87.5	0.1	-0.2	842.4	5.1	898.6	299.5
208 1.0DL+0.75LL+0.75WLX2	261.0	0.30	87.3	20.5	-24.6	8.0	6.0	5.1	347.9
209 1.0DL+0.75LL+0.75WLZ1	242.8	-7.60	121.0	11.0	-13.2	13.8	4.0	8.8	12.8
210 1.0DL+0.75LL+0.75WLZ2	242.8	8.20	54.0	9.8	-11.7	15.6	9.0	9.9	11.8
211 0.6DL+1.0WLX1	116.0	0.10	14.8	-10.9	13.1	6.6	15.7	4.3	463.8
212 0.6DL+1.0WLX2	164.4	0.10	14.9	16.3	-19.5	6.3	22.1	4.0	657.4
213 0.6DL+1.0WLZ1	140.2	-10.50	59.5	3.5	-4.2	25.0	4.7	16.0	5.3
214 0.6DL+1.0WLZ2	140.2	10.70	-29.7	1.9	-2.3	45.7	9.4	29.5	5.2
					Min.=	6.3	4.0	4.0	5.2

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Base pressure check of footing F13

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	5	201	0	24.88	0
Min Px	5	201	0	24.88	0
Max Py	12	209	0	99.50	0
Min Py	7	213	0	0.86	0
Max Pz	5	201	0	24.88	0
Min Pz	5	201	0	24.88	0

Gross Maximum Base pressure, p_{max} = 99.5 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2

Depth of soil above footing = 0.45 m

Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 458.1 KN/m^2

> 99.5 KN/m^2

Hence OK

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Design of foundation raft F13

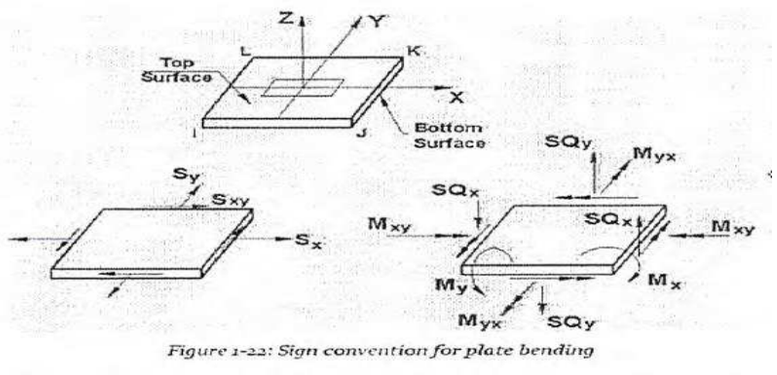
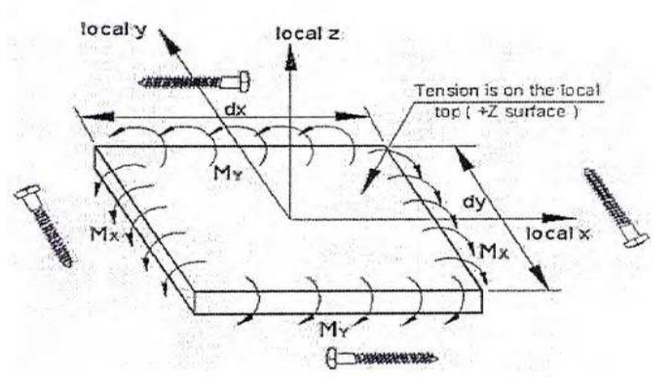


Figure 1-22: Sign convention for plate bending



Sign convention :

Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :

Mx : BM along X direction, +ve means tension on +Z surface of plate

My : BM along Y direction, +ve means tension on +Z surface of plate

For -ve Mx & My it is vice-virsa

Shear force :

SQx : Shear at face perpendicular to X axis

SQy : Shear at face perpendicular to Y axis

Axial force :

+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS : FDN_F13_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	62	0.243	0	0	0	0	-12.29	-0.125	1.254	-13.547	-1.379	102 1.4DL+1.7LL
Min Qx	51	-0.238	-0.036	0	0	0	-17.29	-1.052	0	-17.288	-1.052	102 1.4DL+1.7LL
Max Qy	72	-0.156	0.2	0	0	0	-10.73	14.637	-3.371	-14.105	18.008	102 1.4DL+1.7LL
Min Qy	131	-0.062	-0.174	0	0	0	-5.563	18.518	2.315	-7.878	20.833	102 1.4DL+1.7LL
Max Sx	183	0.004	-0.031	0	0	0	-0.173	6.208	-0.709	-0.882	6.917	101 1.4DL
Min Sx	183	0.004	-0.031	0	0	0	-0.173	6.208	-0.709	-0.882	6.917	101 1.4DL
Max Sy	183	0.004	-0.031	0	0	0	-0.173	6.208	-0.709	-0.882	6.917	101 1.4DL
Min Sy	183	0.004	-0.031	0	0	0	-0.173	6.208	-0.709	-0.882	6.917	101 1.4DL
Max Sxy	183	0.004	-0.031	0	0	0	-0.173	6.208	-0.709	-0.882	6.917	101 1.4DL
Min Sxy	183	0.004	-0.031	0	0	0	-0.173	6.208	-0.709	-0.882	6.917	101 1.4DL
Max Mx	131	0.025	-0.009	0	0	0	2.125	3.424	2.003	4.128	5.427	107 0.9DL+1.6HL+1.6WLX1
Min Mx	60	0.115	-0.064	0	0	0	-32.36	-1.157	1.06	-33.419	-2.217	102 1.4DL+1.7LL
Max My	96	-0.001	0.015	0	0	0	-0.189	68.479	0.489	-0.678	68.968	102 1.4DL+1.7LL
Min My	53	0.011	-0.066	0	0	0	-30.66	-1.41	1.591	-32.255	-3.001	102 1.4DL+1.7LL
Max Mxy	138	0.002	0.12	0	0	0	-3.167	41.874	9.983	-13.15	51.857	102 1.4DL+1.7LL
Min Mxy	178	-0.029	-0.051	0	0	0	0.24	28.749	-5.761	6.001	34.51	104 1.2DL+1.0LL+1.6WLX2

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PLATE CENTER STRESS : FDN_F13_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	64	0.067	0.028	0	0	0	-0.949	0.524	0.825	-1.774	1.349	102 1.4DL+1.7LL
Min Qx	49	-0.082	0.03	0	0	0	-2.358	0.52	0	-2.358	0.52	102 1.4DL+1.7LL
Max Qy	76	-0.024	0.145	0	0	0	-1.702	37.194	-1.336	-3.038	38.53	102 1.4DL+1.7LL
Min Qy	126	-0.008	-0.128	0	0	0	-0.921	35.686	0.339	-1.26	36.025	102 1.4DL+1.7LL
Max Sx	183	0.004	-0.031	0	0	0	-0.173	6.208	-0.709	-0.882	6.917	101 1.4DL
Min Sx	183	0.004	-0.031	0	0	0	-0.173	6.208	-0.709	-0.882	6.917	101 1.4DL
Max Sy	183	0.004	-0.031	0	0	0	-0.173	6.208	-0.709	-0.882	6.917	101 1.4DL
Min Sy	183	0.004	-0.031	0	0	0	-0.173	6.208	-0.709	-0.882	6.917	101 1.4DL
Max Sxy	183	0.004	-0.031	0	0	0	-0.173	6.208	-0.709	-0.882	6.917	101 1.4DL
Min Sxy	183	0.004	-0.031	0	0	0	-0.173	6.208	-0.709	-0.882	6.917	101 1.4DL
Max Mx	26	0.005	-0.006	0	0	0	0.472	1.637	0.771	1.243	2.408	107 0.9DL+1.6HL+1.6WLX1
Min Mx	84	-0.005	0.113	0	0	0	-6.538	55.069	1.627	-8.165	56.696	102 1.4DL+1.7LL
Max My	96	-0.001	0.015	0	0	0	-0.189	68.479	0.489	-0.678	68.968	102 1.4DL+1.7LL
Min My	69	0.002	0.005	0	0	0	-0.112	-0.195	-0.362	-0.474	-0.557	110 0.9DL+1.6HL+1.6WLZ2
Max Mxy	142	0.005	0.068	0	0	0	-2.646	38.605	5.682	-8.328	44.287	104 1.2DL+1.0LL+1.6WLX2
Min Mxy	173	-0.003	-0.058	0	0	0	-1.546	36.472	-3.056	-4.602	39.528	104 1.2DL+1.0LL+1.6WLX2

Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
107 0.9D	4.13	top face	50	344	4.128	0.05	0.0001	38.1	619.2	12	182.72	150	754.3	10.64	13.29	0.075	tension control
102 1.4D	-33.42	bottom face	50	344	33.419	0.38	0.0009	310.4		12	182.72	150	754.3	10.64	13.29	0.075	tension control
102 1.4D	-13.15	bottom face	50	344	13.15	0.15	0.0004	121.7		12	182.72	150	754.3	10.64	13.29	0.075	tension control
104 1.2D	6.00	top face	50	344	6.001	0.07	0.0002	55.4		12	182.72	150	754.3	10.64	13.29	0.075	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.067	1	26.8	1.774	0.002	0.344	392.261	338.444	590.188	338.44	1261	> 26.8	ok	253.8	φVc>Vu	not required
	Min Qx	0.082	1	32.8	2.358			386.98			338.44	1261	> 32.8	ok	253.8	φVc>Vu	not required

Reinforcement Along Z:

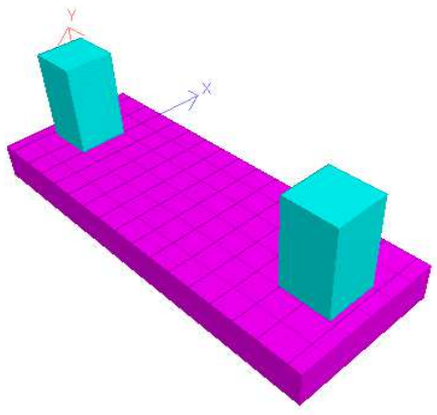
Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
102 1.4D	68.97	top face	50	344	68.968	0.78	0.0019	645.0	619.2	12	175.41	150	754.3	10.64	13.29	0.075	tension control
102 1.4D	-3.00	bottom face	50	344	3.001	0.03	0.0001	27.7		12	182.72	150	754.3	10.64	13.29	0.075	tension control
102 1.4D	51.86	top face	50	344	51.857	0.58	0.0014	483.4		12	182.72	150	754.3	10.64	13.29	0.075	tension control
104 1.2D	34.51	top face	50	344	34.51	0.39	0.0009	320.6		12	182.72	150	754.3	10.64	13.29	0.075	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	102 1.4D	0.145	1	58	38.53	0.002	0.344	332.261	338.444	fc'590.2	332.26	350	> 58	ok	249.2	φVc>Vu	not required
	102 1.4D	0.128	1	51.2	36.025			331.89			331.89	350	> 51.2	ok	248.9	φVc>Vu	not required

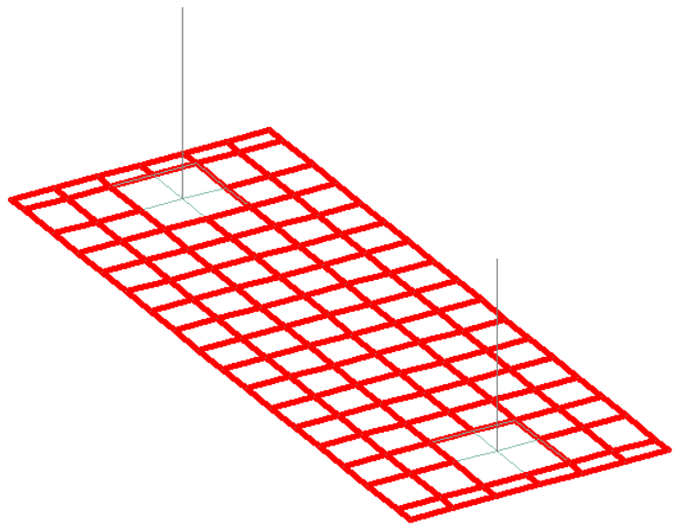
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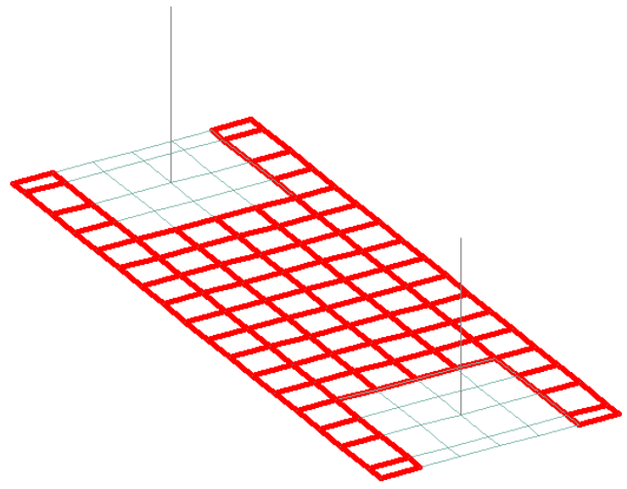
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3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	12ϕT	**150**	c/c
At Top	12ϕT	**150**	c/c
Along Z Direction :			
At Bottom	12ϕT	**150**	c/c
At Top	12ϕT	**150**	c/c

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5.5 DESIGN CALCULATION OF FOOTING F14

Size of Columns at node	Along-Z	Along-X
1	0.50m	0.50m
2	0.50m	0.50m
Size of fdn. raft:	4.00m	1.50m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.35	m
U/S level of fdn:	-1.05	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

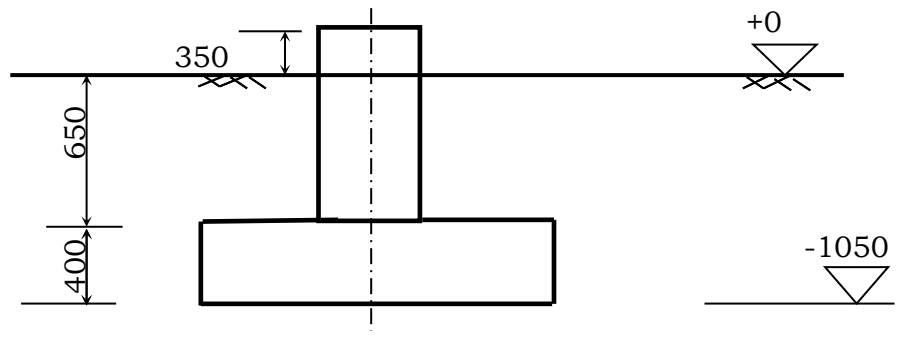
Calculation of Weight of Foundation :

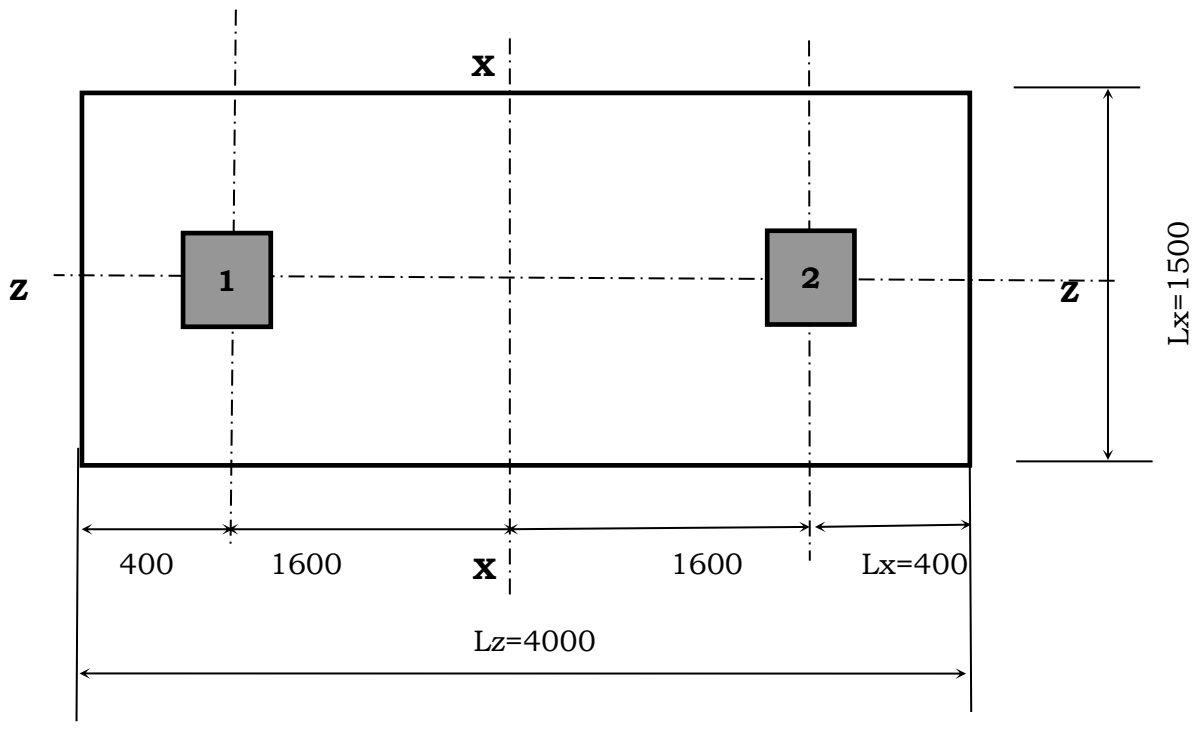
Weight of raft = (4*1.5*0.4*25) = 60.00KN

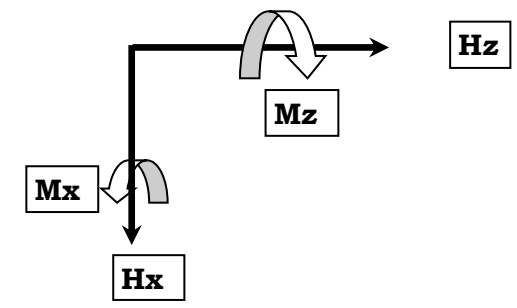
Weight of soil = ((4*1.5)-2*(0.5*0.5))*0.65*18 = 64.35KN

Total = 124.35KN

Submerged wt. = 124.35-4*1.5*1.05*10 = 61.35KN







H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F14

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0FL1	21.0	0.0	0.0	0.0	0.0	16.2	0.0	0.0	0.0	0.0
202 1.0DL+1.0FL2	21.0	0.0	0.0	0.0	0.0	16.2	0.0	0.0	0.0	0.0
203 1.0DL+1.0FL1+1.0LL+1.0TL	55.6	0.0	0.0	0.3	0.4	41.3	0.0	0.0	-0.3	-0.4
204 1.0DL+1.0FL2+1.0LL+1.0TL	55.6	0.0	0.0	0.3	0.4	41.3	0.0	0.0	-0.3	-0.4
205 1.0DL+1.0FL1+0.75LL+0.75TL	47.0	0.0	0.0	0.2	0.3	35.0	0.0	0.0	-0.2	-0.3
206 1.0DL+1.0FL2+0.75LL+0.75TL	47.0	0.0	0.0	0.2	0.3	35.0	0.0	0.0	-0.2	-0.3
207 1.0DL+1.0FL1+1.0PL	70.0	0.0	0.0	0.0	0.0	65.2	0.0	0.0	0.0	0.0
208 1.0DL+1.0FL2+1.0PL	70.0	0.0	0.0	0.0	0.0	65.2	0.0	0.0	0.0	0.0
209 1.0DL+1.0FL1+1.0LL+1.0TL+1.(	104.6	0.0	0.0	0.3	0.4	90.3	0.0	0.0	-0.3	-0.4
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.(	104.6	0.0	0.0	0.3	0.4	90.3	0.0	0.0	-0.3	-0.4
211 1.0DL+1.0FL1+0.75LL+0.75TL+	96.0	0.0	0.0	0.2	0.3	84.0	0.0	0.0	-0.2	-0.3
212 1.0DL+1.0FL2+0.75LL+0.75TL+	96.0	0.0	0.0	0.2	0.3	84.0	0.0	0.0	-0.2	-0.3
213 1.0DL+1.0FL1+1.0WLX1	17.4	-2.7	3.8	0.0	0.0	12.8	-2.7	3.8	0.0	0.0
214 1.0DL+1.0FL1+1.0WLX2	24.6	2.7	-3.8	0.0	0.0	19.6	2.7	-3.8	0.0	0.0
215 1.0DL+1.0FL1+1.0WLZ1	4.3	0.0	0.0	-5.3	-7.4	32.9	0.0	0.0	-5.3	-7.4
216 1.0DL+1.0FL1+1.0WLZ2	37.7	0.0	0.0	5.3	7.4	-0.5	0.0	0.0	5.3	7.4
217 1.0DL+1.0FL2+1.0WLX1	17.4	-2.7	3.8	0.0	0.0	12.8	-2.7	3.8	0.0	0.0
218 1.0DL+1.0FL2+1.0WLX2	24.6	2.7	-3.8	0.0	0.0	19.6	2.7	-3.8	0.0	0.0
219 1.0DL+1.0FL2+1.0WLZ1	4.3	0.0	0.0	-5.3	-7.4	32.9	0.0	0.0	-5.3	-7.4
220 1.0DL+1.0FL2+1.0WLZ2	37.7	0.0	0.0	5.3	7.4	-0.5	0.0	0.0	5.3	7.4
221 1.0DL+1.0FL1+0.7SLX1	19.5	-1.0	1.4	0.1	0.1	14.2	-0.9	1.3	0.1	0.1
222 1.0DL+1.0FL1+0.7SLX2	22.5	1.0	-1.4	-0.1	-0.1	18.2	0.9	-1.3	-0.1	-0.1
223 1.0DL+1.0FL1+0.7SLZ1	16.8	0.0	0.0	-0.9	-1.3	20.4	0.0	0.0	-0.9	-1.3
224 1.0DL+1.0FL1+0.7SLZ2	25.2	0.0	0.0	0.9	1.3	12.0	0.0	0.0	0.9	1.3
225 1.0DL+1.0FL2+0.7SLX1	19.5	-1.0	1.4	0.1	0.1	14.2	-0.9	1.3	0.1	0.1
226 1.0DL+1.0FL2+0.7SLX2	22.5	1.0	-1.4	-0.1	-0.1	18.2	0.9	-1.3	-0.1	-0.1
227 1.0DL+1.0FL2+0.7SLZ1	16.8	0.0	0.0	-0.9	-1.3	20.4	0.0	0.0	-0.9	-1.3
228 1.0DL+1.0FL2+0.7SLZ2	25.2	0.0	0.0	0.9	1.3	12.0	0.0	0.0	0.9	1.3
229 1.0DL+1.0FL1+0.75WLX1+0.75	44.3	-2.0	2.8	0.0	0.0	32.5	-2.0	2.8	0.0	0.0
230 1.0DL+1.0FL1+0.75WLX2+0.75	49.7	2.0	-2.8	0.0	0.0	37.6	2.0	-2.8	0.0	0.0
231 1.0DL+1.0FL1+0.75WLZ1+0.75	34.4	0.0	0.0	-4.0	-5.6	47.6	0.0	0.0	-4.0	-5.6
232 1.0DL+1.0FL1+0.75WLZ2+0.75	59.5	0.0	0.0	4.0	5.6	22.5	0.0	0.0	4.0	5.6
233 1.0DL+1.0FL2+0.75WLX1+0.75	44.3	-2.0	2.8	0.0	0.0	32.5	-2.0	2.8	0.0	0.0
234 1.0DL+1.0FL2+0.75WLX2+0.75	49.7	2.0	-2.8	0.0	0.0	37.6	2.0	-2.8	0.0	0.0
235 1.0DL+1.0FL2+0.75WLZ1+0.75	34.4	0.0	0.0	-4.0	-5.6	47.6	0.0	0.0	-4.0	-5.6
236 1.0DL+1.0FL2+0.75WLZ2+0.75	59.5	0.0	0.0	4.0	5.6	22.5	0.0	0.0	4.0	5.6
237 1.0DL+1.0FL1+0.525SLX1+0.75	45.8	-0.7	1.0	0.1	0.1	33.5	-0.7	1.0	0.1	0.1
238 1.0DL+1.0FL1+0.525SLX2+0.75	48.1	0.7	-1.0	-0.1	-0.1	36.5	0.7	-1.0	-0.1	-0.1
239 1.0DL+1.0FL1+0.525SLZ1+0.75	43.8	0.0	0.0	-0.7	-1.0	38.2	0.0	0.0	-0.7	-1.0
240 1.0DL+1.0FL1+0.525SLZ2+0.75	50.1	0.0	0.0	0.7	1.0	31.9	0.0	0.0	0.7	1.0
241 1.0DL+1.0FL2+0.525SLX1+0.75	45.8	-0.7	1.0	0.1	0.1	33.5	-0.7	1.0	0.1	0.1
242 1.0DL+1.0FL2+0.525SLX2+0.75	48.1	0.7	-1.0	-0.1	-0.1	36.5	0.7	-1.0	-0.1	-0.1
243 1.0DL+1.0FL2+0.525SLZ1+0.75	43.8	0.0	0.0	-0.7	-1.0	38.2	0.0	0.0	-0.7	-1.0
244 1.0DL+1.0FL2+0.525SLZ2+0.75	50.1	0.0	0.0	0.7	1.0	31.9	0.0	0.0	0.7	1.0
245 0.6DL+1.0WLX1	9.0	-2.7	3.8	0.0	0.0	6.3	-2.7	3.8	0.0	0.0
246 0.6DL+1.0WLX2	16.2	2.7	-3.8	0.0	0.0	13.1	2.7	-3.8	0.0	0.0
247 0.6DL+1.0WLZ1	-4.1	0.0	0.0	-5.3	-7.4	26.4	0.0	0.0	-5.3	-7.4
248 0.6DL+1.0WLZ2	29.3	0.0	0.0	5.3	7.4	-7.0	0.0	0.0	5.3	7.4
249 0.6DL+0.6FL1+0.7SLX1	11.1	-1.0	1.4	0.1	0.1	7.7	-0.9	1.3	0.1	0.1
250 0.6DL+0.6FL1+0.7SLX2	14.1	1.0	-1.4	-0.1	-0.1	11.8	0.9	-1.3	-0.1	-0.1
251 0.6DL+0.6FL1+0.7SLZ1	8.4	0.0	0.0	-0.9	-1.3	13.9	0.0	0.0	-0.9	-1.3
252 0.6DL+0.6FL1+0.7SLZ2	16.8	0.0	0.0	0.9	1.3	5.5	0.0	0.0	0.9	1.3

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL+1.0FL1	161.6	0.00	-7.7	0.0	0.0	NA	42.1	NA	NA
202 1.0DL+1.0FL2	161.6	0.00	-7.7	0.0	0.0	NA	42.1	NA	NA
203 1.0DL+1.0FL1+1.0LL+1.0TL	221.3	0.00	-22.9	0.0	0.0	NA	19.3	NA	NA
204 1.0DL+1.0FL2+1.0LL+1.0TL	221.3	0.00	-22.9	0.0	0.0	NA	19.3	NA	NA
205 1.0DL+1.0FL1+0.75LL+0.75TL	206.4	0.00	-19.2	0.0	0.0	NA	21.5	NA	NA
206 1.0DL+1.0FL2+0.75LL+0.75TL	206.4	0.00	-19.2	0.0	0.0	NA	21.5	NA	NA
207 1.0DL+1.0FL1+1.0PL	259.6	0.00	-7.7	0.0	0.0	NA	67.6	NA	NA
208 1.0DL+1.0FL2+1.0PL	259.6	0.00	-7.7	0.0	0.0	NA	67.6	NA	NA
209 1.0DL+1.0FL1+1.0LL+1.0TL+1.(	319.3	0.00	-22.9	0.0	0.0	NA	27.9	NA	NA
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.(	319.3	0.00	-22.9	0.0	0.0	NA	27.9	NA	NA
211 1.0DL+1.0FL1+0.75LL+0.75TL+	304.4	0.00	-19.2	0.0	0.0	NA	31.7	NA	NA
212 1.0DL+1.0FL2+0.75LL+0.75TL+	304.4	0.00	-19.2	0.0	0.0	NA	31.7	NA	NA
213 1.0DL+1.0FL1+1.0WLX1	154.6	0.00	-7.4	-5.4	7.6	15.3	42.0	11.4	NA
214 1.0DL+1.0FL1+1.0WLX2	168.6	0.00	-8.0	5.4	-7.6	16.6	42.1	12.5	NA
215 1.0DL+1.0FL1+1.0WLZ1	161.6	-10.60	31.0	0.0	0.0	NA	10.4	NA	6.1
216 1.0DL+1.0FL1+1.0WLZ2	161.6	10.60	-46.3	0.0	0.0	NA	7.0	NA	6.1
217 1.0DL+1.0FL2+1.0WLX1	154.6	0.00	-7.4	-5.4	7.6	15.3	42.0	11.4	NA
218 1.0DL+1.0FL2+1.0WLX2	168.6	0.00	-8.0	5.4	-7.6	16.6	42.1	12.5	NA
219 1.0DL+1.0FL2+1.0WLZ1	161.6	-10.60	31.0	0.0	0.0	NA	10.4	NA	6.1
220 1.0DL+1.0FL2+1.0WLZ2	161.6	10.60	-46.3	0.0	0.0	NA	7.0	NA	6.1
221 1.0DL+1.0FL1+0.7SLX1	158.1	0.20	-8.3	-1.9	2.7	43.9	38.2	33.3	316.1
222 1.0DL+1.0FL1+0.7SLX2	165.1	-0.20	-7.1	1.9	-2.7	45.8	46.6	34.7	330.1
223 1.0DL+1.0FL1+0.7SLZ1	161.6	-1.80	3.2	0.0	0.0	NA	102.2	NA	35.9
224 1.0DL+1.0FL1+0.7SLZ2	161.6	1.80	-18.5	0.0	0.0	NA	17.4	NA	35.9
225 1.0DL+1.0FL2+0.7SLX1	158.1	0.20	-8.3	-1.9	2.7	43.9	38.2	33.3	316.1
226 1.0DL+1.0FL2+0.7SLX2	165.1	-0.20	-7.1	1.9	-2.7	45.8	46.6	34.7	330.1
227 1.0DL+1.0FL2+0.7SLZ1	161.6	-1.80	3.2	0.0	0.0	NA	102.2	NA	35.9
228 1.0DL+1.0FL2+0.7SLZ2	161.6	1.80	-18.5	0.0	0.0	NA	17.4	NA	35.9
229 1.0DL+1.0FL1+0.75WLX1+0.75	201.2	0.00	-18.9	-4.0	5.6	26.9	21.3	20.1	NA
230 1.0DL+1.0FL1+0.75WLX2+0.75	211.7	0.00	-19.4	4.0	-5.6	28.3	21.9	21.2	NA
231 1.0DL+1.0FL1+0.75WLZ1+0.75	206.4	-8.00	9.9	0.0	0.0	NA	41.6	NA	10.3
232 1.0DL+1.0FL1+0.75WLZ2+0.75	206.4	8.00	-48.0	0.0	0.0	NA	8.6	NA	10.3
233 1.0DL+1.0FL2+0.75WLX1+0.75	201.2	0.00	-18.9	-4.0	5.6	26.9	21.3	20.1	NA
234 1.0DL+1.0FL2+0.75WLX2+0.75	211.7	0.00	-19.4	4.0	-5.6	28.3	21.9	21.2	NA
235 1.0DL+1.0FL2+0.75WLZ1+0.75	206.4	-8.00	9.9	0.0	0.0	NA	41.6	NA	10.3
236 1.0DL+1.0FL2+0.75WLZ2+0.75	206.4	8.00	-48.0	0.0	0.0	NA	8.6	NA	10.3
237 1.0DL+1.0FL1+0.525SLX1+0.75	203.7	0.20	-19.5	-1.4	2.0	76.4	20.9	58.2	407.3
238 1.0DL+1.0FL1+0.525SLX2+0.75	209.0	-0.20	-18.8	1.4	-2.0	78.4	22.3	59.7	417.9
239 1.0DL+1.0FL1+0.525SLZ1+0.75	206.4	-1.40	-11.0	0.0	0.0	NA	37.7	NA	59.0
240 1.0DL+1.0FL1+0.525SLZ2+0.75	206.4	1.40	-27.1	0.0	0.0	NA	15.2	NA	59.0
241 1.0DL+1.0FL2+0.525SLX1+0.75	203.7	0.20	-19.5	-1.4	2.0	76.4	20.9	58.2	407.3
242 1.0DL+1.0FL2+0.525SLX2+0.75	209.0	-0.20	-18.8	1.4	-2.0	78.4	22.3	59.7	417.9
243 1.0DL+1.0FL2+0.525SLZ1+0.75	206.4	-1.40	-11.0	0.0	0.0	NA	37.7	NA	59.0
244 1.0DL+1.0FL2+0.525SLZ2+0.75	206.4	1.40	-27.1	0.0	0.0	NA	15.2	NA	59.0
245 0.6DL+1.0WLX1	139.7	0.00	-4.3	-5.4	7.6	13.8	64.7	10.3	NA
246 0.6DL+1.0WLX2	153.7	0.00	-5.0	5.4	-7.6	15.2	62.0	11.4	NA
247 0.6DL+1.0WLZ1	146.7	-10.60	34.0	0.0	0.0	NA	8.6	NA	5.5
248 0.6DL+1.0WLZ2	146.7	10.60	-43.3	0.0	0.0	NA	6.8	NA	5.5
249 0.6DL+0.6FL1+0.7SLX1	143.2	0.20	-5.2	-1.9	2.7	39.8	54.6	30.1	286.3
250 0.6DL+0.6FL1+0.7SLX2	150.3	-0.20	-3.9	1.9	-2.7	41.7	77.4	31.6	300.5
251 0.6DL+0.6FL1+0.7SLZ1	146.7	-1.80	6.2	0.0	0.0	NA	47.3	NA	32.6
252 0.6DL+0.6FL1+0.7SLZ2	146.7	1.80	-15.5	0.0	0.0	NA	18.9	NA	32.6
Min.=	13.8		6.8			10.3		5.5	

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Base pressure check of footing F14

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	1	201	0	34.23	0
Min Px	1	201	0	34.23	0
Max Py	3	209	0	74.96	0
Min Py	4	248	0	1.55	0
Max Pz	1	201	0	34.23	0
Min Pz	1	201	0	34.23	0

Gross Maximum Base pressure, p_{max} = 75.0 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.65 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 461.7 KN/m^2
> 75.0 KN/m^2
Hence OK

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Design of foundation raft F14

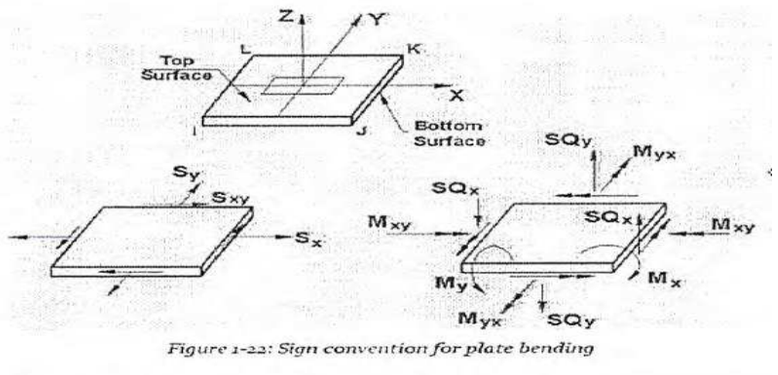
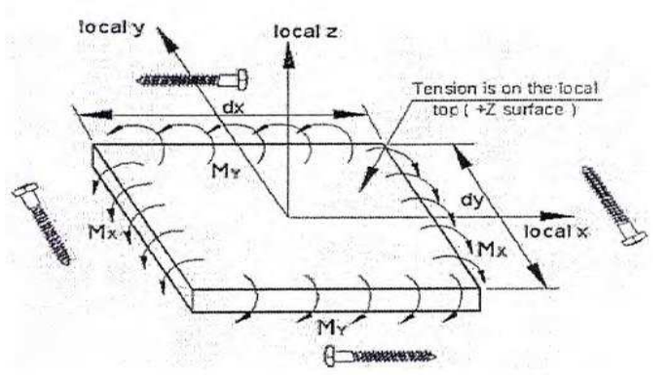


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F14_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	39	0.118	-0.037	0	0	0	-7.195	-2.268	0.318	-7.513	-2.586	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Min Qx	28	-0.118	-0.037	0	0	0	-7.195	-2.268	0	-7.195	-2.268	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Max Qy	137	0.011	0.159	0	0	0	-10.46	17.907	1.731	-12.187	19.638	109 1.2DL+1.2FL1+1.6LL+1.2PL
Min Qy	177	0.013	-0.175	0	0	0	-12.02	18.733	-2.07	-14.088	20.803	109 1.2DL+1.2FL1+1.6LL+1.2PL
Max Sx	183	0.009	-0.018	0	0	0	-0.245	2.019	-0.345	-0.59	2.364	101 1.4DL+1.4FL1
Min Sx	183	0.009	-0.018	0	0	0	-0.245	2.019	-0.345	-0.59	2.364	101 1.4DL+1.4FL1
Max Sy	183	0.009	-0.018	0	0	0	-0.245	2.019	-0.345	-0.59	2.364	101 1.4DL+1.4FL1
Min Sy	183	0.009	-0.018	0	0	0	-0.245	2.019	-0.345	-0.59	2.364	101 1.4DL+1.4FL1
Max Sxy	183	0.009	-0.018	0	0	0	-0.245	2.019	-0.345	-0.59	2.364	101 1.4DL+1.4FL1
Min Sxy	183	0.009	-0.018	0	0	0	-0.245	2.019	-0.345	-0.59	2.364	101 1.4DL+1.4FL1
Max Mx	60	-0.018	0.013	0	0	0	2.138	0.42	-0.384	2.522	0.804	128 0.9DL+1.6WLZ2
Min Mx	43	0.023	0.03	0	0	0	-20.5	-0.324	0.723	-21.218	-1.047	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Max My	157	0	0.003	0	0	0	-0.407	46.854	-0.041	-0.448	46.895	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Min My	137	0.007	0.03	0	0	0	-0.763	-5.09	-0.342	-1.105	-5.432	128 0.9DL+1.6WLZ2
Max Mxy	127	-0.053	-0.127	0	0	0	-4.96	18.531	4.008	-8.968	22.539	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Min Mxy	178	0.053	-0.127	0	0	0	-4.96	18.531	-4.008	-8.968	22.539	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL

PLATE CENTER STRESS : FDN_F14_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	41	0.036	-0.027	0	0	0	-1.21	1.032	-0.335	-1.545	1.367	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Min Qx	26	-0.036	-0.027	0	0	0	-1.21	1.032	0	-1.21	1.032	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Max Qy	141	0.01	0.109	0	0	0	-5.51	28.474	0.816	-6.326	29.29	109 1.2DL+1.2FL1+1.6LL+1.2PL
Min Qy	173	0.011	-0.115	0	0	0	-6.317	29.882	-1.017	-7.334	30.899	109 1.2DL+1.2FL1+1.6LL+1.2PL
Max Sx	183	0.009	-0.018	0	0	0	-0.245	2.019	-0.345	-0.59	2.364	101 1.4DL+1.4FL1
Min Sx	183	0.009	-0.018	0	0	0	-0.245	2.019	-0.345	-0.59	2.364	101 1.4DL+1.4FL1
Max Sy	183	0.009	-0.018	0	0	0	-0.245	2.019	-0.345	-0.59	2.364	101 1.4DL+1.4FL1
Min Sy	183	0.009	-0.018	0	0	0	-0.245	2.019	-0.345	-0.59	2.364	101 1.4DL+1.4FL1
Max Sxy	183	0.009	-0.018	0	0	0	-0.245	2.019	-0.345	-0.59	2.364	101 1.4DL+1.4FL1
Min Sxy	183	0.009	-0.018	0	0	0	-0.245	2.019	-0.345	-0.59	2.364	101 1.4DL+1.4FL1
Max Mx	55	0.001	0.001	0	0	0	0.171	0.45	-0.319	0.49	0.769	125 0.9DL+1.6WLX1
Min Mx	173	0.011	-0.115	0	0	0	-6.317	29.882	-1.017	-7.334	30.899	109 1.2DL+1.2FL1+1.6LL+1.2PL
Max My	157	0	0.003	0	0	0	-0.407	46.854	-0.041	-0.448	46.895	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Min My	136	0.002	0.002	0	0	0	0.09	-3.125	-0.683	0.773	-3.808	128 0.9DL+1.6WLZ2
Max Mxy	122	-0.018	-0.099	0	0	0	-3.814	29.114	2.194	-6.008	31.308	109 1.2DL+1.2FL1+1.6LL+1.2PL
Min Mxy	174	0.018	-0.099	0	0	0	-3.814	29.114	-2.194	-6.008	31.308	109 1.2DL+1.2FL1+1.6LL+1.2PL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

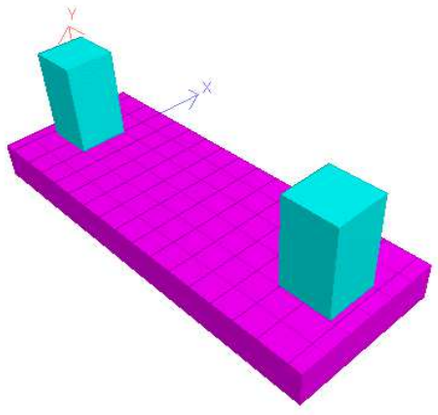
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
128 0.9D	2.52	top face	50	344	2.522	0.03	0.0001	23.3	619.2	12	182.72	150	754.3	10.64	13.29	0.075	tension control
111 1.2D	-21.22	bottom face	50	344	21.218	0.24	0.0006	196.6		12	182.72	150	754.3	10.64	13.29	0.075	tension control
111 1.2D	-8.97	bottom face	50	344	8.968	0.10	0.0002	82.9		12	182.72	150	754.3	10.64	13.29	0.075	tension control
111 1.2D	-8.97	bottom face	50	344	8.968	0.10	0.0002	82.9		12	182.72	150	754.3	10.64	13.29	0.075	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.036	1	14.4	1.545	0.002	0.344	366.735	338.444	590.188	338.44	1261	> 14.4	ok	253.8	φVc>Vu	not required
	Min Qx	0.036	1	14.4	1.21			378.117			338.44	1261	> 14.4	ok	253.8	φVc>Vu	not required

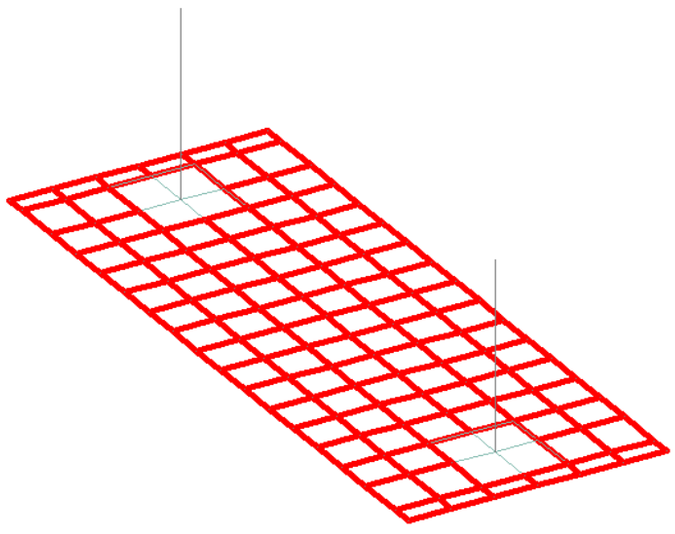
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
111 1.2D	46.90	top face	50	344	46.895	0.53	0.0013	436.7	619.2	12	182.72	150	754.3	10.64	13.29	0.075	tension control
128 0.9D	-5.43	bottom face	50	344	5.432	0.06	0.0001	50.2		12	182.72	150	754.3	10.64	13.29	0.075	tension control
111 1.2D	22.54	top face	50	344	22.539	0.25	0.0006	208.9		12	182.72	150	754.3	10.64	13.29	0.075	tension control
111 1.2D	22.54	top face	50	344	22.539	0.25	0.0006	208.9		12	182.72	150	754.3	10.64	13.29	0.075	tension control

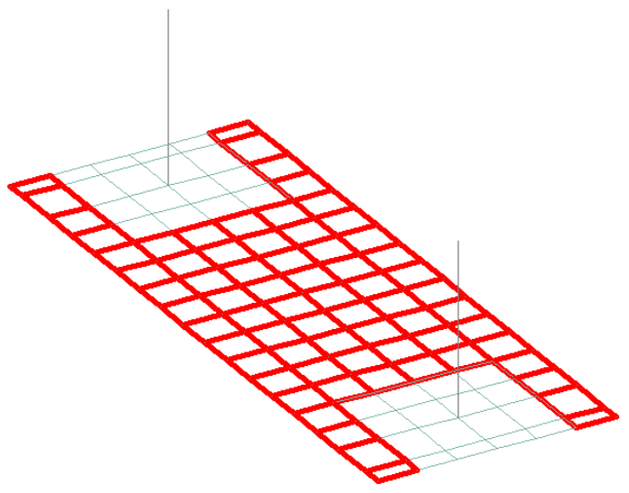
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	109 1.2D	0.109	1	43.6	29.29	0.002	0.344	332.187	338.444	fc'590.2	332.19	350	> 43.6	ok	249.1	φVc>Vu	not required
	109 1.2D	0.115	1	46	30.899			332.188			332.19	350	> 46	ok	249.1	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	12ϕT	**150**	c/c
At Top	12ϕT	**150**	c/c
Along Z Direction :			
At Bottom	12ϕT	**150**	c/c
At Top	12ϕT	**150**	c/c

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5.6 DESIGN CALCULATION OF FOOTING F14A

Size of Columns at node	Along-Z	Along-X
1	0.50m	0.50m
2	0.50m	0.50m
Size of fdn. raft:	4.70m	1.55m
Thickness of raft:	0.50	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.35	m
U/S level of fdn:	-1.05	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

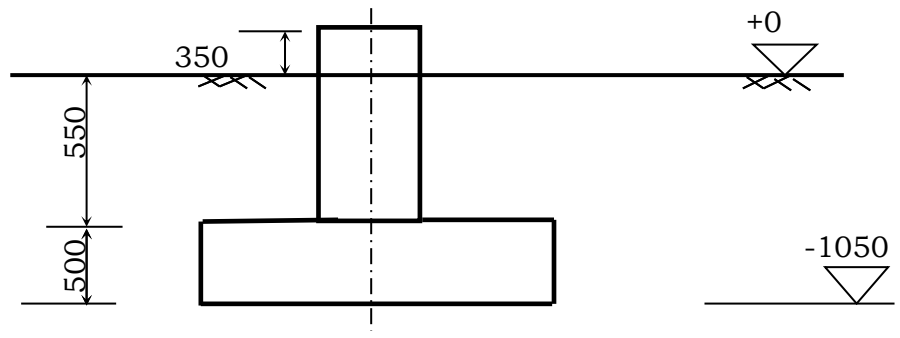
Calculation of Weight of Foundation :

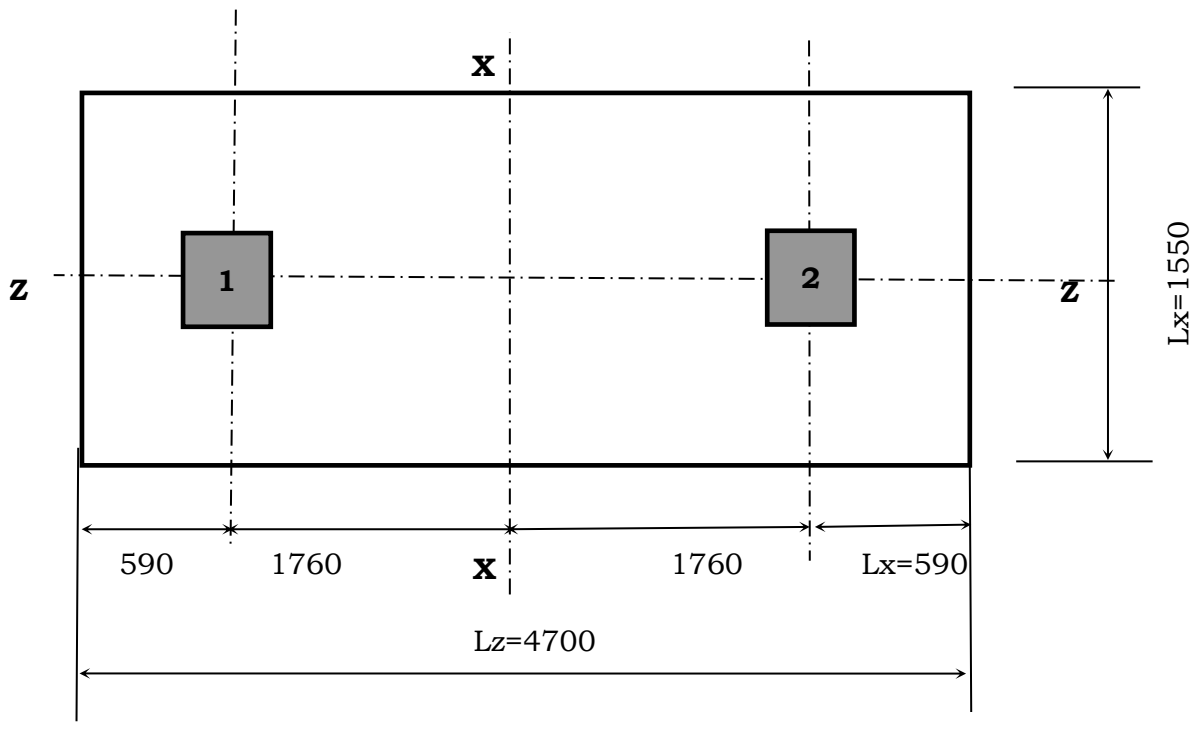
Weight of raft = (4.7*1.55*0.5*25) = 91.06KN

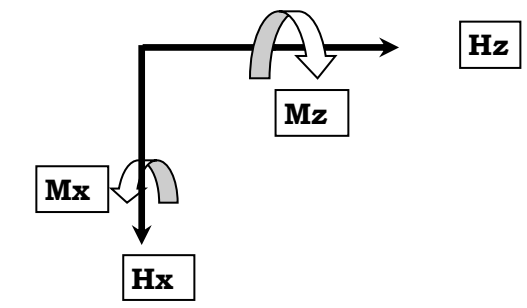
Weight of soil = ((4.7*1.55)-2*(0.5*0.5))*0.55*18 = 67.17KN

Total = 158.23KN

Submerged wt. = 158.23-4.7*1.55*1.05*10 = 81.74KN







H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F14A

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0FL1	19.4	0.0	0.0	1.2	1.7	22.8	3.3	-4.6	-1.1	-1.5
202 1.0DL+1.0FL2	19.4	0.0	0.0	1.2	1.7	22.8	3.3	-4.6	-1.1	-1.5
203 1.0DL+1.0FL1+1.0LL+1.0TL	63.6	0.0	0.0	20.0	28.0	77.2	19.1	-26.7	-20.1	-28.1
204 1.0DL+1.0FL2+1.0LL+1.0TL	63.6	0.0	0.0	20.0	28.0	77.2	19.1	-26.7	-20.1	-28.1
205 1.0DL+1.0FL1+0.75LL+0.75TL	52.6	0.0	0.0	15.3	21.4	63.6	15.2	-21.2	-15.4	-21.5
206 1.0DL+1.0FL2+0.75LL+0.75TL	52.6	0.0	0.0	15.3	21.4	63.6	15.2	-21.2	-15.4	-21.5
207 1.0DL+1.0FL1+1.0PL	68.4	0.0	0.0	5.4	7.6	120.8	3.6	-5.0	-5.3	-7.4
208 1.0DL+1.0FL2+1.0PL	68.4	0.0	0.0	5.4	7.6	120.8	3.6	-5.0	-5.3	-7.4
209 1.0DL+1.0FL1+1.0LL+1.0TL+1.(	112.6	0.0	0.0	24.2	33.9	175.2	19.4	-27.2	-24.3	-34.0
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.(	112.6	0.0	0.0	24.2	33.9	175.2	19.4	-27.2	-24.3	-34.0
211 1.0DL+1.0FL1+0.75LL+0.75TL+	101.6	0.0	0.0	19.5	27.3	161.6	15.5	-21.6	-19.6	-27.4
212 1.0DL+1.0FL2+0.75LL+0.75TL+	101.6	0.0	0.0	19.5	27.3	161.6	15.5	-21.6	-19.6	-27.4
213 1.0DL+1.0FL1+1.0WLX1	23.1	-2.3	3.2	1.6	2.2	25.9	0.8	-1.1	-1.3	-1.8
214 1.0DL+1.0FL1+1.0WLX2	15.7	2.3	-3.2	0.8	1.1	19.7	5.8	-8.1	-0.9	-1.3
215 1.0DL+1.0FL1+1.0WLZ1	9.7	0.0	0.0	-2.8	-3.9	32.4	3.2	-4.5	-6.5	-9.1
216 1.0DL+1.0FL1+1.0WLZ2	29.1	0.0	0.0	5.2	7.3	13.2	3.4	-4.8	4.3	6.0
217 1.0DL+1.0FL2+1.0WLX1	23.1	-2.3	3.2	1.6	2.2	25.9	0.8	-1.1	-1.3	-1.8
218 1.0DL+1.0FL2+1.0WLX2	15.7	2.3	-3.2	0.8	1.1	19.7	5.8	-8.1	-0.9	-1.3
219 1.0DL+1.0FL2+1.0WLZ1	9.7	0.0	0.0	-2.8	-3.9	32.4	3.2	-4.5	-6.5	-9.1
220 1.0DL+1.0FL2+1.0WLZ2	29.1	0.0	0.0	5.2	7.3	13.2	3.4	-4.8	4.3	6.0
221 1.0DL+1.0FL1+0.7SLX1	21.1	-1.0	1.4	1.4	2.0	24.1	1.5	-2.1	-1.2	-1.6
222 1.0DL+1.0FL1+0.7SLX2	17.7	1.0	-1.4	1.0	1.4	21.5	5.1	-7.2	-1.0	-1.4
223 1.0DL+1.0FL1+0.7SLZ1	16.1	0.0	0.0	-0.2	-0.3	26.0	3.2	-4.5	-2.6	-3.7
224 1.0DL+1.0FL1+0.7SLZ2	22.7	0.0	0.0	2.6	3.6	19.6	3.4	-4.7	0.4	0.6
225 1.0DL+1.0FL2+0.7SLX1	21.1	-1.0	1.4	1.4	2.0	24.1	1.5	-2.1	-1.2	-1.6
226 1.0DL+1.0FL2+0.7SLX2	17.7	1.0	-1.4	1.0	1.4	21.5	5.1	-7.2	-1.0	-1.4
227 1.0DL+1.0FL2+0.7SLZ1	16.1	0.0	0.0	-0.2	-0.3	26.0	3.2	-4.5	-2.6	-3.7
228 1.0DL+1.0FL2+0.7SLZ2	22.7	0.0	0.0	2.6	3.6	19.6	3.4	-4.7	0.4	0.6
229 1.0DL+1.0FL1+0.75WLX1+0.75	55.7	-1.7	2.4	2.9	4.1	65.6	12.0	-16.8	-2.3	-3.2
230 1.0DL+1.0FL1+0.75WLX2+0.75	50.2	1.7	-2.4	2.3	3.3	60.9	15.8	-22.1	-2.0	-2.8
231 1.0DL+1.0FL1+0.75WLZ1+0.75	45.7	0.0	0.0	-0.4	-0.5	70.4	13.8	-19.3	-6.2	-8.7
232 1.0DL+1.0FL1+0.75WLZ2+0.75	60.2	0.0	0.0	5.6	7.9	56.0	14.0	-19.5	1.9	2.7
233 1.0DL+1.0FL2+0.75WLX1+0.75	55.7	-1.7	2.4	2.9	4.1	65.6	12.0	-16.8	-2.3	-3.2
234 1.0DL+1.0FL2+0.75WLX2+0.75	50.2	1.7	-2.4	2.3	3.3	60.9	15.8	-22.1	-2.0	-2.8
235 1.0DL+1.0FL2+0.75WLZ1+0.75	45.7	0.0	0.0	-0.4	-0.5	70.4	13.8	-19.3	-6.2	-8.7
236 1.0DL+1.0FL2+0.75WLZ2+0.75	60.2	0.0	0.0	5.6	7.9	56.0	14.0	-19.5	1.9	2.7
237 1.0DL+1.0FL1+0.525SLX1+0.75	54.2	-0.7	1.0	2.8	3.9	64.2	12.5	-17.5	-2.2	-3.1
238 1.0DL+1.0FL1+0.525SLX2+0.75	51.7	0.7	-1.0	2.5	3.5	62.2	15.2	-21.3	-2.1	-2.9
239 1.0DL+1.0FL1+0.525SLZ1+0.75	50.5	0.0	0.0	1.6	2.2	65.6	13.8	-19.4	-3.3	-4.6
240 1.0DL+1.0FL1+0.525SLZ2+0.75	55.4	0.0	0.0	3.7	5.1	60.8	13.9	-19.5	-1.0	-1.4
241 1.0DL+1.0FL2+0.525SLX1+0.75	54.2	-0.7	1.0	2.8	3.9	64.2	12.5	-17.5	-2.2	-3.1
242 1.0DL+1.0FL2+0.525SLX2+0.75	51.7	0.7	-1.0	2.5	3.5	62.2	15.2	-21.3	-2.1	-2.9
243 1.0DL+1.0FL2+0.525SLZ1+0.75	50.5	0.0	0.0	1.6	2.2	65.6	13.8	-19.4	-3.3	-4.6
244 1.0DL+1.0FL2+0.525SLZ2+0.75	55.4	0.0	0.0	3.7	5.1	60.8	13.9	-19.5	-1.0	-1.4
245 0.6DL+1.0WLX1	15.3	-2.3	3.2	1.1	1.6	16.8	-0.5	0.7	-0.9	-1.2
246 0.6DL+1.0WLX2	7.9	2.3	-3.2	0.3	0.4	10.6	4.5	-6.3	-0.5	-0.6
247 0.6DL+1.0WLZ1	1.9	0.0	0.0	-3.3	-4.6	23.3	1.9	-2.6	-6.1	-8.5
248 0.6DL+1.0WLZ2	21.3	0.0	0.0	4.7	6.6	4.1	2.1	-2.9	4.7	6.6
249 0.6DL+0.6FL1+0.7SLX1	13.3	-1.0	1.4	0.9	1.3	15.0	0.2	-0.2	-0.7	-1.0
250 0.6DL+0.6FL1+0.7SLX2	10.0	1.0	-1.4	0.5	0.7	12.4	3.8	-5.3	-0.6	-0.8
251 0.6DL+0.6FL1+0.7SLZ1	8.4	0.0	0.0	-0.7	-1.0	16.9	1.9	-2.7	-2.2	-3.1
252 0.6DL+0.6FL1+0.7SLZ2	14.9	0.0	0.0	2.1	3.0	10.5	2.1	-2.9	0.9	1.2

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL+1.0FL1	200.4	0.10	6.2	3.3	-4.6	33.8	76.2	24.3	801.7
202 1.0DL+1.0FL2	200.4	0.10	6.2	3.3	-4.6	33.8	76.2	24.3	801.7
203 1.0DL+1.0FL1+1.0LL+1.0TL	299.0	-0.10	23.8	19.1	-26.7	8.7	29.5	6.3	1196.1
204 1.0DL+1.0FL2+1.0LL+1.0TL	299.0	-0.10	23.8	19.1	-26.7	8.7	29.5	6.3	1196.1
205 1.0DL+1.0FL1+0.75LL+0.75TL	274.4	-0.10	19.3	15.2	-21.2	10.0	33.5	7.2	1097.7
206 1.0DL+1.0FL2+0.75LL+0.75TL	274.4	-0.10	19.3	15.2	-21.2	10.0	33.5	7.2	1097.7
207 1.0DL+1.0FL1+1.0PL	347.4	0.10	92.4	3.6	-5.0	53.9	8.8	38.6	1389.7
208 1.0DL+1.0FL2+1.0PL	347.4	0.10	92.4	3.6	-5.0	53.9	8.8	38.6	1389.7
209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0	446.0	-0.10	110.1	19.4	-27.2	12.7	9.5	9.2	1784.1
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0	446.0	-0.10	110.1	19.4	-27.2	12.7	9.5	9.2	1784.1
211 1.0DL+1.0FL1+0.75LL+0.75TL+	421.4	-0.10	105.5	15.5	-21.6	15.1	9.4	10.9	1685.7
212 1.0DL+1.0FL2+0.75LL+0.75TL+	421.4	-0.10	105.5	15.5	-21.6	15.1	9.4	10.9	1685.7
213 1.0DL+1.0FL1+1.0WLX1	207.2	0.30	5.3	-1.5	2.1	76.5	91.4	55.3	276.3
214 1.0DL+1.0FL1+1.0WLX2	193.6	-0.10	6.8	8.1	-11.3	13.3	66.5	9.6	774.5
215 1.0DL+1.0FL1+1.0WLZ1	200.3	-9.30	27.0	3.2	-4.5	34.5	17.5	25.0	8.6
216 1.0DL+1.0FL1+1.0WLZ2	200.5	9.50	-14.7	3.4	-4.8	32.4	32.1	23.6	8.4
217 1.0DL+1.0FL2+1.0WLX1	207.2	0.30	5.3	-1.5	2.1	76.5	91.4	55.3	276.3
218 1.0DL+1.0FL2+1.0WLX2	193.6	-0.10	6.8	8.1	-11.3	13.3	66.5	9.6	774.5
219 1.0DL+1.0FL2+1.0WLZ1	200.3	-9.30	27.0	3.2	-4.5	34.5	17.5	25.0	8.6
220 1.0DL+1.0FL2+1.0WLZ2	200.5	9.50	-14.7	3.4	-4.8	32.4	32.1	23.6	8.4
221 1.0DL+1.0FL1+0.7SLX1	203.4	0.20	5.7	0.5	-0.7	225.2	84.2	162.7	406.9
222 1.0DL+1.0FL1+0.7SLX2	197.4	0.00	6.7	6.1	-8.6	17.8	69.4	12.9	NA
223 1.0DL+1.0FL1+0.7SLZ1	200.3	-2.80	13.4	3.2	-4.5	34.5	35.1	25.0	28.6
224 1.0DL+1.0FL1+0.7SLZ2	200.5	3.00	-1.3	3.4	-4.7	33.1	375.2	23.6	26.7
225 1.0DL+1.0FL2+0.7SLX1	203.4	0.20	5.7	0.5	-0.7	225.2	84.2	162.7	406.9
226 1.0DL+1.0FL2+0.7SLX2	197.4	0.00	6.7	6.1	-8.6	17.8	69.4	12.9	NA
227 1.0DL+1.0FL2+0.7SLZ1	200.3	-2.80	13.4	3.2	-4.5	34.5	35.1	25.0	28.6
228 1.0DL+1.0FL2+0.7SLZ2	200.5	3.00	-1.3	3.4	-4.7	33.1	375.2	23.6	26.7
229 1.0DL+1.0FL1+0.75WLX1+0.75	279.5	0.60	18.3	10.3	-14.4	15.0	35.8	10.9	186.4
230 1.0DL+1.0FL1+0.75WLX2+0.75	269.3	0.30	19.3	17.5	-24.5	8.5	32.7	6.2	359.1
231 1.0DL+1.0FL1+0.75WLZ1+0.75	274.3	-6.60	34.3	13.8	-19.3	11.0	18.8	8.0	16.6
232 1.0DL+1.0FL1+0.75WLZ2+0.75	274.4	7.50	3.2	14.0	-19.5	10.9	201.0	7.8	14.6
233 1.0DL+1.0FL2+0.75WLX1+0.75	279.5	0.60	18.3	10.3	-14.4	15.0	35.8	10.9	186.4
234 1.0DL+1.0FL2+0.75WLX2+0.75	269.3	0.30	19.3	17.5	-24.5	8.5	32.7	6.2	359.1
235 1.0DL+1.0FL2+0.75WLZ1+0.75	274.3	-6.60	34.3	13.8	-19.3	11.0	18.8	8.0	16.6
236 1.0DL+1.0FL2+0.75WLZ2+0.75	274.4	7.50	3.2	14.0	-19.5	10.9	201.0	7.8	14.6
237 1.0DL+1.0FL1+0.525SLX1+0.75	276.6	0.60	18.4	11.8	-16.5	13.0	35.3	9.4	184.4
238 1.0DL+1.0FL1+0.525SLX2+0.75	272.1	0.40	19.1	15.9	-22.3	9.5	33.5	6.8	272.1
239 1.0DL+1.0FL1+0.525SLZ1+0.75	274.3	-1.70	24.2	13.8	-19.4	11.0	26.7	8.0	64.5
240 1.0DL+1.0FL1+0.525SLZ2+0.75	274.4	2.70	13.2	13.9	-19.5	10.9	48.8	7.9	40.7
241 1.0DL+1.0FL2+0.525SLX1+0.75	276.6	0.60	18.4	11.8	-16.5	13.0	35.3	9.4	184.4
242 1.0DL+1.0FL2+0.525SLX2+0.75	272.1	0.40	19.1	15.9	-22.3	9.5	33.5	6.8	272.1
243 1.0DL+1.0FL2+0.525SLZ1+0.75	274.3	-1.70	24.2	13.8	-19.4	11.0	26.7	8.0	64.5
244 1.0DL+1.0FL2+0.525SLZ2+0.75	274.4	2.70	13.2	13.9	-19.5	10.9	48.8	7.9	40.7
245 0.6DL+1.0WLX1	190.3	0.20	3.0	-2.8	3.9	37.8	147.1	27.2	380.7
246 0.6DL+1.0WLX2	176.7	-0.20	4.6	6.8	-9.5	14.4	91.2	10.4	353.5
247 0.6DL+1.0WLZ1	183.4	-9.40	24.6	1.9	-2.6	54.7	17.5	38.6	7.8
248 0.6DL+1.0WLZ2	183.6	9.40	-17.1	2.1	-2.9	49.1	25.3	35.0	7.8
249 0.6DL+0.6FL1+0.7SLX1	186.5	0.20	3.3	-0.8	1.2	120.5	133.2	93.3	373.1
250 0.6DL+0.6FL1+0.7SLX2	180.6	-0.10	4.1	4.8	-6.7	20.9	102.9	15.1	722.5
251 0.6DL+0.6FL1+0.7SLZ1	183.5	-2.90	10.9	1.9	-2.7	52.7	39.7	38.6	25.3
252 0.6DL+0.6FL1+0.7SLZ2	183.6	3.00	-3.5	2.1	-2.9	49.1	121.8	35.0	24.5
Min.=						8.5	8.8	6.2	7.8

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Base pressure check of footing F14A

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	129	201	0	26.34	0
Min Px	129	201	0	26.34	0
Max Py	166	209	0	119.27	0
Min Py	138	247	0	6.48	0
Max Pz	129	201	0	26.34	0
Min Pz	129	201	0	26.34	0

Gross Maximum Base pressure, p_{max} = 119.3 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.55 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 459.9 KN/m^2
> 119.3 KN/m^2
Hence OK

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Design of foundation raft F14A

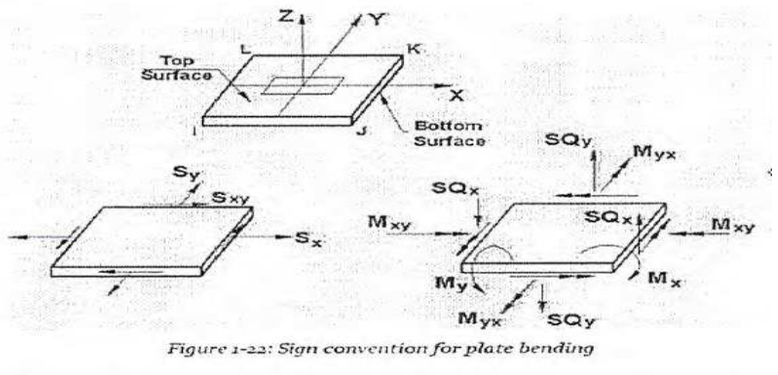
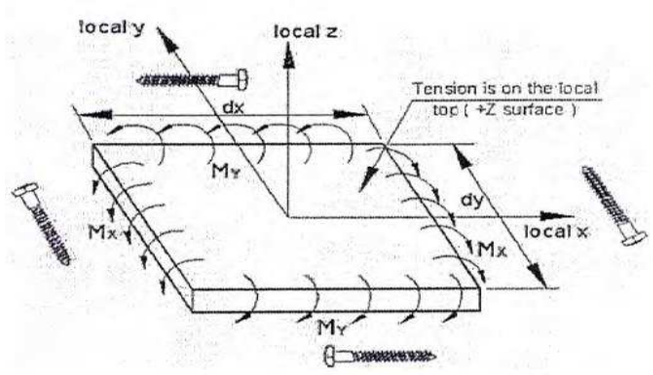


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F14-A_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	303	0.128	0.022	0	0	0	-11.71	-1.014	-0.794	-12.499	-1.808	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Min Qx	311	-0.14	-0.029	0	0	0	-22.14	-1.045	0	-22.144	-1.045	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Max Qy	252	-0.003	0.176	0	0	0	-14.32	39.986	-7.271	-21.594	47.257	109 1.2DL+1.2FL1+1.6LL+1.2PL
Min Qy	337	0.023	-0.168	0	0	0	-14.2	26.292	-5.411	-19.613	31.703	109 1.2DL+1.2FL1+1.6LL+1.2PL
Max Sx	342	0.007	-0.017	0	0	0	-0.191	2.74	-0.444	-0.635	3.184	101 1.4DL+1.4FL1
Min Sx	342	0.007	-0.017	0	0	0	-0.191	2.74	-0.444	-0.635	3.184	101 1.4DL+1.4FL1
Max Sy	342	0.007	-0.017	0	0	0	-0.191	2.74	-0.444	-0.635	3.184	101 1.4DL+1.4FL1
Min Sy	342	0.007	-0.017	0	0	0	-0.191	2.74	-0.444	-0.635	3.184	101 1.4DL+1.4FL1
Max Sxy	342	0.007	-0.017	0	0	0	-0.191	2.74	-0.444	-0.635	3.184	101 1.4DL+1.4FL1
Min Sxy	342	0.007	-0.017	0	0	0	-0.191	2.74	-0.444	-0.635	3.184	101 1.4DL+1.4FL1
Max Mx	340	0.002	-0.007	0	0	0	0.605	1.037	-1.184	1.789	2.221	126 0.9DL+1.6WLX2
Min Mx	313	-0.062	-0.065	0	0	0	-31.71	-1.793	-2.543	-34.248	-4.336	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Max My	269	0	0.018	0	0	0	-0.157	90.709	-2.244	-2.401	92.953	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Min My	299	-0.038	0.108	0	0	0	-22.55	-6.843	-0.688	-23.237	-7.531	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Max Mxy	327	-0.001	-0.079	0	0	0	-1.356	28.634	5.021	-6.377	33.655	105 1.2DL+1.2FL1+1.2TL+1.6LL
Min Mxy	215	-0.02	0.102	0	0	0	-5.121	29.487	-13.206	-18.327	42.693	107 1.4DL+1.4FL1+1.4PL

PLATE CENTER STRESS : FDN_F14-A_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	306	0.041	-0.015	0	0	0	-2.371	0.193	0.123	-2.494	0.316	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Min Qx	307	-0.038	0.011	0	0	0	-1.292	0.345	0	-1.292	0.345	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Max Qy	254	0.013	0.168	0	0	0	-2.779	28.627	-3.914	-6.693	32.541	107 1.4DL+1.4FL1+1.4PL
Min Qy	325	-0.011	-0.135	0	0	0	-0.86	34.918	0.235	-1.095	35.153	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Max Sx	342	0.007	-0.017	0	0	0	-0.191	2.74	-0.444	-0.635	3.184	101 1.4DL+1.4FL1
Min Sx	342	0.007	-0.017	0	0	0	-0.191	2.74	-0.444	-0.635	3.184	101 1.4DL+1.4FL1
Max Sy	342	0.007	-0.017	0	0	0	-0.191	2.74	-0.444	-0.635	3.184	101 1.4DL+1.4FL1
Min Sy	342	0.007	-0.017	0	0	0	-0.191	2.74	-0.444	-0.635	3.184	101 1.4DL+1.4FL1
Max Sxy	342	0.007	-0.017	0	0	0	-0.191	2.74	-0.444	-0.635	3.184	101 1.4DL+1.4FL1
Min Sxy	342	0.007	-0.017	0	0	0	-0.191	2.74	-0.444	-0.635	3.184	101 1.4DL+1.4FL1
Max Mx	267	0.009	-0.005	0	0	0	0.474	55.622	-7.821	8.295	63.443	107 1.4DL+1.4FL1+1.4PL
Min Mx	245	-0.004	-0.124	0	0	0	-7.827	38.711	-1.161	-8.988	39.872	109 1.2DL+1.2FL1+1.6LL+1.2PL
Max My	269	0	0.018	0	0	0	-0.157	90.709	-2.244	-2.401	92.953	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Min My	201	-0.004	0.002	0	0	0	-0.117	-1.412	0.451	-0.568	-1.863	128 0.9DL+1.6WLZ2
Max Mxy	254	0.016	0.041	0	0	0	-2.644	22.057	2.832	-5.476	24.889	103 1.2DL+1.2FL1+1.6LL
Min Mxy	218	-0.003	0.074	0	0	0	-2.914	39.9	-11.802	-14.716	51.702	107 1.4DL+1.4FL1+1.4PL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

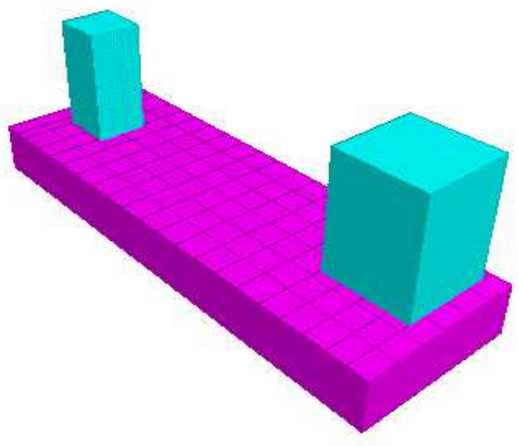
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c\ *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
126 0.9D	1.79	top face	50	444	1.789	0.01	0.0000	12.8	799.2	12	141.57	125	905.2	12.76	15.95	0.08	tension control
111 1.2D	-34.25	bottom face	50	444	34.248	0.23	0.0006	245.8		12	141.57	125	905.2	12.76	15.95	0.08	tension control
105 1.2D	-6.38	bottom face	50	444	6.377	0.04	0.0001	45.6		12	141.57	125	905.2	12.76	15.95	0.08	tension control
107 1.4D	-18.33	bottom face	50	444	18.327	0.12	0.0003	131.3		12	141.57	125	905.2	12.76	15.95	0.08	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.041	1	20.5	2.494	0.002	0.444	476.439	435.667	761.754	435.67	1627	> 20.5	ok	326.8	φVc>Vu	not required
	Min Qx	0.038	1	19	1.292			520.756			435.67	1627	> 19	ok	326.8	φVc>Vu	not required

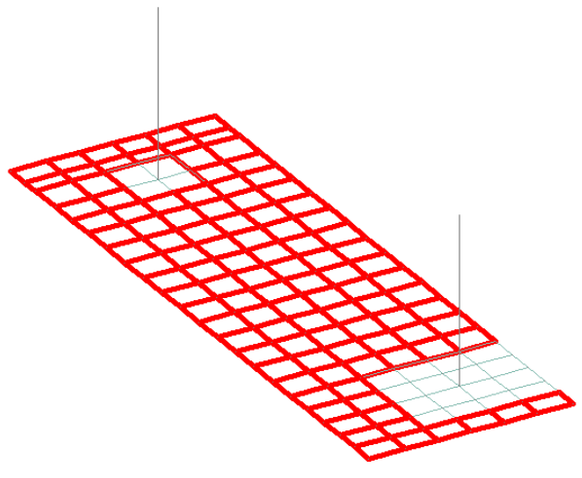
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c\ *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
111 1.2D	92.95	top face	50	444	92.953	0.63	0.0015	671.8	799.2	12	141.57	125	905.2	12.76	15.95	0.08	tension control
111 1.2D	-7.53	bottom face	50	444	7.531	0.05	0.0001	53.9		12	141.57	125	905.2	12.76	15.95	0.08	tension control
105 1.2D	33.66	top face	50	444	33.655	0.23	0.0005	241.6		12	141.57	125	905.2	12.76	15.95	0.08	tension control
107 1.4D	42.69	top face	50	444	42.693	0.29	0.0007	306.8		12	141.57	125	905.2	12.76	15.95	0.08	tension control

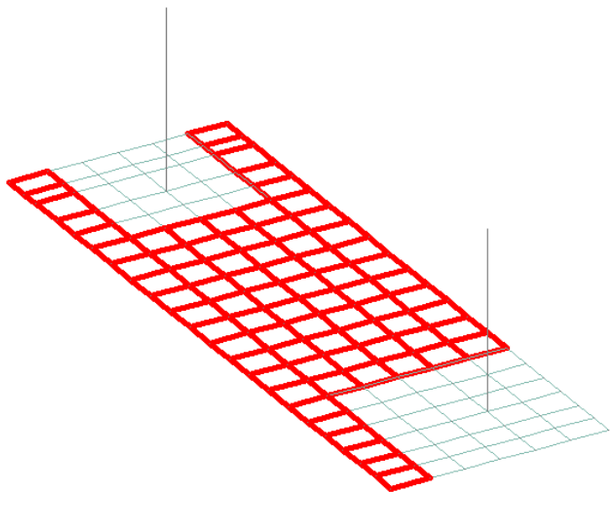
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	107 1.4D	0.168	1	84	32.541	0.002	0.444	437.915	435.667	fc'761.8	435.67	457	> 84	ok	326.8	φVc>Vu	not required
	111 1.2D	0.135	1	67.5	35.153			433.398			433.4	455	> 67.5	ok	325	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	12ϕT	**125**	c/c
At Top	12ϕT	**125**	c/c
Along Z Direction :			
At Bottom	12ϕT	**125**	c/c
At Top	12ϕT	**125**	c/c

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5.7 DESIGN OF CALCULATION OF FOOTING F15

(STAAD Node -7732)

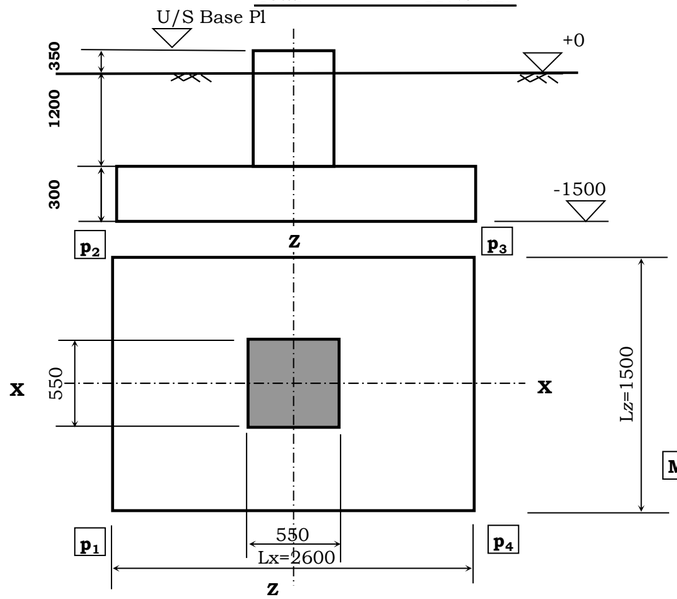
	Along-X	Along-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	2.60m	1.50m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.35 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

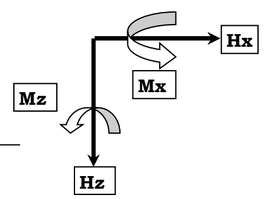
SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.= (2.6*1.5*0.3*24)+(0.55*0.55*1.55*24) = 39.33KN
Weight of soil = ((2.6*1.5)-(0.55*0.55))*1.2*18 = 77.71KN
Total = 117.04KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 244mm = 0.244m

Effective depth of slab along Z direction = 232mm = 0.232m

Clear projection along Z direction = 0.475m

Clear projection along X direction = 1.025m

H$_X$=Force along X-X direction

Hz=Force along Z-Z direction

M$_X$=Moment about X-X direction

Mz=Moment about Z-Z direction

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CHECK FOR OVERTURNING AND SLIDING OF FOOTING **F15**

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	1.0DL+1.0FL1	30.6	-3.0	0.0	-3.0	0.0	147.64	-3.00	-5.55	-3.00	-5.55
202	1.0DL+1.0FL2	30.6	-3.0	0.0	-3.0	0.0	147.64	-3.00	-5.55	-3.00	-5.55
203	1.0DL+1.0FL1+1.0LL+1.0TL	33.9	-27.6	0.0	-3.9	0.0	150.94	-27.60	-51.06	-3.90	-7.22
204	1.0DL+1.0FL2+1.0LL+1.0TL	33.9	-27.6	0.0	-3.9	0.0	150.94	-27.60	-51.06	-3.90	-7.22
205	1.0DL+1.0FL1+0.75LL+0.75TL	33.0	-21.4	0.0	-3.7	0.0	150.04	-21.40	-39.59	-3.70	-6.85
206	1.0DL+1.0FL2+0.75LL+0.75TL	33.1	-21.4	0.0	-3.7	0.0	150.14	-21.40	-39.59	-3.70	-6.85
207	1.0DL+1.0FL1+1.0PL	76.3	-5.5	0.0	-3.2	0.0	193.34	-5.50	-10.18	-3.20	-5.92
208	1.0DL+1.0FL2+1.0PL	76.4	-5.6	0.0	-3.2	0.0	193.44	-5.60	-10.36	-3.20	-5.92
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	79.6	-30.1	0.0	-4.1	0.0	196.64	-30.10	-55.69	-4.10	-7.59
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	79.6	-30.2	0.0	-4.1	0.0	196.64	-30.20	-55.87	-4.10	-7.59
211	1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	78.8	-24.0	0.0	-3.9	0.0	195.84	-24.00	-44.40	-3.90	-7.22
212	1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	78.8	-24.0	0.0	-3.9	0.0	195.84	-24.00	-44.40	-3.90	-7.22
213	1.0DL+1.0FL1+1.0WLX1	32.0	-4.5	0.0	-3.0	0.0	149.04	-4.50	-8.33	-3.00	-5.55
214	1.0DL+1.0FL1+1.0WLX2	29.2	-1.4	0.0	-3.0	0.0	146.24	-1.40	-2.59	-3.00	-5.55
215	1.0DL+1.0FL1+1.0WLZ1	29.0	-1.6	0.0	-4.7	0.0	146.04	-1.60	-2.96	-4.70	-8.70
216	1.0DL+1.0FL1+1.0WLZ2	32.1	-4.4	0.0	-1.2	0.0	149.14	-4.40	-8.14	-1.20	-2.22
217	1.0DL+1.0FL2+1.0WLX1	32.0	-4.5	0.0	-3.0	0.0	149.04	-4.50	-8.33	-3.00	-5.55
218	1.0DL+1.0FL2+1.0WLX2	29.2	-1.4	0.0	-3.0	0.0	146.24	-1.40	-2.59	-3.00	-5.55
219	1.0DL+1.0FL2+1.0WLZ1	29.1	-1.6	0.0	-4.7	0.0	146.14	-1.60	-2.96	-4.70	-8.70
220	1.0DL+1.0FL2+1.0WLZ2	32.2	-4.4	0.0	-1.2	0.0	149.24	-4.40	-8.14	-1.20	-2.22
221	1.0DL+1.0FL1+0.7SLX1	32.4	-4.1	0.0	-3.0	0.0	149.44	-4.10	-7.59	-3.00	-5.55
222	1.0DL+1.0FL1+0.7SLX2	28.8	-1.9	0.0	-3.0	0.0	145.84	-1.90	-3.52	-3.00	-5.55
223	1.0DL+1.0FL1+0.7SLZ1	30.8	-2.9	0.0	-3.2	0.0	147.84	-2.90	-5.37	-3.20	-5.92
224	1.0DL+1.0FL1+0.7SLZ2	30.3	-3.1	0.0	-2.7	0.0	147.34	-3.10	-5.74	-2.70	-5.00
225	1.0DL+1.0FL2+0.7SLX1	32.4	-4.1	0.0	-3.0	0.0	149.44	-4.10	-7.59	-3.00	-5.55
226	1.0DL+1.0FL2+0.7SLX2	28.8	-1.9	0.0	-3.0	0.0	145.84	-1.90	-3.52	-3.00	-5.55
227	1.0DL+1.0FL2+0.7SLZ1	30.9	-2.9	0.0	-3.2	0.0	147.94	-2.90	-5.37	-3.20	-5.92
228	1.0DL+1.0FL2+0.7SLZ2	30.4	-3.1	0.0	-2.7	0.0	147.44	-3.10	-5.74	-2.70	-5.00
229	1.0DL+1.0FL1+0.75WLX1+0.75LL	33.3	-3.9	0.0	-3.5	0.0	150.34	-3.90	-7.22	-3.50	-6.48
230	1.0DL+1.0FL1+0.75WLX2+0.75LL	31.2	-1.6	0.0	-3.5	0.0	148.24	-1.60	-2.96	-3.50	-6.48
231	1.0DL+1.0FL1+0.75WLZ1+0.75LL	31.1	-1.7	0.0	-4.8	0.0	148.14	-1.70	-3.15	-4.80	-8.88

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232 1.0DL+1.0FL1+0.75WLZ2+0.75LL	33.4	-3.8	0.0	-2.2	0.0	150.44	-3.80	-7.03	-2.20	-4.07
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	33.4	-3.9	0.0	-3.5	0.0	150.44	-3.90	-7.22	-3.50	-6.48
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	31.2	-1.6	0.0	-3.5	0.0	148.24	-1.60	-2.96	-3.50	-6.48
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	31.1	-1.7	0.0	-4.8	0.0	148.14	-1.70	-3.15	-4.80	-8.88
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	33.5	-3.8	0.0	-2.2	0.0	150.54	-3.80	-7.03	-2.20	-4.07
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	33.7	-3.6	0.0	-3.5	0.0	150.74	-3.60	-6.66	-3.50	-6.48
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	30.9	-1.9	0.0	-3.5	0.0	147.94	-1.90	-3.52	-3.50	-6.48
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	32.5	-2.7	0.0	-3.7	0.0	149.54	-2.70	-5.00	-3.70	-6.85
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	32.1	-2.8	0.0	-3.3	0.0	149.14	-2.80	-5.18	-3.30	-6.11
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	33.7	-3.6	0.0	-3.5	0.0	150.74	-3.60	-6.66	-3.50	-6.48
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	30.9	-2.0	0.0	-3.5	0.0	147.94	-2.00	-3.70	-3.50	-6.48
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	32.5	-2.7	0.0	-3.7	0.0	149.54	-2.70	-5.00	-3.70	-6.85
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	32.1	-2.9	0.0	-3.3	0.0	149.14	-2.90	-5.37	-3.30	-6.11
245 0.6DL+1.0WLX1	19.7	-3.3	0.0	-1.8	0.0	136.74	-3.30	-6.11	-1.80	-3.33
246 0.6DL+1.0WLX2	16.9	-0.2	0.0	-1.8	0.0	133.94	-0.20	-0.37	-1.80	-3.33
247 0.6DL+1.0WLZ1	16.8	-0.3	0.0	-3.5	0.0	133.84	-0.30	-0.56	-3.50	-6.48
248 0.6DL+1.0WLZ2	19.8	-3.1	0.0	0.0	0.0	136.84	-3.10	-5.74	0.00	0.00
249 0.6DL+0.6FL1+0.7SLX1	20.2	-2.9	0.0	-1.8	0.0	137.24	-2.90	-5.37	-1.80	-3.33
250 0.6DL+0.6FL1+0.7SLX2	16.5	-0.7	0.0	-1.8	0.0	133.54	-0.70	-1.30	-1.80	-3.33
251 0.6DL+0.6FL1+0.7SLZ1	18.6	-1.7	0.0	-2.0	0.0	135.64	-1.70	-3.15	-2.00	-3.70
252 0.6DL+0.6FL1+0.7SLZ2	18.1	-1.9	0.0	-1.6	0.0	135.14	-1.90	-3.52	-1.60	-2.96
Ultimate Conditions										
101 1.4DL+1.4FL1	42.8	-4.2	0.0	-4.2	0.0	159.84	-4.20	-7.77	-4.20	-7.77
102 1.4DL+1.4FL2	42.9	-4.2	0.0	-4.2	0.0	159.94	-4.20	-7.77	-4.20	-7.77
103 1.2DL+1.2FL1+1.6LL	40.3	-3.1	0.0	-4.7	0.0	157.34	-3.10	-5.74	-4.70	-8.70
104 1.2DL+1.2FL2+1.6LL	40.3	-3.1	0.0	-4.7	0.0	157.34	-3.10	-5.74	-4.70	-8.70
105 1.2DL+1.2FL1+1.2TL+1.6LL	41.5	-33.0	0.0	-5.0	0.0	158.54	-33.00	-61.05	-5.00	-9.25
106 1.2DL+1.2FL2+1.2TL+1.6LL	41.6	-33.0	0.0	-5.0	0.0	158.64	-33.00	-61.05	-5.00	-9.25
107 1.4DL+1.4FL1+1.4PL	106.9	-7.8	0.0	-4.5	0.0	223.94	-7.80	-14.43	-4.50	-8.33
108 1.4DL+1.4FL2+1.4PL	106.9	-7.8	0.0	-4.5	0.0	223.94	-7.80	-14.43	-4.50	-8.33
109 1.2DL+1.2FL1+1.6LL+1.2PL	95.2	-6.2	0.0	-5.0	0.0	212.24	-6.20	-11.47	-5.00	-9.25
110 1.2DL+1.2FL2+1.6LL+1.2PL	95.2	-6.2	0.0	-5.0	0.0	212.24	-6.20	-11.47	-5.00	-9.25

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111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	96.4	-36.1	0.0	-5.2	0.0	213.44	-36.10	-66.79	-5.20	-9.62
112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	96.5	-36.1	0.0	-5.2	0.0	213.54	-36.10	-66.79	-5.20	-9.62
113 1.2DL+0.8WLX1	37.7	-4.7	0.0	-3.6	0.0	154.74	-4.70	-8.70	-3.60	-6.66
114 1.2DL+0.8WLX2	35.5	-2.2	0.0	-3.6	0.0	152.54	-2.20	-4.07	-3.60	-6.66
115 1.2DL+0.8WLZ1	35.4	-2.3	0.0	-5.0	0.0	152.44	-2.30	-4.26	-5.00	-9.25
116 1.2DL+0.8WLZ2	37.8	-4.6	0.0	-2.2	0.0	154.84	-4.60	-8.51	-2.20	-4.07
117 1.2DL+1.6WLX1+0.5LL	40.0	-5.8	0.0	-4.0	0.0	157.04	-5.80	-10.73	-4.00	-7.40
118 1.2DL+1.6WLX2+0.5LL	35.5	-0.8	0.0	-3.9	0.0	152.54	-0.80	-1.48	-3.90	-7.22
119 1.2DL+1.6WLZ1+0.5LL	35.3	-1.1	0.0	-6.7	0.0	152.34	-1.10	-2.04	-6.70	-12.40
120 1.2DL+1.6WLZ2+0.5LL	40.2	-5.5	0.0	-1.1	0.0	157.24	-5.50	-10.18	-1.10	-2.04
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	40.4	-4.9	0.0	-3.9	0.0	157.44	-4.90	-9.07	-3.90	-7.22
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	35.2	-1.8	0.0	-3.9	0.0	152.24	-1.80	-3.33	-3.90	-7.22
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	38.1	-3.2	0.0	-4.3	0.0	155.14	-3.20	-5.92	-4.30	-7.96
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	37.4	-3.5	0.0	-3.6	0.0	154.44	-3.50	-6.48	-3.60	-6.66
125 0.9DL+1.6WLX1	29.7	-5.1	0.0	-2.7	0.0	146.74	-5.10	-9.44	-2.70	-5.00
126 0.9DL+1.6WLX2	25.2	-0.1	0.0	-2.7	0.0	142.24	-0.10	-0.19	-2.70	-5.00
127 0.9DL+1.6WLZ1	25.0	-0.4	0.0	-5.5	0.0	142.04	-0.40	-0.74	-5.50	-10.18
128 0.9DL+1.6WLZ2	29.9	-4.8	0.0	0.1	0.0	146.94	-4.80	-8.88	0.10	0.19
129 0.9DL+1.0SLX1	30.1	-4.2	0.0	-2.7	0.0	147.14	-4.20	-7.77	-2.70	-5.00
130 0.9DL+1.0SLX2	24.8	-1.0	0.0	-2.7	0.0	141.84	-1.00	-1.85	-2.70	-5.00
131 0.9DL+1.0SLZ1	27.8	-2.4	0.0	-3.0	0.0	144.84	-2.40	-4.44	-3.00	-5.55
132 0.9DL+1.0SLZ2	27.1	-2.8	0.0	-2.4	0.0	144.14	-2.80	-5.18	-2.40	-4.44

Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL+1.0FL1		147.6	-3.0	-5.6	-3.0	-5.6	34.6	20.0	19.7	19.7
202 1.0DL+1.0FL2		147.6	-3.0	-5.6	-3.0	-5.6	34.6	20.0	19.7	19.7
203 1.0DL+1.0FL1+1.0LL+1.0TL		150.9	-27.6	-51.1	-3.9	-7.2	3.8	15.7	2.2	15.5
204 1.0DL+1.0FL2+1.0LL+1.0TL		150.9	-27.6	-51.1	-3.9	-7.2	3.8	15.7	2.2	15.5
205 1.0DL+1.0FL1+0.75LL+0.75TL		150.0	-21.4	-39.6	-3.7	-6.8	4.9	16.4	2.8	16.2
206 1.0DL+1.0FL2+0.75LL+0.75TL		150.1	-21.4	-39.6	-3.7	-6.8	4.9	16.5	2.8	16.2
207 1.0DL+1.0FL1+1.0PL		193.3	-5.5	-10.2	-3.2	-5.9	24.7	24.5	14.1	24.2
208 1.0DL+1.0FL2+1.0PL		193.4	-5.6	-10.4	-3.2	-5.9	24.3	24.5	13.8	24.2

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209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	196.6	-30.1	-55.7	-4.1	-7.6	4.6	19.4	2.6	19.2
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	196.6	-30.2	-55.9	-4.1	-7.6	4.6	19.4	2.6	19.2
211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	195.8	-24.0	-44.4	-3.9	-7.2	5.7	20.4	3.3	20.1
212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	195.8	-24.0	-44.4	-3.9	-7.2	5.7	20.4	3.3	20.1
213 1.0DL+1.0FL1+1.0WLX1	149.0	-4.5	-8.3	-3.0	-5.6	23.3	20.1	13.2	19.9
214 1.0DL+1.0FL1+1.0WLX2	146.2	-1.4	-2.6	-3.0	-5.6	73.4	19.8	41.8	19.5
215 1.0DL+1.0FL1+1.0WLZ1	146.0	-1.6	-3.0	-4.7	-8.7	64.1	12.6	36.5	12.4
216 1.0DL+1.0FL1+1.0WLZ2	149.1	-4.4	-8.1	-1.2	-2.2	23.8	50.4	13.6	49.7
217 1.0DL+1.0FL2+1.0WLX1	149.0	-4.5	-8.3	-3.0	-5.6	23.3	20.1	13.2	19.9
218 1.0DL+1.0FL2+1.0WLX2	146.2	-1.4	-2.6	-3.0	-5.6	73.4	19.8	41.8	19.5
219 1.0DL+1.0FL2+1.0WLZ1	146.1	-1.6	-3.0	-4.7	-8.7	64.2	12.6	36.5	12.4
220 1.0DL+1.0FL2+1.0WLZ2	149.2	-4.4	-8.1	-1.2	-2.2	23.8	50.4	13.6	49.7
221 1.0DL+1.0FL1+0.7SLX1	149.4	-4.1	-7.6	-3.0	-5.6	25.6	20.2	14.6	19.9
222 1.0DL+1.0FL1+0.7SLX2	145.8	-1.9	-3.5	-3.0	-5.6	53.9	19.7	30.7	19.4
223 1.0DL+1.0FL1+0.7SLZ1	147.8	-2.9	-5.4	-3.2	-5.9	35.8	18.7	20.4	18.5
224 1.0DL+1.0FL1+0.7SLZ2	147.3	-3.1	-5.7	-2.7	-5.0	33.4	22.1	19.0	21.8
225 1.0DL+1.0FL2+0.7SLX1	149.4	-4.1	-7.6	-3.0	-5.6	25.6	20.2	14.6	19.9
226 1.0DL+1.0FL2+0.7SLX2	145.8	-1.9	-3.5	-3.0	-5.6	53.9	19.7	30.7	19.4
227 1.0DL+1.0FL2+0.7SLZ1	147.9	-2.9	-5.4	-3.2	-5.9	35.8	18.7	20.4	18.5
228 1.0DL+1.0FL2+0.7SLZ2	147.4	-3.1	-5.7	-2.7	-5.0	33.4	22.1	19.0	21.8
229 1.0DL+1.0FL1+0.75WLX1+0.75LL	150.3	-3.9	-7.2	-3.5	-6.5	27.1	17.4	15.4	17.2
230 1.0DL+1.0FL1+0.75WLX2+0.75LL	148.2	-1.6	-3.0	-3.5	-6.5	65.1	17.2	37.1	16.9
231 1.0DL+1.0FL1+0.75WLZ1+0.75LL	148.1	-1.7	-3.1	-4.8	-8.9	61.2	12.5	34.9	12.3
232 1.0DL+1.0FL1+0.75WLZ2+0.75LL	150.4	-3.8	-7.0	-2.2	-4.1	27.8	27.7	15.8	27.4
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	150.4	-3.9	-7.2	-3.5	-6.5	27.1	17.4	15.4	17.2
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	148.2	-1.6	-3.0	-3.5	-6.5	65.1	17.2	37.1	16.9
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	148.1	-1.7	-3.1	-4.8	-8.9	61.2	12.5	34.9	12.3
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	150.5	-3.8	-7.0	-2.2	-4.1	27.8	27.7	15.8	27.4
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	150.7	-3.6	-6.7	-3.5	-6.5	29.4	17.5	16.7	17.2
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	147.9	-1.9	-3.5	-3.5	-6.5	54.7	17.1	31.1	16.9
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	149.5	-2.7	-5.0	-3.7	-6.8	38.9	16.4	22.2	16.2
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	149.1	-2.8	-5.2	-3.3	-6.1	37.4	18.3	21.3	18.1
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	150.7	-3.6	-6.7	-3.5	-6.5	29.4	17.5	16.7	17.2
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	147.9	-2.0	-3.7	-3.5	-6.5	52.0	17.1	29.6	16.9
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	149.5	-2.7	-5.0	-3.7	-6.8	38.9	16.4	22.2	16.2
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	149.1	-2.9	-5.4	-3.3	-6.1	36.1	18.3	20.6	18.1
245 0.6DL+1.0WLX1	136.7	-3.3	-6.1	-1.8	-3.3	29.1	30.8	16.6	30.4
246 0.6DL+1.0WLX2	133.9	-0.2	-0.4	-1.8	-3.3	470.6	30.2	267.9	29.8
247 0.6DL+1.0WLZ1	133.8	-0.3	-0.6	-3.5	-6.5	313.5	15.5	178.5	15.3
248 0.6DL+1.0WLZ2	136.8	-3.1	-5.7	0.0	0.0	31.0	NA	17.7	NA
249 0.6DL+0.6FL1+0.7SLX1	137.2	-2.9	-5.4	-1.8	-3.3	33.3	30.9	18.9	30.5
250 0.6DL+0.6FL1+0.7SLX2	133.5	-0.7	-1.3	-1.8	-3.3	134.1	30.1	76.3	29.7
251 0.6DL+0.6FL1+0.7SLZ1	135.6	-1.7	-3.1	-2.0	-3.7	56.1	27.5	31.9	27.1
252 0.6DL+0.6FL1+0.7SLZ2	135.1	-1.9	-3.5	-1.6	-3.0	50.0	34.2	28.5	33.8
					Min.=	3.8	12.5	2.2	12.3
						OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F15

Load Combination & Calculation of Base pressure

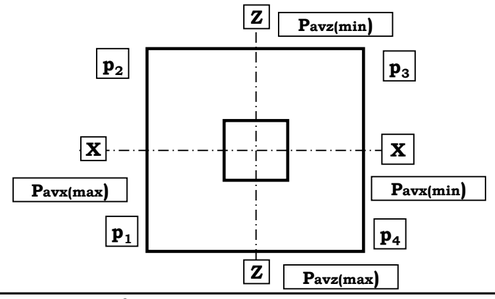
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 450 kN/m^2 For WL/SL case = 562.5 kN/m^2

Gross SBC of soil for DL, LL= 477 kN/m^2 For WL/SL case = 589.5 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	Base Pressure (kN/m^2) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	1.0DL+1.0FL1	147.64	-5.55	-5.55	37.86	3.28	5.69	46.83	35.45	40.26	28.88	19.83	41.14	34.57	43.55	32.17	O.K
202	1.0DL+1.0FL2	147.64	-5.55	-5.55	37.86	3.28	5.69	46.83	35.45	40.26	28.88	19.83	41.14	34.57	43.55	32.17	O.K
203	1.0DL+1.0FL1+1.0LL+1.0TL	150.94	-51.06	-7.22	38.70	30.21	7.40	76.32	61.52	15.89	1.09	49.32	68.92	8.49	46.11	31.31	O.K
204	1.0DL+1.0FL2+1.0LL+1.0TL	150.94	-51.06	-7.22	38.70	30.21	7.40	76.32	61.52	15.89	1.09	49.32	68.92	8.49	46.11	31.31	O.K
205	1.0DL+1.0FL1+0.75LL+0.75TL	150.04	-39.59	-6.85	38.47	23.43	7.02	68.92	54.88	22.07	8.03	41.92	61.90	15.05	45.50	31.46	O.K
206	1.0DL+1.0FL2+0.75LL+0.75TL	150.14	-39.59	-6.85	38.50	23.43	7.02	68.94	54.90	22.09	8.05	41.94	61.92	15.07	45.52	31.48	O.K
207	1.0DL+1.0FL1+1.0PL	193.34	-10.18	-5.92	49.57	6.02	6.07	61.67	49.52	49.63	37.48	34.67	55.60	43.56	55.65	43.50	O.K
208	1.0DL+1.0FL2+1.0PL	193.44	-10.36	-5.92	49.60	6.13	6.07	61.80	49.66	49.54	37.40	34.80	55.73	43.47	55.67	43.53	O.K
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0	196.64	-55.69	-7.59	50.42	32.95	7.78	91.15	75.59	25.25	9.69	64.15	83.37	17.47	58.20	42.64	O.K
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0	196.64	-55.87	-7.59	50.42	33.06	7.78	91.26	75.70	25.14	9.58	64.26	83.48	17.36	58.20	42.64	O.K
211	1.0DL+1.0FL1+0.75LL+0.75TL+1	195.84	-44.40	-7.22	50.22	26.27	7.40	83.89	69.09	31.34	16.54	56.89	76.49	23.94	57.62	42.82	O.K
212	1.0DL+1.0FL2+0.75LL+0.75TL+1	195.84	-44.40	-7.22	50.22	26.27	7.40	83.89	69.09	31.34	16.54	56.89	76.49	23.94	57.62	42.82	O.K
213	1.0DL+1.0FL1+1.0WLX1	149.04	-8.33	-5.55	38.22	4.93	5.69	48.83	37.45	38.98	27.60	21.83	43.14	33.29	43.91	32.53	O.K
214	1.0DL+1.0FL1+1.0WLX2	146.24	-2.59	-5.55	37.50	1.53	5.69	44.72	33.34	41.66	30.27	17.72	39.03	35.97	43.19	31.81	O.K
215	1.0DL+1.0FL1+1.0WLZ1	146.04	-2.96	-8.70	37.45	1.75	8.92	48.12	30.28	44.61	26.78	21.12	39.20	35.70	46.37	28.53	O.K
216	1.0DL+1.0FL1+1.0WLZ2	149.14	-8.14	-2.22	38.24	4.82	2.28	45.33	40.78	35.70	31.15	18.33	43.06	33.43	40.52	35.97	O.K
217	1.0DL+1.0FL2+1.0WLX1	149.04	-8.33	-5.55	38.22	4.93	5.69	48.83	37.45	38.98	27.60	21.83	43.14	33.29	43.91	32.53	O.K
218	1.0DL+1.0FL2+1.0WLX2	146.24	-2.59	-5.55	37.50	1.53	5.69	44.72	33.34	41.66	30.27	17.72	39.03	35.97	43.19	31.81	O.K
219	1.0DL+1.0FL2+1.0WLZ1	146.14	-2.96	-8.70	37.47	1.75	8.92	48.14	30.31	44.64	26.80	21.14	39.23	35.72	46.39	28.56	O.K

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220 1.0DL+1.0FL2+1.0WLZ2	149.24	-8.14	-2.22	38.27	4.82	2.28	45.36	40.81	35.73	31.17	18.36	43.09	33.45	40.55	35.99	O.K
221 1.0DL+1.0FL1+0.7SLX1	149.44	-7.59	-5.55	38.32	4.49	5.69	48.50	37.11	39.52	28.14	21.50	42.81	33.83	44.01	32.63	O.K
222 1.0DL+1.0FL1+0.7SLX2	145.84	-3.52	-5.55	37.39	2.08	5.69	45.17	33.78	41.01	29.62	18.17	39.48	35.32	43.09	31.70	O.K
223 1.0DL+1.0FL1+0.7SLZ1	147.84	-5.37	-5.92	37.91	3.17	6.07	47.15	35.01	40.80	28.66	20.15	41.08	34.73	43.98	31.84	O.K
224 1.0DL+1.0FL1+0.7SLZ2	147.34	-5.74	-5.00	37.78	3.39	5.12	46.30	36.05	39.51	29.26	19.30	41.18	34.39	42.91	32.66	O.K
225 1.0DL+1.0FL2+0.7SLX1	149.44	-7.59	-5.55	38.32	4.49	5.69	48.50	37.11	39.52	28.14	21.50	42.81	33.83	44.01	32.63	O.K
226 1.0DL+1.0FL2+0.7SLX2	145.84	-3.52	-5.55	37.39	2.08	5.69	45.17	33.78	41.01	29.62	18.17	39.48	35.32	43.09	31.70	O.K
227 1.0DL+1.0FL2+0.7SLZ1	147.94	-5.37	-5.92	37.93	3.17	6.07	47.18	35.04	40.83	28.69	20.18	41.11	34.76	44.01	31.87	O.K
228 1.0DL+1.0FL2+0.7SLZ2	147.44	-5.74	-5.00	37.81	3.39	5.12	46.32	36.08	39.53	29.29	19.32	41.20	34.41	42.93	32.69	O.K
229 1.0DL+1.0FL1+0.75WLX1+0.75L	150.34	-7.22	-6.48	38.55	4.27	6.64	49.46	36.18	40.92	27.64	22.46	42.82	34.28	45.19	31.91	O.K
230 1.0DL+1.0FL1+0.75WLX2+0.75L	148.24	-2.96	-6.48	38.01	1.75	6.64	46.40	33.12	42.90	29.62	19.40	39.76	36.26	44.65	31.37	O.K
231 1.0DL+1.0FL1+0.75WLZ1+0.75L	148.14	-3.15	-8.88	37.98	1.86	9.11	48.95	30.74	45.23	27.02	21.95	39.85	36.13	47.09	28.88	O.K
232 1.0DL+1.0FL1+0.75WLZ2+0.75L	150.44	-7.03	-4.07	38.57	4.16	4.17	46.91	38.56	38.59	30.24	19.91	42.74	34.42	42.75	34.40	O.K
233 1.0DL+1.0FL2+0.75WLX1+0.75L	150.44	-7.22	-6.48	38.57	4.27	6.64	49.48	36.20	40.95	27.66	22.48	42.84	34.31	45.22	31.93	O.K
234 1.0DL+1.0FL2+0.75WLX2+0.75L	148.24	-2.96	-6.48	38.01	1.75	6.64	46.40	33.12	42.90	29.62	19.40	39.76	36.26	44.65	31.37	O.K
235 1.0DL+1.0FL2+0.75WLZ1+0.75L	148.14	-3.15	-8.88	37.98	1.86	9.11	48.95	30.74	45.23	27.02	21.95	39.85	36.13	47.09	28.88	O.K
236 1.0DL+1.0FL2+0.75WLZ2+0.75L	150.54	-7.03	-4.07	38.60	4.16	4.17	46.93	38.59	38.61	30.27	19.93	42.76	34.44	42.77	34.43	O.K
237 1.0DL+1.0FL1+0.525SLX1+0.75	150.74	-6.66	-6.48	38.65	3.94	6.64	49.23	35.95	41.35	28.07	22.23	42.59	34.71	45.29	32.01	O.K
238 1.0DL+1.0FL1+0.525SLX2+0.75	147.94	-3.52	-6.48	37.93	2.08	6.64	46.65	33.37	42.49	29.21	19.65	40.01	35.85	44.57	31.29	O.K
239 1.0DL+1.0FL1+0.525SLZ1+0.75	149.54	-5.00	-6.85	38.34	2.96	7.02	48.32	34.28	42.41	28.37	21.32	41.30	35.39	45.37	31.33	O.K
240 1.0DL+1.0FL1+0.525SLZ2+0.75	149.14	-5.18	-6.11	38.24	3.07	6.26	47.57	35.04	41.44	28.91	20.57	41.31	35.18	44.51	31.98	O.K
241 1.0DL+1.0FL2+0.525SLX1+0.75	150.74	-6.66	-6.48	38.65	3.94	6.64	49.23	35.95	41.35	28.07	22.23	42.59	34.71	45.29	32.01	O.K
242 1.0DL+1.0FL2+0.525SLX2+0.75	147.94	-3.70	-6.48	37.93	2.19	6.64	46.76	33.48	42.39	29.10	19.76	40.12	35.75	44.58	31.29	O.K
243 1.0DL+1.0FL2+0.525SLZ1+0.75	149.54	-5.00	-6.85	38.34	2.96	7.02	48.32	34.28	42.41	28.37	21.32	41.30	35.39	45.37	31.33	O.K
244 1.0DL+1.0FL2+0.525SLZ2+0.75	149.14	-5.37	-6.11	38.24	3.17	6.26	47.68	35.15	41.33	28.80	20.68	41.42	35.07	44.51	31.98	O.K
245 0.6DL+1.0WLX1	136.74	-6.11	-3.33	35.06	3.61	3.42	42.09	35.26	34.86	28.03	15.09	38.68	31.45	38.48	31.65	O.K
246 0.6DL+1.0WLX2	133.94	-0.37	-3.33	34.34	0.22	3.42	37.98	31.15	37.54	30.71	10.98	34.57	34.13	37.76	30.93	O.K
247 0.6DL+1.0WLZ1	133.84	-0.56	-6.48	34.32	0.33	6.64	41.29	28.01	40.63	27.35	14.29	34.65	33.99	40.96	27.68	O.K
248 0.6DL+1.0WLZ2	136.84	-5.74	0.00	35.09	3.39	0.00	38.48	38.48	31.69	31.69	11.48	38.48	31.69	35.09	35.09	O.K
249 0.6DL+0.6FL1+0.7SLX1	137.24	-5.37	-3.33	35.19	3.17	3.42	41.78	34.95	35.43	28.60	14.78	38.37	32.02	38.61	31.78	O.K
250 0.6DL+0.6FL1+0.7SLX2	133.54	-1.30	-3.33	34.24	0.77	3.42	38.42	31.59	36.89	30.06	11.42	35.01	33.48	37.66	30.83	O.K
251 0.6DL+0.6FL1+0.7SLZ1	135.64	-3.15	-3.70	34.78	1.86	3.79	40.44	32.85	36.71	29.12	13.44	36.65	32.92	38.58	30.99	O.K
252 0.6DL+0.6FL1+0.7SLZ2	135.14	-3.52	-2.96	34.65	2.08	3.04	39.77	33.70	35.61	29.54	12.77	36.74	32.58	37.69	31.62	O.K

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101 1.4DL+1.4FL1	159.84	-7.77	-7.77	40.98	4.60	7.97	53.55	37.61	44.36	28.42	26.55	45.58	36.39	48.96	33.02	
102 1.4DL+1.4FL2	159.94	-7.77	-7.77	41.01	4.60	7.97	53.58	37.64	44.38	28.44	26.58	45.61	36.41	48.98	33.04	
103 1.2DL+1.2FL1+1.6LL	157.34	-5.74	-8.70	40.34	3.39	8.92	52.66	34.82	45.87	28.03	25.66	43.74	36.95	49.27	31.43	
104 1.2DL+1.2FL2+1.6LL	157.34	-5.74	-8.70	40.34	3.39	8.92	52.66	34.82	45.87	28.03	25.66	43.74	36.95	49.27	31.43	
105 1.2DL+1.2FL1+1.2TL+1.6LL	158.54	-61.05	-9.25	40.65	36.12	9.49	86.26	67.29	14.01	-4.96	59.26	76.78	4.53	50.14	31.17	
106 1.2DL+1.2FL2+1.2TL+1.6LL	158.64	-61.05	-9.25	40.68	36.12	9.49	86.29	67.31	14.04	-4.93	59.29	76.80	4.56	50.17	31.19	
107 1.4DL+1.4FL1+1.4PL	223.94	-14.43	-8.33	57.42	8.54	8.54	74.50	57.42	57.42	40.34	47.50	65.96	48.88	65.96	48.88	
108 1.4DL+1.4FL2+1.4PL	223.94	-14.43	-8.33	57.42	8.54	8.54	74.50	57.42	57.42	40.34	47.50	65.96	48.88	65.96	48.88	
109 1.2DL+1.2FL1+1.6LL+1.2PL	212.24	-11.47	-9.25	54.42	6.79	9.49	70.69	51.72	57.12	38.15	43.69	61.21	47.64	63.91	44.94	
110 1.2DL+1.2FL2+1.6LL+1.2PL	212.24	-11.47	-9.25	54.42	6.79	9.49	70.69	51.72	57.12	38.15	43.69	61.21	47.64	63.91	44.94	
111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2	213.44	-66.79	-9.62	54.73	39.52	9.87	104.11	84.38	25.08	5.34	77.11	94.25	15.21	64.60	44.86	
112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2	213.54	-66.79	-9.62	54.75	39.52	9.87	104.14	84.40	25.10	5.37	77.14	94.27	15.24	64.62	44.89	
113 1.2DL+0.8WLX1	154.74	-8.70	-6.66	39.68	5.14	6.83	51.65	37.99	41.36	27.70	24.65	44.82	34.53	46.51	32.85	
114 1.2DL+0.8WLX2	152.54	-4.07	-6.66	39.11	2.41	6.83	48.35	34.69	43.54	29.87	21.35	41.52	36.71	45.95	32.28	
115 1.2DL+0.8WLZ1	152.44	-4.26	-9.25	39.09	2.52	9.49	51.09	32.12	46.06	27.08	24.09	41.61	36.57	48.58	29.60	
116 1.2DL+0.8WLZ2	154.84	-8.51	-4.07	39.70	5.04	4.17	48.91	40.56	38.84	30.49	21.91	44.74	34.67	43.88	35.53	
117 1.2DL+1.6WLX1+0.5LL	157.04	-10.73	-7.40	40.27	6.35	7.59	54.21	39.03	41.51	26.33	27.21	46.62	33.92	47.86	32.68	
118 1.2DL+1.6WLX2+0.5LL	152.54	-1.48	-7.22	39.11	0.88	7.40	47.39	32.59	45.64	30.84	20.39	39.99	38.24	46.52	31.72	
119 1.2DL+1.6WLZ1+0.5LL	152.34	-2.04	-12.40	39.06	1.20	12.71	52.98	27.55	50.57	25.14	25.98	40.27	37.86	51.78	26.35	
120 1.2DL+1.6WLZ2+0.5LL	157.24	-10.18	-2.04	40.32	6.02	2.09	48.43	44.25	36.38	32.21	21.43	46.34	34.30	42.41	38.23	
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	157.44	-9.07	-7.22	40.37	5.36	7.40	53.13	38.33	42.41	27.61	26.13	45.73	35.01	47.77	32.97	
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	152.24	-3.33	-7.22	39.04	1.97	7.40	48.41	33.61	44.47	29.67	21.41	41.01	37.07	46.44	31.64	
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	155.14	-5.92	-7.96	39.78	3.50	8.16	51.44	35.12	44.44	28.12	24.44	43.28	36.28	47.94	31.62	
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	154.44	-6.48	-6.66	39.60	3.83	6.83	50.26	36.60	42.60	28.94	23.26	43.43	35.77	46.43	32.77	
125 0.9DL+1.6WLX1	146.74	-9.44	-5.00	37.63	5.58	5.12	48.33	38.09	37.17	26.92	21.33	43.21	32.05	42.75	32.51	
126 0.9DL+1.6WLX2	142.24	-0.19	-5.00	36.47	0.11	5.12	41.70	31.46	41.49	31.24	14.70	36.58	36.37	41.60	31.35	
127 0.9DL+1.6WLZ1	142.04	-0.74	-10.18	36.42	0.44	10.44	47.29	26.42	46.42	25.55	20.29	36.86	35.99	46.86	25.99	
128 0.9DL+1.6WLZ2	146.94	-8.88	0.19	37.68	5.25	0.19	43.12	42.74	32.61	32.23	16.12	42.93	32.42	37.87	37.49	
129 0.9DL+1.0SLX1	147.14	-7.77	-5.00	37.73	4.60	5.12	47.45	37.20	38.25	28.01	20.45	42.33	33.13	42.85	32.61	
130 0.9DL+1.0SLX2	141.84	-1.85	-5.00	36.37	1.09	5.12	42.59	32.34	40.40	30.15	15.59	37.47	35.28	41.50	31.25	
131 0.9DL+1.0SLZ1	144.84	-4.44	-5.55	37.14	2.63	5.69	45.46	34.07	40.20	28.82	18.46	39.77	34.51	42.83	31.45	
132 0.9DL+1.0SLZ2	144.14	-5.18	-4.44	36.96	3.07	4.55	44.58	35.47	38.45	29.34	17.58	40.03	33.90	41.52	32.41	
Max :	224	0	0													
Min :	134	-67	-12													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 19.83 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 29.83 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 64.26 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 74.26 kN/m²

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Sample calculation of values of base pressure:

Load case : 202 1.0DL+1.0FL2

Maximum pressure, p_1 (pmax) = 37.86+3.28+5.69= 46.83 kN/m^2

p_2= 37.86+3.28-5.69= 35.45 kN/m^2

p_3= 37.86-(3.28+5.69)= 28.88 kN/m^2

p_4= 37.86-3.28+5.69= 40.26 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 46.83-1.5*18= 19.83 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (46.83+35.45)/2= 41.14 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (28.88+40.26)/2= 34.57 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (46.83+40.26)/2= 43.55 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (28.88+35.45)/2= 32.16 kN/m^2

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DESIGN OF ISOLATED FOOTING F15

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.55+2*0.5*0.232)+(0.55+2*0.5*0.232))*2

= 3.128 m

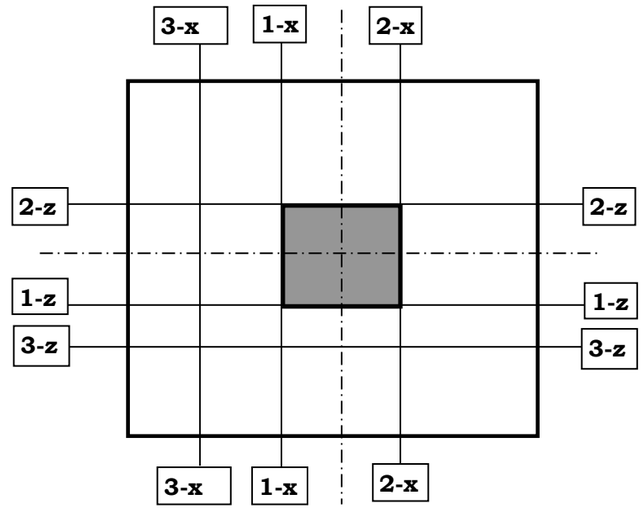
Punching Area= ((0.55+2*0.5*0.232)*(0.55+2*0.5*0.232))

= 0.61 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.3*24+1.2*18)

= 28.80 kN/m^2

Area of footing, A_f= 3.90 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4DL+1.4FL1	41.96	40.01	42.82	8.18	4.62	12.03	40.09
102	1.4DL+1.4FL2	41.98	40.04	42.85	8.20	4.63	12.05	40.17
103	1.2DL+1.2FL1+1.6LL	41.06	39.63	41.70	7.38	4.75	10.87	37.98
104	1.2DL+1.2FL2+1.6LL	41.06	39.63	41.70	7.38	4.75	10.87	37.98
105	1.2DL+1.2FL1+1.2TL+1.6LL	48.30	33.01	55.08	20.22	-7.76	29.00	38.99
106	1.2DL+1.2FL2+1.2TL+1.6LL	48.32	33.04	55.10	20.23	-7.75	29.01	39.08
107	1.4DL+1.4FL1+1.4PL	59.23	55.61	60.83	18.34	11.73	27.02	94.16
108	1.4DL+1.4FL2+1.4PL	59.23	55.61	60.83	18.34	11.73	27.02	94.16
109	1.2DL+1.2FL1+1.6LL+1.2PL	55.86	52.99	57.13	16.09	10.83	23.72	84.29
110	1.2DL+1.2FL2+1.6LL+1.2PL	55.86	52.99	57.13	16.09	10.83	23.72	84.29
111	1.2DL+1.2FL1+1.2TL+1.6LL+1	63.09	46.37	70.51	28.93	-1.68	41.85	85.30
112	1.2DL+1.2FL2+1.2TL+1.6LL+1	63.11	46.40	70.53	28.94	-1.67	41.86	85.39
113	1.2DL+0.8WLX1	40.76	38.59	41.73	7.71	3.72	11.30	35.79

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114 1.2DL+0.8WLX2	39.62	38.61	40.08	6.35	4.49	9.37	33.93
115 1.2DL+0.8WLZ1	39.62	38.56	40.10	6.38	4.43	9.41	33.84
116 1.2DL+0.8WLZ2	40.77	38.64	41.72	7.68	3.78	11.27	35.87
117 1.2DL+1.6WLX1+0.5LL	41.61	38.93	42.81	8.48	3.57	12.43	37.73
118 1.2DL+1.6WLX2+0.5LL	39.30	38.93	39.46	5.76	5.08	8.53	33.93
119 1.2DL+1.6WLZ1+0.5LL	39.32	38.81	39.55	5.86	4.93	8.68	33.76
120 1.2DL+1.6WLZ2+0.5LL	41.59	39.05	42.72	8.38	3.72	12.29	37.89
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	41.50	39.24	42.51	8.15	4.00	11.96	38.06
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	39.46	38.62	39.83	6.14	4.62	9.07	33.68
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	40.52	39.04	41.18	7.12	4.41	10.49	36.12
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	40.41	38.79	41.13	7.16	4.19	10.53	35.53
125 0.9DL+1.6WLX1	38.81	36.45	39.86	6.80	2.48	9.95	29.04
126 0.9DL+1.6WLX2	36.50	36.45	36.52	4.07	3.99	6.05	25.24
127 0.9DL+1.6WLZ1	36.52	36.33	36.60	4.17	3.84	6.19	25.07
128 0.9DL+1.6WLZ2	38.79	36.56	39.77	6.70	2.63	9.80	29.21
129 0.9DL+1.0SLX1	38.70	36.76	39.57	6.47	2.91	9.49	29.37
130 0.9DL+1.0SLX2	36.61	36.14	36.81	4.40	3.56	6.51	24.90
131 0.9DL+1.0SLZ1	37.70	36.58	38.19	5.40	3.36	7.95	27.43
132 0.9DL+1.0SLZ2	37.61	36.32	38.19	5.48	3.10	8.05	26.84

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Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1x} = 45.58-(45.58-36.39)*1.025/2.6 = 41.96 kN/m^2 Overhang = 1.025 m

p_{2x} = 36.39+(45.58-36.39)*1.025/2.6 = 40.01 kN/m^2 Effective depth = 0.244 m

p_{3x} = 45.58-(45.58-36.39)*(1.025-0.244)/2.6 = 42.82 kN/m^2

BM_{1x} = (41.96-28.8)*1.025*1.025/2+0.5*(45.58-41.96)*1.025*1.025*0.67 = 8.18 kN-m/m

BM_{2x} = (40.01-28.8)*1.025*1.025/2-0.5*(-36.39+40.01)*1.025*1.025*0.67 = 4.62 kN-m/m

SF_{3x} = (42.82-28.8)*(1.025-0.244)+0.5*(45.58-42.82)*(1.025-0.244) = 12.03 kN/m

$S.F_{(punching)}$ = (3.9-0.61)*(40.98-28.8) = 40.09 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	28.94	0.49	0.65	0.002	380.68	439.20	12ϕT	257.51	300	200	565.49
At top	-7.76	-0.13	0.17	0.000	101.28	219.60	12ϕT	515.02	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 41.86 kN/m

Shear stress , τ_v= (41.86*1000)/(1000*244)

= 0.17 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 94.16 kN

Strength of Concrete in shear Vc1 = 2257.67 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1339.59 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1505.11 kN **Table 22.6.5.2_ (c)**

Design Vc= 1339.59 kN

Allowable shear stress, (ϕ Vc)/bd = 1.32 N/mm^2

Design punching shear stress, τ = (94.16*1000)/(3128*244)

= 0.12 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4DL+1.4FL1	43.91	38.07	46.38	2.09	0.66	4.59
102	1.4DL+1.4FL2	43.93	40.20	46.40	2.09	1.54	4.59
103	1.2DL+1.2FL1+1.6LL	43.62	39.11	46.38	2.10	1.50	4.62
104	1.2DL+1.2FL2+1.6LL	43.62	39.91	46.38	2.10	1.46	4.62
105	1.2DL+1.2FL1+1.2TL+1.6LL	44.13	39.73	47.07	2.18	1.45	4.81
106	1.2DL+1.2FL2+1.2TL+1.6LL	44.16	39.56	47.10	2.19	1.46	4.82
107	1.4DL+1.4FL1+1.4PL	60.55	51.65	63.19	3.99	1.91	8.69
108	1.4DL+1.4FL2+1.4PL	60.55	47.31	63.19	3.99	0.97	8.69
109	1.2DL+1.2FL1+1.6LL+1.2PL	57.90	44.39	60.84	3.74	0.81	8.16
110	1.2DL+1.2FL2+1.6LL+1.2PL	57.90	45.39	60.84	3.74	0.60	8.16
111	1.2DL+1.2FL1+1.2TL+1.6LL+1	58.35	43.49	61.40	3.81	1.09	8.31
112	1.2DL+1.2FL2+1.2TL+1.6LL+1	58.37	44.58	61.42	3.81	0.87	8.32
113	1.2DL+0.8WLX1	42.18	36.12	44.30	1.84	0.50	4.03
114	1.2DL+0.8WLX2	41.62	36.75	43.74	1.77	0.28	3.90
115	1.2DL+0.8WLZ1	42.57	33.07	45.51	2.01	0.70	4.43
116	1.2DL+0.8WLZ2	41.24	38.22	42.53	1.60	0.64	3.50
117	1.2DL+1.6WLX1+0.5LL	43.05	35.98	45.40	1.97	0.49	4.33
118	1.2DL+1.6WLX2+0.5LL	41.83	35.60	44.12	1.82	0.48	4.01
119	1.2DL+1.6WLZ1+0.5LL	43.73	31.59	47.66	2.29	0.40	5.08
120	1.2DL+1.6WLZ2+0.5LL	41.09	40.06	41.73	1.49	0.71	3.22

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121 1.2DL+1.0SLX1+0.5LL+0.5FL1	43.08	36.17	45.37	1.97	0.49	4.32
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	41.75	35.56	44.04	1.82	0.48	4.00
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	42.77	35.20	45.30	1.97	0.53	4.33
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	42.10	36.70	44.22	1.83	0.53	4.02
125 0.9DL+1.6WLX1	39.51	36.35	41.09	1.45	0.48	3.19
126 0.9DL+1.6WLX2	38.35	36.33	39.94	1.32	0.29	2.91
127 0.9DL+1.6WLZ1	40.25	31.30	43.48	1.79	0.52	3.98
128 0.9DL+1.6WLZ2	37.75	39.94	37.81	1.02	0.65	2.20
129 0.9DL+1.0SLX1	39.61	36.42	41.19	1.46	0.48	3.21
130 0.9DL+1.0SLX2	38.25	36.27	39.84	1.31	0.28	2.88
131 0.9DL+1.0SLZ1	39.23	35.03	40.99	1.45	0.66	3.19
132 0.9DL+1.0SLZ2	38.64	36.49	40.04	1.33	0.53	2.91

Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1z} = 48.96-(48.96-33.02)*0.475/1.5= 43.91 kN/m^2

p_{2z} = 33.02+(48.96-33.02)*0.475/1.5= 38.07 kN/m^2

p_{3z} = 48.96-(48.96-33.02)*(0.475-0.232)/1.5= 46.38 kN/m^2

Overhang = 0.475 m

Effective depth = 0.232 m

BM_{1z} = (43.91-28.8)*0.475*0.475/2+0.5*(48.96-43.91)*0.475*0.475*0.67 = 2.09 kN-m/m

BM_{2z} = (38.07-28.8)*0.475*0.475/2-0.5*(-33.02+38.07)*0.475*0.475*0.67 = 0.66 kN-m/m

SF_{3z} = (46.38-28.8)*(0.475-0.232)+0.5*(48.96-46.38)*(0.475-0.232) = 4.59 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	3.99	0.074	0.099	0.0002	57.53	417.60	12ϕT	270.83	300	200	565.5
At top	0.00	0.000	0.000	0.0000	0.00	208.80	12ϕT	541.65	250	250	452.4

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Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 8.69 kN/m

Shear stress , τ_v= (8.69*1000)/(1000*232)

= 0.04 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	12ϕT	**200**	c/c
	At Top	12ϕT	**250**	c/c
Along Z Direction :				
	At Bottom	12ϕT	**200**	c/c
	At Top	12ϕT	**250**	c/c

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5.8 DESIGN OF CALCULATION OF FOOTING F16

(STAAD Node -7428)

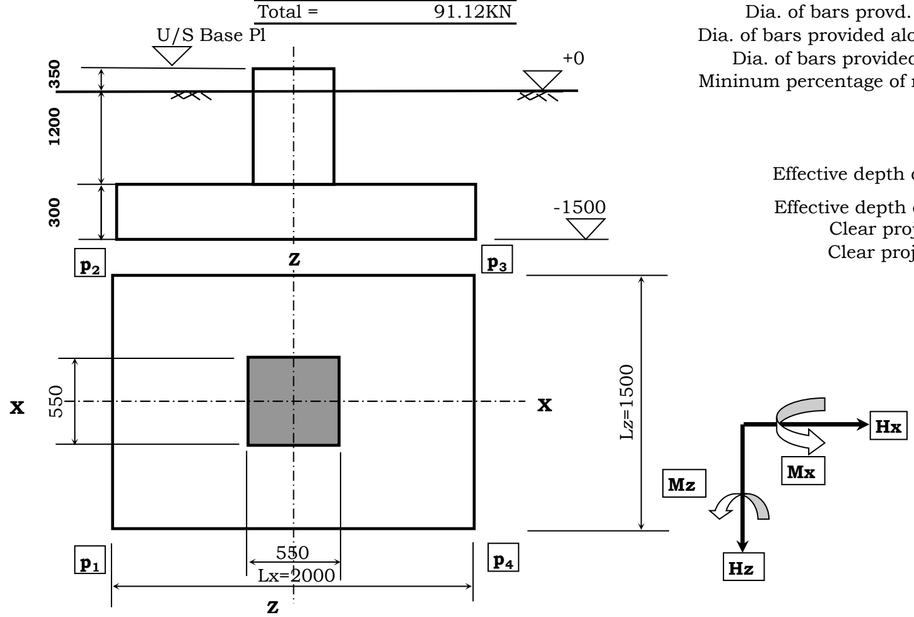
	Along-X	Along-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	2.00m	1.50m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.35 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.= (2*1.5*0.3*24)+(0.55*0.55*1.55*24) = 32.85KN
Weight of soil = ((2*1.5)-(0.55*0.55))*1.2*18 = 58.27KN
Total = 91.12KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 244mm = 0.244m

Effective depth of slab along Z direction = 232mm = 0.232m
Clear projection along Z direction = 0.475m
Clear projection along X direction = 0.725m

H$_X$=Force along X-X direction

Hz=Force along Z-Z direction

M$_X$=Moment about X-X direction

Mz=Moment about Z-Z direction

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CHECK FOR OVERTURNING AND SLIDING OF FOOTING **F16**

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	1.0DL+1.0FL1	38.9	0.0	0.0	-0.3	0.0	130.02	0.00	0.00	-0.30	-0.56
202	1.0DL+1.0FL2	37.8	0.0	0.0	-0.3	0.0	128.92	0.00	0.00	-0.30	-0.56
203	1.0DL+1.0FL1+1.0LL+1.0TL	100.9	0.3	0.0	-1.5	0.0	192.02	0.30	0.56	-1.50	-2.78
204	1.0DL+1.0FL2+1.0LL+1.0TL	99.9	0.3	0.0	-1.5	0.0	191.02	0.30	0.56	-1.50	-2.78
205	1.0DL+1.0FL1+0.75LL+0.75TL	85.4	0.2	0.0	-1.2	0.0	176.52	0.20	0.37	-1.20	-2.22
206	1.0DL+1.0FL2+0.75LL+0.75TL	84.4	0.2	0.0	-1.2	0.0	175.52	0.20	0.37	-1.20	-2.22
207	1.0DL+1.0FL1+1.0PL	141.0	0.0	0.0	0.4	0.0	232.12	0.00	0.00	0.40	0.74
208	1.0DL+1.0FL2+1.0PL	139.9	0.0	0.0	0.4	0.0	231.02	0.00	0.00	0.40	0.74
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	203.0	0.3	0.0	-0.9	0.0	294.12	0.30	0.56	-0.90	-1.67
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	201.9	0.3	0.0	-0.9	0.0	293.02	0.30	0.56	-0.90	-1.67
211	1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	187.5	0.3	0.0	-0.6	0.0	278.62	0.30	0.56	-0.60	-1.11
212	1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	186.4	0.3	0.0	-0.6	0.0	277.52	0.30	0.56	-0.60	-1.11
213	1.0DL+1.0FL1+1.0WLX1	-10.6	-10.9	0.0	-0.2	0.0	80.52	-10.90	-20.17	-0.20	-0.37
214	1.0DL+1.0FL1+1.0WLX2	88.4	10.9	0.0	-0.3	0.0	179.52	10.90	20.17	-0.30	-0.56
215	1.0DL+1.0FL1+1.0WLZ1	88.8	0.0	0.0	-0.1	0.0	179.92	0.00	0.00	-0.10	-0.19
216	1.0DL+1.0FL1+1.0WLZ2	-11.0	0.0	0.0	-0.4	0.0	80.12	0.00	0.00	-0.40	-0.74
217	1.0DL+1.0FL2+1.0WLX1	-11.6	-10.9	0.0	-0.2	0.0	79.52	-10.90	-20.17	-0.20	-0.37
218	1.0DL+1.0FL2+1.0WLX2	87.3	10.9	0.0	-0.3	0.0	178.42	10.90	20.17	-0.30	-0.56
219	1.0DL+1.0FL2+1.0WLZ1	87.8	0.0	0.0	-0.1	0.0	178.92	0.00	0.00	-0.10	-0.19
220	1.0DL+1.0FL2+1.0WLZ2	-12.1	0.0	0.0	-0.4	0.0	79.02	0.00	0.00	-0.40	-0.74
221	1.0DL+1.0FL1+0.7SLX1	30.3	-1.8	0.0	-0.2	0.0	121.42	-1.80	-3.33	-0.20	-0.37
222	1.0DL+1.0FL1+0.7SLX2	47.5	1.8	0.0	-0.3	0.0	138.62	1.80	3.33	-0.30	-0.56
223	1.0DL+1.0FL1+0.7SLZ1	52.4	0.0	0.0	-0.2	0.0	143.52	0.00	0.00	-0.20	-0.37
224	1.0DL+1.0FL1+0.7SLZ2	25.4	0.0	0.0	-0.3	0.0	116.52	0.00	0.00	-0.30	-0.56
225	1.0DL+1.0FL2+0.7SLX1	29.3	-1.8	0.0	-0.2	0.0	120.42	-1.80	-3.33	-0.20	-0.37
226	1.0DL+1.0FL2+0.7SLX2	46.4	1.8	0.0	-0.3	0.0	137.52	1.80	3.33	-0.30	-0.56
227	1.0DL+1.0FL2+0.7SLZ1	51.3	0.0	0.0	-0.2	0.0	142.42	0.00	0.00	-0.20	-0.37
228	1.0DL+1.0FL2+0.7SLZ2	24.4	0.0	0.0	-0.3	0.0	115.52	0.00	0.00	-0.30	-0.56
229	1.0DL+1.0FL1+0.75WLX1+0.75LL	52.0	-8.2	0.0	-0.6	0.0	143.12	-8.20	-15.17	-0.60	-1.11
230	1.0DL+1.0FL1+0.75WLX2+0.75LL	126.2	8.2	0.0	-0.7	0.0	217.32	8.20	15.17	-0.70	-1.30
231	1.0DL+1.0FL1+0.75WLZ1+0.75LL	126.6	0.0	0.0	-0.5	0.0	217.72	0.00	0.00	-0.50	-0.93

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232 1.0DL+1.0FL1+0.75WLZ2+0.75LL	51.7	0.0	0.0	-0.8	0.0	142.82	0.00	0.00	-0.80	-1.48
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	51.0	-8.2	0.0	-0.6	0.0	142.12	-8.20	-15.17	-0.60	-1.11
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	125.2	8.2	0.0	-0.7	0.0	216.32	8.20	15.17	-0.70	-1.30
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	125.5	0.0	0.0	-0.5	0.0	216.62	0.00	0.00	-0.50	-0.93
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	50.6	0.0	0.0	-0.8	0.0	141.72	0.00	0.00	-0.80	-1.48
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	82.7	-1.3	0.0	-0.6	0.0	173.82	-1.30	-2.41	-0.60	-1.11
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	95.6	1.3	0.0	-0.7	0.0	186.72	1.30	2.41	-0.70	-1.30
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	99.3	0.0	0.0	-0.6	0.0	190.42	0.00	0.00	-0.60	-1.11
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	79.0	0.0	0.0	-0.7	0.0	170.12	0.00	0.00	-0.70	-1.30
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	81.7	-1.3	0.0	-0.6	0.0	172.82	-1.30	-2.41	-0.60	-1.11
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	94.5	1.3	0.0	-0.7	0.0	185.62	1.30	2.41	-0.70	-1.30
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	98.2	0.0	0.0	-0.6	0.0	189.32	0.00	0.00	-0.60	-1.11
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	78.0	0.0	0.0	-0.7	0.0	169.12	0.00	0.00	-0.70	-1.30
245 0.6DL+1.0WLX1	-23.3	-10.9	0.0	-0.1	0.0	67.82	-10.90	-20.17	-0.10	-0.19
246 0.6DL+1.0WLX2	75.6	10.9	0.0	-0.2	0.0	166.72	10.90	20.17	-0.20	-0.37
247 0.6DL+1.0WLZ1	76.1	0.0	0.0	0.0	0.0	167.22	0.00	0.00	0.00	0.00
248 0.6DL+1.0WLZ2	-23.8	0.0	0.0	-0.3	0.0	67.32	0.00	0.00	-0.30	-0.56
249 0.6DL+0.6FL1+0.7SLX1	14.8	-1.8	0.0	-0.1	0.0	105.92	-1.80	-3.33	-0.10	-0.19
250 0.6DL+0.6FL1+0.7SLX2	31.9	1.8	0.0	-0.2	0.0	123.02	1.80	3.33	-0.20	-0.37
251 0.6DL+0.6FL1+0.7SLZ1	36.8	0.0	0.0	-0.1	0.0	127.92	0.00	0.00	-0.10	-0.19
252 0.6DL+0.6FL1+0.7SLZ2	9.8	0.0	0.0	-0.2	0.0	100.92	0.00	0.00	-0.20	-0.37
Ultimate Conditions										
101 1.4DL+1.4FL1	54.4	0.0	0.0	-0.4	0.0	145.52	0.00	0.00	-0.40	-0.74
102 1.4DL+1.4FL2	53.0	0.0	0.0	-0.4	0.0	144.12	0.00	0.00	-0.40	-0.74
103 1.2DL+1.2FL1+1.6LL	153.9	0.0	0.0	-1.1	0.0	245.02	0.00	0.00	-1.10	-2.04
104 1.2DL+1.2FL2+1.6LL	152.6	0.0	0.0	-1.2	0.0	243.72	0.00	0.00	-1.20	-2.22
105 1.2DL+1.2FL1+1.2TL+1.6LL	147.9	0.4	0.0	-2.0	0.0	239.02	0.40	0.74	-2.00	-3.70
106 1.2DL+1.2FL2+1.2TL+1.6LL	146.7	0.4	0.0	-2.1	0.0	237.82	0.40	0.74	-2.10	-3.89
107 1.4DL+1.4FL1+1.4PL	197.3	0.0	0.0	0.5	0.0	288.42	0.00	0.00	0.50	0.93
108 1.4DL+1.4FL2+1.4PL	195.9	0.0	0.0	0.5	0.0	287.02	0.00	0.00	0.50	0.93
109 1.2DL+1.2FL1+1.6LL+1.2PL	276.4	0.0	0.0	-0.4	0.0	367.52	0.00	0.00	-0.40	-0.74
110 1.2DL+1.2FL2+1.6LL+1.2PL	275.1	0.0	0.0	-0.4	0.0	366.22	0.00	0.00	-0.40	-0.74

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111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	270.4	0.4	0.0	-1.3	0.0	361.52	0.40	0.74	-1.30	-2.41
112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	269.1	0.4	0.0	-1.3	0.0	360.22	0.40	0.74	-1.30	-2.41
113 1.2DL+0.8WLX1	12.7	-8.7	0.0	-0.2	0.0	103.82	-8.70	-16.10	-0.20	-0.37
114 1.2DL+0.8WLX2	91.9	8.7	0.0	-0.3	0.0	183.02	8.70	16.10	-0.30	-0.56
115 1.2DL+0.8WLZ1	92.2	0.0	0.0	-0.2	0.0	183.32	0.00	0.00	-0.20	-0.37
116 1.2DL+0.8WLZ2	12.3	0.0	0.0	-0.4	0.0	103.42	0.00	0.00	-0.40	-0.74
117 1.2DL+1.6WLX1+0.5LL	6.6	-17.4	0.0	-0.5	0.0	97.72	-17.40	-32.19	-0.50	-0.93
118 1.2DL+1.6WLX2+0.5LL	164.9	17.4	0.0	-0.6	0.0	256.02	17.40	32.19	-0.60	-1.11
119 1.2DL+1.6WLZ1+0.5LL	165.7	0.0	0.0	-0.3	0.0	256.82	0.00	0.00	-0.30	-0.56
120 1.2DL+1.6WLZ2+0.5LL	5.9	0.0	0.0	-0.8	0.0	97.02	0.00	0.00	-0.80	-1.48
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	71.2	-2.5	0.0	-0.5	0.0	162.32	-2.50	-4.63	-0.50	-0.93
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	95.7	2.5	0.0	-0.6	0.0	186.82	2.50	4.63	-0.60	-1.11
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	102.7	0.0	0.0	-0.5	0.0	193.82	0.00	0.00	-0.50	-0.93
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	64.2	0.0	0.0	-0.6	0.0	155.32	0.00	0.00	-0.60	-1.11
125 0.9DL+1.6WLX1	-39.9	-17.4	0.0	-0.1	0.0	51.22	-17.40	-32.19	-0.10	-0.19
126 0.9DL+1.6WLX2	118.4	17.4	0.0	-0.3	0.0	209.52	17.40	32.19	-0.30	-0.56
127 0.9DL+1.6WLZ1	119.1	0.0	0.0	0.0	0.0	210.22	0.00	0.00	0.00	0.00
128 0.9DL+1.6WLZ2	-40.7	0.0	0.0	-0.4	0.0	50.42	0.00	0.00	-0.40	-0.74
129 0.9DL+1.0SLX1	27.0	-2.5	0.0	-0.2	0.0	118.12	-2.50	-4.63	-0.20	-0.37
130 0.9DL+1.0SLX2	51.4	2.5	0.0	-0.2	0.0	142.52	2.50	4.63	-0.20	-0.37
131 0.9DL+1.0SLZ1	58.5	0.0	0.0	-0.2	0.0	149.62	0.00	0.00	-0.20	-0.37
132 0.9DL+1.0SLZ2	19.9	0.0	0.0	-0.2	0.0	111.02	0.00	0.00	-0.20	-0.37

Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	1.0DL+1.0FL1	130.0	0.0	0.0	-0.3	-0.6	NA	175.7	NA	173.4
202	1.0DL+1.0FL2	128.9	0.0	0.0	-0.3	-0.6	NA	174.2	NA	171.9
203	1.0DL+1.0FL1+1.0LL+1.0TL	192.0	0.3	0.6	-1.5	-2.8	449.8	51.9	256.0	51.2
204	1.0DL+1.0FL2+1.0LL+1.0TL	191.0	0.3	0.6	-1.5	-2.8	447.4	51.6	254.7	50.9
205	1.0DL+1.0FL1+0.75LL+0.75TL	176.5	0.2	0.4	-1.2	-2.2	620.2	59.6	353.0	58.8
206	1.0DL+1.0FL2+0.75LL+0.75TL	175.5	0.2	0.4	-1.2	-2.2	616.7	59.3	351.0	58.5
207	1.0DL+1.0FL1+1.0PL	232.1	0.0	0.0	0.4	0.7	NA	235.3	NA	232.1
208	1.0DL+1.0FL2+1.0PL	231.0	0.0	0.0	0.4	0.7	NA	234.1	NA	231.0

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209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	294.1	0.3	0.6	-0.9	-1.7	688.9	132.5	392.2	130.7
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	293.0	0.3	0.6	-0.9	-1.7	686.4	132.0	390.7	130.2
211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	278.6	0.3	0.6	-0.6	-1.1	652.6	188.3	371.5	185.7
212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	277.5	0.3	0.6	-0.6	-1.1	650.0	187.5	370.0	185.0
213 1.0DL+1.0FL1+1.0WLX1	80.5	-10.9	-20.2	-0.2	-0.4	5.2	163.2	3.0	161.0
214 1.0DL+1.0FL1+1.0WLX2	179.5	10.9	20.2	-0.3	-0.6	11.6	242.6	6.6	239.4
215 1.0DL+1.0FL1+1.0WLZ1	179.9	0.0	0.0	-0.1	-0.2	NA	729.4	NA	719.7
216 1.0DL+1.0FL1+1.0WLZ2	80.1	0.0	0.0	-0.4	-0.7	NA	81.2	NA	80.1
217 1.0DL+1.0FL2+1.0WLX1	79.5	-10.9	-20.2	-0.2	-0.4	5.1	161.2	2.9	159.0
218 1.0DL+1.0FL2+1.0WLX2	178.4	10.9	20.2	-0.3	-0.6	11.5	241.1	6.5	237.9
219 1.0DL+1.0FL2+1.0WLZ1	178.9	0.0	0.0	-0.1	-0.2	NA	725.4	NA	715.7
220 1.0DL+1.0FL2+1.0WLZ2	79.0	0.0	0.0	-0.4	-0.7	NA	80.1	NA	79.0
221 1.0DL+1.0FL1+0.7SLX1	121.4	-1.8	-3.3	-0.2	-0.4	47.4	246.1	27.0	242.8
222 1.0DL+1.0FL1+0.7SLX2	138.6	1.8	3.3	-0.3	-0.6	54.1	187.3	30.8	184.8
223 1.0DL+1.0FL1+0.7SLZ1	143.5	0.0	0.0	-0.2	-0.4	NA	290.9	NA	287.0
224 1.0DL+1.0FL1+0.7SLZ2	116.5	0.0	0.0	-0.3	-0.6	NA	157.5	NA	155.4
225 1.0DL+1.0FL2+0.7SLX1	120.4	-1.8	-3.3	-0.2	-0.4	47.0	244.1	26.8	240.8
226 1.0DL+1.0FL2+0.7SLX2	137.5	1.8	3.3	-0.3	-0.6	53.7	185.8	30.6	183.4
227 1.0DL+1.0FL2+0.7SLZ1	142.4	0.0	0.0	-0.2	-0.4	NA	288.7	NA	284.8
228 1.0DL+1.0FL2+0.7SLZ2	115.5	0.0	0.0	-0.3	-0.6	NA	156.1	NA	154.0
229 1.0DL+1.0FL1+0.75WLX1+0.75LL	143.1	-8.2	-15.2	-0.6	-1.1	12.3	96.7	7.0	95.4
230 1.0DL+1.0FL1+0.75WLX2+0.75LL	217.3	8.2	15.2	-0.7	-1.3	18.6	125.9	10.6	124.2
231 1.0DL+1.0FL1+0.75WLZ1+0.75LL	217.7	0.0	0.0	-0.5	-0.9	NA	176.5	NA	174.2
232 1.0DL+1.0FL1+0.75WLZ2+0.75LL	142.8	0.0	0.0	-0.8	-1.5	NA	72.4	NA	71.4
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	142.1	-8.2	-15.2	-0.6	-1.1	12.2	96.0	6.9	94.7
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	216.3	8.2	15.2	-0.7	-1.3	18.5	125.3	10.6	123.6
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	216.6	0.0	0.0	-0.5	-0.9	NA	175.6	NA	173.3
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	141.7	0.0	0.0	-0.8	-1.5	NA	71.8	NA	70.9
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	173.8	-1.3	-2.4	-0.6	-1.1	94.0	117.4	53.5	115.9
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	186.7	1.3	2.4	-0.7	-1.3	100.9	108.1	57.5	106.7
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	190.4	0.0	0.0	-0.6	-1.1	NA	128.7	NA	126.9
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	170.1	0.0	0.0	-0.7	-1.3	NA	98.5	NA	97.2
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	172.8	-1.3	-2.4	-0.6	-1.1	93.4	116.8	53.2	115.2
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	185.6	1.3	2.4	-0.7	-1.3	100.3	107.5	57.1	106.1
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	189.3	0.0	0.0	-0.6	-1.1	NA	127.9	NA	126.2
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	169.1	0.0	0.0	-0.7	-1.3	NA	97.9	NA	96.6
245 0.6DL+1.0WLX1	67.8	-10.9	-20.2	-0.1	-0.2	4.4	274.9	2.5	271.3
246 0.6DL+1.0WLX2	166.7	10.9	20.2	-0.2	-0.4	10.7	337.9	6.1	333.4
247 0.6DL+1.0WLZ1	167.2	0.0	0.0	0.0	0.0	NA	NA	NA	NA
248 0.6DL+1.0WLZ2	67.3	0.0	0.0	-0.3	-0.6	NA	91.0	NA	89.8
249 0.6DL+0.6FL1+0.7SLX1	105.9	-1.8	-3.3	-0.1	-0.2	41.4	429.4	23.5	423.7
250 0.6DL+0.6FL1+0.7SLX2	123.0	1.8	3.3	-0.2	-0.4	48.0	249.4	27.3	246.0
251 0.6DL+0.6FL1+0.7SLZ1	127.9	0.0	0.0	-0.1	-0.2	NA	518.6	NA	511.7
252 0.6DL+0.6FL1+0.7SLZ2	100.9	0.0	0.0	-0.2	-0.4	NA	204.6	NA	201.8
					Min.=	4.4	51.6	2.5	50.9
						OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F16

Load Combination & Calculation of Base pressure

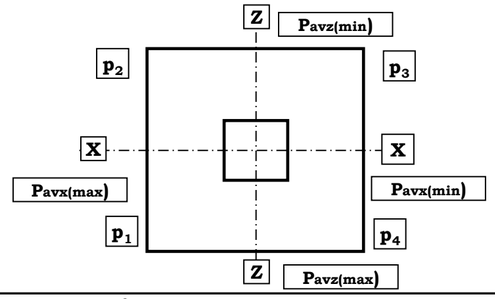
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 450 kN/m^2 For WL/SL case = 562.5 kN/m^2

Gross SBC of soil for DL, LL= 477 kN/m^2 For WL/SL case = 589.5 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	Base Pressure (kN/m^2) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	1.0DL+1.0FL1	130.02	0.00	-0.56	43.34	0.00	0.74	44.08	42.60	44.08	42.60	17.08	43.34	43.34	44.08	42.60	**O.K**
202	1.0DL+1.0FL2	128.92	0.00	-0.56	42.97	0.00	0.74	43.71	42.23	43.71	42.23	16.71	42.97	42.97	43.71	42.23	**O.K**
203	1.0DL+1.0FL1+1.0LL+1.0TL	192.02	0.56	-2.78	64.01	0.56	3.70	68.26	60.86	67.15	59.75	41.26	64.56	63.45	67.71	60.31	**O.K**
204	1.0DL+1.0FL2+1.0LL+1.0TL	191.02	0.56	-2.78	63.67	0.56	3.70	67.93	60.53	66.82	59.42	40.93	64.23	63.12	67.38	59.98	**O.K**
205	1.0DL+1.0FL1+0.75LL+0.75TL	176.52	0.37	-2.22	58.84	0.37	2.96	62.17	56.25	61.43	55.51	35.17	59.21	58.47	61.80	55.88	**O.K**
206	1.0DL+1.0FL2+0.75LL+0.75TL	175.52	0.37	-2.22	58.51	0.37	2.96	61.84	55.92	61.10	55.18	34.84	58.88	58.14	61.47	55.55	**O.K**
207	1.0DL+1.0FL1+1.0PL	232.12	0.00	0.74	77.37	0.00	0.99	78.36	76.39	78.36	76.39	51.36	77.38	77.38	78.36	76.39	**O.K**
208	1.0DL+1.0FL2+1.0PL	231.02	0.00	0.74	77.01	0.00	0.99	77.99	76.02	77.99	76.02	50.99	77.01	77.01	77.99	76.02	**O.K**
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0	294.12	0.56	-1.67	98.04	0.56	2.22	100.82	96.38	99.71	95.27	73.82	98.60	97.49	100.27	95.83	**O.K**
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0	293.02	0.56	-1.67	97.67	0.56	2.22	100.45	96.01	99.34	94.90	73.45	98.23	97.12	99.90	95.46	**O.K**
211	1.0DL+1.0FL1+0.75LL+0.75TL+1	278.62	0.56	-1.11	92.87	0.56	1.48	94.91	91.95	93.80	90.84	67.91	93.43	92.32	94.36	91.40	**O.K**
212	1.0DL+1.0FL2+0.75LL+0.75TL+1	277.52	0.56	-1.11	92.51	0.56	1.48	94.54	91.58	93.43	90.47	67.54	93.06	91.95	93.99	91.03	**O.K**
213	1.0DL+1.0FL1+1.0WLX1	80.52	-20.17	-0.37	26.84	20.17	0.49	47.50	46.51	7.17	6.18	20.50	47.01	6.68	27.34	26.35	**O.K**
214	1.0DL+1.0FL1+1.0WLX2	179.52	20.17	-0.56	59.84	20.17	0.74	80.75	79.27	40.42	38.94	53.75	80.01	39.68	60.59	59.11	**O.K**
215	1.0DL+1.0FL1+1.0WLZ1	179.92	0.00	-0.19	59.97	0.00	0.25	60.22	59.73	60.22	59.73	33.22	59.98	59.98	60.22	59.73	**O.K**
216	1.0DL+1.0FL1+1.0WLZ2	80.12	0.00	-0.74	26.71	0.00	0.99	27.69	25.72	27.69	25.72	0.69	26.71	26.71	27.69	25.72	**O.K**
217	1.0DL+1.0FL2+1.0WLX1	79.52	-20.17	-0.37	26.51	20.17	0.49	47.17	46.18	6.84	5.85	20.17	46.68	6.35	27.01	26.02	**O.K**
218	1.0DL+1.0FL2+1.0WLX2	178.42	20.17	-0.56	59.47	20.17	0.74	80.38	78.90	40.05	38.57	53.38	79.64	39.31	60.22	58.74	**O.K**
219	1.0DL+1.0FL2+1.0WLZ1	178.92	0.00	-0.19	59.64	0.00	0.25	59.89	59.39	59.89	59.39	32.89	59.64	59.64	59.89	59.39	**O.K**

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220 1.0DL+1.0FL2+1.0WLZ2	79.02	0.00	-0.74	26.34	0.00	0.99	27.33	25.35	27.33	25.35	0.33	26.34	26.34	27.33	25.35	O.K
221 1.0DL+1.0FL1+0.7SLX1	121.42	-3.33	-0.37	40.47	3.33	0.49	44.30	43.31	37.64	36.65	17.30	43.81	37.15	40.97	39.98	O.K
222 1.0DL+1.0FL1+0.7SLX2	138.62	3.33	-0.56	46.21	3.33	0.74	50.28	48.80	43.62	42.14	23.28	49.54	42.88	46.95	45.47	O.K
223 1.0DL+1.0FL1+0.7SLZ1	143.52	0.00	-0.37	47.84	0.00	0.49	48.33	47.35	48.33	47.35	21.33	47.84	47.84	48.33	47.35	O.K
224 1.0DL+1.0FL1+0.7SLZ2	116.52	0.00	-0.56	38.84	0.00	0.74	39.58	38.10	39.58	38.10	12.58	38.84	38.84	39.58	38.10	O.K
225 1.0DL+1.0FL2+0.7SLX1	120.42	-3.33	-0.37	40.14	3.33	0.49	43.96	42.98	37.30	36.32	16.96	43.47	36.81	40.63	39.65	O.K
226 1.0DL+1.0FL2+0.7SLX2	137.52	3.33	-0.56	45.84	3.33	0.74	49.91	48.43	43.25	41.77	22.91	49.17	42.51	46.58	45.10	O.K
227 1.0DL+1.0FL2+0.7SLZ1	142.42	0.00	-0.37	47.47	0.00	0.49	47.97	46.98	47.97	46.98	20.97	47.48	47.48	47.97	46.98	O.K
228 1.0DL+1.0FL2+0.7SLZ2	115.52	0.00	-0.56	38.51	0.00	0.74	39.25	37.77	39.25	37.77	12.25	38.51	38.51	39.25	37.77	O.K
229 1.0DL+1.0FL1+0.75WLX1+0.75L	143.12	-15.17	-1.11	47.71	15.17	1.48	64.36	61.40	34.02	31.06	37.36	62.88	32.54	49.19	46.23	O.K
230 1.0DL+1.0FL1+0.75WLX2+0.75L	217.32	15.17	-1.30	72.44	15.17	1.73	89.34	85.88	59.00	55.54	62.34	87.61	57.27	74.17	70.71	O.K
231 1.0DL+1.0FL1+0.75WLZ1+0.75L	217.72	0.00	-0.93	72.57	0.00	1.23	73.81	71.34	73.81	71.34	46.81	72.58	72.58	73.81	71.34	O.K
232 1.0DL+1.0FL1+0.75WLZ2+0.75L	142.82	0.00	-1.48	47.61	0.00	1.97	49.58	45.63	49.58	45.63	22.58	47.61	47.61	49.58	45.63	O.K
233 1.0DL+1.0FL2+0.75WLX1+0.75L	142.12	-15.17	-1.11	47.37	15.17	1.48	64.02	61.06	33.68	30.72	37.02	62.54	32.20	48.85	45.89	O.K
234 1.0DL+1.0FL2+0.75WLX2+0.75L	216.32	15.17	-1.30	72.11	15.17	1.73	89.00	85.55	58.66	55.21	62.00	87.28	56.94	73.83	70.38	O.K
235 1.0DL+1.0FL2+0.75WLZ1+0.75L	216.62	0.00	-0.93	72.21	0.00	1.23	73.44	70.97	73.44	70.97	46.44	72.21	72.21	73.44	70.97	O.K
236 1.0DL+1.0FL2+0.75WLZ2+0.75L	141.72	0.00	-1.48	47.24	0.00	1.97	49.21	45.27	49.21	45.27	22.21	47.24	47.24	49.21	45.27	O.K
237 1.0DL+1.0FL1+0.525SLX1+0.75	173.82	-2.41	-1.11	57.94	2.41	1.48	61.83	58.87	57.02	54.06	34.83	60.35	55.54	59.43	56.47	O.K
238 1.0DL+1.0FL1+0.525SLX2+0.75	186.72	2.41	-1.30	62.24	2.41	1.73	66.37	62.92	61.56	58.11	39.37	64.65	59.84	63.97	60.52	O.K
239 1.0DL+1.0FL1+0.525SLZ1+0.75	190.42	0.00	-1.11	63.47	0.00	1.48	64.95	61.99	64.95	61.99	37.95	63.47	63.47	64.95	61.99	O.K
240 1.0DL+1.0FL1+0.525SLZ2+0.75	170.12	0.00	-1.30	56.71	0.00	1.73	58.43	54.98	58.43	54.98	31.43	56.71	56.71	58.43	54.98	O.K
241 1.0DL+1.0FL2+0.525SLX1+0.75	172.82	-2.41	-1.11	57.61	2.41	1.48	61.49	58.53	56.68	53.72	34.49	60.01	55.20	59.09	56.13	O.K
242 1.0DL+1.0FL2+0.525SLX2+0.75	185.62	2.41	-1.30	61.87	2.41	1.73	66.01	62.55	61.20	57.74	39.01	64.28	59.47	63.61	60.15	O.K
243 1.0DL+1.0FL2+0.525SLZ1+0.75	189.32	0.00	-1.11	63.11	0.00	1.48	64.59	61.63	64.59	61.63	37.59	63.11	63.11	64.59	61.63	O.K
244 1.0DL+1.0FL2+0.525SLZ2+0.75	169.12	0.00	-1.30	56.37	0.00	1.73	58.10	54.65	58.10	54.65	31.10	56.38	56.38	58.10	54.65	O.K
245 0.6DL+1.0WLX1	67.82	-20.17	-0.19	22.61	20.17	0.25	43.02	42.53	2.69	2.20	16.02	42.78	2.45	22.86	22.37	O.K
246 0.6DL+1.0WLX2	166.72	20.17	-0.37	55.57	20.17	0.49	76.23	75.25	35.90	34.92	49.23	75.74	35.41	56.07	55.09	O.K
247 0.6DL+1.0WLZ1	167.22	0.00	0.00	55.74	0.00	0.00	55.74	55.74	55.74	55.74	28.74	55.74	55.74	55.74	55.74	O.K
248 0.6DL+1.0WLZ2	67.32	0.00	-0.56	22.44	0.00	0.74	23.18	21.70	23.18	21.70	-3.82	22.44	22.44	23.18	21.70	O.K
249 0.6DL+0.6FL1+0.7SLX1	105.92	-3.33	-0.19	35.31	3.33	0.25	38.88	38.39	32.22	31.73	11.88	38.64	31.98	35.55	35.06	O.K
250 0.6DL+0.6FL1+0.7SLX2	123.02	3.33	-0.37	41.01	3.33	0.49	44.83	43.84	38.17	37.18	17.83	44.34	37.68	41.50	40.51	O.K
251 0.6DL+0.6FL1+0.7SLZ1	127.92	0.00	-0.19	42.64	0.00	0.25	42.89	42.39	42.89	42.39	15.89	42.64	42.64	42.89	42.39	O.K
252 0.6DL+0.6FL1+0.7SLZ2	100.92	0.00	-0.37	33.64	0.00	0.49	34.13	33.15	34.13	33.15	7.13	33.64	33.64	34.13	33.15	O.K

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101 1.4DL+1.4FL1	145.52	0.00	-0.74	48.51	0.00	0.99	49.49	47.52	49.49	47.52	22.49	48.51	48.51	49.49	47.52	
102 1.4DL+1.4FL2	144.12	0.00	-0.74	48.04	0.00	0.99	49.03	47.05	49.03	47.05	22.03	48.04	48.04	49.03	47.05	
103 1.2DL+1.2FL1+1.6LL	245.02	0.00	-2.04	81.67	0.00	2.71	84.39	78.96	84.39	78.96	57.39	81.68	81.68	84.39	78.96	
104 1.2DL+1.2FL2+1.6LL	243.72	0.00	-2.22	81.24	0.00	2.96	84.20	78.28	84.20	78.28	57.20	81.24	81.24	84.20	78.28	
105 1.2DL+1.2FL1+1.2TL+1.6LL	239.02	0.74	-3.70	79.67	0.74	4.93	85.35	75.48	83.87	74.00	58.35	80.42	78.94	84.61	74.74	
106 1.2DL+1.2FL2+1.2TL+1.6LL	237.82	0.74	-3.89	79.27	0.74	5.18	85.19	74.83	83.71	73.35	58.19	80.01	78.53	84.45	74.09	
107 1.4DL+1.4FL1+1.4PL	288.42	0.00	0.93	96.14	0.00	1.23	97.37	94.91	97.37	94.91	70.37	96.14	96.14	97.37	94.91	
108 1.4DL+1.4FL2+1.4PL	287.02	0.00	0.93	95.67	0.00	1.23	96.91	94.44	96.91	94.44	69.91	95.68	95.68	96.91	94.44	
109 1.2DL+1.2FL1+1.6LL+1.2PL	367.52	0.00	-0.74	122.51	0.00	0.99	123.49	121.52	123.49	121.52	96.49	122.51	122.51	123.49	121.52	
110 1.2DL+1.2FL2+1.6LL+1.2PL	366.22	0.00	-0.74	122.07	0.00	0.99	123.06	121.09	123.06	121.09	96.06	122.08	122.08	123.06	121.09	
111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2	361.52	0.74	-2.41	120.51	0.74	3.21	124.45	118.04	122.97	116.56	97.45	121.25	119.77	123.71	117.30	
112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2	360.22	0.74	-2.41	120.07	0.74	3.21	124.02	117.61	122.54	116.13	97.02	120.82	119.34	123.28	116.87	
113 1.2DL+0.8WLX1	103.82	-16.10	-0.37	34.61	16.10	0.49	51.20	50.21	19.01	18.02	24.20	50.71	18.52	35.11	34.12	
114 1.2DL+0.8WLX2	183.02	16.10	-0.56	61.01	16.10	0.74	77.84	76.36	45.65	44.17	50.84	77.10	44.91	61.75	60.27	
115 1.2DL+0.8WLZ1	183.32	0.00	-0.37	61.11	0.00	0.49	61.60	60.61	61.60	60.61	34.60	61.11	61.11	61.60	60.61	
116 1.2DL+0.8WLZ2	103.42	0.00	-0.74	34.47	0.00	0.99	35.46	33.49	35.46	33.49	8.46	34.48	34.48	35.46	33.49	
117 1.2DL+1.6WLX1+0.5LL	97.72	-32.19	-0.93	32.57	32.19	1.23	66.00	63.53	1.62	-0.85	39.00	64.77	0.39	33.81	31.34	
118 1.2DL+1.6WLX2+0.5LL	256.02	32.19	-1.11	85.34	32.19	1.48	119.01	116.05	54.63	51.67	92.01	117.53	53.15	86.82	83.86	
119 1.2DL+1.6WLZ1+0.5LL	256.82	0.00	-0.56	85.61	0.00	0.74	86.35	84.87	86.35	84.87	59.35	85.61	85.61	86.35	84.87	
120 1.2DL+1.6WLZ2+0.5LL	97.02	0.00	-1.48	32.34	0.00	1.97	34.31	30.37	34.31	30.37	7.31	32.34	32.34	34.31	30.37	
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	162.32	-4.63	-0.93	54.11	4.63	1.23	59.97	57.50	50.72	48.25	32.97	58.74	49.49	55.35	52.88	
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	186.82	4.63	-1.11	62.27	4.63	1.48	68.38	65.42	59.13	56.17	41.38	66.90	57.65	63.76	60.80	
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	193.82	0.00	-0.93	64.61	0.00	1.23	65.84	63.37	65.84	63.37	38.84	64.61	64.61	65.84	63.37	
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	155.32	0.00	-1.11	51.77	0.00	1.48	53.25	50.29	53.25	50.29	26.25	51.77	51.77	53.25	50.29	
125 0.9DL+1.6WLX1	51.22	-32.19	-0.19	17.07	32.19	0.25	49.51	49.02	-14.87	-15.36	22.51	49.27	-15.12	17.32	16.83	
126 0.9DL+1.6WLX2	209.52	32.19	-0.56	69.84	32.19	0.74	102.77	101.29	38.39	36.91	75.77	102.03	37.65	70.58	69.10	
127 0.9DL+1.6WLZ1	210.22	0.00	0.00	70.07	0.00	0.00	70.07	70.07	70.07	70.07	43.07	70.07	70.07	70.07	70.07	
128 0.9DL+1.6WLZ2	50.42	0.00	-0.74	16.81	0.00	0.99	17.79	15.82	17.79	15.82	-9.21	16.81	16.81	17.79	15.82	
129 0.9DL+1.0SLX1	118.12	-4.63	-0.37	39.37	4.63	0.49	44.49	43.51	35.24	34.26	17.49	44.00	34.75	39.87	38.89	
130 0.9DL+1.0SLX2	142.52	4.63	-0.37	47.51	4.63	0.49	52.63	51.64	43.38	42.39	25.63	52.14	42.89	48.01	47.02	
131 0.9DL+1.0SLZ1	149.62	0.00	-0.37	49.87	0.00	0.49	50.37	49.38	50.37	49.38	23.37	49.88	49.88	50.37	49.38	
132 0.9DL+1.0SLZ2	111.02	0.00	-0.37	37.01	0.00	0.49	37.50	36.51	37.50	36.51	10.50	37.01	37.01	37.50	36.51	
Max :	368	32	1													
Min :	50	-32	-4													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 17.08 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 27.08 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 73.82 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 83.82 kN/m²

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Sample calculation of values of base pressure:

Load case : 202 1.0DL+1.0FL2

Maximum pressure, p_1 (pmax) = 42.97+0.00+0.74= 43.71 kN/m^2

p_2= 42.97+0.00-0.74= 42.23 kN/m^2

p_3= 42.97-(0.00+0.74)= 42.23 kN/m^2

p_4= 42.97-0.00+0.74= 43.71 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 43.71-1.5*18= 16.71 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (43.71+42.23)/2= 42.97 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (42.23+43.71)/2= 42.97 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (43.71+43.71)/2= 43.71 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (42.23+42.23)/2= 42.23 kN/m^2

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DESIGN OF ISOLATED FOOTING F16

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

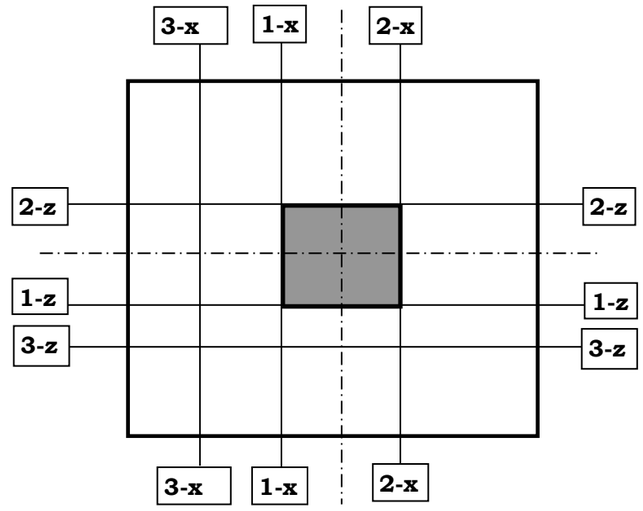
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.55+2*0.5*0.232)+(0.55+2*0.5*0.232))*2

= 3.128 m

Punching Area= ((0.55+2*0.5*0.232)*(0.55+2*0.5*0.232))

= 0.61 m^2 Punching shear stress to be checked



Downward pressure due to self weight of raft and soil = (0.3*24+1.2*18)

= 28.80 kN/m^2

Area of footing, A_f= 3.00 m^2

Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4DL+1.4FL1	48.51	48.51	48.51	5.18	5.18	9.48	47.10
102	1.4DL+1.4FL2	48.04	48.04	48.04	5.06	5.06	9.25	45.98
103	1.2DL+1.2FL1+1.6LL	81.68	81.68	81.68	13.90	13.90	25.44	126.37
104	1.2DL+1.2FL2+1.6LL	81.24	81.24	81.24	13.78	13.78	25.22	125.33
105	1.2DL+1.2FL1+1.2TL+1.6LL	79.88	79.48	80.06	13.52	13.22	24.74	121.59
106	1.2DL+1.2FL2+1.2TL+1.6LL	79.47	79.07	79.65	13.41	13.12	24.55	120.63
107	1.4DL+1.4FL1+1.4PL	96.14	96.14	96.14	17.70	17.70	32.39	160.94
108	1.4DL+1.4FL2+1.4PL	95.68	95.68	95.68	17.58	17.58	32.17	159.83
109	1.2DL+1.2FL1+1.6LL+1.2PL	122.51	122.51	122.51	24.63	24.63	45.07	223.96
110	1.2DL+1.2FL2+1.6LL+1.2PL	122.08	122.08	122.08	24.52	24.52	44.87	222.92
111	1.2DL+1.2FL1+1.2TL+1.6LL+1	120.71	120.31	120.89	24.25	23.96	44.38	219.18
112	1.2DL+1.2FL2+1.2TL+1.6LL+1	120.28	119.88	120.46	24.14	23.84	44.18	218.14
113	1.2DL+0.8WLX1	39.04	30.19	42.97	4.74	-1.68	8.68	13.88

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114 1.2DL+0.8WLX2	65.43	56.58	69.36	11.67	5.26	21.37	76.97
115 1.2DL+0.8WLZ1	61.11	61.11	61.11	8.49	8.49	15.54	77.21
116 1.2DL+0.8WLZ2	34.48	34.48	34.48	1.49	1.49	2.73	13.56
117 1.2DL+1.6WLX1+0.5LL	41.43	23.73	49.29	7.41	-5.42	13.58	9.02
118 1.2DL+1.6WLX2+0.5LL	94.19	76.49	102.05	21.27	8.44	38.96	135.13
119 1.2DL+1.6WLZ1+0.5LL	85.61	85.61	85.61	14.93	14.93	27.33	135.77
120 1.2DL+1.6WLZ2+0.5LL	32.34	32.34	32.34	0.93	0.93	1.70	8.46
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	55.39	52.84	56.52	7.57	5.73	13.87	60.48
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	63.55	61.00	64.68	9.72	7.88	17.79	80.00
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	64.61	64.61	64.61	9.41	9.41	17.22	85.58
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	51.77	51.77	51.77	6.04	6.04	11.05	54.91
125 0.9DL+1.6WLX1	25.93	8.22	33.78	3.33	-9.50	6.12	-28.03
126 0.9DL+1.6WLX2	78.69	60.99	86.55	17.20	4.37	31.50	98.09
127 0.9DL+1.6WLZ1	70.07	70.07	70.07	10.85	10.85	19.85	98.64
128 0.9DL+1.6WLZ2	16.81	16.81	16.81	-3.15	-3.15	-5.77	-28.66
129 0.9DL+1.0SLX1	40.65	38.10	41.78	3.70	1.86	6.78	25.27
130 0.9DL+1.0SLX2	48.79	46.24	49.92	5.84	4.00	10.69	44.71
131 0.9DL+1.0SLZ1	49.88	49.88	49.88	5.54	5.54	10.14	50.37
132 0.9DL+1.0SLZ2	37.01	37.01	37.01	2.16	2.16	3.95	19.61

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Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1x} = 48.51-(48.51-48.51)*0.725/2 = 48.51 kN/m^2 Overhang = 0.725 m

p_{2x} = 48.51+(48.51-48.51)*0.725/2 = 48.51 kN/m^2 Effective depth = 0.244 m

p_{3x} = 48.51-(48.51-48.51)*(0.725-0.244)/2 = 48.51 kN/m^2

BM_{1x} = (48.51-28.8)*0.725*0.725/2+0.5*(48.51-48.51)*0.725*0.725*0.67 = 5.18 kN-m/m

BM_{2x} = (48.51-28.8)*0.725*0.725/2-0.5*(-48.51+48.51)*0.725*0.725*0.67 = 5.18 kN-m/m

SF_{3x} = (48.51-28.8)*(0.725-0.244)+0.5*(48.51-48.51)*(0.725-0.244) = 9.48 kN/m

$S.F_{(punching)}$ = (3-0.61)*(48.51-28.8) = 47.10 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	24.63	0.41	0.55	0.001	323.46	439.20	12ϕT	257.51	300	200	565.49
At top	-9.50	-0.16	0.21	0.001	124.02	219.60	12ϕT	515.02	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 45.07 kN/m

Shear stress , τ_v= (45.07*1000)/(1000*244)

= 0.18 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 223.96 kN

Strength of Concrete in shear Vc1 = 2257.67 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1339.59 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1505.11 kN **Table 22.6.5.2_ (c)**

Design Vc= 1339.59 kN

Allowable shear stress, (ϕ Vc)/bd = 1.32 N/mm^2

Design punching shear stress, τ = (223.96*1000)/(3128*244)

= 0.29 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4DL+1.4FL1	48.87	48.14	49.17	2.31	2.14	4.99
102	1.4DL+1.4FL2	48.40	56.96	48.71	2.26	4.65	4.88
103	1.2DL+1.2FL1+1.6LL	82.67	78.65	83.51	6.21	5.43	13.40
104	1.2DL+1.2FL2+1.6LL	82.33	85.24	83.24	6.18	7.17	13.35
105	1.2DL+1.2FL1+1.2TL+1.6LL	81.48	82.71	83.01	6.18	7.05	13.37
106	1.2DL+1.2FL2+1.2TL+1.6LL	81.17	80.51	82.77	6.16	6.66	13.32
107	1.4DL+1.4FL1+1.4PL	96.59	94.62	96.97	7.71	7.15	16.61
108	1.4DL+1.4FL2+1.4PL	96.13	73.19	96.51	7.65	1.47	16.50
109	1.2DL+1.2FL1+1.6LL+1.2PL	122.87	102.23	123.17	10.66	5.02	22.97
110	1.2DL+1.2FL2+1.6LL+1.2PL	122.44	101.81	122.74	10.61	5.06	22.87
111	1.2DL+1.2FL1+1.2TL+1.6LL+1	121.68	88.92	122.67	10.63	2.01	22.94
112	1.2DL+1.2FL2+1.2TL+1.6LL+1	121.25	88.41	122.24	10.58	2.01	22.83
113	1.2DL+0.8WLX1	34.80	42.39	34.95	0.70	2.77	1.51
114	1.2DL+0.8WLX2	61.28	60.15	61.51	3.70	3.48	7.98
115	1.2DL+0.8WLZ1	61.29	50.07	61.44	3.69	0.53	7.95
116	1.2DL+0.8WLZ2	34.84	35.86	35.14	0.73	1.11	1.58
117	1.2DL+1.6WLX1+0.5LL	33.03	36.28	33.41	0.54	1.54	1.17
118	1.2DL+1.6WLX2+0.5LL	85.88	72.61	86.34	6.51	3.03	14.04
119	1.2DL+1.6WLZ1+0.5LL	85.88	70.53	86.11	6.47	2.26	13.96
120	1.2DL+1.6WLZ2+0.5LL	33.06	33.62	33.67	0.58	1.00	1.26

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121 1.2DL+1.0SLX1+0.5LL+0.5FL1	54.57	50.89	54.95	2.97	2.05	6.40
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	62.82	56.74	63.28	3.91	2.41	8.44
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	65.06	55.73	65.44	4.15	1.68	8.95
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	52.31	49.94	52.77	2.72	2.10	5.88
125 0.9DL+1.6WLX1	17.16	34.99	17.24	-1.30	3.40	-2.80
126 0.9DL+1.6WLX2	70.11	70.59	70.34	4.70	4.77	10.12
127 0.9DL+1.6WLZ1	70.07	63.58	70.07	4.66	2.57	10.03
128 0.9DL+1.6WLZ2	17.17	26.28	17.47	-1.27	1.20	-2.71
129 0.9DL+1.0SLX1	39.56	49.95	39.71	1.24	3.93	2.67
130 0.9DL+1.0SLX2	47.70	55.39	47.85	2.16	4.18	4.65
131 0.9DL+1.0SLZ1	50.06	49.33	50.21	2.42	2.01	5.22
132 0.9DL+1.0SLZ2	37.19	43.77	37.34	0.97	2.65	2.09

Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1z} = 49.49-(49.49-47.52)*0.475/1.5= 48.87 kN/m^2

p_{2z} = 47.52+(49.49-47.52)*0.475/1.5= 48.14 kN/m^2

p_{3z} = 49.49-(49.49-47.52)*(0.475-0.232)/1.5= 49.17 kN/m^2

Overhang = 0.475 m

Effective depth = 0.232 m

BM_{1z} = (48.87-28.8)*0.475*0.475/2+0.5*(49.49-48.87)*0.475*0.475*0.67 = 2.31 kN-m/m

BM_{2z} = (48.14-28.8)*0.475*0.475/2-0.5*(-47.52+48.14)*0.475*0.475*0.67 = 2.14 kN-m/m

SF_{3z} = (49.17-28.8)*(0.475-0.232)+0.5*(49.49-49.17)*(0.475-0.232) = 4.99 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	10.66	0.198	0.264	0.0006	154.08	417.60	12ϕT	270.83	300	200	565.5
At top	-1.30	-0.024	-0.032	-0.0001	-0.18	208.80	12ϕT	541.65	300	250	452.4

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Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 22.97 kN/m

Shear stress , τ_v= (22.97*1000)/(1000*232)

= 0.10 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	12ϕT	**200**	c/c
	At Top	12ϕT	**250**	c/c
Along Z Direction :				
	At Bottom	12ϕT	**200**	c/c
	At Top	12ϕT	**250**	c/c

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5.9 DESIGN OF CALCULATION OF FOOTING F17

(STAAD Node -7430)

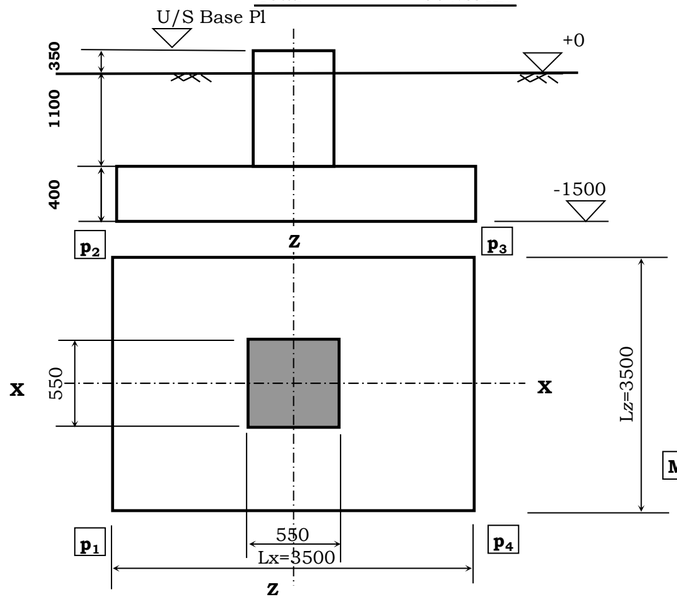
	Along-X	Along-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	3.50m	3.50m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.35 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

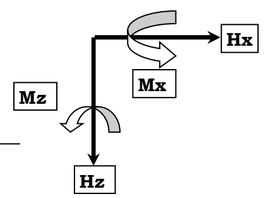
SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.= (3.5*3.5*0.4*24)+(0.55*0.55*1.45*24) = 128.13KN
Weight of soil = ((3.5*3.5)-(0.55*0.55))*1.1*18 = 236.56KN
Total = 364.69KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 344mm = 0.344m

Effective depth of slab along Z direction = 332mm = 0.332m
Clear projection along Z direction = 1.475m
Clear projection along X direction = 1.475m

H$_X$=Force along X-X direction

Hz=Force along Z-Z direction

M$_X$=Moment about X-X direction

Mz=Moment about Z-Z direction

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CHECK FOR OVERTURNING AND SLIDING OF FOOTING **F17**

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	1.0DL+1.0FL1	29.9	0.0	0.0	1.1	0.0	394.59	0.00	0.00	1.10	2.04
202	1.0DL+1.0FL2	29.0	0.0	0.0	1.6	0.0	393.69	0.00	0.00	1.60	2.96
203	1.0DL+1.0FL1+1.0LL+1.0TL	128.0	-0.4	0.0	-25.9	0.0	492.69	-0.40	-0.74	-25.90	-47.92
204	1.0DL+1.0FL2+1.0LL+1.0TL	127.0	-0.4	0.0	-25.4	0.0	491.69	-0.40	-0.74	-25.40	-46.99
205	1.0DL+1.0FL1+0.75LL+0.75TL	103.5	-0.3	0.0	-19.2	0.0	468.19	-0.30	-0.56	-19.20	-35.52
206	1.0DL+1.0FL2+0.75LL+0.75TL	102.5	-0.3	0.0	-18.6	0.0	467.19	-0.30	-0.56	-18.60	-34.41
207	1.0DL+1.0FL1+1.0PL	224.1	0.0	0.0	-76.8	0.0	588.79	0.00	0.00	-76.80	-142.08
208	1.0DL+1.0FL2+1.0PL	223.1	0.0	0.0	-76.2	0.0	587.79	0.00	0.00	-76.20	-140.97
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	322.1	-0.4	0.0	-103.7	0.0	686.79	-0.40	-0.74	-103.70	-191.85
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	321.2	-0.4	0.0	-103.2	0.0	685.89	-0.40	-0.74	-103.20	-190.92
211	1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	297.6	-0.3	0.0	-97.0	0.0	662.29	-0.30	-0.56	-97.00	-179.45
212	1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	296.7	-0.3	0.0	-96.4	0.0	661.39	-0.30	-0.56	-96.40	-178.34
213	1.0DL+1.0FL1+1.0WLX1	75.7	-10.9	0.0	3.2	0.0	440.39	-10.90	-20.17	3.20	5.92
214	1.0DL+1.0FL1+1.0WLX2	-15.8	10.9	0.0	-1.1	0.0	348.89	10.90	20.17	-1.10	-2.04
215	1.0DL+1.0FL1+1.0WLZ1	137.1	0.0	0.0	-63.2	0.0	501.79	0.00	0.00	-63.20	-116.92
216	1.0DL+1.0FL1+1.0WLZ2	-77.2	0.0	0.0	65.3	0.0	287.49	0.00	0.00	65.30	120.81
217	1.0DL+1.0FL2+1.0WLX1	74.8	-10.9	0.0	3.8	0.0	439.49	-10.90	-20.17	3.80	7.03
218	1.0DL+1.0FL2+1.0WLX2	-16.8	10.9	0.0	-0.6	0.0	347.89	10.90	20.17	-0.60	-1.11
219	1.0DL+1.0FL2+1.0WLZ1	136.1	0.0	0.0	-62.6	0.0	500.79	0.00	0.00	-62.60	-115.81
220	1.0DL+1.0FL2+1.0WLZ2	-78.2	0.0	0.0	65.9	0.0	286.49	0.00	0.00	65.90	121.92
221	1.0DL+1.0FL1+0.7SLX1	38.1	-1.7	0.0	1.2	0.0	402.79	-1.70	-3.15	1.20	2.22
222	1.0DL+1.0FL1+0.7SLX2	21.7	1.7	0.0	0.9	0.0	386.39	1.70	3.15	0.90	1.67
223	1.0DL+1.0FL1+0.7SLZ1	84.8	0.0	0.0	-29.4	0.0	449.49	0.00	0.00	-29.40	-54.39
224	1.0DL+1.0FL1+0.7SLZ2	-25.0	0.0	0.0	31.5	0.0	339.69	0.00	0.00	31.50	58.28
225	1.0DL+1.0FL2+0.7SLX1	37.2	-1.7	0.0	1.8	0.0	401.89	-1.70	-3.15	1.80	3.33
226	1.0DL+1.0FL2+0.7SLX2	20.7	1.7	0.0	1.5	0.0	385.39	1.70	3.15	1.50	2.78
227	1.0DL+1.0FL2+0.7SLZ1	83.8	0.0	0.0	-28.8	0.0	448.49	0.00	0.00	-28.80	-53.28
228	1.0DL+1.0FL2+0.7SLZ2	-25.9	0.0	0.0	32.0	0.0	338.79	0.00	0.00	32.00	59.20
229	1.0DL+1.0FL1+0.75WLX1+0.75LL	107.2	-8.1	0.0	0.1	0.0	471.89	-8.10	-14.99	0.10	0.19
230	1.0DL+1.0FL1+0.75WLX2+0.75LL	38.6	8.1	0.0	-3.1	0.0	403.29	8.10	14.99	-3.10	-5.74
231	1.0DL+1.0FL1+0.75WLZ1+0.75LL	153.3	0.0	0.0	-49.7	0.0	517.99	0.00	0.00	-49.70	-91.95

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232 1.0DL+1.0FL1+0.75WLZ2+0.75LL	-7.5	0.0	0.0	46.7	0.0	357.19	0.00	0.00	46.70	86.40
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	106.3	-8.1	0.0	0.7	0.0	470.99	-8.10	-14.99	0.70	1.30
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	37.6	8.1	0.0	-2.6	0.0	402.29	8.10	14.99	-2.60	-4.81
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	152.3	0.0	0.0	-49.1	0.0	516.99	0.00	0.00	-49.10	-90.84
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	-8.5	0.0	0.0	47.2	0.0	356.19	0.00	0.00	47.20	87.32
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	79.1	-1.3	0.0	-1.4	0.0	443.79	-1.30	-2.41	-1.40	-2.59
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	66.7	1.3	0.0	-1.6	0.0	431.39	1.30	2.41	-1.60	-2.96
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	114.1	0.0	0.0	-24.3	0.0	478.79	0.00	0.00	-24.30	-44.96
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	31.7	0.0	0.0	21.3	0.0	396.39	0.00	0.00	21.30	39.41
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	78.1	-1.3	0.0	-0.8	0.0	442.79	-1.30	-2.41	-0.80	-1.48
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	65.8	1.3	0.0	-1.1	0.0	430.49	1.30	2.41	-1.10	-2.04
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	113.1	0.0	0.0	-23.8	0.0	477.79	0.00	0.00	-23.80	-44.03
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	30.7	0.0	0.0	21.9	0.0	395.39	0.00	0.00	21.90	40.52
245 0.6DL+1.0WLX1	66.9	-10.9	0.0	1.1	0.0	431.59	-10.90	-20.17	1.10	2.04
246 0.6DL+1.0WLX2	-24.7	10.9	0.0	-3.3	0.0	339.99	10.90	20.17	-3.30	-6.11
247 0.6DL+1.0WLZ1	128.3	0.0	0.0	-65.3	0.0	492.99	0.00	0.00	-65.30	-120.81
248 0.6DL+1.0WLZ2	-86.0	0.0	0.0	63.2	0.0	278.69	0.00	0.00	63.20	116.92
249 0.6DL+0.6FL1+0.7SLX1	26.2	-1.7	0.0	0.8	0.0	390.89	-1.70	-3.15	0.80	1.48
250 0.6DL+0.6FL1+0.7SLX2	9.7	1.7	0.0	0.5	0.0	374.39	1.70	3.15	0.50	0.93
251 0.6DL+0.6FL1+0.7SLZ1	72.8	0.0	0.0	-29.8	0.0	437.49	0.00	0.00	-29.80	-55.13
252 0.6DL+0.6FL1+0.7SLZ2	-36.9	0.0	0.0	31.1	0.0	327.79	0.00	0.00	31.10	57.54
Ultimate Conditions										
101 1.4DL+1.4FL1	41.9	0.0	0.0	1.5	0.0	406.59	0.00	0.00	1.50	2.78
102 1.4DL+1.4FL2	40.5	0.0	0.0	2.2	0.0	405.19	0.00	0.00	2.20	4.07
103 1.2DL+1.2FL1+1.6LL	127.6	0.0	0.0	-4.2	0.0	492.29	0.00	0.00	-4.20	-7.77
104 1.2DL+1.2FL2+1.6LL	126.4	0.0	0.0	-3.5	0.0	491.09	0.00	0.00	-3.50	-6.48
105 1.2DL+1.2FL1+1.2TL+1.6LL	176.5	-0.5	0.0	-32.4	0.0	541.19	-0.50	-0.93	-32.40	-59.94
106 1.2DL+1.2FL2+1.2TL+1.6LL	175.4	-0.5	0.0	-31.8	0.0	540.09	-0.50	-0.93	-31.80	-58.83
107 1.4DL+1.4FL1+1.4PL	313.7	0.0	0.0	-107.5	0.0	678.39	0.00	0.00	-107.50	-198.88
108 1.4DL+1.4FL2+1.4PL	312.3	0.0	0.0	-106.7	0.0	676.99	0.00	0.00	-106.70	-197.40
109 1.2DL+1.2FL1+1.6LL+1.2PL	360.5	0.0	0.0	-97.5	0.0	725.19	0.00	0.00	-97.50	-180.38
110 1.2DL+1.2FL2+1.6LL+1.2PL	359.4	0.0	0.0	-96.9	0.0	724.09	0.00	0.00	-96.90	-179.27

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111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	409.5	-0.5	0.0	-125.8	0.0	774.19	-0.50	-0.93	-125.80	-232.73
112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	408.3	-0.5	0.0	-125.2	0.0	772.99	-0.50	-0.93	-125.20	-231.62
113 1.2DL+0.8WLX1	78.9	-8.7	0.0	-0.5	0.0	443.59	-8.70	-16.10	-0.50	-0.93
114 1.2DL+0.8WLX2	5.6	8.7	0.0	-3.9	0.0	370.29	8.70	16.10	-3.90	-7.22
115 1.2DL+0.8WLZ1	127.9	0.0	0.0	-53.6	0.0	492.59	0.00	0.00	-53.60	-99.16
116 1.2DL+0.8WLZ2	-43.5	0.0	0.0	49.2	0.0	321.19	0.00	0.00	49.20	91.02
117 1.2DL+1.6WLX1+0.5LL	144.1	-17.4	0.0	-0.5	0.0	508.79	-17.40	-32.19	-0.50	-0.93
118 1.2DL+1.6WLX2+0.5LL	-2.4	17.4	0.0	-7.4	0.0	362.29	17.40	32.19	-7.40	-13.69
119 1.2DL+1.6WLZ1+0.5LL	242.3	0.0	0.0	-106.7	0.0	606.99	0.00	0.00	-106.70	-197.40
120 1.2DL+1.6WLZ2+0.5LL	-100.6	0.0	0.0	98.9	0.0	264.09	0.00	0.00	98.90	182.97
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	80.0	-2.5	0.0	-2.2	0.0	444.69	-2.50	-4.63	-2.20	-4.07
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	56.5	2.5	0.0	-2.7	0.0	421.19	2.50	4.63	-2.70	-5.00
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	146.6	0.0	0.0	-45.9	0.0	511.29	0.00	0.00	-45.90	-84.92
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	-10.2	0.0	0.0	41.0	0.0	354.49	0.00	0.00	41.00	75.85
125 0.9DL+1.6WLX1	105.0	-17.4	0.0	1.8	0.0	469.69	-17.40	-32.19	1.80	3.33
126 0.9DL+1.6WLX2	-41.6	17.4	0.0	-5.1	0.0	323.09	17.40	32.19	-5.10	-9.44
127 0.9DL+1.6WLZ1	203.1	0.0	0.0	-104.4	0.0	567.79	0.00	0.00	-104.40	-193.14
128 0.9DL+1.6WLZ2	-139.8	0.0	0.0	101.2	0.0	224.89	0.00	0.00	101.20	187.22
129 0.9DL+1.0SLX1	43.4	-2.5	0.0	-1.4	0.0	408.09	-2.50	-4.63	-1.40	-2.59
130 0.9DL+1.0SLX2	19.9	2.5	0.0	-1.9	0.0	384.59	2.50	4.63	-1.90	-3.52
131 0.9DL+1.0SLZ1	110.1	0.0	0.0	-45.1	0.0	474.79	0.00	0.00	-45.10	-83.44
132 0.9DL+1.0SLZ2	-46.7	0.0	0.0	41.8	0.0	317.99	0.00	0.00	41.80	77.33

Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	1.0DL+1.0FL1	394.6	0.0	0.0	1.1	2.0	NA	145.4	NA	143.5
202	1.0DL+1.0FL2	393.7	0.0	0.0	1.6	3.0	NA	99.8	NA	98.4
203	1.0DL+1.0FL1+1.0LL+1.0TL	492.7	-0.4	-0.7	-25.9	-47.9	865.5	7.7	492.7	7.6
204	1.0DL+1.0FL2+1.0LL+1.0TL	491.7	-0.4	-0.7	-25.4	-47.0	863.8	7.8	491.7	7.7
205	1.0DL+1.0FL1+0.75LL+0.75TL	468.2	-0.3	-0.6	-19.2	-35.5	1096.7	9.9	624.3	9.8
206	1.0DL+1.0FL2+0.75LL+0.75TL	467.2	-0.3	-0.6	-18.6	-34.4	1094.3	10.2	622.9	10.0
207	1.0DL+1.0FL1+1.0PL	588.8	0.0	0.0	-76.8	-142.1	NA	3.1	NA	3.1
208	1.0DL+1.0FL2+1.0PL	587.8	0.0	0.0	-76.2	-141.0	NA	3.1	NA	3.1

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209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	686.8	-0.4	-0.7	-103.7	-191.8	1206.5	2.7	686.8	2.6
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	685.9	-0.4	-0.7	-103.2	-190.9	1204.9	2.7	685.9	2.7
211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	662.3	-0.3	-0.6	-97.0	-179.5	1551.3	2.8	883.1	2.7
212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	661.4	-0.3	-0.6	-96.4	-178.3	1549.2	2.8	881.9	2.7
213 1.0DL+1.0FL1+1.0WLX1	440.4	-10.9	-20.2	3.2	5.9	28.4	55.8	16.2	55.0
214 1.0DL+1.0FL1+1.0WLX2	348.9	10.9	20.2	-1.1	-2.0	22.5	128.6	12.8	126.9
215 1.0DL+1.0FL1+1.0WLZ1	501.8	0.0	0.0	-63.2	-116.9	NA	3.2	NA	3.2
216 1.0DL+1.0FL1+1.0WLZ2	287.5	0.0	0.0	65.3	120.8	NA	1.8	NA	1.8
217 1.0DL+1.0FL2+1.0WLX1	439.5	-10.9	-20.2	3.8	7.0	28.3	46.9	16.1	46.3
218 1.0DL+1.0FL2+1.0WLX2	347.9	10.9	20.2	-0.6	-1.1	22.4	235.1	12.8	231.9
219 1.0DL+1.0FL2+1.0WLZ1	500.8	0.0	0.0	-62.6	-115.8	NA	3.2	NA	3.2
220 1.0DL+1.0FL2+1.0WLZ2	286.5	0.0	0.0	65.9	121.9	NA	1.8	NA	1.7
221 1.0DL+1.0FL1+0.7SLX1	402.8	-1.7	-3.1	1.2	2.2	166.5	136.1	94.8	134.3
222 1.0DL+1.0FL1+0.7SLX2	386.4	1.7	3.1	0.9	1.7	159.7	174.0	90.9	171.7
223 1.0DL+1.0FL1+0.7SLZ1	449.5	0.0	0.0	-29.4	-54.4	NA	6.2	NA	6.1
224 1.0DL+1.0FL1+0.7SLZ2	339.7	0.0	0.0	31.5	58.3	NA	4.4	NA	4.3
225 1.0DL+1.0FL2+0.7SLX1	401.9	-1.7	-3.1	1.8	3.3	166.1	90.5	94.6	89.3
226 1.0DL+1.0FL2+0.7SLX2	385.4	1.7	3.1	1.5	2.8	159.3	104.2	90.7	102.8
227 1.0DL+1.0FL2+0.7SLZ1	448.5	0.0	0.0	-28.8	-53.3	NA	6.3	NA	6.2
228 1.0DL+1.0FL2+0.7SLZ2	338.8	0.0	0.0	32.0	59.2	NA	4.3	NA	4.2
229 1.0DL+1.0FL1+0.75WLX1+0.75LL	471.9	-8.1	-15.0	0.1	0.2	40.9	1913.1	23.3	1887.6
230 1.0DL+1.0FL1+0.75WLX2+0.75LL	403.3	8.1	15.0	-3.1	-5.7	35.0	52.7	19.9	52.0
231 1.0DL+1.0FL1+0.75WLZ1+0.75LL	518.0	0.0	0.0	-49.7	-91.9	NA	4.2	NA	4.2
232 1.0DL+1.0FL1+0.75WLZ2+0.75LL	357.2	0.0	0.0	46.7	86.4	NA	3.1	NA	3.1
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	471.0	-8.1	-15.0	0.7	1.3	40.9	272.8	23.3	269.1
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	402.3	8.1	15.0	-2.6	-4.8	34.9	62.7	19.9	61.9
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	517.0	0.0	0.0	-49.1	-90.8	NA	4.3	NA	4.2
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	356.2	0.0	0.0	47.2	87.3	NA	3.1	NA	3.0
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	443.8	-1.3	-2.4	-1.4	-2.6	239.9	128.5	136.6	126.8
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	431.4	1.3	2.4	-1.6	-3.0	233.2	109.3	132.7	107.8
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	478.8	0.0	0.0	-24.3	-45.0	NA	8.0	NA	7.9
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	396.4	0.0	0.0	21.3	39.4	NA	7.5	NA	7.4
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	442.8	-1.3	-2.4	-0.8	-1.5	239.3	224.4	136.2	221.4
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	430.5	1.3	2.4	-1.1	-2.0	232.7	158.7	132.5	156.5
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	477.8	0.0	0.0	-23.8	-44.0	NA	8.1	NA	8.0
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	395.4	0.0	0.0	21.9	40.5	NA	7.3	NA	7.2
245 0.6DL+1.0WLX1	431.6	-10.9	-20.2	1.1	2.0	27.8	159.1	15.8	156.9
246 0.6DL+1.0WLX2	340.0	10.9	20.2	-3.3	-6.1	21.9	41.8	12.5	41.2
247 0.6DL+1.0WLZ1	493.0	0.0	0.0	-65.3	-120.8	NA	3.1	NA	3.0
248 0.6DL+1.0WLZ2	278.7	0.0	0.0	63.2	116.9	NA	1.8	NA	1.8
249 0.6DL+0.6FL1+0.7SLX1	390.9	-1.7	-3.1	0.8	1.5	161.6	198.1	92.0	195.4
250 0.6DL+0.6FL1+0.7SLX2	374.4	1.7	3.1	0.5	0.9	154.8	303.6	88.1	299.5
251 0.6DL+0.6FL1+0.7SLZ1	437.5	0.0	0.0	-29.8	-55.1	NA	6.0	NA	5.9
252 0.6DL+0.6FL1+0.7SLZ2	327.8	0.0	0.0	31.1	57.5	NA	4.3	NA	4.2
					Min.=	21.9	1.8	12.5	1.7
						OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F17

Load Combination & Calculation of Base pressure

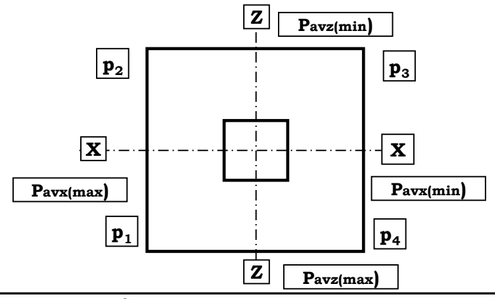
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 450 kN/m^2 For WL/SL case = 562.5 kN/m^2

Gross SBC of soil for DL, LL= 477 kN/m^2 For WL/SL case = 589.5 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	Base Pressure (kN/m^2) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	1.0DL+1.0FL1	394.59	0.00	2.04	32.21	0.00	0.28	32.50	31.93	32.50	31.93	5.50	32.22	32.22	32.50	31.93	**O.K**
202	1.0DL+1.0FL2	393.69	0.00	2.96	32.14	0.00	0.41	32.55	31.72	32.55	31.72	5.55	32.14	32.14	32.55	31.72	**O.K**
203	1.0DL+1.0FL1+1.0LL+1.0TL	492.69	-0.74	-47.92	40.22	0.10	6.71	47.03	33.62	46.82	33.41	20.03	40.33	40.12	46.93	33.52	**O.K**
204	1.0DL+1.0FL2+1.0LL+1.0TL	491.69	-0.74	-46.99	40.14	0.10	6.58	46.82	33.67	46.61	33.46	19.82	40.25	40.04	46.72	33.57	**O.K**
205	1.0DL+1.0FL1+0.75LL+0.75TL	468.19	-0.56	-35.52	38.22	0.08	4.97	43.27	33.33	43.11	33.17	16.27	38.30	38.14	43.19	33.25	**O.K**
206	1.0DL+1.0FL2+0.75LL+0.75TL	467.19	-0.56	-34.41	38.14	0.08	4.82	43.03	33.40	42.88	33.24	16.03	38.22	38.06	42.96	33.32	**O.K**
207	1.0DL+1.0FL1+1.0PL	588.79	0.00	-142.08	48.06	0.00	19.88	67.95	28.18	67.95	28.18	40.95	48.07	48.07	67.95	28.18	**O.K**
208	1.0DL+1.0FL2+1.0PL	587.79	0.00	-140.97	47.98	0.00	19.73	67.71	28.26	67.71	28.26	40.71	47.99	47.99	67.71	28.26	**O.K**
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0	686.79	-0.74	-191.85	56.06	0.10	26.85	83.02	29.32	82.81	29.11	56.02	56.17	55.96	82.92	29.22	**O.K**
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0	685.89	-0.74	-190.92	55.99	0.10	26.72	82.81	29.38	82.61	29.17	55.81	56.10	55.89	82.71	29.28	**O.K**
211	1.0DL+1.0FL1+0.75LL+0.75TL+1	662.29	-0.56	-179.45	54.06	0.08	25.11	79.25	29.03	79.10	28.87	52.25	54.14	53.99	79.18	28.95	**O.K**
212	1.0DL+1.0FL2+0.75LL+0.75TL+1	661.39	-0.56	-178.34	53.99	0.08	24.96	79.03	29.11	78.87	28.96	52.03	54.07	53.92	78.95	29.04	**O.K**
213	1.0DL+1.0FL1+1.0WLX1	440.39	-20.17	5.92	35.95	2.82	0.83	39.60	37.94	33.96	32.30	12.60	38.77	33.13	36.78	35.12	**O.K**
214	1.0DL+1.0FL1+1.0WLX2	348.89	20.17	-2.04	28.48	2.82	0.28	31.59	31.02	25.94	25.37	4.59	31.31	25.66	28.77	28.20	**O.K**
215	1.0DL+1.0FL1+1.0WLZ1	501.79	0.00	-116.92	40.96	0.00	16.36	57.32	24.60	57.32	24.60	30.32	40.96	40.96	57.32	24.60	**O.K**
216	1.0DL+1.0FL1+1.0WLZ2	287.49	0.00	120.81	23.47	0.00	16.91	40.37	6.56	40.37	6.56	13.37	23.47	23.47	40.37	6.56	**O.K**
217	1.0DL+1.0FL2+1.0WLX1	439.49	-20.17	7.03	35.88	2.82	0.98	39.68	37.71	34.04	32.07	12.68	38.70	33.06	36.86	34.89	**O.K**
218	1.0DL+1.0FL2+1.0WLX2	347.89	20.17	-1.11	28.40	2.82	0.16	31.38	31.07	25.73	25.42	4.38	31.23	25.58	28.56	28.25	**O.K**
219	1.0DL+1.0FL2+1.0WLZ1	500.79	0.00	-115.81	40.88	0.00	16.21	57.09	24.67	57.09	24.67	30.09	40.88	40.88	57.09	24.67	**O.K**

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220 1.0DL+1.0FL2+1.0WLZ2	286.49	0.00	121.92	23.39	0.00	17.06	40.45	6.33	40.45	6.33	13.45	23.39	23.39	40.45	6.33	O.K
221 1.0DL+1.0FL1+0.7SLX1	402.79	-3.15	2.22	32.88	0.44	0.31	33.63	33.01	32.75	32.13	6.63	33.32	32.44	33.19	32.57	O.K
222 1.0DL+1.0FL1+0.7SLX2	386.39	3.15	1.67	31.54	0.44	0.23	32.22	31.75	31.33	30.87	5.22	31.99	31.10	31.78	31.31	O.K
223 1.0DL+1.0FL1+0.7SLZ1	449.49	0.00	-54.39	36.69	0.00	7.61	44.30	29.08	44.30	29.08	17.30	36.69	36.69	44.30	29.08	O.K
224 1.0DL+1.0FL1+0.7SLZ2	339.69	0.00	58.28	27.73	0.00	8.16	35.88	19.57	35.88	19.57	8.88	27.73	27.73	35.88	19.57	O.K
225 1.0DL+1.0FL2+0.7SLX1	401.89	-3.15	3.33	32.81	0.44	0.47	33.71	32.78	32.83	31.90	6.71	33.25	32.37	33.27	32.34	O.K
226 1.0DL+1.0FL2+0.7SLX2	385.39	3.15	2.78	31.46	0.44	0.39	32.29	31.51	31.41	30.63	5.29	31.90	31.02	31.85	31.07	O.K
227 1.0DL+1.0FL2+0.7SLZ1	448.49	0.00	-53.28	36.61	0.00	7.46	44.07	29.16	44.07	29.16	17.07	36.62	36.62	44.07	29.16	O.K
228 1.0DL+1.0FL2+0.7SLZ2	338.79	0.00	59.20	27.66	0.00	8.28	35.94	19.37	35.94	19.37	8.94	27.66	27.66	35.94	19.37	O.K
229 1.0DL+1.0FL1+0.75WLX1+0.75L	471.89	-14.99	0.19	38.52	2.10	0.03	40.64	40.59	36.45	36.40	13.64	40.62	36.43	38.55	38.50	O.K
230 1.0DL+1.0FL1+0.75WLX2+0.75L	403.29	14.99	-5.74	32.92	2.10	0.80	35.82	34.22	31.63	30.02	8.82	35.02	30.83	33.73	32.12	O.K
231 1.0DL+1.0FL1+0.75WLZ1+0.75L	517.99	0.00	-91.95	42.28	0.00	12.87	55.15	29.42	55.15	29.42	28.15	42.29	42.29	55.15	29.42	O.K
232 1.0DL+1.0FL1+0.75WLZ2+0.75L	357.19	0.00	86.40	29.16	0.00	12.09	41.25	17.07	41.25	17.07	14.25	29.16	29.16	41.25	17.07	O.K
233 1.0DL+1.0FL2+0.75WLX1+0.75L	470.99	-14.99	1.30	38.45	2.10	0.18	40.73	40.36	36.53	36.17	13.73	40.55	36.35	38.63	38.27	O.K
234 1.0DL+1.0FL2+0.75WLX2+0.75L	402.29	14.99	-4.81	32.84	2.10	0.67	35.61	34.26	31.42	30.07	8.61	34.94	30.75	33.52	32.17	O.K
235 1.0DL+1.0FL2+0.75WLZ1+0.75L	516.99	0.00	-90.84	42.20	0.00	12.71	54.91	29.49	54.91	29.49	27.91	42.20	42.20	54.91	29.49	O.K
236 1.0DL+1.0FL2+0.75WLZ2+0.75L	356.19	0.00	87.32	29.08	0.00	12.22	41.30	16.86	41.30	16.86	14.30	29.08	29.08	41.30	16.86	O.K
237 1.0DL+1.0FL1+0.525SLX1+0.75	443.79	-2.41	-2.59	36.23	0.34	0.36	36.93	36.20	36.25	35.53	9.93	36.57	35.89	36.59	35.87	O.K
238 1.0DL+1.0FL1+0.525SLX2+0.75	431.39	2.41	-2.96	35.22	0.34	0.41	35.97	35.14	35.29	34.46	8.97	35.56	34.88	35.63	34.80	O.K
239 1.0DL+1.0FL1+0.525SLZ1+0.75	478.79	0.00	-44.96	39.08	0.00	6.29	45.38	32.79	45.38	32.79	18.38	39.09	39.09	45.38	32.79	O.K
240 1.0DL+1.0FL1+0.525SLZ2+0.75	396.39	0.00	39.41	32.36	0.00	5.51	37.87	26.84	37.87	26.84	10.87	32.36	32.36	37.87	26.84	O.K
241 1.0DL+1.0FL2+0.525SLX1+0.75	442.79	-2.41	-1.48	36.15	0.34	0.21	36.69	36.28	36.02	35.60	9.69	36.49	35.81	36.36	35.94	O.K
242 1.0DL+1.0FL2+0.525SLX2+0.75	430.49	2.41	-2.04	35.14	0.34	0.28	35.76	35.19	35.09	34.52	8.76	35.48	34.81	35.43	34.86	O.K
243 1.0DL+1.0FL2+0.525SLZ1+0.75	477.79	0.00	-44.03	39.00	0.00	6.16	45.16	32.84	45.16	32.84	18.16	39.00	39.00	45.16	32.84	O.K
244 1.0DL+1.0FL2+0.525SLZ2+0.75	395.39	0.00	40.52	32.28	0.00	5.67	37.95	26.61	37.95	26.61	10.95	32.28	32.28	37.95	26.61	O.K
245 0.6DL+1.0WLX1	431.59	-20.17	2.04	35.23	2.82	0.28	38.34	37.77	32.69	32.13	11.34	38.06	32.41	35.52	34.95	O.K
246 0.6DL+1.0WLX2	339.99	20.17	-6.11	27.75	2.82	0.85	31.43	29.72	25.79	24.08	4.43	30.58	24.94	28.61	26.90	O.K
247 0.6DL+1.0WLZ1	492.99	0.00	-120.81	40.24	0.00	16.91	57.15	23.34	57.15	23.34	30.15	40.25	40.25	57.15	23.34	O.K
248 0.6DL+1.0WLZ2	278.69	0.00	116.92	22.75	0.00	16.36	39.11	6.39	39.11	6.39	12.11	22.75	22.75	39.11	6.39	O.K
249 0.6DL+0.6FL1+0.7SLX1	390.89	-3.15	1.48	31.91	0.44	0.21	32.56	32.14	31.68	31.26	5.56	32.35	31.47	32.12	31.70	O.K
250 0.6DL+0.6FL1+0.7SLX2	374.39	3.15	0.93	30.56	0.44	0.13	31.13	30.87	30.25	29.99	4.13	31.00	30.12	30.69	30.43	O.K
251 0.6DL+0.6FL1+0.7SLZ1	437.49	0.00	-55.13	35.71	0.00	7.71	43.43	28.00	43.43	28.00	16.43	35.72	35.72	43.43	28.00	O.K
252 0.6DL+0.6FL1+0.7SLZ2	327.79	0.00	57.54	26.76	0.00	8.05	34.81	18.71	34.81	18.71	7.81	26.76	26.76	34.81	18.71	O.K

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Ultimate Conditions

101 1.4DL+1.4FL1	406.59	0.00	2.78	33.19	0.00	0.39	33.58	32.80	33.58	32.80	6.58	33.19	33.19	33.58	32.80	
102 1.4DL+1.4FL2	405.19	0.00	4.07	33.08	0.00	0.57	33.65	32.51	33.65	32.51	6.65	33.08	33.08	33.65	32.51	
103 1.2DL+1.2FL1+1.6LL	492.29	0.00	-7.77	40.19	0.00	1.09	41.27	39.10	41.27	39.10	14.27	40.19	40.19	41.27	39.10	
104 1.2DL+1.2FL2+1.6LL	491.09	0.00	-6.48	40.09	0.00	0.91	41.00	39.18	41.00	39.18	14.00	40.09	40.09	41.00	39.18	
105 1.2DL+1.2FL1+1.2TL+1.6LL	541.19	-0.93	-59.94	44.18	0.13	8.39	52.70	35.92	52.44	35.66	25.70	44.31	44.05	52.57	35.79	
106 1.2DL+1.2FL2+1.2TL+1.6LL	540.09	-0.93	-58.83	44.09	0.13	8.23	52.45	35.99	52.19	35.73	25.45	44.22	43.96	52.32	35.86	
107 1.4DL+1.4FL1+1.4PL	678.39	0.00	-198.88	55.38	0.00	27.83	83.21	27.55	83.21	27.55	56.21	55.38	55.38	83.21	27.55	
108 1.4DL+1.4FL2+1.4PL	676.99	0.00	-197.40	55.26	0.00	27.62	82.89	27.64	82.89	27.64	55.89	55.27	55.27	82.89	27.64	
109 1.2DL+1.2FL1+1.6LL+1.2PL	725.19	0.00	-180.38	59.20	0.00	25.24	84.44	33.96	84.44	33.96	57.44	59.20	59.20	84.44	33.96	
110 1.2DL+1.2FL2+1.6LL+1.2PL	724.09	0.00	-179.27	59.11	0.00	25.09	84.20	34.02	84.20	34.02	57.20	59.11	59.11	84.20	34.02	
111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2	774.19	-0.93	-232.73	63.20	0.13	32.57	95.90	30.76	95.64	30.50	68.90	63.33	63.07	95.77	30.63	
112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2	772.99	-0.93	-231.62	63.10	0.13	32.41	95.64	30.82	95.39	30.56	68.64	63.23	62.98	95.52	30.69	
113 1.2DL+0.8WLX1	443.59	-16.10	-0.93	36.21	2.25	0.13	38.59	38.33	34.09	33.83	11.59	38.46	33.96	36.34	36.08	
114 1.2DL+0.8WLX2	370.29	16.10	-7.22	30.23	2.25	1.01	33.49	31.47	28.99	26.97	6.49	32.48	27.98	31.24	29.22	
115 1.2DL+0.8WLZ1	492.59	0.00	-99.16	40.21	0.00	13.88	54.09	26.33	54.09	26.33	27.09	40.21	40.21	54.09	26.33	
116 1.2DL+0.8WLZ2	321.19	0.00	91.02	26.22	0.00	12.74	38.96	13.48	38.96	13.48	11.96	26.22	26.22	38.96	13.48	
117 1.2DL+1.6WLX1+0.5LL	508.79	-32.19	-0.93	41.53	4.50	0.13	46.17	45.91	37.16	36.90	19.17	46.04	37.03	41.67	41.41	
118 1.2DL+1.6WLX2+0.5LL	362.29	32.19	-13.69	29.57	4.50	1.92	36.00	32.16	26.99	23.15	9.00	34.08	25.07	31.50	27.66	
119 1.2DL+1.6WLZ1+0.5LL	606.99	0.00	-197.40	49.55	0.00	27.62	77.17	21.93	77.17	21.93	50.17	49.55	49.55	77.17	21.93	
120 1.2DL+1.6WLZ2+0.5LL	264.09	0.00	182.97	21.56	0.00	25.60	47.16	-4.05	47.16	-4.05	20.16	21.56	21.56	47.16	-4.05	
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	444.69	-4.63	-4.07	36.30	0.65	0.57	37.52	36.38	36.22	35.08	10.52	36.95	35.65	36.87	35.73	
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	421.19	4.63	-5.00	34.38	0.65	0.70	35.73	34.33	34.43	33.04	8.73	35.03	33.74	35.08	33.69	
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	511.29	0.00	-84.92	41.74	0.00	11.88	53.62	29.85	53.62	29.85	26.62	41.74	41.74	53.62	29.85	
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	354.49	0.00	75.85	28.94	0.00	10.61	39.55	18.32	39.55	18.32	12.55	28.94	28.94	39.55	18.32	
125 0.9DL+1.6WLX1	469.69	-32.19	3.33	38.34	4.50	0.47	43.31	42.38	34.30	33.37	16.31	42.85	33.84	38.81	37.88	
126 0.9DL+1.6WLX2	323.09	32.19	-9.44	26.37	4.50	1.32	32.20	29.56	23.19	20.55	5.20	30.88	21.87	27.70	25.06	
127 0.9DL+1.6WLZ1	567.79	0.00	-193.14	46.35	0.00	27.03	73.38	19.32	73.38	19.32	46.38	46.35	46.35	73.38	19.32	
128 0.9DL+1.6WLZ2	224.89	0.00	187.22	18.36	0.00	26.20	44.56	-7.84	44.56	-7.84	17.56	18.36	18.36	44.56	-7.84	
129 0.9DL+1.0SLX1	408.09	-4.63	-2.59	33.31	0.65	0.36	34.32	33.60	33.03	32.30	7.32	33.96	32.67	33.68	32.95	
130 0.9DL+1.0SLX2	384.59	4.63	-3.52	31.40	0.65	0.49	32.53	31.55	31.24	30.26	5.53	32.04	30.75	31.89	30.91	
131 0.9DL+1.0SLZ1	474.79	0.00	-83.44	38.76	0.00	11.68	50.43	27.08	50.43	27.08	23.43	38.76	38.76	50.43	27.08	
132 0.9DL+1.0SLZ2	317.99	0.00	77.33	25.96	0.00	10.82	36.78	15.14	36.78	15.14	9.78	25.96	25.96	36.78	15.14	
Max :	774	32	187													
Min :	225	-32	-233													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 5.50 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 15.50 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 56.02 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 66.02 kN/m²

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Sample calculation of values of base pressure:

Load case : 202 1.0DL+1.0FL2

Maximum pressure, p_1 (pmax) = 32.14+0.00+0.41= 32.55 kN/m^2

p_2= 32.14+0.00-0.41= 31.72 kN/m^2

p_3= 32.14-(0.00+0.41)= 31.72 kN/m^2

p_4= 32.14-0.00+0.41= 32.55 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 32.55-1.5*18= 5.55 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (32.55+31.72)/2= 32.14 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (31.72+32.55)/2= 32.14 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (32.55+32.55)/2= 32.55 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (31.72+31.72)/2= 31.72 kN/m^2

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DESIGN OF ISOLATED FOOTING F17

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.55+2*0.5*0.332)+(0.55+2*0.5*0.332))*2

= 3.528 m

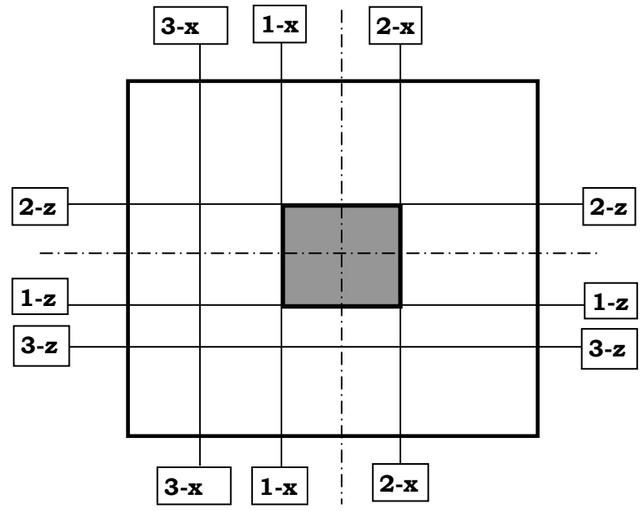
Punching Area= ((0.55+2*0.5*0.332)*(0.55+2*0.5*0.332))

= 0.78 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.1*18)

= 29.40 kN/m^2

Area of footing, A_f= 12.25 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4DL+1.4FL1	33.19	33.19	33.19	4.12	4.12	4.29	43.48
102	1.4DL+1.4FL2	33.08	33.08	33.08	4.00	4.00	4.16	42.17
103	1.2DL+1.2FL1+1.6LL	40.19	40.19	40.19	11.74	11.74	12.20	123.73
104	1.2DL+1.2FL2+1.6LL	40.09	40.09	40.09	11.63	11.63	12.09	122.60
105	1.2DL+1.2FL1+1.2TL+1.6LL	44.20	44.16	44.23	16.18	15.98	16.82	169.51
106	1.2DL+1.2FL2+1.2TL+1.6LL	44.11	44.07	44.14	16.08	15.88	16.71	168.48
107	1.4DL+1.4FL1+1.4PL	55.38	55.38	55.38	28.26	28.26	29.38	297.98
108	1.4DL+1.4FL2+1.4PL	55.27	55.27	55.27	28.14	28.14	29.26	296.67
109	1.2DL+1.2FL1+1.6LL+1.2PL	59.20	59.20	59.20	32.42	32.42	33.70	341.80
110	1.2DL+1.2FL2+1.6LL+1.2PL	59.11	59.11	59.11	32.32	32.32	33.60	340.77
111	1.2DL+1.2FL1+1.2TL+1.6LL+1	63.22	63.18	63.25	36.87	36.67	38.33	387.68
112	1.2DL+1.2FL2+1.2TL+1.6LL+1	63.12	63.09	63.15	36.76	36.57	38.22	386.55
113	1.2DL+0.8WLX1	36.56	35.86	37.01	9.17	5.65	9.42	78.13

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114 1.2DL+0.8WLX2	30.58	29.88	31.03	2.66	-0.86	2.66	9.49
115 1.2DL+0.8WLZ1	40.21	40.21	40.21	11.76	11.76	12.23	124.01
116 1.2DL+0.8WLZ2	26.22	26.22	26.22	-3.46	-3.46	-3.60	-36.48
117 1.2DL+1.6WLX1+0.5LL	42.24	40.83	43.13	16.72	9.68	17.17	139.18
118 1.2DL+1.6WLX2+0.5LL	30.28	28.87	31.17	3.71	-3.33	3.65	2.00
119 1.2DL+1.6WLZ1+0.5LL	49.55	49.55	49.55	21.92	21.92	22.79	231.12
120 1.2DL+1.6WLZ2+0.5LL	21.56	21.56	21.56	-8.53	-8.53	-8.87	-89.94
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	36.40	36.20	36.53	8.01	7.00	8.30	79.16
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	34.49	34.28	34.61	5.93	4.92	6.13	57.15
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	41.74	41.74	41.74	13.42	13.42	13.96	141.52
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	28.94	28.94	28.94	-0.50	-0.50	-0.52	-5.30
125 0.9DL+1.6WLX1	39.05	37.64	39.94	13.25	6.21	13.57	102.57
126 0.9DL+1.6WLX2	27.08	25.67	27.97	0.23	-6.81	0.03	-34.70
127 0.9DL+1.6WLZ1	46.35	46.35	46.35	18.44	18.44	19.17	194.42
128 0.9DL+1.6WLZ2	18.36	18.36	18.36	-12.01	-12.01	-12.49	-126.65
129 0.9DL+1.0SLX1	33.42	33.21	33.54	4.76	3.75	4.92	44.89
130 0.9DL+1.0SLX2	31.50	31.29	31.62	2.67	1.67	2.75	22.88
131 0.9DL+1.0SLZ1	38.76	38.76	38.76	10.18	10.18	10.59	107.34
132 0.9DL+1.0SLZ2	25.96	25.96	25.96	-3.74	-3.74	-3.89	-39.48

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Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1x} = 33.19-(33.19-33.19)*1.475/3.5 = 33.19 kN/m^2 Overhang = 1.475 m

p_{2x} = 33.19+(33.19-33.19)*1.475/3.5 = 33.19 kN/m^2 Effective depth = 0.344 m

p_{3x} = 33.19-(33.19-33.19)*(1.475-0.344)/3.5 = 33.19 kN/m^2

BM_{1x} = (33.19-29.4)*1.475*1.475/2+0.5*(33.19-33.19)*1.475*1.475*0.67 = 4.12 kN-m/m

BM_{2x} = (33.19-29.4)*1.475*1.475/2-0.5*(-33.19+33.19)*1.475*1.475*0.67 = 4.12 kN-m/m

SF_{3x} = (33.19-29.4)*(1.475-0.344)+0.5*(33.19-33.19)*(1.475-0.344) = 4.29 kN/m

$S.F_{(punching)}$ = (12.25-0.78)*(33.19-29.4) = 43.48 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	36.87	0.31	0.42	0.001	342.66	619.20	12ϕT	182.65	300	150	753.98
At top	-12.01	-0.10	0.14	0.000	111.08	309.60	12ϕT	365.30	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 38.33 kN/m

Shear stress , τ_v= (38.33*1000)/(1000*344)

= 0.11 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 387.68 kN

Strength of Concrete in shear Vc1 = 3589.97 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 2363.47 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2393.31 kN **Table 22.6.5.2_ (c)**

Design Vc= 2363.47 kN

Allowable shear stress, (ϕ Vc)/bd = 1.46 N/mm^2

Design punching shear stress, τ = (387.68*1000)/(3528*344)

= 0.32 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4DL+1.4FL1	33.25	33.13	33.33	4.43	3.82	4.63
102	1.4DL+1.4FL2	33.17	47.45	33.28	4.45	5.59	4.64
103	1.2DL+1.2FL1+1.6LL	40.36	51.16	40.56	12.58	6.98	13.16
104	1.2DL+1.2FL2+1.6LL	40.23	57.61	40.41	12.34	10.00	12.92
105	1.2DL+1.2FL1+1.2TL+1.6LL	45.50	55.56	47.09	22.67	9.30	23.35
106	1.2DL+1.2FL2+1.2TL+1.6LL	45.38	54.12	46.94	22.44	8.54	23.13
107	1.4DL+1.4FL1+1.4PL	59.75	49.21	65.03	50.11	6.85	51.12
108	1.4DL+1.4FL2+1.4PL	59.61	31.49	64.85	49.83	4.92	50.83
109	1.2DL+1.2FL1+1.6LL+1.2PL	63.17	31.77	67.95	52.24	-0.02	53.49
110	1.2DL+1.2FL2+1.6LL+1.2PL	63.05	43.84	67.81	52.02	1.68	53.27
111	1.2DL+1.2FL1+1.2TL+1.6LL+1	68.32	34.73	74.50	62.34	-14.73	63.70
112	1.2DL+1.2FL2+1.2TL+1.6LL+1	68.20	33.29	74.35	62.12	5.40	63.48
113	1.2DL+0.8WLX1	36.23	32.91	36.26	7.51	0.42	7.88
114	1.2DL+0.8WLX2	30.39	40.97	30.58	1.70	0.70	1.73
115	1.2DL+0.8WLZ1	42.39	32.28	45.02	22.66	-15.78	23.04
116	1.2DL+0.8WLZ2	28.22	21.79	30.64	6.54	-0.42	6.17
117	1.2DL+1.6WLX1+0.5LL	41.56	37.35	41.59	13.31	4.25	13.98
118	1.2DL+1.6WLX2+0.5LL	29.88	34.67	30.25	1.70	1.66	1.68
119	1.2DL+1.6WLZ1+0.5LL	53.89	27.81	59.13	43.61	-7.74	44.29
120	1.2DL+1.6WLZ2+0.5LL	25.58	11.68	30.44	11.57	-4.22	10.74

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121 1.2DL+1.0SLX1+0.5LL+0.5FL1	36.39	34.09	36.50	7.95	2.90	8.33
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	34.49	38.06	34.63	5.97	2.94	6.23
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	43.60	32.42	45.86	22.75	-6.23	23.25
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	30.60	26.85	32.62	7.83	5.72	7.64
125 0.9DL+1.6WLX1	38.42	36.13	38.51	10.10	4.40	10.58
126 0.9DL+1.6WLX2	26.59	37.74	26.84	-2.25	3.01	-2.44
127 0.9DL+1.6WLZ1	50.60	28.56	55.73	39.66	-9.29	40.18
128 0.9DL+1.6WLZ2	22.48	11.74	27.45	8.56	0.13	7.55
129 0.9DL+1.0SLX1	33.37	33.19	33.44	4.55	3.38	4.76
130 0.9DL+1.0SLX2	31.48	41.02	31.57	2.56	4.24	2.66
131 0.9DL+1.0SLZ1	40.59	33.07	42.80	19.34	-7.82	19.68
132 0.9DL+1.0SLZ2	27.66	24.18	29.71	4.75	2.84	4.40

Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1z} = 33.58-(33.58-32.8)*1.475/3.5= 33.25 kN/m^2

p_{2z} = 32.8+(33.58-32.8)*1.475/3.5= 33.13 kN/m^2

p_{3z} = 33.58-(33.58-32.8)*(1.475-0.332)/3.5= 33.33 kN/m^2

Overhang = 1.475 m

Effective depth = 0.332 m

BM_{1z} = (33.25-29.4)*1.475*1.475/2+0.5*(33.58-33.25)*1.475*1.475*0.67 = 4.43 kN-m/m

BM_{2z} = (33.13-29.4)*1.475*1.475/2-0.5*(-32.8+33.13)*1.475*1.475*0.67 = 3.82 kN-m/m

SF_{3z} = (33.33-29.4)*(1.475-0.332)+0.5*(33.58-33.33)*(1.475-0.332) = 4.63 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	62.34	0.566	0.754	0.0018	625.71	597.60	12ϕT	180.75	300	150	754.0
At top	-15.78	-0.143	-0.191	-0.0005	-1.50	298.80	12ϕT	378.51	300	250	452.4

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Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 63.70 kN/m

Shear stress , τ_v= (63.7*1000)/(1000*332)

= 0.19 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	12ϕT	**150**	c/c
	At Top	12ϕT	**250**	c/c
Along Z Direction :				
	At Bottom	12ϕT	**150**	c/c
	At Top	12ϕT	**250**	c/c

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5.10 Design of column C7&C7A

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 550 mm

Depth parallel to Z-Z axis, b = 550 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.7

Y_b = (b-d')/b = 0.7

Unsupported length of column parallel to X-X axis, l_{ux} = 1.85 m kl_{ux}/r_x = 13.45 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.85 m kl_{uz}/r_z = 13.45

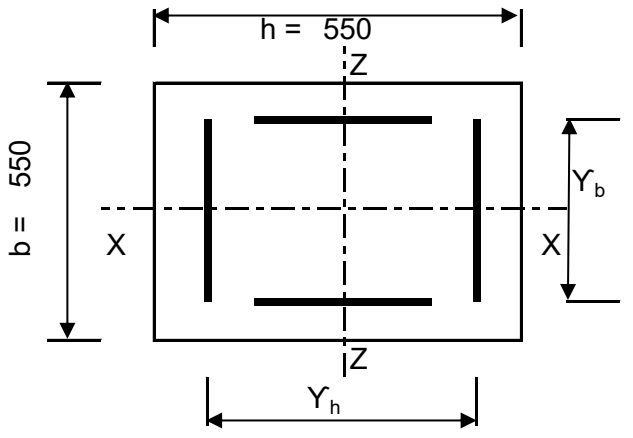
Radius of gyration r_x = 165 mm

Radius of gyration r_z = 165 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 8377 KN safe

Factored Force Value				Design force Value			$K_n = P_n/f_c'bh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx} = M_{ux}/φ$	$M_{nz} = M_{uz}/φ$		$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
106	385.4	0.0	2.8	593	0.0	4.3	0.056	0.050	291.16	0.00	0.050	291.2	0.01		0.00	safe
126	162.2	0.0	0.0	250	0.0	0.0	0.024	0.050	291.16	0.00	0.050	291.2	0.00		0.00	safe
117	53.2	0.0	97.3	82	0.0	149.7	0.008	0.050	291.16	0.00	0.050	291.2	0.51		0.34	safe
126	271.1	0.0	100.2	417	0.0	154.2	0.039	0.050	291.16	0.00	0.050	291.2	0.53		0.36	safe
120	72.3	7.4	29.2	111	11.4	45.0	0.011	0.050	291.16	0.04	0.050	291.2	0.15		0.06	safe
119	64.0	7.4	0.2	98	11.4	0.3	0.009	0.050	291.16	0.04	0.050	291.2	0.00		0.01	safe
101	96.4	0.0	1.4	148	0.0	2.2	0.014	0.050	291.16	0.00	0.050	291.2	0.01	1.609	0.00	safe
101	96.4	0.0	1.4	148	0.0	2.2	0.014	0.050	291.16	0.00	0.050	291.2	0.01		0.00	safe
119	64.0	7.4	0.2	98	11.4	0.3	0.009	0.050	291.16	0.04	0.050	291.2	0.00		0.01	safe
120	72.3	7.4	56.3	111	11.4	86.7	0.011	0.050	291.16	0.04	0.050	291.2	0.30		0.15	safe
117	53.2	0.0	97.3	82	0.0	149.7	0.008	0.050	291.16	0.00	0.050	291.2	0.51		0.34	safe
126	271.1	0.0	100.2	417	0.0	154.2	0.039	0.050	291.16	0.00	0.050	291.2	0.53		0.36	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
3025	4	20	+	8	20	3772	1.25	10	320	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

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5.11 Design of column C8

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1.5 % i.e. ρ_g = 0.015

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 750 mm

Depth parallel to Z-Z axis, b = 750 mm

Clear Cover = 50 mm

d' = 72.5 mm

Y_h = (h-d')/h = 0.81

Y_b = (b-d')/b = 0.81

Unsupported length of column parallel to X-X axis, l_{ux} = 1.85 m kl_{ux}/r_x = 8.222 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.85 m kl_{uz}/r_z = 8.222

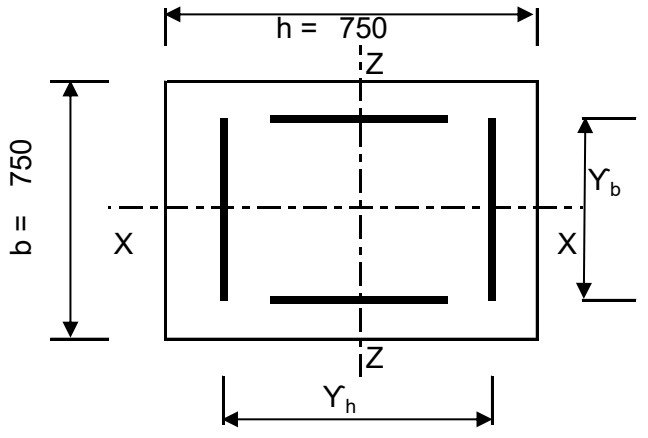
Radius of gyration r_x = 225 mm

Radius of gyration r_z = 225 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 16453.93 KN safe

Factored Force Value				Design force Value			$K_n = P_n/f_cbh$	Uniaxial moment capacity						Check for Interaction Ratio		
								about X-X axis			about ZZ axis					
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx} = M_{ux}/φ$	$M_{nz} = M_{uz}/φ$		$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
112	1889.6	309.9	636.2	2907	476.8	978.8	0.148	0.095	1402.73	0.34	0.095	1403	0.70		0.737	safe
132	49.7	0.0	0.0	76	0.0	0.0	0.004	0.050	738.281	0.00	0.050	738.3	0.00		0	safe
112	1889.6	309.9	636.2	2907	476.8	978.8	0.148	0.095	1402.73	0.34	0.095	1403	0.70		0.737	safe
130	79.3	15.3	71.0	122	23.5	109.2	0.006	0.050	738.281	0.03	0.050	738.3	0.15		0.051	safe
106	1860.7	325.1	631.4	2863	500.2	971.4	0.145	0.095	1402.73	0.36	0.095	1403	0.69		0.744	safe
112	1463.0	354.0	675.0	2251	544.6	1038.5	0.114	0.095	1402.73	0.39	0.095	1403	0.74		0.835	safe
101	683.8	0.0	174.2	1052	0.0	268.0	0.053	0.060	885.938	0.00	0.060	885.9	0.30	1.609	0.147	safe
101	683.8	0.0	174.2	1052	0.0	268.0	0.053	0.060	885.938	0.00	0.060	885.9	0.30		0.147	safe
112	1463.0	354.0	675.0	2251	544.6	1038.5	0.114	0.095	1402.73	0.39	0.095	1403	0.74		0.835	safe
106	1860.7	325.1	631.4	2863	500.2	971.4	0.145	0.095	1402.73	0.36	0.095	1403	0.69		0.744	safe
112	1463.0	354.0	675.0	2251	544.6	1038.5	0.114	0.095	1402.73	0.39	0.095	1403	0.74		0.835	safe
130	79.3	15.3	71.0	122	23.5	109.2	0.006	0.050	738.281	0.03	0.050	738.3	0.15		0.051	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db				A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
8438	4	25 +	16	25	9822	1.75	10	400	300



P_n : Nominal axial load

M_n : Nominal moment

P_u : Factored design axial load

M_u : Factored design moment

Use interaction diagram R35-420.8

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5.12 Design of column C9

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.7

Y_b = (b-d')/b = 0.7

Unsupported length of column parallel to X-X axis, l_{ux} = 1.2 m kl_{ux}/r_x = 9.6 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.2 m kl_{uz}/r_z = 9.6

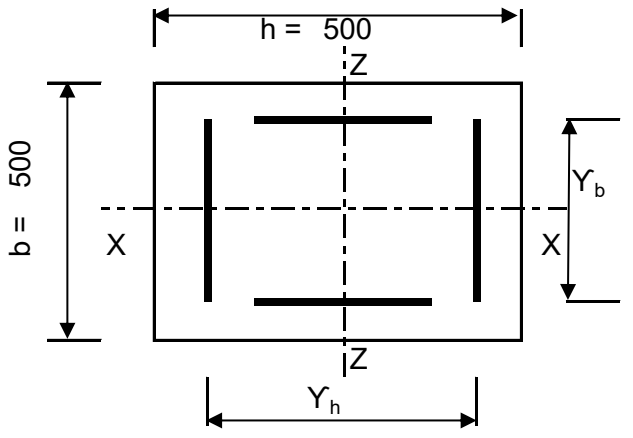
Radius of gyration r_x = 150 mm

Radius of gyration r_z = 150 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 6845 KN safe

Factored Force Value				Design force Value			$K_n = P_n/f_c'bh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx} = M_{ux}/φ$	$M_{nz} = M_{uz}/φ$		$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
102	195.6	33.2	12.3	301	51.1	18.9	0.034	0.050	218.75	0.23	0.050	218.8	0.09		0.12	safe
109	9.4	0.0	0.0	14	0.0	0.0	0.002	0.050	218.75	0.00	0.050	218.8	0.00		0.00	safe
107	6.0	4.8	9.2	9	7.4	14.2	0.001	0.050	218.75	0.03	0.050	218.8	0.06		0.02	safe
104	150.9	21.7	19.1	232	33.4	29.4	0.027	0.050	218.75	0.15	0.050	218.8	0.13		0.09	safe
102	83.8	33.9	29.2	129	52.2	45.0	0.015	0.050	218.75	0.24	0.050	218.8	0.21		0.18	safe
102	195.6	33.2	12.3	301	51.1	18.9	0.034	0.050	218.75	0.23	0.050	218.8	0.09		0.12	safe
101	39.4	7.7	2.7	61	11.8	4.2	0.007	0.050	218.75	0.05	0.050	218.8	0.02	1.609	0.01	safe
101	39.4	7.7	2.7	61	11.8	4.2	0.007	0.050	218.75	0.05	0.050	218.8	0.02		0.01	safe
102	195.6	33.2	12.3	301	51.1	18.9	0.034	0.050	218.75	0.23	0.050	218.8	0.09		0.12	safe
102	83.8	33.9	56.3	129	52.2	86.7	0.015	0.050	218.75	0.24	0.050	218.8	0.40		0.33	safe
107	6.0	4.8	9.2	9	7.4	14.2	0.001	0.050	218.75	0.03	0.050	218.8	0.06		0.02	safe
104	150.9	21.7	19.1	232	33.4	29.4	0.027	0.050	218.75	0.15	0.050	218.8	0.13		0.09	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
2500	4	20	+	8	16	2867	1.15	10	256	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

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5.13 Design of column C10

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.7

Y_b = (b-d')/b = 0.7

Unsupported length of column parallel to X-X axis, l_{ux} = 1.4 m kl_{ux}/r_x = 11.2 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.4 m kl_{uz}/r_z = 11.2

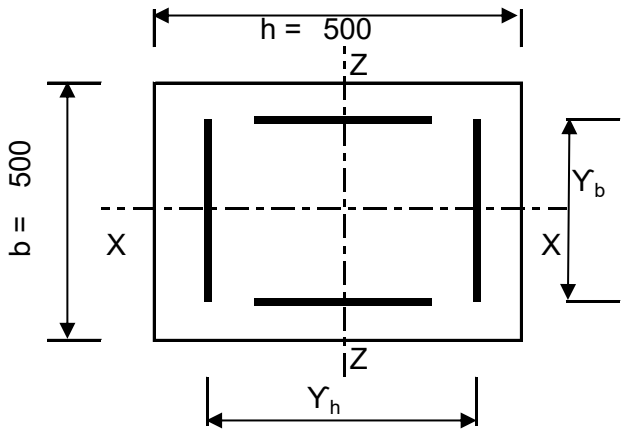
Radius of gyration r_x = 150 mm

Radius of gyration r_z = 150 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 6845 KN safe

Factored Force Value				Design force Value			$K_n = P_n/f_c'bh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx} = M_{ux}/φ$	$M_{nz} = M_{uz}/φ$		$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
109	162.6	11.4	0.0	250	17.5	0.0	0.029	0.050	218.75	0.08	0.050	218.8	0.00		0.02	safe
128	12.1	0.0	0.0	19	0.0	0.0	0.002	0.050	218.75	0.00	0.050	218.8	0.00		0.00	safe
117	45.7	0.0	5.2	70	0.0	8.0	0.008	0.050	218.75	0.00	0.050	218.8	0.04		0.01	safe
118	57.3	0.0	5.2	88	0.0	8.0	0.010	0.050	218.75	0.00	0.050	218.8	0.04		0.01	safe
111	162.0	35.8	29.2	249	55.1	45.0	0.028	0.050	218.75	0.25	0.050	218.8	0.21		0.19	safe
119	24.8	10.2	0.0	38	15.7	0.0	0.004	0.050	218.75	0.07	0.050	218.8	0.00		0.02	safe
101	39.9	0.0	0.0	61	0.0	0.0	0.007	0.050	218.75	0.00	0.050	218.8	0.00	1.609	0.00	safe
101	39.9	0.0	0.0	61	0.0	0.0	0.007	0.050	218.75	0.00	0.050	218.8	0.00		0.00	safe
119	24.8	10.2	0.0	38	15.7	0.0	0.004	0.050	218.75	0.07	0.050	218.8	0.00		0.02	safe
111	162.0	35.8	56.3	249	55.1	86.7	0.028	0.050	218.75	0.25	0.050	218.8	0.40		0.34	safe
117	45.7	0.0	5.2	70	0.0	8.0	0.008	0.050	218.75	0.00	0.050	218.8	0.04		0.01	safe
118	57.3	0.0	5.2	88	0.0	8.0	0.010	0.050	218.75	0.00	0.050	218.8	0.04		0.01	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
2500	4	20	+	8	16	2867	1.15	10	256	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

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5.14 Design of column C10A

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 825 mm

Depth parallel to Z-Z axis, b = 825 mm

Clear Cover = 50 mm

d' = 72.5 mm

Y_h = (h-d')/h = 0.8

Y_b = (b-d')/b = 0.8

Unsupported length of column parallel to X-X axis, l_{ux} = 1.4 m kl_{ux}/r_x = 6.788 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.4 m kl_{uz}/r_z = 6.788

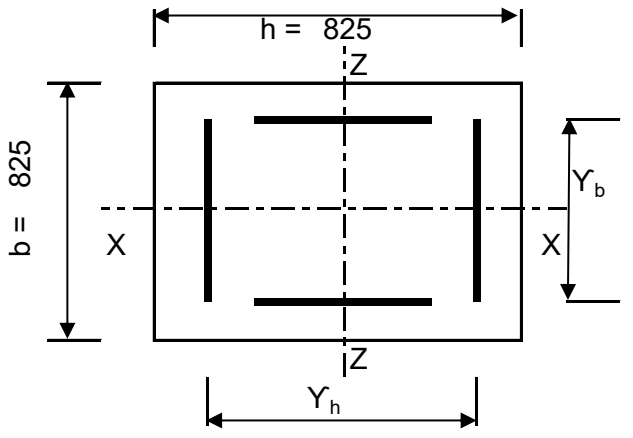
Radius of gyration r_x = 248 mm

Radius of gyration r_z = 248 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 18382 KN safe

Factored Force Value				Design force Value			$K_n = P_n/f_c'bh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx} = M_{ux}/φ$	$M_{nz} = M_{uz}/φ$		$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
111	256.3	35.7	34.7	394	54.9	53.4	0.017	0.050	982.65	0.06	0.050	982.7	0.05		0.02	safe
128	5.2	0.0	0.0	8	0.0	0.0	0.000	0.050	982.65	0.00	0.050	982.7	0.00		0.00	safe
125	43.9	1.6	1.2	68	2.5	1.8	0.003	0.050	982.65	0.00	0.050	982.7	0.00		0.00	safe
111	256.3	35.7	34.7	394	54.9	53.4	0.017	0.050	982.65	0.06	0.050	982.7	0.05		0.02	safe
128	23.5	9.2	29.2	36	14.2	45.0	0.002	0.050	982.65	0.01	0.050	982.7	0.05		0.01	safe
111	256.3	35.7	34.7	394	54.9	53.4	0.017	0.050	982.65	0.06	0.050	982.7	0.05		0.02	safe
101	60.5	1.8	5.5	93	2.8	8.5	0.004	0.050	982.65	0.00	0.050	982.7	0.01	1.609	0.00	safe
101	60.5	1.8	5.5	93	2.8	8.5	0.004	0.050	982.65	0.00	0.050	982.7	0.01		0.00	safe
111	256.3	35.7	34.7	394	54.9	53.4	0.017	0.050	982.65	0.06	0.050	982.7	0.05		0.02	safe
128	23.5	9.2	56.3	36	14.2	86.7	0.002	0.050	982.65	0.01	0.050	982.7	0.09		0.02	safe
125	43.9	1.6	1.2	68	2.5	1.8	0.003	0.050	982.65	0.00	0.050	982.7	0.00		0.00	safe
111	256.3	35.7	34.7	394	54.9	53.4	0.017	0.050	982.65	0.06	0.050	982.7	0.05		0.02	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db				A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
6806	4	25	+	16	20	6993	1.03	10	320	250



P_n : Nominal axial load

M_n : Nominal moment

P_u : Factored design axial load

M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

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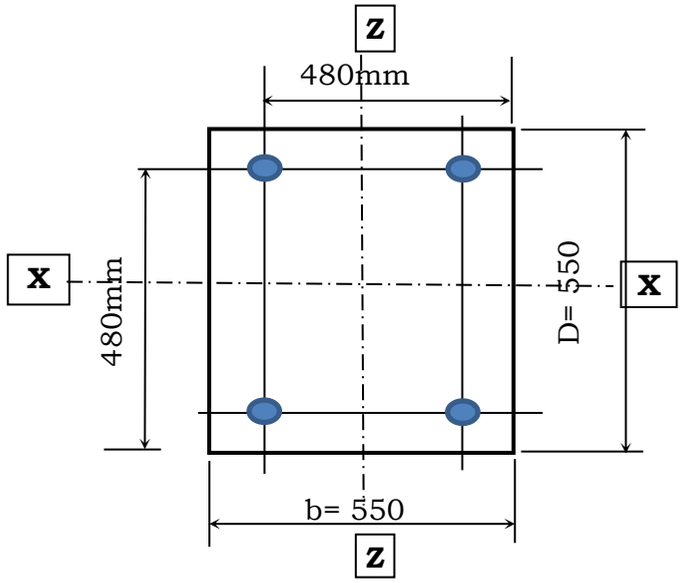
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MD-216-3310-EG-CV-CAL-1001	Mansourah Massarah Gold Project	Page 98 of (219)
900707_GAU01_ZF01_CED01_ 33108_OU100677094_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF MISCELLANEOUS FOUNDATION	Date: 09.11.2023
O19045_450_C_K_24358_001_R2	AS BUILT	Rev: D01

5.15 DESIGN OF COLUMN C11

STAAD Node 7732 7428 7430

Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	20mm	
Dia. of link=	10mm	
Effective cover =	70mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	1.00%	9.6 AC1 318M -14
Max. Steel Ratio	4.00%	
Self weight of Column =	1.343kN	



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1850mm
Unsupproted length of column about Z (l_{uz}) =	1850mm

For Columns not braced against sideway, Where, Effective length factor K= 1.2

Slenderness ratio,	K l_{ux} / r_x =	13.99	< 22 SECTION 6.2.5a	Radius of gyration r_x=	159	mm
	K l_{uz} / r_z =	13.99	< 22 SECTION 6.2.5a	Radius of gyration r_z=	159	mm

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900707_GAU01_ZF01_CED01_ 33108_OU100677094_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF MISCELLANEOUS FOUNDATION	Date: 09.11.2023
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γ_y = 0.78

γ_z = 0.78

	Node	Load Combination	Factored Load From STAAD			Load at Column Base			Nominal Load/Moment			Kn = Pn/f'_cbD (N/mm^2)
			Pu (T)	Hux (T)	Huz (T)	Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
Max Fx	7428	118 1.2DL+1.6WLX2+0.5L	16.80	1.8	-0.1	16.93	3.3	0.2	26.1	5.1	0.3	0.02
Min Fx	7732	112 1.2DL+1.2FL2+1.2TL	9.80	-3.7	-0.5	9.93	6.8	0.9	15.3	10.5	1.4	0.01
Max Fy	7430	111 1.2DL+1.2FL1+1.2TL	41.80	0.0	-12.8	41.93	0.0	23.7	64.5	0.0	36.4	0.06
Min Fy	7430	128 0.9DL+1.6WLZ2	14.30	0.0	10.3	14.43	0.0	19.1	22.2	0.0	29.3	0.02
Max Fz	7430	128 0.9DL+1.6WLZ2	14.30	0.0	10.3	14.43	0.0	19.1	22.2	0.0	29.3	0.02
Min Fz	7430	111 1.2DL+1.2FL1+1.2TL	41.80	0.0	-12.8	41.93	0.0	23.7	64.5	0.0	36.4	0.06
Max Mx	7428	101 1.4DL+1.4FL1	5.60	0.0	0.0	5.73	0.0	0.0	8.8	0.0	0.0	0.01
Min Mx	7428	101 1.4DL+1.4FL1	5.60	0.0	0.0	5.73	0.0	0.0	8.8	0.0	0.0	0.01
Max My	7428	101 1.4DL+1.4FL1	5.60	0.0	0.0	5.73	0.0	0.0	8.8	0.0	0.0	0.01
Min My	7428	101 1.4DL+1.4FL1	5.60	0.0	0.0	5.73	0.0	0.0	8.8	0.0	0.0	0.01
Max Mz	7428	101 1.4DL+1.4FL1	5.60	0.0	0.0	5.73	0.0	0.0	8.8	0.0	0.0	0.01
Min Mz	7428	101 1.4DL+1.4FL1	5.60	0.0	0.0	5.73	0.0	0.0	8.8	0.0	0.0	0.01

Reinforcement: Provide,

4-Nos	20 mm Dia
4-Nos	20 mm Dia
4-Nos	20 mm Dia

1257.14 Sq mm (At Corner)
1257.14 Sq mm (Along Z axis)
1257.14 Sq mm (Along Y axis)

A_{sc} pvd. = 3771.43 Sq mm

Total A_{sc} = 1.25 %

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MD-216-3310-EG-CV-CAL-1001	Mansourah Massarah Gold Project	Page 100 of (219)
900707_GAU01_ZF01_CED01_ 33108_OU100677094_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF MISCELLANEOUS FOUNDATION	Date: 09.11.2023
O19045_450_C_K_24358_001_R2	AS BUILT	Rev: D01

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1047.118 T SECTION 22.4.2.2

Load Comb	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
118 1.2D	0.070	0.070	40.8	40.8	0.55	1.16	0.09	O.K
112 1.2D	0.070	0.070	40.8	40.8	0.55	1.16	0.23	O.K
111 1.2D	0.070	0.070	40.8	40.8	0.55	1.16	0.88	O.K
128 0.9D	0.070	0.070	40.8	40.8	0.55	1.16	0.68	O.K
128 0.9D	0.070	0.070	40.8	40.8	0.55	1.16	0.68	O.K
111 1.2D	0.070	0.070	40.8	40.8	0.55	1.16	0.88	O.K
101 1.4D	0.070	0.070	40.8	40.8	0.55	1.16	0.00	O.K
101 1.4D	0.070	0.070	40.8	40.8	0.55	1.16	0.00	O.K
101 1.4D	0.070	0.070	40.8	40.8	0.55	1.16	0.00	O.K
101 1.4D	0.070	0.070	40.8	40.8	0.55	1.16	0.00	O.K
101 1.4D	0.070	0.070	40.8	40.8	0.55	1.16	0.00	O.K
101 1.4D	0.070	0.070	40.8	40.8	0.55	1.16	0.00	O.K

Links, 10mm dia @ 300 C/C

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST- CRT-1001) Clause 4.1

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ATTACHMENT 1 STAAD INPUT & OUTPUT FILE FOR F12 DESIGN

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  7, 2020                *
*           Time=    11:33:11                    *
*                                                *
*      USER ID: LT Construction                 *
**************************************************
```

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     1. STAAD SPACE
INPUT FILE: F12 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 26-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 1.705 0 0; 21 0 1.85 0; 22 1.705 1.85 0; 49 -0.889071 0 -1.1
     9. 52 0 0 -1.1; 85 1.705 0 -1.1; 114 2.71107 0 -1.1; 133 -0.889071 0 1.1
    10. 138 0 0 1.1; 167 1.705 0 1.1; 187 2.71107 0 1.1; 1001 0 0.25 0
    11. 1002 1.705 0.25 0; 1003 -0.889071 0 0; 1004 2.71107 0 0; 1005 -0.666803 0 0
    12. 1006 -0.666803 0 -0.275; 1007 -0.889071 0 -0.275; 1008 -0.394535 0 0
    13. 1009 -0.394535 0 -0.275; 1010 -0.222268 0 0; 1011 -0.222268 0 -0.275
    14. 1012 0 0 -0.275; 1013 -0.666803 0 -0.375; 1014 -0.889071 0 -0.375
    15. 1015 -0.394535 0 -0.375; 1016 -0.222268 0 -0.375; 1017 0 0 -0.375
    16. 1018 -0.666803 0 -0.825; 1019 -0.889071 0 -0.825; 1020 -0.394535 0 -0.825
    17. 1021 -0.222268 0 -0.825; 1022 0 0 -0.825; 1023 -0.666803 0 -1.1
    18. 1024 -0.394535 0 -1.1; 1025 -0.222268 0 -1.1; 1026 1.95652 0 0
    19. 1027 1.95652 0 -0.275; 1028 1.705 0 -0.275; 1029 2.20803 0 0
    20. 1030 2.20803 0 -0.275; 1031 2.45955 0 0; 1032 2.45955 0 -0.275
    21. 1033 2.71107 0 -0.275; 1034 1.95652 0 -0.375; 1035 1.705 0 -0.375
    22. 1036 2.20803 0 -0.375; 1037 2.45955 0 -0.375; 1038 2.71107 0 -0.375
    23. 1039 1.95652 0 -0.825; 1040 1.705 0 -0.825; 1041 2.20803 0 -0.825
    24. 1042 2.45955 0 -0.825; 1043 2.71107 0 -0.825; 1044 1.95652 0 -1.1
    25. 1045 2.20803 0 -1.1; 1046 2.45955 0 -1.1; 1047 -0.666803 0 1.1
    26. 1048 -0.666803 0 0.825; 1049 -0.889071 0 0.825; 1050 -0.394535 0 1.1
    27. 1051 -0.394535 0 0.825; 1052 -0.222268 0 1.1; 1053 -0.222268 0 0.825
    28. 1054 0 0 0.825; 1055 -0.666803 0 0.375; 1056 -0.889071 0 0.375
    29. 1057 -0.394535 0 0.375; 1058 -0.222268 0 0.375; 1059 0 0 0.375
    30. 1060 -0.666803 0 0.275; 1061 -0.889071 0 0.275; 1062 -0.394535 0 0.275
    31. 1063 -0.222268 0 0.275; 1064 0 0 0.275; 1065 1.95652 0 1.1
    32. 1066 1.95652 0 0.825; 1067 1.705 0 0.825; 1068 2.20803 0 1.1
    33. 1069 2.20803 0 0.825; 1070 2.45955 0 1.1; 1071 2.45955 0 0.825
    34. 1072 2.71107 0 0.825; 1073 1.95652 0 0.375; 1074 1.705 0 0.375
    35. 1075 2.20803 0 0.375; 1076 2.45955 0 0.375; 1077 2.71107 0 0.375
    36. 1078 1.95652 0 0.275; 1079 1.705 0 0.275; 1080 2.20803 0 0.275
    37. 1081 2.45955 0 0.275; 1082 2.71107 0 0.275; 1083 0.243571 0 0
    38. 1084 0.243571 0 -0.275; 1085 0.387143 0 0; 1086 0.387143 0 -0.275
```

```
39. 1087 0.730714 0 0; 1088 0.730714 0 -0.275; 1089 0.974286 0 0
40. 1090 0.974286 0 -0.275; 1091 1.21786 0 0; 1092 1.21786 0 -0.275
41. 1093 1.46143 0 0; 1094 1.46143 0 -0.275; 1095 0.243571 0 -0.375
42. 1096 0.387143 0 -0.375; 1097 0.730714 0 -0.375; 1098 0.974286 0 -0.375
43. 1099 1.21786 0 -0.375; 1100 1.46143 0 -0.375; 1101 0.243571 0 -0.825
44. 1102 0.387143 0 -0.825; 1103 0.730714 0 -0.825; 1104 0.974286 0 -0.825
45. 1105 1.21786 0 -0.825; 1106 1.46143 0 -0.825; 1107 0.243571 0 -1.1
46. 1108 0.387143 0 -1.1; 1109 0.730714 0 -1.1; 1110 0.974286 0 -1.1
47. 1111 1.21786 0 -1.1; 1112 1.46143 0 -1.1; 1113 0.243571 0 1.1
48. 1114 0.243571 0 0.825; 1115 0.387143 0 1.1; 1116 0.387143 0 0.825
49. 1117 0.730714 0 1.1; 1118 0.730714 0 0.825; 1119 0.974286 0 1.1
50. 1120 0.974286 0 0.825; 1121 1.21786 0 1.1; 1122 1.21786 0 0.825
51. 1123 1.46143 0 1.1; 1124 1.46143 0 0.825; 1125 0.243571 0 0.375
52. 1126 0.387143 0 0.375; 1127 0.730714 0 0.375; 1128 0.974286 0 0.375
53. 1129 1.21786 0 0.375; 1130 1.46143 0 0.375; 1131 0.243571 0 0.275
54. 1132 0.387143 0 0.275; 1133 0.730714 0 0.275; 1134 0.974286 0 0.275
55. 1135 1.21786 0 0.275; 1136 1.46143 0 0.275; 1137 -0.666803 0 -0.6
56. 1138 -0.889071 0 -0.6; 1139 -0.394535 0 -0.6; 1140 -0.222268 0 -0.6
57. 1141 0 0 -0.6; 1142 1.95652 0 -0.6; 1143 1.705 0 -0.6; 1144 2.20803 0 -0.6
58. 1145 2.45955 0 -0.6; 1146 2.71107 0 -0.6; 1147 -0.666803 0 0.6
59. 1148 -0.889071 0 0.6; 1149 -0.394535 0 0.6; 1150 -0.222268 0 0.6; 1151 0 0 0.6
60. 1152 1.95652 0 0.6; 1153 1.705 0 0.6; 1154 2.20803 0 0.6; 1155 2.45955 0 0.6
61. 1156 2.71107 0 0.6; 1157 0.243571 0 -0.6; 1158 0.387143 0 -0.6
62. 1159 0.730714 0 -0.6; 1160 0.974286 0 -0.6; 1161 1.21786 0 -0.6
63. 1162 1.46143 0 -0.6; 1163 0.243571 0 0.6; 1164 0.387143 0 0.6
64. 1165 0.730714 0 0.6; 1166 0.974286 0 0.6; 1167 1.21786 0 0.6
65. 1168 1.46143 0 0.6; 1169 0.558929 0 0; 1170 0.558929 0 -0.275
66. 1171 0.558929 0 -0.375; 1172 0.558929 0 -0.825; 1173 0.558929 0 -1.1
67. 1174 0.558929 0 1.1; 1175 0.558929 0 0.825; 1176 0.558929 0 0.375
68. 1177 0.558929 0 0.275; 1178 0.558929 0 -0.6; 1179 0.558929 0 0.6
69. MEMBER INCIDENCES
70. 23 1 1001; 24 2 1002; 987 1001 21; 988 1002 22
71. ELEMENT INCIDENCES SHELL
72. 1086 1003 1005 1006 1007; 1088 1005 1008 1009 1006; 1090 1008 1010 1011 1009
73. 1092 1010 1 1012 1011; 1094 1007 1006 1013 1014; 1095 1006 1009 1015 1013
74. 1096 1009 1011 1016 1015; 1098 1011 1012 1017 1016; 1106 1019 1018 1023 49
75. 1108 1018 1020 1024 1023; 1110 1020 1021 1025 1024; 1111 1021 1022 52 1025
76. 1114 2 1026 1027 1028; 1116 1026 1029 1030 1027; 1118 1029 1031 1032 1030
77. 1120 1031 1004 1033 1032; 1122 1028 1027 1034 1035; 1123 1027 1030 1036 1034
78. 1124 1030 1032 1037 1036; 1126 1032 1033 1038 1037; 1134 1040 1039 1044 85
79. 1136 1039 1041 1045 1044; 1138 1041 1042 1046 1045; 1139 1042 1043 114 1046
80. 1142 133 1047 1048 1049; 1144 1047 1050 1051 1048; 1146 1050 1052 1053 1051
81. 1148 1052 138 1054 1053; 1156 1056 1055 1060 1061; 1157 1055 1057 1062 1060
82. 1158 1057 1058 1063 1062; 1160 1058 1059 1064 1063; 1161 1061 1060 1005 1003
83. 1162 1060 1062 1008 1005; 1163 1062 1063 1010 1008; 1164 1063 1064 1 1010
84. 1167 167 1065 1066 1067; 1169 1065 1068 1069 1066; 1171 1068 1070 1071 1069
85. 1173 1070 187 1072 1071; 1181 1074 1073 1078 1079; 1182 1073 1075 1080 1078
86. 1183 1075 1076 1081 1080; 1185 1076 1077 1082 1081; 1186 1079 1078 1026 2
87. 1187 1078 1080 1029 1026; 1188 1080 1081 1031 1029; 1189 1081 1082 1004 1031
88. 1191 1 1083 1084 1012; 1193 1083 1085 1086 1084; 1197 1087 1089 1090 1088
89. 1199 1089 1091 1092 1090; 1201 1091 1093 1094 1092; 1202 1093 2 1028 1094
90. 1203 1012 1084 1095 1017; 1204 1084 1086 1096 1095; 1206 1088 1090 1098 1097
91. 1207 1090 1092 1099 1098; 1208 1092 1094 1100 1099; 1209 1094 1028 1035 1100
92. 1218 1022 1101 1107 52; 1220 1101 1102 1108 1107; 1224 1103 1104 1110 1109
93. 1226 1104 1105 1111 1110; 1228 1105 1106 1112 1111; 1229 1106 1040 85 1112
94. 1231 138 1113 1114 1054; 1233 1113 1115 1116 1114; 1237 1117 1119 1120 1118
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 95. 1239 1119 1121 1122 1120; 1241 1121 1123 1124 1122; 1242 1123 167 1067 1124
 96. 1250 1059 1125 1131 1064; 1251 1125 1126 1132 1131; 1253 1127 1128 1134 1133
 97. 1254 1128 1129 1135 1134; 1255 1129 1130 1136 1135; 1256 1130 1074 1079 1136
 98. 1257 1064 1131 1083 1; 1258 1131 1132 1085 1083; 1260 1133 1134 1089 1087
 99. 1261 1134 1135 1091 1089; 1262 1135 1136 1093 1091; 1263 1136 1079 2 1093
100. 1264 1014 1013 1137 1138; 1265 1138 1137 1018 1019; 1266 1013 1015 1139 1137
101. 1267 1137 1139 1020 1018; 1268 1015 1016 1140 1139; 1269 1139 1140 1021 1020
102. 1270 1016 1017 1141 1140; 1271 1140 1141 1022 1021; 1272 1035 1034 1142 1143
103. 1273 1143 1142 1039 1040; 1274 1034 1036 1144 1142; 1275 1142 1144 1041 1039
104. 1276 1036 1037 1145 1144; 1277 1144 1145 1042 1041; 1278 1037 1038 1146 1145
105. 1279 1145 1146 1043 1042; 1280 1049 1048 1147 1148; 1281 1148 1147 1055 1056
106. 1282 1048 1051 1149 1147; 1283 1147 1149 1057 1055; 1284 1051 1053 1150 1149
107. 1285 1149 1150 1058 1057; 1286 1053 1054 1151 1150; 1287 1150 1151 1059 1058
108. 1288 1067 1066 1152 1153; 1289 1153 1152 1073 1074; 1290 1066 1069 1154 1152
109. 1291 1152 1154 1075 1073; 1292 1069 1071 1155 1154; 1293 1154 1155 1076 1075
110. 1294 1071 1072 1156 1155; 1295 1155 1156 1077 1076; 1296 1017 1095 1157 1141
111. 1297 1141 1157 1101 1022; 1298 1095 1096 1158 1157; 1299 1157 1158 1102 1101
112. 1302 1097 1098 1160 1159; 1303 1159 1160 1104 1103; 1304 1098 1099 1161 1160
113. 1305 1160 1161 1105 1104; 1306 1099 1100 1162 1161; 1307 1161 1162 1106 1105
114. 1308 1100 1035 1143 1162; 1309 1162 1143 1040 1106; 1310 1054 1114 1163 1151
115. 1311 1151 1163 1125 1059; 1312 1114 1116 1164 1163; 1313 1163 1164 1126 1125
116. 1316 1118 1120 1166 1165; 1317 1165 1166 1128 1127; 1318 1120 1122 1167 1166
117. 1319 1166 1167 1129 1128; 1320 1122 1124 1168 1167; 1321 1167 1168 1130 1129
118. 1322 1124 1067 1153 1168; 1323 1168 1153 1074 1130; 1324 1085 1169 1170 1086
119. 1325 1169 1087 1088 1170; 1326 1086 1170 1171 1096; 1327 1170 1088 1097 1171
120. 1328 1102 1172 1173 1108; 1329 1172 1103 1109 1173; 1330 1115 1174 1175 1116
121. 1331 1174 1117 1118 1175; 1332 1126 1176 1177 1132; 1333 1176 1127 1133 1177
122. 1334 1132 1177 1169 1085; 1335 1177 1133 1087 1169; 1336 1096 1171 1178 1158
123. 1337 1171 1097 1159 1178; 1338 1158 1178 1172 1102; 1339 1178 1159 1103 1172
124. 1340 1116 1175 1179 1164; 1341 1175 1118 1165 1179; 1342 1164 1179 1176 1126
125. 1343 1179 1165 1127 1176
126. START GROUP DEFINITION
127. FLOOR
128. ELEMENT
129. _RAFT 1086 1088 1090 1092 1094 TO 1096 1098 1106 1108 1110 1111 1114 1116 -
130. 1118 1120 1122 TO 1124 1126 1134 1136 1138 1139 1142 1144 1146 1148 1156 -
131. 1157 TO 1158 1160 TO 1164 1167 1169 1171 1173 1181 TO 1183 1185 TO 1189 1191 -
132. 1193 1197 1199 1201 TO 1204 1206 TO 1209 1218 1220 1224 1226 1228 1229 1231 -
133. 1233 1237 1239 1241 1242 1250 1251 1253 TO 1258 1260 TO 1299 1302 TO 1313 -
134. 1316 TO 1343
135. _RAFT_MOM 1086 1088 1094 1095 1106 1108 1110 1111 1116 1118 1120 1122 TO 1124 -
136. 1126 1134 1136 1138 1139 1142 1144 1146 1148 1156 1157 1161 1162 1167 1169 -
137. 1171 1173 1181 TO 1183 1185 1187 TO 1189 1197 1199 1201 1206 TO 1209 1218 -
138. 1220 1224 1226 1228 1229 1231 1233 1237 1239 1241 1242 1253 TO 1256 1260 -
139. 1261 TO 1262 1264 TO 1299 1302 TO 1313 1316 TO 1343
140. _RAFT_SHEAR 1086 1094 1106 1108 1110 1111 1118 1120 1124 1126 1134 1136 1138 -
141. 1139 1142 1144 1146 1148 1156 1161 1167 1169 1171 1173 1183 1185 1188 1189 -
142. 1199 1207 1218 1220 1224 1226 1228 1229 1231 1233 1237 1239 1241 1242 1254 -
143. 1261 1264 1265 1267 1269 1271 1273 1275 TO 1282 1284 1286 1288 1290 1292 -
144. 1293 TO 1295 1297 1299 1303 TO 1305 1307 1309 1310 1312 1316 1318 TO 1320 -
145. 1322 1328 TO 1331 1338 TO 1341
146. MEMBER
147. _C8 987
148. _C7 988
149. END GROUP DEFINITION
150. ELEMENT PROPERTY
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151. 1086 1088 1090 1092 1094 TO 1096 1098 1106 1108 1110 1111 1114 1116 1118 1120 -
152. 1122 TO 1124 1126 1134 1136 1138 1139 1142 1144 1146 1148 1156 TO 1158 1160 -
153. 1161 TO 1164 1167 1169 1171 1173 1181 TO 1183 1185 TO 1189 1191 1193 1197 -
154. 1199 1201 TO 1204 1206 TO 1209 1218 1220 1224 1226 1228 1229 1231 1233 1237 -
155. 1239 1241 1242 1250 1251 1253 TO 1258 1260 TO 1299 1302 TO 1313 1316 TO 1342 -
156. 1343 THICKNESS 0.5
157. DEFINE MATERIAL START
158. ISOTROPIC CONCRETE
159. E 2.17185E+007
160. POISSON 0.17
161. DENSITY 25
162. ALPHA 1E-005
163. DAMP 0.05
164. TYPE CONCRETE
165. STRENGTH FCU 27579
166. END DEFINE MATERIAL
167. MEMBER PROPERTY AMERICAN
168. 23 987 PRIS YD 0.75 ZD 0.75
169. MEMBER PROPERTY AMERICAN
170. 24 988 PRIS YD 0.55 ZD 0.55
171. CONSTANTS
172. MATERIAL CONCRETE ALL
173. ************************************************************
174. SUPPORTS
175. 1086 1088 1090 1092 1094 TO 1096 1098 1106 1108 1110 1111 1114 1116 1118 1120 -
176. 1122 TO 1124 1126 1134 1136 1138 1139 1142 1144 1146 1148 1156 TO 1158 1160 -
177. 1161 TO 1164 1167 1169 1171 1173 1181 TO 1183 1185 TO 1189 1191 1193 1197 -
178. 1199 1201 TO 1204 1206 TO 1209 1218 1220 1224 1226 1228 1229 1231 1233 1237 -
179. 1239 1241 1242 1250 1251 1253 TO 1258 1260 TO 1299 1302 TO 1313 1316 TO 1342 -
180. 1343 PLATE MAT DIRECT Y SUBGRADE 40500 PRINT
```

JOINT	INFLUENCE AREA
1	0.064053
2	0.068075
49	0.015281
52	0.032026
85	0.034037
114	0.017292
133	0.015281
138	0.032026
167	0.034037
187	0.017292
1003	0.030562
1004	0.034584
1005	0.067999
1006	0.046363
1007	0.020838
1008	0.061124
1009	0.041675
1010	0.054249
1011	0.036988
1012	0.043672
1013	0.040181
1014	0.018059

1015	0.036118
1016	0.032056
1017	0.037849
1018	0.061817
1019	0.027784
1020	0.055567
1021	0.049317
1022	0.058230
1023	0.033999
1024	0.030562
1025	0.027124
1026	0.069167
1027	0.047159
1028	0.046415
1029	0.069167
1030	0.047159
1031	0.069168
1032	0.047160
1033	0.023580
1034	0.040871
1035	0.040226
1036	0.040871
1037	0.040872
1038	0.020436
1039	0.062879
1040	0.061886
1041	0.062879
1042	0.062880
1043	0.031440
1044	0.034583
1045	0.034583
1046	0.034584
1047	0.033999
1048	0.061817
1049	0.027784
1050	0.030562
1051	0.055567
1052	0.027124
1053	0.049317
1054	0.058230
1055	0.040181
1056	0.018059
1057	0.036118
1058	0.032056
1059	0.037849
1060	0.046363
1061	0.020838
1062	0.041675
1063	0.036988
1064	0.043672
1065	0.034583
1066	0.062879
1067	0.061886
1068	0.034583
1069	0.062879
1070	0.034584

1071	0.062880
1072	0.031440
1073	0.040871
1074	0.040226
1075	0.040871
1076	0.040872
1077	0.020436
1078	0.047159
1079	0.046415
1080	0.047159
1081	0.047160
1082	0.023580
1083	0.053232
1084	0.036295
1085	0.043362
1086	0.029565
1087	0.057112
1088	0.038940
1089	0.066983
1090	0.045670
1091	0.066982
1092	0.045670
1093	0.066982
1094	0.045669
1095	0.031455
1096	0.025623
1097	0.033748
1098	0.039581
1099	0.039580
1100	0.039580
1101	0.048393
1102	0.039420
1103	0.051920
1104	0.060893
1105	0.060893
1106	0.060893
1107	0.026616
1108	0.021681
1109	0.028556
1110	0.033491
1111	0.033491
1112	0.033491
1113	0.026616
1114	0.048393
1115	0.021681
1116	0.039420
1117	0.028556
1118	0.051920
1119	0.033491
1120	0.060893
1121	0.033491
1122	0.060893
1123	0.033491
1124	0.060893
1125	0.031455
1126	0.025623

1127	0.033748
1128	0.039581
1129	0.039580
1130	0.039580
1131	0.036295
1132	0.029565
1133	0.038940
1134	0.045670
1135	0.045670
1136	0.045669
1137	0.055635
1138	0.025005
1139	0.050010
1140	0.044385
1141	0.052407
1142	0.056591
1143	0.055698
1144	0.056591
1145	0.056592
1146	0.028296
1147	0.055635
1148	0.025005
1149	0.050010
1150	0.044385
1151	0.052407
1152	0.056591
1153	0.055698
1154	0.056591
1155	0.056592
1156	0.028296
1157	0.043554
1158	0.035478
1159	0.046728
1160	0.054804
1161	0.054804
1162	0.054803
1163	0.043554
1164	0.035478
1165	0.046728
1166	0.054804
1167	0.054804
1168	0.054803
1169	0.047241
1170	0.032210
1171	0.027915
1172	0.042946
1173	0.023621
1174	0.023621
1175	0.042946
1176	0.027915
1177	0.032210
1178	0.038652
1179	0.038652
TOTAL	7.920310

181. LOAD 1 LOADTYPE NONE TITLE DEAD LOAD (DL)
182. JOINT LOAD

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183. 21 FX 14.7 FY -128.8 FZ 0
184. 22 FX 0.1 FY -40 FZ 0
185. ELEMENT LOAD
186. 1086 1088 1090 1092 1094 TO 1096 1098 1106 1108 1110 1111 1114 1116 1118 1120 -
187. 1122 TO 1124 1126 1134 1136 1138 1139 1142 1144 1146 1148 1156 TO 1158 1160 -
188. 1161 TO 1164 1167 1169 1171 1173 1181 TO 1183 1185 TO 1189 1191 1193 1197 -
189. 1199 1201 TO 1204 1206 TO 1209 1218 1220 1224 1226 1228 1229 1231 1233 1237 -
190. 1239 1241 1242 1250 1251 1253 TO 1258 1260 TO 1299 1302 TO 1313 1316 TO 1342 -
191. 1343 PR -18
192. SELFWEIGHT Y -1
193. ***
194. LOAD 2 LOADTYPE NONE  TITLE NORMAL OPERATING FLUID LOAD ( FL1)
195. JOINT LOAD
196. 21 FX 60.7 FY -336.4 FZ 0
197. 22 FX 0.5 FY -16.4 FZ 0
198. LOAD 3 LOADTYPE NONE  TITLE MAX. FLUID LOAD ( FL2)
199. JOINT LOAD
200. 21 FX 68.6 FY -381.1 FZ 0
201. 22 FX 0.6 FY -18.5 FZ 0
202. LOAD 4 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
203. JOINT LOAD
204. 21 FX 1.9 FY -26.3 FZ 0
205. 22 FX 0 FY -10.4 FZ 0
206. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD + X ( SLX1 )
207. JOINT LOAD
208. 21 FX 41.4 FY -126.1 FZ 0
209. 22 FX 0.1 FY -20.6 FZ 0
210. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD - X ( SLX2 )
211. JOINT LOAD
212. 21 FX -41.4 FY 126.1 FZ 0
213. 22 FX -0.1 FY 20.6 FZ 0
214. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD +Z ( SLZ1)
215. JOINT LOAD
216. 21 FX -23.1 FY 72.4 FZ 0.2
217. 22 FX -0.1 FY 8.7 FZ 0.2
218. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD -Z ( SLZ2)
219. JOINT LOAD
220. 21 FX 23.1 FY -72.4 FZ -0.2
221. 22 FX 0.1 FY -8.7 FZ -0.2
222. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
223. JOINT LOAD
224. 21 FX 20.8 FY -54.4 FZ 0
225. 22 FX 0.6 FY -23 FZ 0
226. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
227. JOINT LOAD
228. 21 FX -20.8 FY 54.3 FZ 0
229. 22 FX -0.6 FY 23 FZ 0
230. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
231. JOINT LOAD
232. 21 FX -7.4 FY 28.2 FZ 4.9
233. 22 FX 0 FY 2.6 FZ 2.8
234. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
235. JOINT LOAD
236. 21 FX 7.4 FY -28.2 FZ -4.9
237. 22 FX 0 FY -2.6 FZ -2.8
238. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD ( TL )
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239. JOINT LOAD
240. 21 FX 49.9 FY -11.7 FZ 0
241. 22 FX 1.3 FY -15.5 FZ 0
242. LOAD 14 LOADTYPE NONE  TITLE LATERAL PIPE LOAD ( PL )
243. JOINT LOAD
244. 21 FX 9.4 FY -81.3 FZ -0.4
245. 22 FX 0.1 FY -34.6 FZ 0
246. **********************************************************
247. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
248. LOAD COMB 101 1.4DL+1.4FL1
249. 1 1.4 2 1.4
250. LOAD COMB 102 1.4DL+1.4FL2
251. 1 1.4 3 1.4
252. LOAD COMB 103 1.2DL+1.2FL1+1.6LL
253. 1 1.2 2 1.2 4 1.6
254. LOAD COMB 104 1.2DL+1.2FL2+1.6LL
255. 1 1.2 3 1.2 4 1.6
256. LOAD COMB 105 1.2DL+1.2FL1+1.2TL+1.6LL
257. 1 1.2 2 1.2 4 1.6 13 1.2
258. LOAD COMB 106 1.2DL+1.2FL2+1.2TL+1.6LL
259. 1 1.2 3 1.2 4 1.6 13 1.2
260. LOAD COMB 107 1.4DL+1.4FL1+1.4PL
261. 1 1.4 2 1.4 14 1.4
262. LOAD COMB 108 1.4DL+1.4FL2+1.4PL
263. 1 1.4 3 1.4 14 1.4
264. LOAD COMB 109 1.2DL+1.2FL1+1.6LL+1.2PL
265. 1 1.2 2 1.2 4 1.6 14 1.2
266. LOAD COMB 110 1.2DL+1.2FL2+1.6LL+1.2PL
267. 1 1.2 3 1.2 4 1.6 14 1.2
268. LOAD COMB 111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
269. 1 1.2 2 1.2 4 1.6 13 1.2 14 1.2
270. LOAD COMB 112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
271. 1 1.2 3 1.2 4 1.6 13 1.2 14 1.2
272. LOAD COMB 113 1.2DL+0.8WLX1
273. 1 1.2 9 0.8
274. LOAD COMB 114 1.2DL+0.8WLX2
275. 1 1.2 10 0.8
276. LOAD COMB 115 1.2DL+0.8WLZ1
277. 1 1.2 11 0.8
278. LOAD COMB 116 1.2DL+0.8WLZ2
279. 1 1.2 12 0.8
280. LOAD COMB 117 1.2DL+1.6WLX1+0.5LL
281. 1 1.2 9 1.6 4 0.5
282. LOAD COMB 118 1.2DL+1.6WLX2+0.5LL
283. 1 1.2 10 1.6 4 0.5
284. LOAD COMB 119 1.2DL+1.6WLZ1+0.5LL
285. 1 1.2 11 1.6 4 0.5
286. LOAD COMB 120 1.2DL+1.6WLZ2+0.5LL
287. 1 1.2 12 1.6 4 0.5
288. LOAD COMB 121 1.2DL+1.0SLX1+0.5LL+0.5FL1
289. 1 1.2 5 1.0 4 0.5 2 0.5
290. LOAD COMB 122 1.2DL+1.0SLX2+0.5LL+0.5FL1
291. 1 1.2 6 1.0 4 0.5 2 0.5
292. LOAD COMB 123 1.2DL+1.0SLZ1+0.5LL+0.5FL1
293. 1 1.2 7 1.0 4 0.5 2 0.5
294. LOAD COMB 124 1.2DL+1.0SLZ2+0.5LL+0.5FL1
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295. 1 1.2 8 1.0 4 0.5 2 0.5
296. LOAD COMB 125 0.9DL+1.6WLX1
297. 1 0.9 9 1.6
298. LOAD COMB 126 0.9DL+1.6WLX2
299. 1 0.9 10 1.6
300. LOAD COMB 127 0.9DL+1.6WLZ1
301. 1 0.9 11 1.6
302. LOAD COMB 128 0.9DL+1.6WLZ2
303. 1 0.9 12 1.6
304. LOAD COMB 129 0.9DL+1.0SLX1
305. 1 0.9 5 1.0
306. LOAD COMB 130 0.9DL+1.0SLX2
307. 1 0.9 6 1.0
308. LOAD COMB 131 0.9DL+1.0SLZ1
309. 1 0.9 7 1.0
310. LOAD COMB 132 0.9DL+1.0SLZ2
311. 1 0.9 8 1.0
312. ********************************************************************************
313. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
314. LOAD COMB 201 1.0DL+1.0FL1
315. 1 1.0 2 1.0
316. LOAD COMB 202 1.0DL+1.0FL2
317. 1 1.0 3 1.0
318. LOAD COMB 203 1.0DL+1.0FL1+1.0LL+1.0TL
319. 1 1.0 2 1.0 4 1.0 13 1.0
320. LOAD COMB 204 1.0DL+1.0FL2+1.0LL+1.0TL
321. 1 1.0 3 1.0 4 1.0 13 1.0
322. LOAD COMB 205 1.0DL+1.0FL1+0.75LL+0.75TL
323. 1 1.0 2 1.0 4 0.75 13 0.75
324. LOAD COMB 206 1.0DL+1.0FL2+0.75LL+0.75TL
325. 1 1.0 3 1.0 4 0.75 13 0.75
326. LOAD COMB 207 1.0DL+1.0FL1+1.0PL
327. 1 1.0 2 1.0 14 1.0
328. LOAD COMB 208 1.0DL+1.0FL2+1.0PL
329. 1 1.0 3 1.0 14 1.0
330. LOAD COMB 209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL
331. 1 1.0 2 1.0 4 1.0 13 1.0 14 1.0
332. LOAD COMB 210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL
333. 1 1.0 3 1.0 4 1.0 13 1.0 14 1.0
334. LOAD COMB 211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL
335. 1 1.0 2 1.0 4 0.75 13 0.75 14 1.0
336. LOAD COMB 212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL
337. 1 1.0 3 1.0 4 0.75 13 0.75 14 1.0
338. LOAD COMB 213 1.0DL+1.0FL1+1.0WLX1
339. 1 1.0 2 1.0 9 1.0
340. LOAD COMB 214 1.0DL+1.0FL1+1.0WLX2
341. 1 1.0 2 1.0 10 1.0
342. LOAD COMB 215 1.0DL+1.0FL1+1.0WLZ1
343. 1 1.0 2 1.0 11 1.0
344. LOAD COMB 216 1.0DL+1.0FL1+1.0WLZ2
345. 1 1.0 2 1.0 12 1.0
346. LOAD COMB 217 1.0DL+1.0FL2+1.0WLX1
347. 1 1.0 3 1.0 9 1.0
348. LOAD COMB 218 1.0DL+1.0FL2+1.0WLX2
349. 1 1.0 3 1.0 10 1.0
350. LOAD COMB 219 1.0DL+1.0FL2+1.0WLZ1
```

```
351. 1 1.0 3 1.0 11 1.0
352. LOAD COMB 220 1.0DL+1.0FL2+1.0WLZ2
353. 1 1.0 3 1.0 12 1.0
354. LOAD COMB 221 1.0DL+1.0FL1+0.7SLX1
355. 1 1.0 2 1.0 5 0.7
356. LOAD COMB 222 1.0DL+1.0FL1+0.7SLX2
357. 1 1.0 2 1.0 6 0.7
358. LOAD COMB 223 1.0DL+1.0FL1+0.7SLZ1
359. 1 1.0 2 1.0 7 0.7
360. LOAD COMB 224 1.0DL+1.0FL1+0.7SLZ2
361. 1 1.0 2 1.0 8 0.7
362. LOAD COMB 225 1.0DL+1.0FL2+0.7SLX1
363. 1 1.0 3 1.0 5 0.7
364. LOAD COMB 226 1.0DL+1.0FL2+0.7SLX2
365. 1 1.0 3 1.0 6 0.7
366. LOAD COMB 227 1.0DL+1.0FL2+0.7SLZ1
367. 1 1.0 3 1.0 7 0.7
368. LOAD COMB 228 1.0DL+1.0FL2+0.7SLZ2
369. 1 1.0 3 1.0 8 0.7
370. LOAD COMB 229 1.0DL+1.0FL1+0.75WLX1+0.75LL
371. 1 1.0 2 1.0 4 0.75 9 0.75
372. LOAD COMB 230 1.0DL+1.0FL1+0.75WLX2+0.75LL
373. 1 1.0 2 1.0 4 0.75 10 0.75
374. LOAD COMB 231 1.0DL+1.0FL1+0.75WLZ1+0.75LL
375. 1 1.0 2 1.0 4 0.75 11 0.75
376. LOAD COMB 232 1.0DL+1.0FL1+0.75WLZ2+0.75LL
377. 1 1.0 2 1.0 4 0.75 12 0.75
378. LOAD COMB 233 1.0DL+1.0FL2+0.75WLX1+0.75LL
379. 1 1.0 3 1.0 4 0.75 9 0.75
380. LOAD COMB 234 1.0DL+1.0FL2+0.75WLX2+0.75LL
381. 1 1.0 3 1.0 4 0.75 10 0.75
382. LOAD COMB 235 1.0DL+1.0FL2+0.75WLZ1+0.75LL
383. 1 1.0 3 1.0 4 0.75 11 0.75
384. LOAD COMB 236 1.0DL+1.0FL2+0.75WLZ2+0.75LL
385. 1 1.0 3 1.0 4 0.75 12 0.75
386. LOAD COMB 237 1.0DL+1.0FL1+0.525SLX1+0.75LL
387. 1 1.0 2 1.0 4 0.75 5 0.525
388. LOAD COMB 238 1.0DL+1.0FL1+0.525SLX2+0.75LL
389. 1 1.0 2 1.0 4 0.75 6 0.525
390. LOAD COMB 239 1.0DL+1.0FL1+0.525SLZ1+0.75LL
391. 1 1.0 2 1.0 4 0.75 7 0.525
392. LOAD COMB 240 1.0DL+1.0FL1+0.525SLZ2+0.75LL
393. 1 1.0 2 1.0 4 0.75 8 0.525
394. LOAD COMB 241 1.0DL+1.0FL2+0.525SLX1+0.75LL
395. 1 1.0 3 1.0 4 0.75 5 0.525
396. LOAD COMB 242 1.0DL+1.0FL2+0.525SLX2+0.75LL
397. 1 1.0 3 1.0 4 0.75 6 0.525
398. LOAD COMB 243 1.0DL+1.0FL2+0.525SLZ1+0.75LL
399. 1 1.0 3 1.0 4 0.75 7 0.525
400. LOAD COMB 244 1.0DL+1.0FL2+0.525SLZ2+0.75LL
401. 1 1.0 3 1.0 4 0.75 8 0.525
402. LOAD COMB 245 0.6DL+1.0WLX1
403. 1 0.6 9 1.0
404. LOAD COMB 246 0.6DL+1.0WLX2
405. 1 0.6 10 1.0
406. LOAD COMB 247 0.6DL+1.0WLZ1
```

```
407. 1 0.6 11 1.0
408. LOAD COMB 248 0.6DL+1.0WLZ2
409. 1 0.6 12 1.0
410. LOAD COMB 249 0.6DL+0.6FL1+0.7SLX1
411. 1 0.6 2 0.6 5 0.7
412. LOAD COMB 250 0.6DL+0.6FL1+0.7SLX2
413. 1 0.6 2 0.6 6 0.7
414. LOAD COMB 251 0.6DL+0.6FL1+0.7SLZ1
415. 1 0.6 2 0.6 7 0.7
416. LOAD COMB 252 0.6DL+0.6FL1+0.7SLZ2
417. 1 0.6 2 0.6 8 0.7
418. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**191**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**160**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**187**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 146/ 19/ 60 DOF
TOTAL PRIMARY LOAD CASES = 14, TOTAL DEGREES OF FREEDOM = 585
TOTAL LOAD COMBINATION CASES = 84 SO FAR.
SIZE OF STIFFNESS MATRIX = 36 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.3/ 112122.3 MB

```
419. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 11:33:25 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 2 STAAD INPUT & OUTPUT FILE FOR F12 STABILITY

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    APR  7, 2020                   *
*           Time=     9:28:56                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F12 STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 26-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 21 0 1.85 0; 22 1.705 1.85 0; 1001 0 0.25 0; 1002 1.705 0.25 0
     9. MEMBER INCIDENCES
    10. 987 1001 21; 988 1002 22
    11. DEFINE MATERIAL START
    12. ISOTROPIC CONCRETE
    13. E 2.17185E+007
    14. POISSON 0.17
    15. DENSITY 25
    16. ALPHA 1E-005
    17. DAMP 0.05
    18. TYPE CONCRETE
    19. STRENGTH FCU 27579
    20. END DEFINE MATERIAL
    21. MEMBER PROPERTY
    22. 987 PRIS YD 0.75 ZD 0.75
    23. 988 PRIS YD 0.55 ZD 0.55
    24. CONSTANTS
    25. MATERIAL CONCRETE ALL
    26. ************************************************************
    27. SUPPORTS
    28. 1001 1002 FIXED
    29. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
              MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    30. JOINT LOAD
    31. 21 FX 14.7 FY -128.8 FZ 0
    32. 22 FX 0.1 FY -40 FZ 0
    33. ***
    34. LOAD 2 LOADTYPE NONE  TITLE NORMAL OPERATING FLUID LOAD ( FL1)
    35. JOINT LOAD
    36. 21 FX 60.7 FY -336.4 FZ 0
```

```
37. 22 FX 0.5 FY -16.4 FZ 0
38. LOAD 3 LOADTYPE NONE  TITLE MAX. FLUID LOAD ( FL2)
39. JOINT LOAD
40. 21 FX 68.6 FY -381.1 FZ 0
41. 22 FX 0.6 FY -18.5 FZ 0
42. LOAD 4 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
43. JOINT LOAD
44. 21 FX 1.9 FY -26.3 FZ 0
45. 22 FX 0 FY -10.4 FZ 0
46. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD + X ( SLX1 )
47. JOINT LOAD
48. 21 FX 41.4 FY -126.1 FZ 0
49. 22 FX 0.1 FY -20.6 FZ 0
50. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD - X ( SLX2 )
51. JOINT LOAD
52. 21 FX -41.4 FY 126.1 FZ 0
53. 22 FX -0.1 FY 20.6 FZ 0
54. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD +Z ( SLZ1)
55. JOINT LOAD
56. 21 FX -23.1 FY 72.4 FZ 0.2
57. 22 FX -0.1 FY 8.7 FZ 0.2
58. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD -Z ( SLZ2)
59. JOINT LOAD
60. 21 FX 23.1 FY -72.4 FZ -0.2
61. 22 FX 0.1 FY -8.7 FZ -0.2
62. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
63. JOINT LOAD
64. 21 FX 20.8 FY -54.4 FZ 0
65. 22 FX 0.6 FY -23 FZ 0
66. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
67. JOINT LOAD
68. 21 FX -20.8 FY 54.3 FZ 0
69. 22 FX -0.6 FY 23 FZ 0
70. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
71. JOINT LOAD
72. 21 FX -7.4 FY 28.2 FZ 4.9
73. 22 FX 0 FY 2.6 FZ 2.8
74. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
75. JOINT LOAD
76. 21 FX 7.4 FY -28.2 FZ -4.9
77. 22 FX 0 FY -2.6 FZ -2.8
78. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD ( TL )
79. JOINT LOAD
80. 21 FX 49.9 FY -11.7 FZ 0
81. 22 FX 1.3 FY -15.5 FZ 0
82. LOAD 14 LOADTYPE NONE  TITLE LATERAL PIPE LOAD ( PL )
83. JOINT LOAD
84. 21 FX 9.4 FY -81.3 FZ -0.4
85. 22 FX 0.1 FY -34.6 FZ 0
86. ********************************************************************************
87. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
88. LOAD COMB 201 1.0DL+1.0FL1
89. 1 1.0 2 1.0
90. LOAD COMB 202 1.0DL+1.0FL2
91. 1 1.0 3 1.0
92. LOAD COMB 203 1.0DL+1.0FL1+1.0LL+1.0TL
```

```
 93. 1 1.0 2 1.0 4 1.0 13 1.0
 94. LOAD COMB 204 1.0DL+1.0FL2+1.0LL+1.0TL
 95. 1 1.0 3 1.0 4 1.0 13 1.0
 96. LOAD COMB 205 1.0DL+1.0FL1+0.75LL+0.75TL
 97. 1 1.0 2 1.0 4 0.75 13 0.75
 98. LOAD COMB 206 1.0DL+1.0FL2+0.75LL+0.75TL
 99. 1 1.0 3 1.0 4 0.75 13 0.75
100. LOAD COMB 207 1.0DL+1.0FL1+1.0PL
101. 1 1.0 2 1.0 14 1.0
102. LOAD COMB 208 1.0DL+1.0FL2+1.0PL
103. 1 1.0 3 1.0 14 1.0
104. LOAD COMB 209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL
105. 1 1.0 2 1.0 4 1.0 13 1.0 14 1.0
106. LOAD COMB 210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL
107. 1 1.0 3 1.0 4 1.0 13 1.0 14 1.0
108. LOAD COMB 211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL
109. 1 1.0 2 1.0 4 0.75 13 0.75 14 1.0
110. LOAD COMB 212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL
111. 1 1.0 3 1.0 4 0.75 13 0.75 14 1.0
112. LOAD COMB 213 1.0DL+1.0FL1+1.0WLX1
113. 1 1.0 2 1.0 9 1.0
114. LOAD COMB 214 1.0DL+1.0FL1+1.0WLX2
115. 1 1.0 2 1.0 10 1.0
116. LOAD COMB 215 1.0DL+1.0FL1+1.0WLZ1
117. 1 1.0 2 1.0 11 1.0
118. LOAD COMB 216 1.0DL+1.0FL1+1.0WLZ2
119. 1 1.0 2 1.0 12 1.0
120. LOAD COMB 217 1.0DL+1.0FL2+1.0WLX1
121. 1 1.0 3 1.0 9 1.0
122. LOAD COMB 218 1.0DL+1.0FL2+1.0WLX2
123. 1 1.0 3 1.0 10 1.0
124. LOAD COMB 219 1.0DL+1.0FL2+1.0WLZ1
125. 1 1.0 3 1.0 11 1.0
126. LOAD COMB 220 1.0DL+1.0FL2+1.0WLZ2
127. 1 1.0 3 1.0 12 1.0
128. LOAD COMB 221 1.0DL+1.0FL1+0.7SLX1
129. 1 1.0 2 1.0 5 0.7
130. LOAD COMB 222 1.0DL+1.0FL1+0.7SLX2
131. 1 1.0 2 1.0 6 0.7
132. LOAD COMB 223 1.0DL+1.0FL1+0.7SLZ1
133. 1 1.0 2 1.0 7 0.7
134. LOAD COMB 224 1.0DL+1.0FL1+0.7SLZ2
135. 1 1.0 2 1.0 8 0.7
136. LOAD COMB 225 1.0DL+1.0FL2+0.7SLX1
137. 1 1.0 3 1.0 5 0.7
138. LOAD COMB 226 1.0DL+1.0FL2+0.7SLX2
139. 1 1.0 3 1.0 6 0.7
140. LOAD COMB 227 1.0DL+1.0FL2+0.7SLZ1
141. 1 1.0 3 1.0 7 0.7
142. LOAD COMB 228 1.0DL+1.0FL2+0.7SLZ2
143. 1 1.0 3 1.0 8 0.7
144. LOAD COMB 229 1.0DL+1.0FL1+0.75WLX1+0.75LL
145. 1 1.0 2 1.0 4 0.75 9 0.75
146. LOAD COMB 230 1.0DL+1.0FL1+0.75WLX2+0.75LL
147. 1 1.0 2 1.0 4 0.75 10 0.75
148. LOAD COMB 231 1.0DL+1.0FL1+0.75WLZ1+0.75LL
```

```
149. 1 1.0 2 1.0 4 0.75 11 0.75
150. LOAD COMB 232 1.0DL+1.0FL1+0.75WLZ2+0.75LL
151. 1 1.0 2 1.0 4 0.75 12 0.75
152. LOAD COMB 233 1.0DL+1.0FL2+0.75WLX1+0.75LL
153. 1 1.0 3 1.0 4 0.75 9 0.75
154. LOAD COMB 234 1.0DL+1.0FL2+0.75WLX2+0.75LL
155. 1 1.0 3 1.0 4 0.75 10 0.75
156. LOAD COMB 235 1.0DL+1.0FL2+0.75WLZ1+0.75LL
157. 1 1.0 3 1.0 4 0.75 11 0.75
158. LOAD COMB 236 1.0DL+1.0FL2+0.75WLZ2+0.75LL
159. 1 1.0 3 1.0 4 0.75 12 0.75
160. LOAD COMB 237 1.0DL+1.0FL1+0.525SLX1+0.75LL
161. 1 1.0 2 1.0 4 0.75 5 0.525
162. LOAD COMB 238 1.0DL+1.0FL1+0.525SLX2+0.75LL
163. 1 1.0 2 1.0 4 0.75 6 0.525
164. LOAD COMB 239 1.0DL+1.0FL1+0.525SLZ1+0.75LL
165. 1 1.0 2 1.0 4 0.75 7 0.525
166. LOAD COMB 240 1.0DL+1.0FL1+0.525SLZ2+0.75LL
167. 1 1.0 2 1.0 4 0.75 8 0.525
168. LOAD COMB 241 1.0DL+1.0FL2+0.525SLX1+0.75LL
169. 1 1.0 3 1.0 4 0.75 5 0.525
170. LOAD COMB 242 1.0DL+1.0FL2+0.525SLX2+0.75LL
171. 1 1.0 3 1.0 4 0.75 6 0.525
172. LOAD COMB 243 1.0DL+1.0FL2+0.525SLZ1+0.75LL
173. 1 1.0 3 1.0 4 0.75 7 0.525
174. LOAD COMB 244 1.0DL+1.0FL2+0.525SLZ2+0.75LL
175. 1 1.0 3 1.0 4 0.75 8 0.525
176. LOAD COMB 245 0.6DL+1.0WLX1
177. 1 0.6 9 1.0
178. LOAD COMB 246 0.6DL+1.0WLX2
179. 1 0.6 10 1.0
180. LOAD COMB 247 0.6DL+1.0WLZ1
181. 1 0.6 11 1.0
182. LOAD COMB 248 0.6DL+1.0WLZ2
183. 1 0.6 12 1.0
184. LOAD COMB 249 0.6DL+0.6FL1+0.7SLX1
185. 1 0.6 2 0.6 5 0.7
186. LOAD COMB 250 0.6DL+0.6FL1+0.7SLX2
187. 1 0.6 2 0.6 6 0.7
188. LOAD COMB 251 0.6DL+0.6FL1+0.7SLZ1
189. 1 0.6 2 0.6 7 0.7
190. LOAD COMB 252 0.6DL+0.6FL1+0.7SLZ2
191. 1 0.6 2 0.6 8 0.7
192. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 4 NUMBER OF MEMBERS 2
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 2

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 2/ 2/ 6 DOF
TOTAL PRIMARY LOAD CASES = 14, TOTAL DEGREES OF FREEDOM = 12
TOTAL LOAD COMBINATION CASES = 52 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 112122.4 MB

193. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 9:29: 2 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 3 STAAD INPUT & OUTPUT FILE FOR F12A DESIGN

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  7, 2020                *
*           Time=    11:41:52                    *
*                                                *
*      USER ID: LT Construction                 *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F12A DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 26-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 5 0 0 12.63; 6 1.705 0 12.63; 211 0 1.85 12.63; 212 1.705 1.85 12.63
     9. 417 -1.14679 0 12.63; 418 -1.14679 0 12.355; 419 -0.941429 0 12.63
    10. 420 -0.941429 0 12.355; 421 -0.706071 0 12.63; 422 -0.706071 0 12.355
    11. 423 -0.370714 0 12.63; 424 -0.370714 0 12.355; 425 -0.235357 0 12.63
    12. 426 -0.235357 0 12.355; 427 0 0 12.355; 430 -1.14679 0 12.255
    13. 431 -0.941429 0 12.255; 432 -0.706071 0 12.255; 433 -0.370714 0 12.255
    14. 434 -0.235357 0 12.255; 435 0 0 12.255; 438 -1.14679 0 11.805
    15. 439 -0.941429 0 11.805; 440 -0.706071 0 11.805; 441 -0.370714 0 11.805
    16. 442 -0.235357 0 11.805; 443 0 0 11.805; 446 -1.14679 0 11.38
    17. 447 -0.941429 0 11.38; 448 -0.706071 0 11.38; 449 -0.370714 0 11.38
    18. 450 -0.235357 0 11.38; 451 0 0 11.38; 452 0.243571 0 12.63
    19. 453 0.243571 0 12.355; 454 0.387143 0 12.63; 455 0.387143 0 12.355
    20. 456 0.730714 0 12.63; 457 0.730714 0 12.355; 458 0.974286 0 12.63
    21. 459 0.974286 0 12.355; 460 1.21786 0 12.63; 461 1.21786 0 12.355
    22. 462 1.46143 0 12.63; 463 1.46143 0 12.355; 464 1.705 0 12.355
    23. 465 0.243571 0 12.255; 466 0.387143 0 12.255; 467 0.730714 0 12.255
    24. 468 0.974286 0 12.255; 469 1.21786 0 12.255; 470 1.46143 0 12.255
    25. 471 1.705 0 12.255; 472 0.243571 0 11.805; 473 0.387143 0 11.805
    26. 474 0.730714 0 11.805; 475 0.974286 0 11.805; 476 1.21786 0 11.805
    27. 477 1.46143 0 11.805; 478 1.705 0 11.805; 479 0.243571 0 11.38
    28. 480 0.387143 0 11.38; 481 0.730714 0 11.38; 482 0.974286 0 11.38
    29. 483 1.21786 0 11.38; 484 1.46143 0 11.38; 485 1.705 0 11.38
    30. 486 1.94036 0 12.63; 487 1.94036 0 12.355; 488 2.17571 0 12.63
    31. 489 2.17571 0 12.355; 490 2.41107 0 12.63; 491 2.41107 0 12.355
    32. 492 2.64643 0 12.63; 493 2.64643 0 12.355; 494 2.85279 0 12.63
    33. 495 2.85279 0 12.355; 500 1.94036 0 12.255; 501 2.17571 0 12.255
    34. 502 2.41107 0 12.255; 503 2.64643 0 12.255; 504 2.85279 0 12.255
    35. 507 1.94036 0 11.805; 508 2.17571 0 11.805; 509 2.41107 0 11.805
    36. 510 2.64643 0 11.805; 511 2.85279 0 11.805; 514 1.94036 0 11.38
    37. 515 2.17571 0 11.38; 516 2.41107 0 11.38; 517 2.64643 0 11.38
    38. 518 2.85279 0 11.38; 525 -1.14679 0 13.88; 526 -1.14679 0 13.455
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39. 527 -0.941429 0 13.88; 528 -0.941429 0 13.455; 529 -0.706071 0 13.88
40. 530 -0.706071 0 13.455; 531 -0.370714 0 13.88; 532 -0.370714 0 13.455
41. 533 -0.235357 0 13.88; 534 -0.235357 0 13.455; 535 0 0 13.88; 536 0 0 13.455
42. 539 -1.14679 0 13.005; 540 -0.941429 0 13.005; 541 -0.706071 0 13.005
43. 542 -0.370714 0 13.005; 543 -0.235357 0 13.005; 544 0 0 13.005
44. 547 -1.14679 0 12.905; 548 -0.941429 0 12.905; 549 -0.706071 0 12.905
45. 550 -0.370714 0 12.905; 551 -0.235357 0 12.905; 552 0 0 12.905
46. 553 0.243571 0 13.88; 554 0.243571 0 13.455; 555 0.387143 0 13.88
47. 556 0.387143 0 13.455; 557 0.730714 0 13.88; 558 0.730714 0 13.455
48. 559 0.974286 0 13.88; 560 0.974286 0 13.455; 561 1.21786 0 13.88
49. 562 1.21786 0 13.455; 563 1.46143 0 13.88; 564 1.46143 0 13.455
50. 565 1.705 0 13.88; 566 1.705 0 13.455; 567 0.243571 0 13.005
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54. 577 0.974286 0 12.905; 578 1.21786 0 12.905; 579 1.46143 0 12.905
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58. 589 2.85279 0 13.88; 590 2.85279 0 13.455; 595 1.94036 0 13.005
59. 596 2.17571 0 13.005; 597 2.41107 0 13.005; 598 2.64643 0 13.005
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61. 604 2.41107 0 12.905; 605 2.64643 0 12.905; 606 2.85279 0 12.905
62. 1005 0 0.25 12.63; 1006 1.705 0.25 12.63; 1009 -1.14679 0 12.03
63. 1010 -0.941429 0 12.03; 1011 -0.706071 0 12.03; 1012 -0.370714 0 12.03
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73. 1044 1.94036 0 13.23; 1045 2.17571 0 13.23; 1046 2.41107 0 13.23
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86. 1083 2.64643 0 13.6675; 1084 2.85279 0 13.6675
87. MEMBER INCIDENCES
88. 279 5 1005; 280 6 1006; 991 1005 211; 992 1006 212
89. ELEMENT INCIDENCES SHELL
90. 461 417 419 420 418; 462 419 421 422 420; 463 421 423 424 422
91. 464 423 425 426 424; 465 425 5 427 426; 468 418 420 431 430
92. 469 420 422 432 431; 470 422 424 433 432; 471 424 426 434 433
93. 472 426 427 435 434; 488 5 452 453 427; 490 452 454 455 453
94. 492 454 456 457 455; 494 456 458 459 457; 496 458 460 461 459
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96. 501 453 455 466 465; 502 455 457 467 466; 503 457 459 468 467
97. 504 459 461 469 468; 505 461 463 470 469; 506 463 464 471 470
98. 521 6 486 487 464; 522 486 488 489 487; 523 488 490 491 489
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102. 567 541 542 550 549; 568 542 543 551 550; 569 543 544 552 551
103. 572 547 548 419 417; 573 548 549 421 419; 574 549 550 423 421
104. 575 550 551 425 423; 576 551 552 5 425; 591 544 567 574 552
105. 592 567 568 575 574; 593 568 569 576 575; 594 569 570 577 576
106. 595 570 571 578 577; 596 571 572 579 578; 597 572 573 580 579
107. 598 552 574 452 5; 599 574 575 454 452; 600 575 576 456 454
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109. 604 579 580 6 462; 619 573 595 602 580; 620 595 596 603 602
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113. 998 1009 1010 439 438; 999 431 432 1011 1010; 1000 1010 1011 440 439
114. 1001 432 433 1012 1011; 1002 1011 1012 441 440; 1003 433 434 1013 1012
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116. 1007 435 465 1015 1014; 1008 1014 1015 472 443; 1009 465 466 1016 1015
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118. 1013 467 468 1018 1017; 1014 1017 1018 475 474; 1015 468 469 1019 1018
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132. 1063 566 582 1044 1043; 1064 1043 1044 595 573; 1065 582 584 1045 1044
133. 1066 1044 1045 596 595; 1067 584 586 1046 1045; 1068 1045 1046 597 596
134. 1069 586 588 1047 1046; 1070 1046 1047 598 597; 1071 588 590 1048 1047
135. 1072 1047 1048 599 598; 1073 438 439 1049 1050; 1074 1050 1049 447 446
136. 1075 439 440 1051 1049; 1076 1049 1051 448 447; 1077 440 441 1052 1051
137. 1078 1051 1052 449 448; 1079 441 442 1053 1052; 1080 1052 1053 450 449
138. 1081 442 443 1054 1053; 1082 1053 1054 451 450; 1083 443 472 1055 1054
139. 1084 1054 1055 479 451; 1085 472 473 1056 1055; 1086 1055 1056 480 479
140. 1087 473 474 1057 1056; 1088 1056 1057 481 480; 1089 474 475 1058 1057
141. 1090 1057 1058 482 481; 1091 475 476 1059 1058; 1092 1058 1059 483 482
142. 1093 476 477 1060 1059; 1094 1059 1060 484 483; 1095 477 478 1061 1060
143. 1096 1060 1061 485 484; 1097 478 507 1062 1061; 1098 1061 1062 514 485
144. 1099 507 508 1063 1062; 1100 1062 1063 515 514; 1101 508 509 1064 1063
145. 1102 1063 1064 516 515; 1103 509 510 1065 1064; 1104 1064 1065 517 516
146. 1105 510 511 1066 1065; 1106 1065 1066 518 517; 1107 525 527 1067 1068
147. 1108 1068 1067 528 526; 1109 527 529 1069 1067; 1110 1067 1069 530 528
148. 1111 529 531 1070 1069; 1112 1069 1070 532 530; 1113 531 533 1071 1070
149. 1114 1070 1071 534 532; 1115 533 535 1072 1071; 1116 1071 1072 536 534
150. 1117 535 553 1073 1072; 1118 1072 1073 554 536; 1119 553 555 1074 1073

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151. 1120 1073 1074 556 554; 1121 555 557 1075 1074; 1122 1074 1075 558 556
152. 1123 557 559 1076 1075; 1124 1075 1076 560 558; 1125 559 561 1077 1076
153. 1126 1076 1077 562 560; 1127 561 563 1078 1077; 1128 1077 1078 564 562
154. 1129 563 565 1079 1078; 1130 1078 1079 566 564; 1131 565 581 1080 1079
155. 1132 1079 1080 582 566; 1133 581 583 1081 1080; 1134 1080 1081 584 582
156. 1135 583 585 1082 1081; 1136 1081 1082 586 584; 1137 585 587 1083 1082
157. 1138 1082 1083 588 586; 1139 587 589 1084 1083; 1140 1083 1084 590 588
158. START GROUP DEFINITION
159. FLOOR
160. ELEMENT
161. _RAFT 461 TO 465 468 TO 472 488 490 492 494 496 498 TO 506 521 TO 525 528 -
162. 529 TO 532 565 TO 569 572 TO 576 591 TO 604 619 TO 623 626 TO 630 997 TO 1030 -
163. 1039 TO 1140
164. _RAFT_MOM 461 TO 463 468 TO 470 492 494 496 498 502 TO 506 522 TO 525 528 -
165. 529 TO 532 565 TO 569 572 TO 574 591 TO 597 600 TO 603 619 TO 623 627 TO 630 -
166. 997 TO 1030 1039 TO 1140
167. _RAFT_SHEAR 461 468 524 525 531 532 565 572 622 623 629 630 997 998 -
168. 1027 TO 1030 1039 1040 1069 TO 1140
169. MEMBER
170. _C8 991
171. _C7 992
172. END GROUP DEFINITION
173. ELEMENT PROPERTY
174. 461 TO 465 468 TO 472 488 490 492 494 496 498 TO 506 521 TO 525 528 TO 532 -
175. 565 TO 569 572 TO 576 591 TO 604 619 TO 623 626 TO 630 997 TO 1030 -
176. 1039 TO 1140 THICKNESS 0.75
177. DEFINE MATERIAL START
178. ISOTROPIC CONCRETE
179. E 2.17185E+007
180. POISSON 0.17
181. DENSITY 25
182. ALPHA 1E-005
183. DAMP 0.05
184. TYPE CONCRETE
185. STRENGTH FCU 27579
186. END DEFINE MATERIAL
187. MEMBER PROPERTY AMERICAN
188. 279 991 PRIS YD 0.75 ZD 0.75
189. MEMBER PROPERTY AMERICAN
190. 280 992 PRIS YD 0.55 ZD 0.55
191. CONSTANTS
192. MATERIAL CONCRETE ALL
193. ************************************************************
194. SUPPORTS
195. 461 TO 465 468 TO 472 488 490 492 494 496 498 TO 506 521 TO 525 528 TO 532 -
196. 565 TO 569 572 TO 576 591 TO 604 619 TO 623 626 TO 630 997 TO 1030 -
197. 1039 TO 1140 PLATE MAT DIRECT Y SUBGRADE 40500 PRINT
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JOINT	INFLUENCE AREA
5	0.065853
6	0.065853
417	0.028237
418	0.019253
419	0.060599

420	0.041317
421	0.078473
422	0.053505
423	0.064723
424	0.044129
425	0.050973
426	0.034754
427	0.044900
430	0.016686
431	0.035808
432	0.046371
433	0.038246
434	0.030121
435	0.038913
438	0.022461
439	0.048204
440	0.062422
441	0.051484
442	0.040547
443	0.052383
446	0.010910
447	0.023413
448	0.030319
449	0.025007
450	0.019694
451	0.025443
452	0.053232
453	0.036295
454	0.066982
455	0.045670
456	0.080732
457	0.055045
458	0.066983
459	0.045670
460	0.066982
461	0.045670
462	0.066982
463	0.045669
464	0.044900
465	0.031455
466	0.039580
467	0.047705
468	0.039581
469	0.039580
470	0.039580
471	0.038913
472	0.042344
473	0.053281
474	0.064219
475	0.053282
476	0.053281
477	0.053281
478	0.052383
479	0.020567
480	0.025879
481	0.031192

482	0.025880
483	0.025880
484	0.025879
485	0.025443
486	0.064723
487	0.044129
488	0.064723
489	0.044129
490	0.064724
491	0.044130
492	0.060736
493	0.041411
494	0.028374
495	0.019346
500	0.038245
501	0.038245
502	0.038246
503	0.035890
504	0.016767
507	0.051484
508	0.051484
509	0.051485
510	0.048313
511	0.022571
514	0.025006
515	0.025006
516	0.025007
517	0.023466
518	0.010963
525	0.010910
526	0.022461
527	0.023413
528	0.048204
529	0.030319
530	0.062422
531	0.025007
532	0.051484
533	0.019694
534	0.040547
535	0.025443
536	0.052383
539	0.016686
540	0.035808
541	0.046371
542	0.038246
543	0.030121
544	0.038913
547	0.019253
548	0.041317
549	0.053504
550	0.044129
551	0.034754
552	0.044899
553	0.020567
554	0.042344
555	0.025879

556	0.053281
557	0.031192
558	0.064219
559	0.025880
560	0.053282
561	0.025880
562	0.053281
563	0.025879
564	0.053281
565	0.025443
566	0.052383
567	0.031455
568	0.039580
569	0.047705
570	0.039581
571	0.039580
572	0.039580
573	0.038913
574	0.036295
575	0.045670
576	0.055045
577	0.045670
578	0.045670
579	0.045669
580	0.044900
581	0.025006
582	0.051484
583	0.025006
584	0.051484
585	0.025007
586	0.051485
587	0.023466
588	0.048313
589	0.010963
590	0.022571
595	0.038245
596	0.038245
597	0.038246
598	0.035890
599	0.016767
602	0.044129
603	0.044129
604	0.044130
605	0.041411
606	0.019346
1009	0.023103
1010	0.049581
1011	0.064205
1012	0.052955
1013	0.041705
1014	0.053879
1015	0.043554
1016	0.054804
1017	0.066054
1018	0.054804
1019	0.054804

1020	0.054803
1021	0.053880
1022	0.052955
1023	0.052955
1024	0.052956
1025	0.049693
1026	0.023215
1031	0.023103
1032	0.049581
1033	0.064205
1034	0.052955
1035	0.041705
1036	0.053879
1037	0.043554
1038	0.054804
1039	0.066054
1040	0.054804
1041	0.054804
1042	0.054803
1043	0.053880
1044	0.052955
1045	0.052955
1046	0.052956
1047	0.049694
1048	0.023216
1049	0.046826
1050	0.021820
1051	0.060639
1052	0.050013
1053	0.039388
1054	0.050886
1055	0.041134
1056	0.051759
1057	0.062384
1058	0.051759
1059	0.051759
1060	0.051759
1061	0.050886
1062	0.050013
1063	0.050013
1064	0.050014
1065	0.046933
1066	0.021926
1067	0.046826
1068	0.021820
1069	0.060639
1070	0.050013
1071	0.039388
1072	0.050886
1073	0.041134
1074	0.051759
1075	0.062384
1076	0.051759
1077	0.051759
1078	0.051759
1079	0.050886

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      1080          0.050013
      1081          0.050013
      1082          0.050014
      1083          0.046933
      1084          0.021926
    TOTAL           9.998951
198. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
199. JOINT LOAD
200. 211 FX 42.3 FY -280.2 FZ 26
201. 212 FX 0.3 FY -58.1 FZ 0
202. ELEMENT LOAD
203. 461 TO 465 468 TO 472 488 490 492 494 496 498 TO 506 521 TO 525 528 TO 532 -
204. 565 TO 569 572 TO 576 591 TO 604 619 TO 623 626 TO 630 997 TO 1030 -
205. 1039 TO 1140 PR -13.5
206. SELFWEIGHT Y -1
207. ***
208. LOAD 2 LOADTYPE NONE  TITLE NORMAL OPERATING FLUID LOAD ( FL1)
209. JOINT LOAD
210. 211 FX 105.6 FY -467.7 FZ 72.6
211. 212 FX 0.6 FY -39.4 FZ 0
212. LOAD 3 LOADTYPE NONE  TITLE MAX. FLUID LOAD ( FL2)
213. JOINT LOAD
214. 211 FX 129.9 FY -544.6 FZ 82.6
215. 212 FX 0.6 FY -43.2 FZ 0
216. LOAD 4 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
217. JOINT LOAD
218. 211 FX 21.1 FY -118.6 FZ 9
219. 212 FX 0.2 FY -21.2 FZ 0
220. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD + X ( SLX1 )
221. JOINT LOAD
222. 211 FX 51.4 FY -87.8 FZ 2.6
223. 212 FX 0.1 FY -24.8 FZ 0
224. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD - X ( SLX2 )
225. JOINT LOAD
226. 211 FX -51.4 FY 87.8 FZ -2.6
227. 212 FX -0.1 FY 24.8 FZ 0
228. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD +Z ( SLZ1)
229. JOINT LOAD
230. 211 FX 42.5 FY -301.9 FZ 124.6
231. 212 FX 0.2 FY -24.5 FZ 0.2
232. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD -Z ( SLZ2)
233. JOINT LOAD
234. 211 FX -42.5 FY 301.9 FZ -124.6
235. 212 FX -0.2 FY 24.5 FZ -0.2
236. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
237. JOINT LOAD
238. 211 FX 31.7 FY -37.1 FZ -0.7
239. 212 FX 1.4 FY -23.8 FZ 0
240. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
241. JOINT LOAD
242. 211 FX -32.7 FY 37.1 FZ 0.7
243. 212 FX -1.6 FY 24.3 FZ 0
244. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
245. JOINT LOAD
246. 211 FX 14.1 FY -99 FZ 45.9
247. 212 FX 0.1 FY -5 FZ 2.8
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248. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
249. JOINT LOAD
250. 211 FX -14.1 FY 99.2 FZ -46.1
251. 212 FX -0.1 FY 5 FZ -2.8
252. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD ( TL )
253. JOINT LOAD
254. 211 FX 153.5 FY -130.9 FZ 46.8
255. 212 FX 1.2 FY -62.4 FZ 0
256. LOAD 14 LOADTYPE NONE  TITLE LATERAL PIPE LOAD ( PL )
257. JOINT LOAD
258. 211 FX 3.3 FY -83.2 FZ 19.9
259. 212 FX 0.1 FY -17.1 FZ 0
260. ******************************************************
261. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
262. LOAD COMB 101 1.4DL+1.4FL1
263. 1 1.4 2 1.4
264. LOAD COMB 102 1.4DL+1.4FL2
265. 1 1.4 3 1.4
266. LOAD COMB 103 1.2DL+1.2FL1+1.6LL
267. 1 1.2 2 1.2 4 1.6
268. LOAD COMB 104 1.2DL+1.2FL2+1.6LL
269. 1 1.2 3 1.2 4 1.6
270. LOAD COMB 105 1.2DL+1.2FL1+1.2TL+1.6LL
271. 1 1.2 2 1.2 4 1.6 13 1.2
272. LOAD COMB 106 1.2DL+1.2FL2+1.2TL+1.6LL
273. 1 1.2 3 1.2 4 1.6 13 1.2
274. LOAD COMB 107 1.4DL+1.4FL1+1.4PL
275. 1 1.4 2 1.4 14 1.4
276. LOAD COMB 108 1.4DL+1.4FL2+1.4PL
277. 1 1.4 3 1.4 14 1.4
278. LOAD COMB 109 1.2DL+1.2FL1+1.6LL+1.2PL
279. 1 1.2 2 1.2 4 1.6 14 1.2
280. LOAD COMB 110 1.2DL+1.2FL2+1.6LL+1.2PL
281. 1 1.2 3 1.2 4 1.6 14 1.2
282. LOAD COMB 111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
283. 1 1.2 2 1.2 4 1.6 13 1.2 14 1.2
284. LOAD COMB 112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
285. 1 1.2 3 1.2 4 1.6 13 1.2 14 1.2
286. LOAD COMB 113 1.2DL+0.8WLX1
287. 1 1.2 9 0.8
288. LOAD COMB 114 1.2DL+0.8WLX2
289. 1 1.2 10 0.8
290. LOAD COMB 115 1.2DL+0.8WLZ1
291. 1 1.2 11 0.8
292. LOAD COMB 116 1.2DL+0.8WLZ2
293. 1 1.2 12 0.8
294. LOAD COMB 117 1.2DL+1.6WLX1+0.5LL
295. 1 1.2 9 1.6 4 0.5
296. LOAD COMB 118 1.2DL+1.6WLX2+0.5LL
297. 1 1.2 10 1.6 4 0.5
298. LOAD COMB 119 1.2DL+1.6WLZ1+0.5LL
299. 1 1.2 11 1.6 4 0.5
300. LOAD COMB 120 1.2DL+1.6WLZ2+0.5LL
301. 1 1.2 12 1.6 4 0.5
302. LOAD COMB 121 1.2DL+1.0SLX1+0.5LL+0.5FL1
303. 1 1.2 5 1.0 4 0.5 2 0.5
```

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304. LOAD COMB 122 1.2DL+1.0SLX2+0.5LL+0.5FL1
305. 1 1.2 6 1.0 4 0.5 2 0.5
306. LOAD COMB 123 1.2DL+1.0SLZ1+0.5LL+0.5FL1
307. 1 1.2 7 1.0 4 0.5 2 0.5
308. LOAD COMB 124 1.2DL+1.0SLZ2+0.5LL+0.5FL1
309. 1 1.2 8 1.0 4 0.5 2 0.5
310. LOAD COMB 125 0.9DL+1.6WLX1
311. 1 0.9 9 1.6
312. LOAD COMB 126 0.9DL+1.6WLX2
313. 1 0.9 10 1.6
314. LOAD COMB 127 0.9DL+1.6WLZ1
315. 1 0.9 11 1.6
316. LOAD COMB 128 0.9DL+1.6WLZ2
317. 1 0.9 12 1.6
318. LOAD COMB 129 0.9DL+1.0SLX1
319. 1 0.9 5 1.0
320. LOAD COMB 130 0.9DL+1.0SLX2
321. 1 0.9 6 1.0
322. LOAD COMB 131 0.9DL+1.0SLZ1
323. 1 0.9 7 1.0
324. LOAD COMB 132 0.9DL+1.0SLZ2
325. 1 0.9 8 1.0
326. ********************************************************************************
327. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
328. LOAD COMB 201 1.0DL+1.0FL1
329. 1 1.0 2 1.0
330. LOAD COMB 202 1.0DL+1.0FL2
331. 1 1.0 3 1.0
332. LOAD COMB 203 1.0DL+1.0FL1+1.0LL+1.0TL
333. 1 1.0 2 1.0 4 1.0 13 1.0
334. LOAD COMB 204 1.0DL+1.0FL2+1.0LL+1.0TL
335. 1 1.0 3 1.0 4 1.0 13 1.0
336. LOAD COMB 205 1.0DL+1.0FL1+0.75LL+0.75TL
337. 1 1.0 2 1.0 4 0.75 13 0.75
338. LOAD COMB 206 1.0DL+1.0FL2+0.75LL+0.75TL
339. 1 1.0 3 1.0 4 0.75 13 0.75
340. LOAD COMB 207 1.0DL+1.0FL1+1.0PL
341. 1 1.0 2 1.0 14 1.0
342. LOAD COMB 208 1.0DL+1.0FL2+1.0PL
343. 1 1.0 3 1.0 14 1.0
344. LOAD COMB 209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL
345. 1 1.0 2 1.0 4 1.0 13 1.0 14 1.0
346. LOAD COMB 210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL
347. 1 1.0 3 1.0 4 1.0 13 1.0 14 1.0
348. LOAD COMB 211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL
349. 1 1.0 2 1.0 4 0.75 13 0.75 14 1.0
350. LOAD COMB 212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL
351. 1 1.0 3 1.0 4 0.75 13 0.75 14 1.0
352. LOAD COMB 213 1.0DL+1.0FL1+1.0WLX1
353. 1 1.0 2 1.0 9 1.0
354. LOAD COMB 214 1.0DL+1.0FL1+1.0WLX2
355. 1 1.0 2 1.0 10 1.0
356. LOAD COMB 215 1.0DL+1.0FL1+1.0WLZ1
357. 1 1.0 2 1.0 11 1.0
358. LOAD COMB 216 1.0DL+1.0FL1+1.0WLZ2
359. 1 1.0 2 1.0 12 1.0
```

```
360. LOAD COMB 217 1.0DL+1.0FL2+1.0WLX1
361. 1 1.0 3 1.0 9 1.0
362. LOAD COMB 218 1.0DL+1.0FL2+1.0WLX2
363. 1 1.0 3 1.0 10 1.0
364. LOAD COMB 219 1.0DL+1.0FL2+1.0WLZ1
365. 1 1.0 3 1.0 11 1.0
366. LOAD COMB 220 1.0DL+1.0FL2+1.0WLZ2
367. 1 1.0 3 1.0 12 1.0
368. LOAD COMB 221 1.0DL+1.0FL1+0.7SLX1
369. 1 1.0 2 1.0 5 0.7
370. LOAD COMB 222 1.0DL+1.0FL1+0.7SLX2
371. 1 1.0 2 1.0 6 0.7
372. LOAD COMB 223 1.0DL+1.0FL1+0.7SLZ1
373. 1 1.0 2 1.0 7 0.7
374. LOAD COMB 224 1.0DL+1.0FL1+0.7SLZ2
375. 1 1.0 2 1.0 8 0.7
376. LOAD COMB 225 1.0DL+1.0FL2+0.7SLX1
377. 1 1.0 3 1.0 5 0.7
378. LOAD COMB 226 1.0DL+1.0FL2+0.7SLX2
379. 1 1.0 3 1.0 6 0.7
380. LOAD COMB 227 1.0DL+1.0FL2+0.7SLZ1
381. 1 1.0 3 1.0 7 0.7
382. LOAD COMB 228 1.0DL+1.0FL2+0.7SLZ2
383. 1 1.0 3 1.0 8 0.7
384. LOAD COMB 229 1.0DL+1.0FL1+0.75WLX1+0.75LL
385. 1 1.0 2 1.0 4 0.75 9 0.75
386. LOAD COMB 230 1.0DL+1.0FL1+0.75WLX2+0.75LL
387. 1 1.0 2 1.0 4 0.75 10 0.75
388. LOAD COMB 231 1.0DL+1.0FL1+0.75WLZ1+0.75LL
389. 1 1.0 2 1.0 4 0.75 11 0.75
390. LOAD COMB 232 1.0DL+1.0FL1+0.75WLZ2+0.75LL
391. 1 1.0 2 1.0 4 0.75 12 0.75
392. LOAD COMB 233 1.0DL+1.0FL2+0.75WLX1+0.75LL
393. 1 1.0 3 1.0 4 0.75 9 0.75
394. LOAD COMB 234 1.0DL+1.0FL2+0.75WLX2+0.75LL
395. 1 1.0 3 1.0 4 0.75 10 0.75
396. LOAD COMB 235 1.0DL+1.0FL2+0.75WLZ1+0.75LL
397. 1 1.0 3 1.0 4 0.75 11 0.75
398. LOAD COMB 236 1.0DL+1.0FL2+0.75WLZ2+0.75LL
399. 1 1.0 3 1.0 4 0.75 12 0.75
400. LOAD COMB 237 1.0DL+1.0FL1+0.525SLX1+0.75LL
401. 1 1.0 2 1.0 4 0.75 5 0.525
402. LOAD COMB 238 1.0DL+1.0FL1+0.525SLX2+0.75LL
403. 1 1.0 2 1.0 4 0.75 6 0.525
404. LOAD COMB 239 1.0DL+1.0FL1+0.525SLZ1+0.75LL
405. 1 1.0 2 1.0 4 0.75 7 0.525
406. LOAD COMB 240 1.0DL+1.0FL1+0.525SLZ2+0.75LL
407. 1 1.0 2 1.0 4 0.75 8 0.525
408. LOAD COMB 241 1.0DL+1.0FL2+0.525SLX1+0.75LL
409. 1 1.0 3 1.0 4 0.75 5 0.525
410. LOAD COMB 242 1.0DL+1.0FL2+0.525SLX2+0.75LL
411. 1 1.0 3 1.0 4 0.75 6 0.525
412. LOAD COMB 243 1.0DL+1.0FL2+0.525SLZ1+0.75LL
413. 1 1.0 3 1.0 4 0.75 7 0.525
414. LOAD COMB 244 1.0DL+1.0FL2+0.525SLZ2+0.75LL
415. 1 1.0 3 1.0 4 0.75 8 0.525
```

```
416. LOAD COMB 245 0.6DL+1.0WLX1
417. 1 0.6 9 1.0
418. LOAD COMB 246 0.6DL+1.0WLX2
419. 1 0.6 10 1.0
420. LOAD COMB 247 0.6DL+1.0WLZ1
421. 1 0.6 11 1.0
422. LOAD COMB 248 0.6DL+1.0WLZ2
423. 1 0.6 12 1.0
424. LOAD COMB 249 0.6DL+0.6FL1+0.7SLX1
425. 1 0.6 2 0.6 5 0.7
426. LOAD COMB 250 0.6DL+0.6FL1+0.7SLX2
427. 1 0.6 2 0.6 6 0.7
428. LOAD COMB 251 0.6DL+0.6FL1+0.7SLZ1
429. 1 0.6 2 0.6 7 0.7
430. LOAD COMB 252 0.6DL+0.6FL1+0.7SLZ2
431. 1 0.6 2 0.6 8 0.7
432. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**238**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**204**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**234**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 182/ 22/ 69 DOF
TOTAL PRIMARY LOAD CASES = 14, TOTAL DEGREES OF FREEDOM = 726
TOTAL LOAD COMBINATION CASES = 84 SO FAR.
SIZE OF STIFFNESS MATRIX = 51 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.7/ 112115.6 MB

```
433. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 11:42: 7 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 4 STAAD INPUT & OUTPUT FILE FOR F12A STABILITY

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    APR  7, 2020                   *
*           Time=     9:37:14                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F12A STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 26-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 211 0 1.85 12.63; 212 1.705 1.85 12.63; 1005 0 0.25 12.63
     9. 1006 1.705 0.25 12.63
    10. MEMBER INCIDENCES
    11. 991 1005 211; 992 1006 212
    12. DEFINE MATERIAL START
    13. ISOTROPIC CONCRETE
    14. E 2.17185E+007
    15. POISSON 0.17
    16. DENSITY 25
    17. ALPHA 1E-005
    18. DAMP 0.05
    19. TYPE CONCRETE
    20. STRENGTH FCU 27579
    21. END DEFINE MATERIAL
    22. MEMBER PROPERTY
    23. 992 PRIS YD 0.55 ZD 0.55
    24. 991 PRIS YD 0.75 ZD 0.75
    25. CONSTANTS
    26. MATERIAL CONCRETE ALL
    27. ************************************************************
    28. SUPPORTS
    29. 1005 1006 FIXED
    30. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
              MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    31. JOINT LOAD
    32. 211 FX 42.3 FY -280.2 FZ 26
    33. 212 FX 0.3 FY -58.1 FZ 0
    34. ***
    35. LOAD 2 LOADTYPE NONE  TITLE NORMAL OPERATING FLUID LOAD ( FL1)
    36. JOINT LOAD
```

```
37. 211 FX 105.6 FY -467.7 FZ 72.6
38. 212 FX 0.6 FY -39.4 FZ 0
39. LOAD 3 LOADTYPE NONE  TITLE MAX. FLUID LOAD ( FL2)
40. JOINT LOAD
41. 211 FX 129.9 FY -544.6 FZ 82.6
42. 212 FX 0.6 FY -43.2 FZ 0
43. LOAD 4 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
44. JOINT LOAD
45. 211 FX 21.1 FY -118.6 FZ 9
46. 212 FX 0.2 FY -21.2 FZ 0
47. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD + X ( SLX1 )
48. JOINT LOAD
49. 211 FX 51.4 FY -87.8 FZ 2.6
50. 212 FX 0.1 FY -24.8 FZ 0
51. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD - X ( SLX2 )
52. JOINT LOAD
53. 211 FX -51.4 FY 87.8 FZ -2.6
54. 212 FX -0.1 FY 24.8 FZ 0
55. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD +Z ( SLZ1)
56. JOINT LOAD
57. 211 FX 42.5 FY -301.9 FZ 124.6
58. 212 FX 0.2 FY -24.5 FZ 0.2
59. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD -Z ( SLZ2)
60. JOINT LOAD
61. 211 FX -42.5 FY 301.9 FZ -124.6
62. 212 FX -0.2 FY 24.5 FZ -0.2
63. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
64. JOINT LOAD
65. 211 FX 31.7 FY -37.1 FZ -0.7
66. 212 FX 1.4 FY -23.8 FZ 0
67. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
68. JOINT LOAD
69. 211 FX -32.7 FY 37.1 FZ 0.7
70. 212 FX -1.6 FY 24.3 FZ 0
71. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
72. JOINT LOAD
73. 211 FX 14.1 FY -99 FZ 45.9
74. 212 FX 0.1 FY -5 FZ 2.8
75. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
76. JOINT LOAD
77. 211 FX -14.1 FY 99.2 FZ -46.1
78. 212 FX -0.1 FY 5 FZ -2.8
79. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD ( TL )
80. JOINT LOAD
81. 211 FX 153.5 FY -130.9 FZ 46.8
82. 212 FX 1.2 FY -62.4 FZ 0
83. LOAD 14 LOADTYPE NONE  TITLE LATERAL PIPE LOAD ( PL )
84. JOINT LOAD
85. 211 FX 3.3 FY -83.2 FZ 19.9
86. 212 FX 0.1 FY -17.1 FZ 0
87. ********************************************************
88. ********************************************************************************
89. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
90. LOAD COMB 201 1.0DL+1.0FL1
91. 1 1.0 2 1.0
92. LOAD COMB 202 1.0DL+1.0FL2
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 93. 1 1.0 3 1.0
 94. LOAD COMB 203 1.0DL+1.0FL1+1.0LL+1.0TL
 95. 1 1.0 2 1.0 4 1.0 13 1.0
 96. LOAD COMB 204 1.0DL+1.0FL2+1.0LL+1.0TL
 97. 1 1.0 3 1.0 4 1.0 13 1.0
 98. LOAD COMB 205 1.0DL+1.0FL1+0.75LL+0.75TL
 99. 1 1.0 2 1.0 4 0.75 13 0.75
100. LOAD COMB 206 1.0DL+1.0FL2+0.75LL+0.75TL
101. 1 1.0 3 1.0 4 0.75 13 0.75
102. LOAD COMB 207 1.0DL+1.0FL1+1.0PL
103. 1 1.0 2 1.0 14 1.0
104. LOAD COMB 208 1.0DL+1.0FL2+1.0PL
105. 1 1.0 3 1.0 14 1.0
106. LOAD COMB 209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL
107. 1 1.0 2 1.0 4 1.0 13 1.0 14 1.0
108. LOAD COMB 210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL
109. 1 1.0 3 1.0 4 1.0 13 1.0 14 1.0
110. LOAD COMB 211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL
111. 1 1.0 2 1.0 4 0.75 13 0.75 14 1.0
112. LOAD COMB 212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL
113. 1 1.0 3 1.0 4 0.75 13 0.75 14 1.0
114. LOAD COMB 213 1.0DL+1.0FL1+1.0WLX1
115. 1 1.0 2 1.0 9 1.0
116. LOAD COMB 214 1.0DL+1.0FL1+1.0WLX2
117. 1 1.0 2 1.0 10 1.0
118. LOAD COMB 215 1.0DL+1.0FL1+1.0WLZ1
119. 1 1.0 2 1.0 11 1.0
120. LOAD COMB 216 1.0DL+1.0FL1+1.0WLZ2
121. 1 1.0 2 1.0 12 1.0
122. LOAD COMB 217 1.0DL+1.0FL2+1.0WLX1
123. 1 1.0 3 1.0 9 1.0
124. LOAD COMB 218 1.0DL+1.0FL2+1.0WLX2
125. 1 1.0 3 1.0 10 1.0
126. LOAD COMB 219 1.0DL+1.0FL2+1.0WLZ1
127. 1 1.0 3 1.0 11 1.0
128. LOAD COMB 220 1.0DL+1.0FL2+1.0WLZ2
129. 1 1.0 3 1.0 12 1.0
130. LOAD COMB 221 1.0DL+1.0FL1+0.7SLX1
131. 1 1.0 2 1.0 5 0.7
132. LOAD COMB 222 1.0DL+1.0FL1+0.7SLX2
133. 1 1.0 2 1.0 6 0.7
134. LOAD COMB 223 1.0DL+1.0FL1+0.7SLZ1
135. 1 1.0 2 1.0 7 0.7
136. LOAD COMB 224 1.0DL+1.0FL1+0.7SLZ2
137. 1 1.0 2 1.0 8 0.7
138. LOAD COMB 225 1.0DL+1.0FL2+0.7SLX1
139. 1 1.0 3 1.0 5 0.7
140. LOAD COMB 226 1.0DL+1.0FL2+0.7SLX2
141. 1 1.0 3 1.0 6 0.7
142. LOAD COMB 227 1.0DL+1.0FL2+0.7SLZ1
143. 1 1.0 3 1.0 7 0.7
144. LOAD COMB 228 1.0DL+1.0FL2+0.7SLZ2
145. 1 1.0 3 1.0 8 0.7
146. LOAD COMB 229 1.0DL+1.0FL1+0.75WLX1+0.75LL
147. 1 1.0 2 1.0 4 0.75 9 0.75
148. LOAD COMB 230 1.0DL+1.0FL1+0.75WLX2+0.75LL
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149. 1 1.0 2 1.0 4 0.75 10 0.75
150. LOAD COMB 231 1.0DL+1.0FL1+0.75WLZ1+0.75LL
151. 1 1.0 2 1.0 4 0.75 11 0.75
152. LOAD COMB 232 1.0DL+1.0FL1+0.75WLZ2+0.75LL
153. 1 1.0 2 1.0 4 0.75 12 0.75
154. LOAD COMB 233 1.0DL+1.0FL2+0.75WLX1+0.75LL
155. 1 1.0 3 1.0 4 0.75 9 0.75
156. LOAD COMB 234 1.0DL+1.0FL2+0.75WLX2+0.75LL
157. 1 1.0 3 1.0 4 0.75 10 0.75
158. LOAD COMB 235 1.0DL+1.0FL2+0.75WLZ1+0.75LL
159. 1 1.0 3 1.0 4 0.75 11 0.75
160. LOAD COMB 236 1.0DL+1.0FL2+0.75WLZ2+0.75LL
161. 1 1.0 3 1.0 4 0.75 12 0.75
162. LOAD COMB 237 1.0DL+1.0FL1+0.525SLX1+0.75LL
163. 1 1.0 2 1.0 4 0.75 5 0.525
164. LOAD COMB 238 1.0DL+1.0FL1+0.525SLX2+0.75LL
165. 1 1.0 2 1.0 4 0.75 6 0.525
166. LOAD COMB 239 1.0DL+1.0FL1+0.525SLZ1+0.75LL
167. 1 1.0 2 1.0 4 0.75 7 0.525
168. LOAD COMB 240 1.0DL+1.0FL1+0.525SLZ2+0.75LL
169. 1 1.0 2 1.0 4 0.75 8 0.525
170. LOAD COMB 241 1.0DL+1.0FL2+0.525SLX1+0.75LL
171. 1 1.0 3 1.0 4 0.75 5 0.525
172. LOAD COMB 242 1.0DL+1.0FL2+0.525SLX2+0.75LL
173. 1 1.0 3 1.0 4 0.75 6 0.525
174. LOAD COMB 243 1.0DL+1.0FL2+0.525SLZ1+0.75LL
175. 1 1.0 3 1.0 4 0.75 7 0.525
176. LOAD COMB 244 1.0DL+1.0FL2+0.525SLZ2+0.75LL
177. 1 1.0 3 1.0 4 0.75 8 0.525
178. LOAD COMB 245 0.6DL+1.0WLX1
179. 1 0.6 9 1.0
180. LOAD COMB 246 0.6DL+1.0WLX2
181. 1 0.6 10 1.0
182. LOAD COMB 247 0.6DL+1.0WLZ1
183. 1 0.6 11 1.0
184. LOAD COMB 248 0.6DL+1.0WLZ2
185. 1 0.6 12 1.0
186. LOAD COMB 249 0.6DL+0.6FL1+0.7SLX1
187. 1 0.6 2 0.6 5 0.7
188. LOAD COMB 250 0.6DL+0.6FL1+0.7SLX2
189. 1 0.6 2 0.6 6 0.7
190. LOAD COMB 251 0.6DL+0.6FL1+0.7SLZ1
191. 1 0.6 2 0.6 7 0.7
192. LOAD COMB 252 0.6DL+0.6FL1+0.7SLZ2
193. 1 0.6 2 0.6 8 0.7
194. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 4 NUMBER OF MEMBERS 2
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 2

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 2/ 2/ 6 DOF
TOTAL PRIMARY LOAD CASES = 14, TOTAL DEGREES OF FREEDOM = 12
TOTAL LOAD COMBINATION CASES = 52 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 112122.3 MB

195. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 9:37:20 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                     *
*************************************************************
```

ATTACHMENT 5 STAAD INPUT & OUTPUT FILE FOR F12B DESIGN

PROPRIETARY & CONFIDENTIAL

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  7, 2020                *
*           Time=    11:45: 1                    *
*                                                *
*      USER ID: LT Construction                  *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F12B DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 26-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 7 0 0 16.59; 8 1.705 0 16.59; 213 0 1.85 16.59; 214 1.705 1.85 16.59
     9. 613 -1.14679 0 16.59; 614 -1.14679 0 16.315; 615 -0.941429 0 16.59
    10. 616 -0.941429 0 16.315; 617 -0.706071 0 16.59; 618 -0.706071 0 16.315
    11. 619 -0.470714 0 16.59; 620 -0.470714 0 16.315; 621 -0.235357 0 16.59
    12. 622 -0.235357 0 16.315; 623 0 0 16.315; 626 -1.14679 0 16.04
    13. 627 -0.941429 0 16.04; 628 -0.706071 0 16.04; 629 -0.470714 0 16.04
    14. 630 -0.235357 0 16.04; 631 0 0 16.04; 634 -1.14679 0 15.765
    15. 635 -0.941429 0 15.765; 636 -0.706071 0 15.765; 637 -0.470714 0 15.765
    16. 638 -0.235357 0 15.765; 639 0 0 15.765; 642 -1.14679 0 15.34
    17. 643 -0.941429 0 15.34; 644 -0.706071 0 15.34; 645 -0.470714 0 15.34
    18. 646 -0.235357 0 15.34; 647 0 0 15.34; 648 0.243571 0 16.59
    19. 649 0.243571 0 16.315; 650 0.487143 0 16.59; 651 0.487143 0 16.315
    20. 652 0.730714 0 16.59; 653 0.730714 0 16.315; 654 0.974286 0 16.59
    21. 655 0.974286 0 16.315; 656 1.21786 0 16.59; 657 1.21786 0 16.315
    22. 658 1.46143 0 16.59; 659 1.46143 0 16.315; 660 1.705 0 16.315
    23. 661 0.243571 0 16.04; 662 0.487143 0 16.04; 663 0.730714 0 16.04
    24. 664 0.974286 0 16.04; 665 1.21786 0 16.04; 666 1.46143 0 16.04
    25. 667 1.705 0 16.04; 668 0.243571 0 15.765; 669 0.487143 0 15.765
    26. 670 0.730714 0 15.765; 671 0.974286 0 15.765; 672 1.21786 0 15.765
    27. 673 1.46143 0 15.765; 674 1.705 0 15.765; 675 0.243571 0 15.34
    28. 676 0.487143 0 15.34; 677 0.730714 0 15.34; 678 0.974286 0 15.34
    29. 679 1.21786 0 15.34; 680 1.46143 0 15.34; 681 1.705 0 15.34
    30. 682 1.94036 0 16.59; 683 1.94036 0 16.315; 684 2.17571 0 16.59
    31. 685 2.17571 0 16.315; 686 2.41107 0 16.59; 687 2.41107 0 16.315
    32. 688 2.64643 0 16.59; 689 2.64643 0 16.315; 690 2.85279 0 16.59
    33. 691 2.85279 0 16.315; 696 1.94036 0 16.04; 697 2.17571 0 16.04
    34. 698 2.41107 0 16.04; 699 2.64643 0 16.04; 700 2.85279 0 16.04
    35. 703 1.94036 0 15.765; 704 2.17571 0 15.765; 705 2.41107 0 15.765
    36. 706 2.64643 0 15.765; 707 2.85279 0 15.765; 710 1.94036 0 15.34
    37. 711 2.17571 0 15.34; 712 2.41107 0 15.34; 713 2.64643 0 15.34
    38. 714 2.85279 0 15.34; 721 -1.14679 0 17.84; 722 -1.14679 0 17.415
```

```
39. 723 -0.941429 0 17.84; 724 -0.941429 0 17.415; 725 -0.706071 0 17.84
40. 726 -0.706071 0 17.415; 727 -0.470714 0 17.84; 728 -0.470714 0 17.415
41. 729 -0.235357 0 17.84; 730 -0.235357 0 17.415; 731 0 0 17.84; 732 0 0 17.415
42. 735 -1.14679 0 17.14; 736 -0.941429 0 17.14; 737 -0.706071 0 17.14
43. 738 -0.470714 0 17.14; 739 -0.235357 0 17.14; 740 0 0 17.14
44. 743 -1.14679 0 16.865; 744 -0.941429 0 16.865; 745 -0.706071 0 16.865
45. 746 -0.470714 0 16.865; 747 -0.235357 0 16.865; 748 0 0 16.865
46. 749 0.243571 0 17.84; 750 0.243571 0 17.415; 751 0.487143 0 17.84
47. 752 0.487143 0 17.415; 753 0.730714 0 17.84; 754 0.730714 0 17.415
48. 755 0.974286 0 17.84; 756 0.974286 0 17.415; 757 1.21786 0 17.84
49. 758 1.21786 0 17.415; 759 1.46143 0 17.84; 760 1.46143 0 17.415
50. 761 1.705 0 17.84; 762 1.705 0 17.415; 763 0.243571 0 17.14
51. 764 0.487143 0 17.14; 765 0.730714 0 17.14; 766 0.974286 0 17.14
52. 767 1.21786 0 17.14; 768 1.46143 0 17.14; 769 1.705 0 17.14
53. 770 0.243571 0 16.865; 771 0.487143 0 16.865; 772 0.730714 0 16.865
54. 773 0.974286 0 16.865; 774 1.21786 0 16.865; 775 1.46143 0 16.865
55. 776 1.705 0 16.865; 777 1.94036 0 17.84; 778 1.94036 0 17.415
56. 779 2.17571 0 17.84; 780 2.17571 0 17.415; 781 2.41107 0 17.84
57. 782 2.41107 0 17.415; 783 2.64643 0 17.84; 784 2.64643 0 17.415
58. 785 2.85279 0 17.84; 786 2.85279 0 17.415; 791 1.94036 0 17.14
59. 792 2.17571 0 17.14; 793 2.41107 0 17.14; 794 2.64643 0 17.14
60. 795 2.85279 0 17.14; 798 1.94036 0 16.865; 799 2.17571 0 16.865
61. 800 2.41107 0 16.865; 801 2.64643 0 16.865; 802 2.85279 0 16.865
62. 1007 0 0.25 16.59; 1008 1.705 0.25 16.59; 1009 -0.941429 0 15.5525
63. 1010 -1.14679 0 15.5525; 1011 -0.706071 0 15.5525; 1012 -0.470714 0 15.5525
64. 1013 -0.235357 0 15.5525; 1014 0 0 15.5525; 1015 0.243571 0 15.5525
65. 1016 0.487143 0 15.5525; 1017 0.730714 0 15.5525; 1018 0.974286 0 15.5525
66. 1019 1.21786 0 15.5525; 1020 1.46143 0 15.5525; 1021 1.705 0 15.5525
67. 1022 1.94036 0 15.5525; 1023 2.17571 0 15.5525; 1024 2.41107 0 15.5525
68. 1025 2.64643 0 15.5525; 1026 2.85279 0 15.5525; 1027 -0.941429 0 17.6275
69. 1028 -1.14679 0 17.6275; 1029 -0.706071 0 17.6275; 1030 -0.470714 0 17.6275
70. 1031 -0.235357 0 17.6275; 1032 0 0 17.6275; 1033 0.243571 0 17.6275
71. 1034 0.487143 0 17.6275; 1035 0.730714 0 17.6275; 1036 0.974286 0 17.6275
72. 1037 1.21786 0 17.6275; 1038 1.46143 0 17.6275; 1039 1.705 0 17.6275
73. 1040 1.94036 0 17.6275; 1041 2.17571 0 17.6275; 1042 2.41107 0 17.6275
74. 1043 2.64643 0 17.6275; 1044 2.85279 0 17.6275
75. MEMBER INCIDENCES
76. 281 7 1007; 282 8 1008; 993 1007 213; 994 1008 214
77. ELEMENT INCIDENCES SHELL
78. 635 613 615 616 614; 636 615 617 618 616; 637 617 619 620 618
79. 638 619 621 622 620; 639 621 7 623 622; 642 614 616 627 626
80. 643 616 618 628 627; 644 618 620 629 628; 645 620 622 630 629
81. 646 622 623 631 630; 649 626 627 635 634; 650 627 628 636 635
82. 651 628 629 637 636; 652 629 630 638 637; 653 630 631 639 638
83. 662 7 648 649 623; 664 648 650 651 649; 666 650 652 653 651
84. 668 652 654 655 653; 670 654 656 657 655; 672 656 658 659 657
85. 673 658 8 660 659; 674 623 649 661 631; 675 649 651 662 661
86. 676 651 653 663 662; 677 653 655 664 663; 678 655 657 665 664
87. 679 657 659 666 665; 680 659 660 667 666; 681 631 661 668 639
88. 682 661 662 669 668; 683 662 663 670 669; 684 663 664 671 670
89. 685 664 665 672 671; 686 665 666 673 672; 687 666 667 674 673
90. 695 8 682 683 660; 696 682 684 685 683; 697 684 686 687 685
91. 698 686 688 689 687; 699 688 690 691 689; 702 660 683 696 667
92. 703 683 685 697 696; 704 685 687 698 697; 705 687 689 699 698
93. 706 689 691 700 699; 709 667 696 703 674; 710 696 697 704 703
94. 711 697 698 705 704; 712 698 699 706 705; 713 699 700 707 706
```

```
 95. 732 722 724 736 735; 733 724 726 737 736; 734 726 728 738 737
 96. 735 728 730 739 738; 736 730 732 740 739; 739 735 736 744 743
 97. 740 736 737 745 744; 741 737 738 746 745; 742 738 739 747 746
 98. 743 739 740 748 747; 746 743 744 615 613; 747 744 745 617 615
 99. 748 745 746 619 617; 749 746 747 621 619; 750 747 748 7 621
100. 758 732 750 763 740; 759 750 752 764 763; 760 752 754 765 764
101. 761 754 756 766 765; 762 756 758 767 766; 763 758 760 768 767
102. 764 760 762 769 768; 765 740 763 770 748; 766 763 764 771 770
103. 767 764 765 772 771; 768 765 766 773 772; 769 766 767 774 773
104. 770 767 768 775 774; 771 768 769 776 775; 772 748 770 648 7
105. 773 770 771 650 648; 774 771 772 652 650; 775 772 773 654 652
106. 776 773 774 656 654; 777 774 775 658 656; 778 775 776 8 658
107. 786 762 778 791 769; 787 778 780 792 791; 788 780 782 793 792
108. 789 782 784 794 793; 790 784 786 795 794; 793 769 791 798 776
109. 794 791 792 799 798; 795 792 793 800 799; 796 793 794 801 800
110. 797 794 795 802 801; 800 776 798 682 8; 801 798 799 684 682
111. 802 799 800 686 684; 803 800 801 688 686; 804 801 802 690 688
112. 995 634 635 1009 1010; 996 1010 1009 643 642; 997 635 636 1011 1009
113. 998 1009 1011 644 643; 999 636 637 1012 1011; 1000 1011 1012 645 644
114. 1001 637 638 1013 1012; 1002 1012 1013 646 645; 1003 638 639 1014 1013
115. 1004 1013 1014 647 646; 1005 639 668 1015 1014; 1006 1014 1015 675 647
116. 1007 668 669 1016 1015; 1008 1015 1016 676 675; 1009 669 670 1017 1016
117. 1010 1016 1017 677 676; 1011 670 671 1018 1017; 1012 1017 1018 678 677
118. 1013 671 672 1019 1018; 1014 1018 1019 679 678; 1015 672 673 1020 1019
119. 1016 1019 1020 680 679; 1017 673 674 1021 1020; 1018 1020 1021 681 680
120. 1019 674 703 1022 1021; 1020 1021 1022 710 681; 1021 703 704 1023 1022
121. 1022 1022 1023 711 710; 1023 704 705 1024 1023; 1024 1023 1024 712 711
122. 1025 705 706 1025 1024; 1026 1024 1025 713 712; 1027 706 707 1026 1025
123. 1028 1025 1026 714 713; 1029 721 723 1027 1028; 1030 1028 1027 724 722
124. 1031 723 725 1029 1027; 1032 1027 1029 726 724; 1033 725 727 1030 1029
125. 1034 1029 1030 728 726; 1035 727 729 1031 1030; 1036 1030 1031 730 728
126. 1037 729 731 1032 1031; 1038 1031 1032 732 730; 1039 731 749 1033 1032
127. 1040 1032 1033 750 732; 1041 749 751 1034 1033; 1042 1033 1034 752 750
128. 1043 751 753 1035 1034; 1044 1034 1035 754 752; 1045 753 755 1036 1035
129. 1046 1035 1036 756 754; 1047 755 757 1037 1036; 1048 1036 1037 758 756
130. 1049 757 759 1038 1037; 1050 1037 1038 760 758; 1051 759 761 1039 1038
131. 1052 1038 1039 762 760; 1053 761 777 1040 1039; 1054 1039 1040 778 762
132. 1055 777 779 1041 1040; 1056 1040 1041 780 778; 1057 779 781 1042 1041
133. 1058 1041 1042 782 780; 1059 781 783 1043 1042; 1060 1042 1043 784 782
134. 1061 783 785 1044 1043; 1062 1043 1044 786 784
135. START GROUP DEFINITION
136. FLOOR
137. ELEMENT
138. _RAFT 635 TO 639 642 TO 646 649 TO 653 662 664 666 668 670 672 TO 687 695 -
139. 696 TO 699 702 TO 706 709 TO 713 732 TO 736 739 TO 743 746 TO 750 758 TO 778 -
140. 786 TO 790 793 TO 797 800 TO 804 995 TO 1062
141. _RAFT_MOM 635 TO 638 642 TO 646 649 TO 653 664 666 668 670 672 674 TO 687 -
142. 696 TO 699 702 TO 706 709 TO 713 732 TO 736 739 TO 743 746 TO 749 -
143. 758 TO 771 773 TO 777 786 TO 790 793 TO 797 801 TO 804 995 TO 1062
144. _RAFT_SHEAR 635 636 642 643 649 650 668 677 684 698 699 705 706 712 713 732 -
145. 733 739 740 746 747 761 768 775 789 790 796 797 803 804 995 TO 1062
146. MEMBER
147. _C7 993 994
148. END GROUP DEFINITION
149. ELEMENT PROPERTY
150. 635 TO 639 642 TO 646 649 TO 653 662 664 666 668 670 672 TO 687 695 TO 699 -
```

```
151. 702 TO 706 709 TO 713 732 TO 736 739 TO 743 746 TO 750 758 TO 778 -
152. 786 TO 790 793 TO 797 800 TO 804 995 TO 1062 THICKNESS 0.4
153. DEFINE MATERIAL START
154. ISOTROPIC CONCRETE
155. E 2.17185E+007
156. POISSON 0.17
157. DENSITY 25
158. ALPHA 1E-005
159. DAMP 0.05
160. TYPE CONCRETE
161. STRENGTH FCU 27579
162. END DEFINE MATERIAL
163. MEMBER PROPERTY AMERICAN
164. 281 282 993 994 PRIS YD 0.55 ZD 0.55
165. CONSTANTS
166. MATERIAL CONCRETE ALL
167. ******************************************************
168. SUPPORTS
169. 635 TO 639 642 TO 646 649 TO 653 662 664 666 668 670 672 TO 687 695 TO 699 -
170. 702 TO 706 709 TO 713 732 TO 736 739 TO 743 746 TO 750 758 TO 778 -
171. 786 TO 790 793 TO 797 800 TO 804 995 TO 1061 -
172. 1062 PLATE MAT DIRECT Y SUBGRADE 40500 PRINT
```

JOINT	INFLUENCE AREA
7	0.065853
8	0.065853
613	0.028237
614	0.028237
615	0.060599
616	0.060599
617	0.064723
618	0.064723
619	0.064723
620	0.064723
621	0.064723
622	0.064723
623	0.065852
626	0.028237
627	0.060599
628	0.064723
629	0.064723
630	0.064723
631	0.065852
634	0.025028
635	0.053713
636	0.057368
637	0.057368
638	0.057368
639	0.058369
642	0.010910
643	0.023413
644	0.025007
645	0.025007
646	0.025007

647	0.025443
648	0.066982
649	0.066982
650	0.066982
651	0.066982
652	0.066982
653	0.066982
654	0.066983
655	0.066982
656	0.066982
657	0.066982
658	0.066982
659	0.066982
660	0.065853
661	0.066982
662	0.066982
663	0.066982
664	0.066982
665	0.066982
666	0.066982
667	0.065853
668	0.059371
669	0.059371
670	0.059371
671	0.059371
672	0.059371
673	0.059370
674	0.058370
675	0.025879
676	0.025879
677	0.025879
678	0.025880
679	0.025880
680	0.025879
681	0.025443
682	0.064723
683	0.064723
684	0.064723
685	0.064723
686	0.064724
687	0.064724
688	0.060737
689	0.060736
690	0.028375
691	0.028374
696	0.064723
697	0.064723
698	0.064724
699	0.060736
700	0.028374
703	0.057368
704	0.057368
705	0.057369
706	0.053835
707	0.025150
710	0.025006

711	0.025006
712	0.025007
713	0.023466
714	0.010963
721	0.010910
722	0.025028
723	0.023413
724	0.053713
725	0.025007
726	0.057368
727	0.025007
728	0.057368
729	0.025007
730	0.057368
731	0.025443
732	0.058369
735	0.028237
736	0.060599
737	0.064723
738	0.064723
739	0.064723
740	0.065853
743	0.028237
744	0.060599
745	0.064723
746	0.064723
747	0.064723
748	0.065853
749	0.025879
750	0.059371
751	0.025879
752	0.059371
753	0.025879
754	0.059371
755	0.025879
756	0.059371
757	0.025879
758	0.059371
759	0.025879
760	0.059370
761	0.025443
762	0.058370
763	0.066982
764	0.066982
765	0.066982
766	0.066983
767	0.066982
768	0.066982
769	0.065853
770	0.066982
771	0.066982
772	0.066982
773	0.066983
774	0.066982
775	0.066982
776	0.065853

777	0.025006
778	0.057368
779	0.025006
780	0.057368
781	0.025007
782	0.057369
783	0.023466
784	0.053835
785	0.010963
786	0.025150
791	0.064723
792	0.064723
793	0.064724
794	0.060737
795	0.028375
798	0.064723
799	0.064723
800	0.064724
801	0.060737
802	0.028375
1009	0.046826
1010	0.021820
1011	0.050014
1012	0.050013
1013	0.050013
1014	0.050886
1015	0.051759
1016	0.051759
1017	0.051759
1018	0.051759
1019	0.051759
1020	0.051759
1021	0.050886
1022	0.050013
1023	0.050013
1024	0.050014
1025	0.046933
1026	0.021926
1027	0.046826
1028	0.021820
1029	0.050013
1030	0.050013
1031	0.050013
1032	0.050886
1033	0.051759
1034	0.051759
1035	0.051759
1036	0.051759
1037	0.051759
1038	0.051758
1039	0.050886
1040	0.050013
1041	0.050013
1042	0.050014
1043	0.046933
1044	0.021926

```
    TOTAL           9.998947
173. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
174. JOINT LOAD
175. 213 FX 0.8 FY -81.1 FZ 0
176. 214 FX 0.9 FY -75.7 FZ 0
177. ELEMENT LOAD
178. 635 TO 639 642 TO 646 649 TO 653 662 664 666 668 670 672 TO 687 695 TO 699 -
179. 702 TO 706 709 TO 713 732 TO 736 739 TO 743 746 TO 750 758 TO 778 -
180. 786 TO 790 793 TO 797 800 TO 804 995 TO 1062 PR -19.8
181. SELFWEIGHT Y -1
182. ***
183. LOAD 2 LOADTYPE NONE  TITLE NORMAL OPERATING FLUID LOAD ( FL1)
184. JOINT LOAD
185. 213 FX 0.8 FY -1.6 FZ 0
186. 214 FX 0 FY -7.2 FZ 0
187. LOAD 3 LOADTYPE NONE  TITLE MAX. FLUID LOAD ( FL2)
188. JOINT LOAD
189. 213 FX 1 FY -1.4 FZ 0
190. 214 FX 0 FY -8.4 FZ 0
191. LOAD 4 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
192. JOINT LOAD
193. 213 FX 3.4 FY -30.7 FZ 0
194. 214 FX 0.4 FY -31.7 FZ 0
195. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD + X ( SLX1 )
196. JOINT LOAD
197. 213 FX 9.4 FY 48.1 FZ 0
198. 214 FX -0.4 FY -56.3 FZ 0
199. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD - X ( SLX2 )
200. JOINT LOAD
201. 213 FX -9.4 FY -48.1 FZ 0
202. 214 FX 0.4 FY 56.3 FZ 0
203. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD +Z ( SLZ1)
204. JOINT LOAD
205. 213 FX 0.9 FY 2.7 FZ -0.1
206. 214 FX 0 FY -7.9 FZ 0.2
207. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD -Z ( SLZ2)
208. JOINT LOAD
209. 213 FX -0.9 FY -2.7 FZ 0.1
210. 214 FX 0 FY 7.9 FZ -0.2
211. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
212. JOINT LOAD
213. 213 FX 35.2 FY 113 FZ 0
214. 214 FX 0.7 FY -125 FZ 0
215. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
216. JOINT LOAD
217. 213 FX -38.4 FY -116.8 FZ 0
218. 214 FX -1 FY 128.6 FZ 0
219. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
220. JOINT LOAD
221. 213 FX 3.3 FY 17.5 FZ 0.9
222. 214 FX -0.1 FY -23.7 FZ 2.8
223. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
224. JOINT LOAD
225. 213 FX -3.3 FY -17.5 FZ -0.9
226. 214 FX 0.1 FY 23.7 FZ -2.8
227. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD ( TL )
```

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228. JOINT LOAD
229. 213 FX 28.5 FY 164.8 FZ 0.2
230. 214 FX 0 FY -182.3 FZ 0
231. LOAD 14 LOADTYPE NONE  TITLE LATERAL PIPE LOAD ( PL )
232. JOINT LOAD
233. 213 FX -14.7 FY -113.9 FZ -0.6
234. 214 FX 1 FY 39.5 FZ 0
235. ******************************************************
236. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
237. LOAD COMB 101 1.4DL+1.4FL1
238. 1 1.4 2 1.4
239. LOAD COMB 102 1.4DL+1.4FL2
240. 1 1.4 3 1.4
241. LOAD COMB 103 1.2DL+1.2FL1+1.6LL
242. 1 1.2 2 1.2 4 1.6
243. LOAD COMB 104 1.2DL+1.2FL2+1.6LL
244. 1 1.2 3 1.2 4 1.6
245. LOAD COMB 105 1.2DL+1.2FL1+1.2TL+1.6LL
246. 1 1.2 2 1.2 4 1.6 13 1.2
247. LOAD COMB 106 1.2DL+1.2FL2+1.2TL+1.6LL
248. 1 1.2 3 1.2 4 1.6 13 1.2
249. LOAD COMB 107 1.4DL+1.4FL1+1.4PL
250. 1 1.4 2 1.4 14 1.4
251. LOAD COMB 108 1.4DL+1.4FL2+1.4PL
252. 1 1.4 3 1.4 14 1.4
253. LOAD COMB 109 1.2DL+1.2FL1+1.6LL+1.2PL
254. 1 1.2 2 1.2 4 1.6 14 1.2
255. LOAD COMB 110 1.2DL+1.2FL2+1.6LL+1.2PL
256. 1 1.2 3 1.2 4 1.6 14 1.2
257. LOAD COMB 111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
258. 1 1.2 2 1.2 4 1.6 13 1.2 14 1.2
259. LOAD COMB 112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
260. 1 1.2 3 1.2 4 1.6 13 1.2 14 1.2
261. LOAD COMB 113 1.2DL+0.8WLX1
262. 1 1.2 9 0.8
263. LOAD COMB 114 1.2DL+0.8WLX2
264. 1 1.2 10 0.8
265. LOAD COMB 115 1.2DL+0.8WLZ1
266. 1 1.2 11 0.8
267. LOAD COMB 116 1.2DL+0.8WLZ2
268. 1 1.2 12 0.8
269. LOAD COMB 117 1.2DL+1.6WLX1+0.5LL
270. 1 1.2 9 1.6 4 0.5
271. LOAD COMB 118 1.2DL+1.6WLX2+0.5LL
272. 1 1.2 10 1.6 4 0.5
273. LOAD COMB 119 1.2DL+1.6WLZ1+0.5LL
274. 1 1.2 11 1.6 4 0.5
275. LOAD COMB 120 1.2DL+1.6WLZ2+0.5LL
276. 1 1.2 12 1.6 4 0.5
277. LOAD COMB 121 1.2DL+1.0SLX1+0.5LL+0.5FL1
278. 1 1.2 5 1.0 4 0.5 2 0.5
279. LOAD COMB 122 1.2DL+1.0SLX2+0.5LL+0.5FL1
280. 1 1.2 6 1.0 4 0.5 2 0.5
281. LOAD COMB 123 1.2DL+1.0SLZ1+0.5LL+0.5FL1
282. 1 1.2 7 1.0 4 0.5 2 0.5
283. LOAD COMB 124 1.2DL+1.0SLZ2+0.5LL+0.5FL1
```

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284. 1 1.2 8 1.0 4 0.5 2 0.5
285. LOAD COMB 125 0.9DL+1.6WLX1
286. 1 0.9 9 1.6
287. LOAD COMB 126 0.9DL+1.6WLX2
288. 1 0.9 10 1.6
289. LOAD COMB 127 0.9DL+1.6WLZ1
290. 1 0.9 11 1.6
291. LOAD COMB 128 0.9DL+1.6WLZ2
292. 1 0.9 12 1.6
293. LOAD COMB 129 0.9DL+1.0SLX1
294. 1 0.9 5 1.0
295. LOAD COMB 130 0.9DL+1.0SLX2
296. 1 0.9 6 1.0
297. LOAD COMB 131 0.9DL+1.0SLZ1
298. 1 0.9 7 1.0
299. LOAD COMB 132 0.9DL+1.0SLZ2
300. 1 0.9 8 1.0
301. ********************************************************************************
302. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
303. LOAD COMB 201 1.0DL+1.0FL1
304. 1 1.0 2 1.0
305. LOAD COMB 202 1.0DL+1.0FL2
306. 1 1.0 3 1.0
307. LOAD COMB 203 1.0DL+1.0FL1+1.0LL+1.0TL
308. 1 1.0 2 1.0 4 1.0 13 1.0
309. LOAD COMB 204 1.0DL+1.0FL2+1.0LL+1.0TL
310. 1 1.0 3 1.0 4 1.0 13 1.0
311. LOAD COMB 205 1.0DL+1.0FL1+0.75LL+0.75TL
312. 1 1.0 2 1.0 4 0.75 13 0.75
313. LOAD COMB 206 1.0DL+1.0FL2+0.75LL+0.75TL
314. 1 1.0 3 1.0 4 0.75 13 0.75
315. LOAD COMB 207 1.0DL+1.0FL1+1.0PL
316. 1 1.0 2 1.0 14 1.0
317. LOAD COMB 208 1.0DL+1.0FL2+1.0PL
318. 1 1.0 3 1.0 14 1.0
319. LOAD COMB 209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL
320. 1 1.0 2 1.0 4 1.0 13 1.0 14 1.0
321. LOAD COMB 210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL
322. 1 1.0 3 1.0 4 1.0 13 1.0 14 1.0
323. LOAD COMB 211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL
324. 1 1.0 2 1.0 4 0.75 13 0.75 14 1.0
325. LOAD COMB 212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL
326. 1 1.0 3 1.0 4 0.75 13 0.75 14 1.0
327. LOAD COMB 213 1.0DL+1.0FL1+1.0WLX1
328. 1 1.0 2 1.0 9 1.0
329. LOAD COMB 214 1.0DL+1.0FL1+1.0WLX2
330. 1 1.0 2 1.0 10 1.0
331. LOAD COMB 215 1.0DL+1.0FL1+1.0WLZ1
332. 1 1.0 2 1.0 11 1.0
333. LOAD COMB 216 1.0DL+1.0FL1+1.0WLZ2
334. 1 1.0 2 1.0 12 1.0
335. LOAD COMB 217 1.0DL+1.0FL2+1.0WLX1
336. 1 1.0 3 1.0 9 1.0
337. LOAD COMB 218 1.0DL+1.0FL2+1.0WLX2
338. 1 1.0 3 1.0 10 1.0
339. LOAD COMB 219 1.0DL+1.0FL2+1.0WLZ1
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340. 1 1.0 3 1.0 11 1.0
341. LOAD COMB 220 1.0DL+1.0FL2+1.0WLZ2
342. 1 1.0 3 1.0 12 1.0
343. LOAD COMB 221 1.0DL+1.0FL1+0.7SLX1
344. 1 1.0 2 1.0 5 0.7
345. LOAD COMB 222 1.0DL+1.0FL1+0.7SLX2
346. 1 1.0 2 1.0 6 0.7
347. LOAD COMB 223 1.0DL+1.0FL1+0.7SLZ1
348. 1 1.0 2 1.0 7 0.7
349. LOAD COMB 224 1.0DL+1.0FL1+0.7SLZ2
350. 1 1.0 2 1.0 8 0.7
351. LOAD COMB 225 1.0DL+1.0FL2+0.7SLX1
352. 1 1.0 3 1.0 5 0.7
353. LOAD COMB 226 1.0DL+1.0FL2+0.7SLX2
354. 1 1.0 3 1.0 6 0.7
355. LOAD COMB 227 1.0DL+1.0FL2+0.7SLZ1
356. 1 1.0 3 1.0 7 0.7
357. LOAD COMB 228 1.0DL+1.0FL2+0.7SLZ2
358. 1 1.0 3 1.0 8 0.7
359. LOAD COMB 229 1.0DL+1.0FL1+0.75WLX1+0.75LL
360. 1 1.0 2 1.0 4 0.75 9 0.75
361. LOAD COMB 230 1.0DL+1.0FL1+0.75WLX2+0.75LL
362. 1 1.0 2 1.0 4 0.75 10 0.75
363. LOAD COMB 231 1.0DL+1.0FL1+0.75WLZ1+0.75LL
364. 1 1.0 2 1.0 4 0.75 11 0.75
365. LOAD COMB 232 1.0DL+1.0FL1+0.75WLZ2+0.75LL
366. 1 1.0 2 1.0 4 0.75 12 0.75
367. LOAD COMB 233 1.0DL+1.0FL2+0.75WLX1+0.75LL
368. 1 1.0 3 1.0 4 0.75 9 0.75
369. LOAD COMB 234 1.0DL+1.0FL2+0.75WLX2+0.75LL
370. 1 1.0 3 1.0 4 0.75 10 0.75
371. LOAD COMB 235 1.0DL+1.0FL2+0.75WLZ1+0.75LL
372. 1 1.0 3 1.0 4 0.75 11 0.75
373. LOAD COMB 236 1.0DL+1.0FL2+0.75WLZ2+0.75LL
374. 1 1.0 3 1.0 4 0.75 12 0.75
375. LOAD COMB 237 1.0DL+1.0FL1+0.525SLX1+0.75LL
376. 1 1.0 2 1.0 4 0.75 5 0.525
377. LOAD COMB 238 1.0DL+1.0FL1+0.525SLX2+0.75LL
378. 1 1.0 2 1.0 4 0.75 6 0.525
379. LOAD COMB 239 1.0DL+1.0FL1+0.525SLZ1+0.75LL
380. 1 1.0 2 1.0 4 0.75 7 0.525
381. LOAD COMB 240 1.0DL+1.0FL1+0.525SLZ2+0.75LL
382. 1 1.0 2 1.0 4 0.75 8 0.525
383. LOAD COMB 241 1.0DL+1.0FL2+0.525SLX1+0.75LL
384. 1 1.0 3 1.0 4 0.75 5 0.525
385. LOAD COMB 242 1.0DL+1.0FL2+0.525SLX2+0.75LL
386. 1 1.0 3 1.0 4 0.75 6 0.525
387. LOAD COMB 243 1.0DL+1.0FL2+0.525SLZ1+0.75LL
388. 1 1.0 3 1.0 4 0.75 7 0.525
389. LOAD COMB 244 1.0DL+1.0FL2+0.525SLZ2+0.75LL
390. 1 1.0 3 1.0 4 0.75 8 0.525
391. LOAD COMB 245 0.6DL+1.0WLX1
392. 1 0.6 9 1.0
393. LOAD COMB 246 0.6DL+1.0WLX2
394. 1 0.6 10 1.0
395. LOAD COMB 247 0.6DL+1.0WLZ1
```

```
396. 1 0.6 11 1.0
397. LOAD COMB 248 0.6DL+1.0WLZ2
398. 1 0.6 12 1.0
399. LOAD COMB 249 0.6DL+0.6FL1+0.7SLX1
400. 1 0.6 2 0.6 5 0.7
401. LOAD COMB 250 0.6DL+0.6FL1+0.7SLX2
402. 1 0.6 2 0.6 6 0.7
403. LOAD COMB 251 0.6DL+0.6FL1+0.7SLZ1
404. 1 0.6 2 0.6 7 0.7
405. LOAD COMB 252 0.6DL+0.6FL1+0.7SLZ2
406. 1 0.6 2 0.6 8 0.7
407. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

---------- ---------- ----------

NUMBER OF JOINTS	**202**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**170**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**198**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 164/ 20/ 63 DOF
TOTAL PRIMARY LOAD CASES = 14, TOTAL DEGREES OF FREEDOM = 618
TOTAL LOAD COMBINATION CASES = 84 SO FAR.
SIZE OF STIFFNESS MATRIX = 39 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.4/ 112106.6 MB

```
408. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 11:45:16 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 6 STAAD INPUT & OUTPUT FILE FOR F12B STABILITY

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    APR  7, 2020                   *
*           Time=     9:43:57                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F12B STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 26-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 213 0 1.85 16.59; 214 1.705 1.85 16.59; 1007 0 0.25 16.59
     9. 1008 1.705 0.25 16.59
    10. MEMBER INCIDENCES
    11. 993 1007 213; 994 1008 214
    12. DEFINE MATERIAL START
    13. ISOTROPIC CONCRETE
    14. E 2.17185E+007
    15. POISSON 0.17
    16. DENSITY 25
    17. ALPHA 1E-005
    18. DAMP 0.05
    19. TYPE CONCRETE
    20. STRENGTH FCU 27579
    21. END DEFINE MATERIAL
    22. MEMBER PROPERTY
    23. 993 994 PRIS YD 0.55 ZD 0.55
    24. CONSTANTS
    25. MATERIAL CONCRETE ALL
    26. ************************************************************
    27. SUPPORTS
    28. 1007 1008 FIXED
    29. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
              MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    30. JOINT LOAD
    31. 213 FX 0.8 FY -81.1 FZ 0
    32. 214 FX 0.9 FY -75.7 FZ 0
    33. SELFWEIGHT Y -1
    34. ***
    35. LOAD 2 LOADTYPE NONE  TITLE NORMAL OPERATING FLUID LOAD ( FL1)
    36. JOINT LOAD
```

```
37. 213 FX 0.8 FY -1.6 FZ 0
38. 214 FX 0 FY -7.2 FZ 0
39. LOAD 3 LOADTYPE NONE  TITLE MAX. FLUID LOAD ( FL2)
40. JOINT LOAD
41. 213 FX 1 FY -1.4 FZ 0
42. 214 FX 0 FY -8.4 FZ 0
43. LOAD 4 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
44. JOINT LOAD
45. 213 FX 3.4 FY -30.7 FZ 0
46. 214 FX 0.4 FY -31.7 FZ 0
47. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD + X ( SLX1 )
48. JOINT LOAD
49. 213 FX 9.4 FY 48.1 FZ 0
50. 214 FX -0.4 FY -56.3 FZ 0
51. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD - X ( SLX2 )
52. JOINT LOAD
53. 213 FX -9.4 FY -48.1 FZ 0
54. 214 FX 0.4 FY 56.3 FZ 0
55. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD +Z ( SLZ1)
56. JOINT LOAD
57. 213 FX 0.9 FY 2.7 FZ -0.1
58. 214 FX 0 FY -7.9 FZ 0.2
59. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD -Z ( SLZ2)
60. JOINT LOAD
61. 213 FX -0.9 FY -2.7 FZ 0.1
62. 214 FX 0 FY 7.9 FZ -0.2
63. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
64. JOINT LOAD
65. 213 FX 35.2 FY 113 FZ 0
66. 214 FX 0.7 FY -125 FZ 0
67. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
68. JOINT LOAD
69. 213 FX -38.4 FY -116.8 FZ 0
70. 214 FX -1 FY 128.6 FZ 0
71. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
72. JOINT LOAD
73. 213 FX 3.3 FY 17.5 FZ 0.9
74. 214 FX -0.1 FY -23.7 FZ 2.8
75. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
76. JOINT LOAD
77. 213 FX -3.3 FY -17.5 FZ -0.9
78. 214 FX 0.1 FY 23.7 FZ -2.8
79. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD ( TL )
80. JOINT LOAD
81. 213 FX 28.5 FY 164.8 FZ 0.2
82. 214 FX 0 FY -182.3 FZ 0
83. LOAD 14 LOADTYPE NONE  TITLE LATERAL PIPE LOAD ( PL )
84. JOINT LOAD
85. 213 FX -14.7 FY -113.9 FZ -0.6
86. 214 FX 1 FY 39.5 FZ 0
87. ******************************************************
88. ********************************************************************************
89. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
90. LOAD COMB 201 1.0DL+1.0FL1
91. 1 1.0 2 1.0
92. LOAD COMB 202 1.0DL+1.0FL2
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 93. 1 1.0 3 1.0
 94. LOAD COMB 203 1.0DL+1.0FL1+1.0LL+1.0TL
 95. 1 1.0 2 1.0 4 1.0 13 1.0
 96. LOAD COMB 204 1.0DL+1.0FL2+1.0LL+1.0TL
 97. 1 1.0 3 1.0 4 1.0 13 1.0
 98. LOAD COMB 205 1.0DL+1.0FL1+0.75LL+0.75TL
 99. 1 1.0 2 1.0 4 0.75 13 0.75
100. LOAD COMB 206 1.0DL+1.0FL2+0.75LL+0.75TL
101. 1 1.0 3 1.0 4 0.75 13 0.75
102. LOAD COMB 207 1.0DL+1.0FL1+1.0PL
103. 1 1.0 2 1.0 14 1.0
104. LOAD COMB 208 1.0DL+1.0FL2+1.0PL
105. 1 1.0 3 1.0 14 1.0
106. LOAD COMB 209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL
107. 1 1.0 2 1.0 4 1.0 13 1.0 14 1.0
108. LOAD COMB 210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL
109. 1 1.0 3 1.0 4 1.0 13 1.0 14 1.0
110. LOAD COMB 211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL
111. 1 1.0 2 1.0 4 0.75 13 0.75 14 1.0
112. LOAD COMB 212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL
113. 1 1.0 3 1.0 4 0.75 13 0.75 14 1.0
114. LOAD COMB 213 1.0DL+1.0FL1+1.0WLX1
115. 1 1.0 2 1.0 9 1.0
116. LOAD COMB 214 1.0DL+1.0FL1+1.0WLX2
117. 1 1.0 2 1.0 10 1.0
118. LOAD COMB 215 1.0DL+1.0FL1+1.0WLZ1
119. 1 1.0 2 1.0 11 1.0
120. LOAD COMB 216 1.0DL+1.0FL1+1.0WLZ2
121. 1 1.0 2 1.0 12 1.0
122. LOAD COMB 217 1.0DL+1.0FL2+1.0WLX1
123. 1 1.0 3 1.0 9 1.0
124. LOAD COMB 218 1.0DL+1.0FL2+1.0WLX2
125. 1 1.0 3 1.0 10 1.0
126. LOAD COMB 219 1.0DL+1.0FL2+1.0WLZ1
127. 1 1.0 3 1.0 11 1.0
128. LOAD COMB 220 1.0DL+1.0FL2+1.0WLZ2
129. 1 1.0 3 1.0 12 1.0
130. LOAD COMB 221 1.0DL+1.0FL1+0.7SLX1
131. 1 1.0 2 1.0 5 0.7
132. LOAD COMB 222 1.0DL+1.0FL1+0.7SLX2
133. 1 1.0 2 1.0 6 0.7
134. LOAD COMB 223 1.0DL+1.0FL1+0.7SLZ1
135. 1 1.0 2 1.0 7 0.7
136. LOAD COMB 224 1.0DL+1.0FL1+0.7SLZ2
137. 1 1.0 2 1.0 8 0.7
138. LOAD COMB 225 1.0DL+1.0FL2+0.7SLX1
139. 1 1.0 3 1.0 5 0.7
140. LOAD COMB 226 1.0DL+1.0FL2+0.7SLX2
141. 1 1.0 3 1.0 6 0.7
142. LOAD COMB 227 1.0DL+1.0FL2+0.7SLZ1
143. 1 1.0 3 1.0 7 0.7
144. LOAD COMB 228 1.0DL+1.0FL2+0.7SLZ2
145. 1 1.0 3 1.0 8 0.7
146. LOAD COMB 229 1.0DL+1.0FL1+0.75WLX1+0.75LL
147. 1 1.0 2 1.0 4 0.75 9 0.75
148. LOAD COMB 230 1.0DL+1.0FL1+0.75WLX2+0.75LL
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149. 1 1.0 2 1.0 4 0.75 10 0.75
150. LOAD COMB 231 1.0DL+1.0FL1+0.75WLZ1+0.75LL
151. 1 1.0 2 1.0 4 0.75 11 0.75
152. LOAD COMB 232 1.0DL+1.0FL1+0.75WLZ2+0.75LL
153. 1 1.0 2 1.0 4 0.75 12 0.75
154. LOAD COMB 233 1.0DL+1.0FL2+0.75WLX1+0.75LL
155. 1 1.0 3 1.0 4 0.75 9 0.75
156. LOAD COMB 234 1.0DL+1.0FL2+0.75WLX2+0.75LL
157. 1 1.0 3 1.0 4 0.75 10 0.75
158. LOAD COMB 235 1.0DL+1.0FL2+0.75WLZ1+0.75LL
159. 1 1.0 3 1.0 4 0.75 11 0.75
160. LOAD COMB 236 1.0DL+1.0FL2+0.75WLZ2+0.75LL
161. 1 1.0 3 1.0 4 0.75 12 0.75
162. LOAD COMB 237 1.0DL+1.0FL1+0.525SLX1+0.75LL
163. 1 1.0 2 1.0 4 0.75 5 0.525
164. LOAD COMB 238 1.0DL+1.0FL1+0.525SLX2+0.75LL
165. 1 1.0 2 1.0 4 0.75 6 0.525
166. LOAD COMB 239 1.0DL+1.0FL1+0.525SLZ1+0.75LL
167. 1 1.0 2 1.0 4 0.75 7 0.525
168. LOAD COMB 240 1.0DL+1.0FL1+0.525SLZ2+0.75LL
169. 1 1.0 2 1.0 4 0.75 8 0.525
170. LOAD COMB 241 1.0DL+1.0FL2+0.525SLX1+0.75LL
171. 1 1.0 3 1.0 4 0.75 5 0.525
172. LOAD COMB 242 1.0DL+1.0FL2+0.525SLX2+0.75LL
173. 1 1.0 3 1.0 4 0.75 6 0.525
174. LOAD COMB 243 1.0DL+1.0FL2+0.525SLZ1+0.75LL
175. 1 1.0 3 1.0 4 0.75 7 0.525
176. LOAD COMB 244 1.0DL+1.0FL2+0.525SLZ2+0.75LL
177. 1 1.0 3 1.0 4 0.75 8 0.525
178. LOAD COMB 245 0.6DL+1.0WLX1
179. 1 0.6 9 1.0
180. LOAD COMB 246 0.6DL+1.0WLX2
181. 1 0.6 10 1.0
182. LOAD COMB 247 0.6DL+1.0WLZ1
183. 1 0.6 11 1.0
184. LOAD COMB 248 0.6DL+1.0WLZ2
185. 1 0.6 12 1.0
186. LOAD COMB 249 0.6DL+0.6FL1+0.7SLX1
187. 1 0.6 2 0.6 5 0.7
188. LOAD COMB 250 0.6DL+0.6FL1+0.7SLX2
189. 1 0.6 2 0.6 6 0.7
190. LOAD COMB 251 0.6DL+0.6FL1+0.7SLZ1
191. 1 0.6 2 0.6 7 0.7
192. LOAD COMB 252 0.6DL+0.6FL1+0.7SLZ2
193. 1 0.6 2 0.6 8 0.7
194. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 4 NUMBER OF MEMBERS 2
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 2

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 2/ 2/ 6 DOF
TOTAL PRIMARY LOAD CASES = 14, TOTAL DEGREES OF FREEDOM = 12
TOTAL LOAD COMBINATION CASES = 52 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 112121.6 MB

195. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 9:44: 4 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                     *
*************************************************************
```

ATTACHMENT 7 STAAD INPUT & OUTPUT FILE FOR F13 DESIGN

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5           *
*           Version  20.07.10.64                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    APR  7, 2020                 *
*           Time=    11:47:21                     *
*                                                *
*      USER ID: LT Construction                  *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F13 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 31-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 0 0 3.15; 3 0 0 -0.425; 4 0 0 3.575; 5 0.75 0 0; 6 0.75 0 3.15
     9. 7 0.75 0 -0.425; 8 0.75 0 3.575; 9 -0.75 0 0; 10 -0.75 0 3.15
    10. 11 -0.75 0 -0.425; 12 -0.75 0 3.575; 13 -0.5 0 0; 14 -0.5 0 -0.3
    11. 15 -0.75 0 -0.3; 16 -0.25 0 0; 17 -0.25 0 -0.3; 18 0 0 -0.3; 19 -0.5 0 -0.425
    12. 20 -0.25 0 -0.425; 21 0.25 0 0; 22 0.25 0 -0.3; 23 0.5 0 0; 24 0.5 0 -0.3
    13. 25 0.75 0 -0.3; 26 0.25 0 -0.425; 27 0.5 0 -0.425; 28 -0.5 0 3.575
    14. 29 -0.5 0 3.45; 30 -0.75 0 3.45; 31 -0.25 0 3.575; 32 -0.25 0 3.45
    15. 33 0 0 3.45; 34 -0.5 0 3.15; 35 -0.25 0 3.15; 36 0.25 0 3.575; 37 0.25 0 3.45
    16. 38 0.5 0 3.575; 39 0.5 0 3.45; 40 0.75 0 3.45; 41 0.25 0 3.15; 42 0.5 0 3.15
    17. 43 -0.5 0 2.90769; 44 -0.75 0 2.90769; 45 -0.25 0 2.90769; 46 0 0 2.90769
    18. 47 -0.5 0 2.66538; 48 -0.75 0 2.66538; 49 -0.25 0 2.66538; 50 0 0 2.66538
    19. 51 -0.5 0 2.42308; 52 -0.75 0 2.42308; 53 -0.25 0 2.42308; 54 0 0 2.42308
    20. 55 -0.5 0 2.18077; 56 -0.75 0 2.18077; 57 -0.25 0 2.18077; 58 0 0 2.18077
    21. 59 -0.5 0 1.93846; 60 -0.75 0 1.93846; 61 -0.25 0 1.93846; 62 0 0 1.93846
    22. 63 -0.5 0 1.69615; 64 -0.75 0 1.69615; 65 -0.25 0 1.69615; 66 0 0 1.69615
    23. 67 -0.5 0 1.45385; 68 -0.75 0 1.45385; 69 -0.25 0 1.45385; 70 0 0 1.45385
    24. 71 -0.5 0 1.21154; 72 -0.75 0 1.21154; 73 -0.25 0 1.21154; 74 0 0 1.21154
    25. 75 -0.5 0 0.969231; 76 -0.75 0 0.969231; 77 -0.25 0 0.969231; 78 0 0 0.969231
    26. 79 -0.5 0 0.726923; 80 -0.75 0 0.726923; 81 -0.25 0 0.726923; 82 0 0 0.726923
    27. 83 -0.5 0 0.484615; 84 -0.75 0 0.484615; 85 -0.25 0 0.484615; 86 0 0 0.484615
    28. 87 -0.5 0 0.242308; 88 -0.75 0 0.242308; 89 -0.25 0 0.242308; 90 0 0 0.242308
    29. 91 0.25 0 2.90769; 92 0.5 0 2.90769; 93 0.75 0 2.90769; 94 0.25 0 2.66538
    30. 95 0.5 0 2.66538; 96 0.75 0 2.66538; 97 0.25 0 2.42308; 98 0.5 0 2.42308
    31. 99 0.75 0 2.42308; 100 0.25 0 2.18077; 101 0.5 0 2.18077; 102 0.75 0 2.18077
    32. 103 0.25 0 1.93846; 104 0.5 0 1.93846; 105 0.75 0 1.93846; 106 0.25 0 1.69615
    33. 107 0.5 0 1.69615; 108 0.75 0 1.69615; 109 0.25 0 1.45385; 110 0.5 0 1.45385
    34. 111 0.75 0 1.45385; 112 0.25 0 1.21154; 113 0.5 0 1.21154; 114 0.75 0 1.21154
    35. 115 0.25 0 0.969231; 116 0.5 0 0.969231; 117 0.75 0 0.969231
    36. 118 0.25 0 0.726923; 119 0.5 0 0.726923; 120 0.75 0 0.726923
    37. 121 0.25 0 0.484615; 122 0.5 0 0.484615; 123 0.75 0 0.484615
    38. 124 0.25 0 0.242308; 125 0.5 0 0.242308; 126 0.75 0 0.242308; 127 0 1.2 0
```

```
39. 128 0 1.2 3.15; 129 0 0.2 0; 130 0 0.2 3.15
40. MEMBER INCIDENCES
41. 184 1 129; 185 2 130; 186 129 127; 187 130 128
42. ELEMENT INCIDENCES SHELL
43. 26 9 13 14 15; 28 13 16 17 14; 30 16 1 18 17; 32 15 14 19 11; 34 14 17 20 19
44. 35 17 18 3 20; 37 1 21 22 18; 39 21 23 24 22; 41 23 5 25 24; 43 18 22 26 3
45. 45 22 24 27 26; 46 24 25 7 27; 49 12 28 29 30; 51 28 31 32 29; 53 31 4 33 32
46. 55 30 29 34 10; 57 29 32 35 34; 58 32 33 2 35; 60 4 36 37 33; 62 36 38 39 37
47. 64 38 8 40 39; 66 33 37 41 2; 68 37 39 42 41; 69 39 40 6 42; 71 10 34 43 44
48. 72 34 35 45 43; 74 35 2 46 45; 76 44 43 47 48; 77 43 45 49 47; 79 45 46 50 49
49. 81 48 47 51 52; 82 47 49 53 51; 84 49 50 54 53; 86 52 51 55 56; 87 51 53 57 55
50. 89 53 54 58 57; 91 56 55 59 60; 92 55 57 61 59; 94 57 58 62 61; 96 60 59 63 64
51. 97 59 61 65 63; 99 61 62 66 65; 101 64 63 67 68; 102 63 65 69 67
52. 104 65 66 70 69; 106 68 67 71 72; 107 67 69 73 71; 109 69 70 74 73
53. 111 72 71 75 76; 112 71 73 77 75; 114 73 74 78 77; 116 76 75 79 80
54. 117 75 77 81 79; 119 77 78 82 81; 121 80 79 83 84; 122 79 81 85 83
55. 124 81 82 86 85; 126 84 83 87 88; 127 83 85 89 87; 129 85 86 90 89
56. 130 88 87 13 9; 131 87 89 16 13; 132 89 90 1 16; 133 2 41 91 46
57. 134 41 42 92 91; 136 42 6 93 92; 137 46 91 94 50; 138 91 92 95 94
58. 140 92 93 96 95; 141 50 94 97 54; 142 94 95 98 97; 144 95 96 99 98
59. 145 54 97 100 58; 146 97 98 101 100; 148 98 99 102 101; 149 58 100 103 62
60. 150 100 101 104 103; 152 101 102 105 104; 153 62 103 106 66
61. 154 103 104 107 106; 156 104 105 108 107; 157 66 106 109 70
62. 158 106 107 110 109; 160 107 108 111 110; 161 70 109 112 74
63. 162 109 110 113 112; 164 110 111 114 113; 165 74 112 115 78
64. 166 112 113 116 115; 168 113 114 117 116; 169 78 115 118 82
65. 170 115 116 119 118; 172 116 117 120 119; 173 82 118 121 86
66. 174 118 119 122 121; 176 119 120 123 122; 177 86 121 124 90
67. 178 121 122 125 124; 180 122 123 126 125; 181 90 124 21 1; 182 124 125 23 21
68. 183 125 126 5 23
69. START GROUP DEFINITION
70. ELEMENT
71. _RAFT 26 28 30 32 34 35 37 39 41 43 45 46 49 51 53 55 57 58 60 62 64 66 68 -
72. 69 71 72 74 76 77 79 81 82 84 86 87 89 91 92 94 96 97 99 101 102 104 106 -
73. 107 109 111 112 114 116 117 119 121 122 124 126 127 129 TO 134 136 TO 138 -
74. 140 TO 142 144 TO 146 148 TO 150 152 TO 154 156 TO 158 160 TO 162 -
75. 164 TO 166 168 TO 170 172 TO 174 176 TO 178 180 TO 183
76. _RAFT_MOM 26 28 32 34 35 39 41 43 45 46 49 51 53 55 57 60 62 64 68 69 71 72 -
77. 76 77 79 81 82 84 86 87 89 91 92 94 96 97 99 101 102 104 106 107 109 111 -
78. 112 114 116 117 119 121 122 124 126 127 129 TO 131 134 136 TO 138 -
79. 140 TO 142 144 TO 146 148 TO 150 152 TO 154 156 TO 158 160 TO 162 -
80. 164 TO 166 168 TO 170 172 TO 174 176 TO 178 180 182 183
81. _RAFT_SHEAR 26 32 41 46 49 55 64 69 71 76 81 82 84 86 87 89 91 92 94 96 97 -
82. 99 101 102 104 106 107 109 111 112 114 116 117 119 121 122 124 126 130 136 -
83. 140 TO 142 144 TO 146 148 TO 150 152 TO 154 156 TO 158 160 TO 162 -
84. 164 TO 166 168 TO 170 172 TO 174 176 180 183
85. MEMBER
86. _C9 186 187
87. END GROUP DEFINITION
88. ELEMENT PROPERTY
89. 26 28 30 32 34 35 37 39 41 43 45 46 49 51 53 55 57 58 60 62 64 66 68 69 71 -
90. 72 74 76 77 79 81 82 84 86 87 89 91 92 94 96 97 99 101 102 104 106 107 109 -
91. 111 112 114 116 117 119 121 122 124 126 127 129 TO 134 136 TO 138 -
92. 140 TO 142 144 TO 146 148 TO 150 152 TO 154 156 TO 158 160 TO 162 -
93. 164 TO 166 168 TO 170 172 TO 174 176 TO 178 180 TO 183 THICKNESS 0.4
94. DEFINE MATERIAL START
```

```
 95. ISOTROPIC CONCRETE
 96. E 2.17185E+007
 97. POISSON 0.17
 98. DENSITY 25
 99. ALPHA 1E-005
100. DAMP 0.05
101. TYPE CONCRETE
102. STRENGTH FCU 27579
103. END DEFINE MATERIAL
104. MEMBER PROPERTY AMERICAN
105. 184 TO 187 PRIS YD 0.5 ZD 0.5
106. CONSTANTS
107. MATERIAL CONCRETE ALL
108. ******************************************************
109. SUPPORTS
110. 26 28 30 32 34 35 37 39 41 43 45 46 49 51 53 55 57 58 60 62 64 66 68 69 71 -
111. 72 74 76 77 79 81 82 84 86 87 89 91 92 94 96 97 99 101 102 104 106 107 109 -
112. 111 112 114 116 117 119 121 122 124 126 127 129 TO 134 136 TO 138 -
113. 140 TO 142 144 TO 146 148 TO 150 152 TO 154 156 TO 158 160 TO 162 -
114. 164 TO 166 168 TO 170 172 TO 174 176 TO 178 180 TO 182 -
115. 183 PLATE MAT DIRECT Y SUBGRADE 40500 PRINT
```

JOINT	INFLUENCE AREA
1	0.067789
2	0.067789
3	0.015625
4	0.015625
5	0.033894
6	0.033894
7	0.007813
8	0.007813
9	0.033894
10	0.033894
11	0.007813
12	0.007813
13	0.067789
14	0.053125
15	0.026563
16	0.067789
17	0.053125
18	0.053125
19	0.015625
20	0.015625
21	0.067789
22	0.053125
23	0.067789
24	0.053125
25	0.026563
26	0.015625
27	0.015625
28	0.015625
29	0.053125
30	0.026563
31	0.015625

32	0.053125
33	0.053125
34	0.067789
35	0.067789
36	0.015625
37	0.053125
38	0.015625
39	0.053125
40	0.026563
41	0.067789
42	0.067789
43	0.060578
44	0.030289
45	0.060578
46	0.060578
47	0.060576
48	0.030288
49	0.060576
50	0.060576
51	0.060576
52	0.030288
53	0.060576
54	0.060576
55	0.060578
56	0.030289
57	0.060578
58	0.060578
59	0.060578
60	0.030289
61	0.060578
62	0.060578
63	0.060576
64	0.030288
65	0.060576
66	0.060576
67	0.060576
68	0.030288
69	0.060576
70	0.060576
71	0.060577
72	0.030289
73	0.060577
74	0.060577
75	0.060577
76	0.030289
77	0.060577
78	0.060577
79	0.060577
80	0.030289
81	0.060577
82	0.060577
83	0.060577
84	0.030288
85	0.060577
86	0.060577
87	0.060577

88	0.030288
89	0.060577
90	0.060577
91	0.060578
92	0.060578
93	0.030289
94	0.060576
95	0.060576
96	0.030288
97	0.060576
98	0.060576
99	0.030288
100	0.060578
101	0.060578
102	0.030289
103	0.060578
104	0.060578
105	0.030289
106	0.060576
107	0.060576
108	0.030288
109	0.060576
110	0.060576
111	0.030288
112	0.060577
113	0.060577
114	0.030289
115	0.060577
116	0.060577
117	0.030289
118	0.060577
119	0.060577
120	0.030289
121	0.060577
122	0.060577
123	0.030288
124	0.060577
125	0.060577
126	0.030288
TOTAL	5.999999

```
116. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
117. JOINT LOAD
118. 127 FX -1.9 FY -21.9 FZ -5.5
119. 128 FX -2.6 FY -37.5 FZ 5.4
120. SELFWEIGHT Y -1
121. ***
122. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
123. JOINT LOAD
124. 127 FX -2.7 FY -26.1 FZ -15.4
125. 128 FX -5.1 FY -79 FZ 15.1
126. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
127. JOINT LOAD
128. 127 FX 6.8 FY 12.1 FZ 0.1
129. 128 FX 6.8 FY 12.1 FZ -0.1
130. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
131. JOINT LOAD
```

```
132. 127 FX -6.8 FY -12.1 FZ -0.1
133. 128 FX -6.8 FY -12.1 FZ 0.1
134. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
135. JOINT LOAD
136. 127 FX 0.7 FY 18.2 FZ 5.3
137. 128 FX -1.5 FY -18.2 FZ 5.3
138. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
139. JOINT LOAD
140. 127 FX -0.7 FY -18.2 FZ -5.3
141. 128 FX 1.5 FY 18.2 FZ -5.3
142. LOAD 7 LOADTYPE SOIL  TITLE SOIL LOAD (HL)
143. ELEMENT LOAD
144. 26 28 30 32 34 35 37 39 41 43 45 46 49 51 53 55 57 58 60 62 64 66 68 69 71 -
145. 72 74 76 77 79 81 82 84 86 87 89 91 92 94 96 97 99 101 102 104 106 107 109 -
146. 111 112 114 116 117 119 121 122 124 126 127 129 TO 134 136 TO 138 -
147. 140 TO 142 144 TO 146 148 TO 150 152 TO 154 156 TO 158 160 TO 162 -
148. 164 TO 166 168 TO 170 172 TO 174 176 TO 178 180 TO 183 PR -8.1
149. ******************************************************
150. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
151. LOAD COMB 101 1.4DL
152. 1 1.4
153. LOAD COMB 102 1.4DL+1.7LL
154. 1 1.4 2 1.7
155. LOAD COMB 103 1.2DL+1.0LL+1.6WLX1
156. 1 1.2 2 1.0 3 1.6
157. LOAD COMB 104 1.2DL+1.0LL+1.6WLX2
158. 1 1.2 2 1.0 4 1.6
159. LOAD COMB 105 1.2DL+1.0LL+1.6WLZ1
160. 1 1.2 2 1.0 5 1.6
161. LOAD COMB 106 1.2DL+1.0LL+1.6WLZ2
162. 1 1.2 2 1.0 6 1.6
163. LOAD COMB 107 0.9DL+1.6HL+1.6WLX1
164. 1 0.9 7 1.6 3 1.6
165. LOAD COMB 108 0.9DL+1.6HL+1.6WLX2
166. 1 0.9 7 1.6 4 1.6
167. LOAD COMB 109 0.9DL+1.6HL+1.6WLZ1
168. 1 0.9 7 1.6 5 1.6
169. LOAD COMB 110 0.9DL+1.6HL+1.6WLZ2
170. 1 0.9 7 1.6 6 1.6
171. ***************
172. ********************************************************************************
173. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
174. ************************************************************
175. LOAD COMB 201 1.0DL+1.0HL+1.0LL
176. 1 1.0 7 1.0 2 1.0
177. LOAD COMB 202 1.0DL+1.0HL+0.75LL
178. 1 1.0 7 1.0 2 0.75
179. LOAD COMB 203 1.0DL+1.0HL+1.0WLX1
180. 1 1.0 7 1.0 3 1.0
181. LOAD COMB 204 1.0DL+1.0HL+1.0WLX2
182. 1 1.0 7 1.0 4 1.0
183. LOAD COMB 205 1.0DL+1.0HL+1.0WLZ1
184. 1 1.0 7 1.0 5 1.0
185. LOAD COMB 206 1.0DL+1.0HL+1.0WLZ2
186. 1 1.0 7 1.0 6 1.0
187. LOAD COMB 207 1.0DL+1.0HL+0.75LL+0.75WLX1
```

```
188. 1 1.0 7 1.0 2 0.75 3 0.75
189. LOAD COMB 208 1.0DL+1.0HL+0.75LL+0.75WLX2
190. 1 1.0 7 1.0 2 0.75 4 0.75
191. LOAD COMB 209 1.0DL+1.0HL+0.75LL+0.75WLZ1
192. 1 1.0 7 1.0 2 0.75 5 0.75
193. LOAD COMB 210 1.0DL+1.0HL+0.75LL+0.75WLZ2
194. 1 1.0 7 1.0 2 0.75 6 0.75
195. LOAD COMB 211 0.6DL+1.0HL+1.0WLX1
196. 1 0.6 7 1.0 3 1.0
197. LOAD COMB 212 0.6DL+1.0HL+1.0WLX2
198. 1 0.6 7 1.0 4 1.0
199. LOAD COMB 213 0.6DL+1.0HL+1.0WLZ1
200. 1 0.6 7 1.0 5 1.0
201. LOAD COMB 214 0.6DL+1.0HL+1.0WLX1
202. 1 0.6 7 1.0 6 1.0
203. ********
204. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**130**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**102**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**126**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 128/ 13/ 45 DOF
TOTAL PRIMARY LOAD CASES = 7, TOTAL DEGREES OF FREEDOM = 402
TOTAL LOAD COMBINATION CASES = 24 SO FAR.
SIZE OF STIFFNESS MATRIX = 19 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.8/ 112099.0 MB

```
205. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 11:47:29 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 8 STAAD INPUT & OUTPUT FILE FOR F13 STABILITY

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5           *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  7, 2020                *
*           Time=     9:48: 1                    *
*                                                *
*      USER ID: LT Construction                  *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F13 STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 31-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 0 0 3.15; 127 0 1.2 0; 128 0 1.2 3.15
     9. MEMBER INCIDENCES
    10. 184 1 127; 185 2 128
    11. DEFINE MATERIAL START
    12. ISOTROPIC CONCRETE
    13. E 2.17185E+007
    14. POISSON 0.17
    15. DENSITY 25
    16. ALPHA 1E-005
    17. DAMP 0.05
    18. TYPE CONCRETE
    19. STRENGTH FCU 27579
    20. END DEFINE MATERIAL
    21. MEMBER PROPERTY AMERICAN
    22. 184 185 PRIS YD 0.5 ZD 0.5
    23. CONSTANTS
    24. MATERIAL CONCRETE ALL
    25. ************************************************************
    26. SUPPORTS
    27. 1 2 FIXED
    28. ****************************
    29. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
              MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    30. JOINT LOAD
    31. 127 FX -1.9 FY -21.9 FZ -5.5
    32. 128 FX -2.6 FY -37.5 FZ 5.4
    33. ***
    34. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
    35. JOINT LOAD
    36. 127 FX -2.7 FY -26.1 FZ -15.4
```

```
37. 128 FX -5.1 FY -79 FZ 15.1
38. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
39. JOINT LOAD
40. 127 FX 6.8 FY 12.1 FZ 0.1
41. 128 FX 6.8 FY 12.1 FZ -0.1
42. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
43. JOINT LOAD
44. 127 FX -6.8 FY -12.1 FZ -0.1
45. 128 FX -6.8 FY -12.1 FZ 0.1
46. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
47. JOINT LOAD
48. 127 FX 0.7 FY 18.2 FZ 5.3
49. 128 FX -1.5 FY -18.2 FZ 5.3
50. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
51. JOINT LOAD
52. 127 FX -0.7 FY -18.2 FZ -5.3
53. 128 FX 1.5 FY 18.2 FZ -5.3
54. ********************************************************************************
55. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
56. ************************************************************
57. LOAD COMB 201 1.0DL+1.0LL
58. 1 1.0 2 1.0
59. LOAD COMB 202 1.0DL+0.75LL
60. 1 1.0 2 0.75
61. LOAD COMB 203 1.0DL+1.0WLX1
62. 1 1.0 3 1.0
63. LOAD COMB 204 1.0DL+1.0WLX2
64. 1 1.0 4 1.0
65. LOAD COMB 205 1.0DL+1.0WLZ1
66. 1 1.0 5 1.0
67. LOAD COMB 206 1.0DL+1.0WLZ2
68. 1 1.0 6 1.0
69. LOAD COMB 207 1.0DL+0.75LL+0.75WLX1
70. 1 1.0 2 0.75 3 0.75
71. LOAD COMB 208 1.0DL+0.75LL+0.75WLX2
72. 1 1.0 2 0.75 4 0.75
73. LOAD COMB 209 1.0DL+0.75LL+0.75WLZ1
74. 1 1.0 2 0.75 5 0.75
75. LOAD COMB 210 1.0DL+0.75LL+0.75WLZ2
76. 1 1.0 2 0.75 6 0.75
77. LOAD COMB 211 0.6DL+1.0WLX1
78. 1 0.6 3 1.0
79. LOAD COMB 212 0.6DL+1.0WLX2
80. 1 0.6 4 1.0
81. LOAD COMB 213 0.6DL+1.0WLZ1
82. 1 0.6 5 1.0
83. LOAD COMB 214 0.6DL+1.0WLZ2
84. 1 0.6 6 1.0
85. ********
86. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 4 NUMBER OF MEMBERS 2
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 2

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 2/ 2/ 6 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 12
TOTAL LOAD COMBINATION CASES = 14 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 112121.5 MB

87. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 9:48: 7 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                     *
*************************************************************
```

ATTACHMENT 9 STAAD INPUT & OUTPUT FILE FOR F14 DESIGN

PROPRIETARY & CONFIDENTIAL

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5           *
*           Version  20.07.10.64                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    APR  7, 2020                 *
*           Time=    11:54:49                     *
*                                                *
*      USER ID: LT Construction                  *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F14 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 31-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0 -0.025; 2 0 0 3.175; 3 0 0 -0.425; 4 0 0 3.575; 5 0.75 0 -0.025
     9. 6 0.75 0 3.175; 7 0.75 0 -0.425; 8 0.75 0 3.575; 9 -0.75 0 -0.025
    10. 10 -0.75 0 3.175; 11 -0.75 0 -0.425; 12 -0.75 0 3.575; 13 -0.5 0 -0.025
    11. 14 -0.5 0 -0.3; 15 -0.75 0 -0.3; 16 -0.25 0 -0.025; 17 -0.25 0 -0.3
    12. 18 0 0 -0.3; 19 -0.5 0 -0.425; 20 -0.25 0 -0.425; 21 0.25 0 -0.025
    13. 22 0.25 0 -0.3; 23 0.5 0 -0.025; 24 0.5 0 -0.3; 25 0.75 0 -0.3
    14. 26 0.25 0 -0.425; 27 0.5 0 -0.425; 28 -0.5 0 3.575; 29 -0.5 0 3.45
    15. 30 -0.75 0 3.45; 31 -0.25 0 3.575; 32 -0.25 0 3.45; 33 0 0 3.45
    16. 34 -0.5 0 3.175; 35 -0.25 0 3.175; 36 0.25 0 3.575; 37 0.25 0 3.45
    17. 38 0.5 0 3.575; 39 0.5 0 3.45; 40 0.75 0 3.45; 41 0.25 0 3.175; 42 0.5 0 3.175
    18. 43 -0.5 0 2.90769; 44 -0.75 0 2.90769; 45 -0.25 0 2.90769; 46 0 0 2.90769
    19. 47 -0.5 0 2.66538; 48 -0.75 0 2.66538; 49 -0.25 0 2.66538; 50 0 0 2.66538
    20. 51 -0.5 0 2.42308; 52 -0.75 0 2.42308; 53 -0.25 0 2.42308; 54 0 0 2.42308
    21. 55 -0.5 0 2.18077; 56 -0.75 0 2.18077; 57 -0.25 0 2.18077; 58 0 0 2.18077
    22. 59 -0.5 0 1.93846; 60 -0.75 0 1.93846; 61 -0.25 0 1.93846; 62 0 0 1.93846
    23. 63 -0.5 0 1.69615; 64 -0.75 0 1.69615; 65 -0.25 0 1.69615; 66 0 0 1.69615
    24. 67 -0.5 0 1.45385; 68 -0.75 0 1.45385; 69 -0.25 0 1.45385; 70 0 0 1.45385
    25. 71 -0.5 0 1.21154; 72 -0.75 0 1.21154; 73 -0.25 0 1.21154; 74 0 0 1.21154
    26. 75 -0.5 0 0.969231; 76 -0.75 0 0.969231; 77 -0.25 0 0.969231; 78 0 0 0.969231
    27. 79 -0.5 0 0.726923; 80 -0.75 0 0.726923; 81 -0.25 0 0.726923; 82 0 0 0.726923
    28. 83 -0.5 0 0.484615; 84 -0.75 0 0.484615; 85 -0.25 0 0.484615; 86 0 0 0.484615
    29. 87 -0.5 0 0.242308; 88 -0.75 0 0.242308; 89 -0.25 0 0.242308; 90 0 0 0.242308
    30. 91 0.25 0 2.90769; 92 0.5 0 2.90769; 93 0.75 0 2.90769; 94 0.25 0 2.66538
    31. 95 0.5 0 2.66538; 96 0.75 0 2.66538; 97 0.25 0 2.42308; 98 0.5 0 2.42308
    32. 99 0.75 0 2.42308; 100 0.25 0 2.18077; 101 0.5 0 2.18077; 102 0.75 0 2.18077
    33. 103 0.25 0 1.93846; 104 0.5 0 1.93846; 105 0.75 0 1.93846; 106 0.25 0 1.69615
    34. 107 0.5 0 1.69615; 108 0.75 0 1.69615; 109 0.25 0 1.45385; 110 0.5 0 1.45385
    35. 111 0.75 0 1.45385; 112 0.25 0 1.21154; 113 0.5 0 1.21154; 114 0.75 0 1.21154
    36. 115 0.25 0 0.969231; 116 0.5 0 0.969231; 117 0.75 0 0.969231
    37. 118 0.25 0 0.726923; 119 0.5 0 0.726923; 120 0.75 0 0.726923
    38. 121 0.25 0 0.484615; 122 0.5 0 0.484615; 123 0.75 0 0.484615
```

```
39. 124 0.25 0 0.242308; 125 0.5 0 0.242308; 126 0.75 0 0.242308; 127 0 1.4 -0.025
40. 128 0 1.4 3.175; 258 0 0.2 -0.025; 259 0 0.2 3.175
41. MEMBER INCIDENCES
42. 184 1 258; 185 2 259; 291 258 127; 292 259 128
43. ELEMENT INCIDENCES SHELL
44. 26 9 13 14 15; 28 13 16 17 14; 30 16 1 18 17; 32 15 14 19 11; 34 14 17 20 19
45. 35 17 18 3 20; 37 1 21 22 18; 39 21 23 24 22; 41 23 5 25 24; 43 18 22 26 3
46. 45 22 24 27 26; 46 24 25 7 27; 49 12 28 29 30; 51 28 31 32 29; 53 31 4 33 32
47. 55 30 29 34 10; 57 29 32 35 34; 58 32 33 2 35; 60 4 36 37 33; 62 36 38 39 37
48. 64 38 8 40 39; 66 33 37 41 2; 68 37 39 42 41; 69 39 40 6 42; 71 10 34 43 44
49. 72 34 35 45 43; 74 35 2 46 45; 76 44 43 47 48; 77 43 45 49 47; 79 45 46 50 49
50. 81 48 47 51 52; 82 47 49 53 51; 84 49 50 54 53; 86 52 51 55 56; 87 51 53 57 55
51. 89 53 54 58 57; 91 56 55 59 60; 92 55 57 61 59; 94 57 58 62 61; 96 60 59 63 64
52. 97 59 61 65 63; 99 61 62 66 65; 101 64 63 67 68; 102 63 65 69 67
53. 104 65 66 70 69; 106 68 67 71 72; 107 67 69 73 71; 109 69 70 74 73
54. 111 72 71 75 76; 112 71 73 77 75; 114 73 74 78 77; 116 76 75 79 80
55. 117 75 77 81 79; 119 77 78 82 81; 121 80 79 83 84; 122 79 81 85 83
56. 124 81 82 86 85; 126 84 83 87 88; 127 83 85 89 87; 129 85 86 90 89
57. 130 88 87 13 9; 131 87 89 16 13; 132 89 90 1 16; 133 2 41 91 46
58. 134 41 42 92 91; 136 42 6 93 92; 137 46 91 94 50; 138 91 92 95 94
59. 140 92 93 96 95; 141 50 94 97 54; 142 94 95 98 97; 144 95 96 99 98
60. 145 54 97 100 58; 146 97 98 101 100; 148 98 99 102 101; 149 58 100 103 62
61. 150 100 101 104 103; 152 101 102 105 104; 153 62 103 106 66
62. 154 103 104 107 106; 156 104 105 108 107; 157 66 106 109 70
63. 158 106 107 110 109; 160 107 108 111 110; 161 70 109 112 74
64. 162 109 110 113 112; 164 110 111 114 113; 165 74 112 115 78
65. 166 112 113 116 115; 168 113 114 117 116; 169 78 115 118 82
66. 170 115 116 119 118; 172 116 117 120 119; 173 82 118 121 86
67. 174 118 119 122 121; 176 119 120 123 122; 177 86 121 124 90
68. 178 121 122 125 124; 180 122 123 126 125; 181 90 124 21 1; 182 124 125 23 21
69. 183 125 126 5 23
70. START GROUP DEFINITION
71. ELEMENT
72. _RAFT 26 28 30 32 34 35 37 39 41 43 45 46 49 51 53 55 57 58 60 62 64 66 68 -
73. 69 71 72 74 76 77 79 81 82 84 86 87 89 91 92 94 96 97 99 101 102 104 106 -
74. 107 109 111 112 114 116 117 119 121 122 124 126 127 129 TO 134 136 TO 138 -
75. 140 TO 142 144 TO 146 148 TO 150 152 TO 154 156 TO 158 160 TO 162 -
76. 164 TO 166 168 TO 170 172 TO 174 176 TO 178 180 TO 183
77. _RAFT_MOM 26 28 32 34 35 39 41 43 45 46 49 51 53 55 57 60 62 64 68 69 71 72 -
78. 76 77 79 81 82 84 86 87 89 91 92 94 96 97 99 101 102 104 106 107 109 111 -
79. 112 114 116 117 119 121 122 124 126 127 129 TO 131 134 136 TO 138 -
80. 140 TO 142 144 TO 146 148 TO 150 152 TO 154 156 TO 158 160 TO 162 -
81. 164 TO 166 168 TO 170 172 TO 174 176 TO 178 180 182 183
82. _RAFT_SHEAR 26 32 41 46 49 55 64 69 71 76 81 82 84 86 87 89 91 92 94 96 97 -
83. 99 101 102 104 106 107 109 111 112 114 116 117 119 121 122 124 126 130 136 -
84. 140 TO 142 144 TO 146 148 TO 150 152 TO 154 156 TO 158 160 TO 162 -
85. 164 TO 166 168 TO 170 172 TO 174 176 180 183
86. MEMBER
87. _C10 291 292
88. END GROUP DEFINITION
89. ELEMENT PROPERTY
90. 26 28 30 32 34 35 37 39 41 43 45 46 49 51 53 55 57 58 60 62 64 66 68 69 71 -
91. 72 74 76 77 79 81 82 84 86 87 89 91 92 94 96 97 99 101 102 104 106 107 109 -
92. 111 112 114 116 117 119 121 122 124 126 127 129 TO 134 136 TO 138 -
93. 140 TO 142 144 TO 146 148 TO 150 152 TO 154 156 TO 158 160 TO 162 -
94. 164 TO 166 168 TO 170 172 TO 174 176 TO 178 180 TO 183 THICKNESS 0.4
```

```
 95. DEFINE MATERIAL START
 96. ISOTROPIC CONCRETE
 97. E 2.17185E+007
 98. POISSON 0.17
 99. DENSITY 25
100. ALPHA 1E-005
101. DAMP 0.05
102. TYPE CONCRETE
103. STRENGTH FCU 27579
104. END DEFINE MATERIAL
105. MEMBER PROPERTY AMERICAN
106. 184 185 291 292 PRIS YD 0.5 ZD 0.5
107. CONSTANTS
108. MATERIAL CONCRETE ALL
109. ******************************************************
110. SUPPORTS
111. 26 28 30 32 34 35 37 39 41 43 45 46 49 51 53 55 57 58 60 62 64 66 68 69 71 -
112. 72 74 76 77 79 81 82 84 86 87 89 91 92 94 96 97 99 101 102 104 106 107 109 -
113. 111 112 114 116 117 119 121 122 124 126 127 129 TO 134 136 TO 138 -
114. 140 TO 142 144 TO 146 148 TO 150 152 TO 154 156 TO 158 160 TO 162 -
115. 164 TO 166 168 TO 170 172 TO 174 176 TO 178 180 TO 182 -
116. 183 PLATE MAT DIRECT Y SUBGRADE 40500 PRINT
```

JOINT	INFLUENCE AREA
1	0.067789
2	0.067789
3	0.015625
4	0.015625
5	0.033894
6	0.033894
7	0.007813
8	0.007813
9	0.033894
10	0.033894
11	0.007813
12	0.007813
13	0.067789
14	0.050000
15	0.025000
16	0.067789
17	0.050000
18	0.050000
19	0.015625
20	0.015625
21	0.067789
22	0.050000
23	0.067789
24	0.050000
25	0.025000
26	0.015625
27	0.015625
28	0.015625
29	0.050000
30	0.025000

31	0.015625
32	0.050000
33	0.050000
34	0.067789
35	0.067789
36	0.015625
37	0.050000
38	0.015625
39	0.050000
40	0.025000
41	0.067789
42	0.067789
43	0.063703
44	0.031851
45	0.063703
46	0.063703
47	0.060576
48	0.030288
49	0.060576
50	0.060576
51	0.060576
52	0.030288
53	0.060576
54	0.060576
55	0.060578
56	0.030289
57	0.060578
58	0.060578
59	0.060578
60	0.030289
61	0.060578
62	0.060578
63	0.060576
64	0.030288
65	0.060576
66	0.060576
67	0.060576
68	0.030288
69	0.060576
70	0.060576
71	0.060577
72	0.030289
73	0.060577
74	0.060577
75	0.060577
76	0.030289
77	0.060577
78	0.060577
79	0.060577
80	0.030289
81	0.060577
82	0.060577
83	0.060577
84	0.030288
85	0.060577
86	0.060577

87	0.063702
88	0.031851
89	0.063702
90	0.063702
91	0.063703
92	0.063703
93	0.031851
94	0.060576
95	0.060576
96	0.030288
97	0.060576
98	0.060576
99	0.030288
100	0.060578
101	0.060578
102	0.030289
103	0.060578
104	0.060578
105	0.030289
106	0.060576
107	0.060576
108	0.030288
109	0.060576
110	0.060576
111	0.030288
112	0.060577
113	0.060577
114	0.030289
115	0.060577
116	0.060577
117	0.030289
118	0.060577
119	0.060577
120	0.030289
121	0.060577
122	0.060577
123	0.030288
124	0.063702
125	0.063702
126	0.031851
TOTAL	6.000000

```
117. **************************************************
118. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
119. JOINT LOAD
120. 127 FX 0 FY -21 FZ 0
121. 128 FX 0 FY -16.2 FZ 0
122. SELFWEIGHT Y -1
123. ELEMENT LOAD
124. 26 28 30 32 34 35 37 39 41 43 45 46 49 51 53 55 57 58 60 62 64 66 68 69 71 -
125. 72 74 76 77 79 81 82 84 86 87 89 91 92 94 96 97 99 101 102 104 106 107 109 -
126. 111 112 114 116 117 119 121 122 124 126 127 129 TO 134 136 TO 138 -
127. 140 TO 142 144 TO 146 148 TO 150 152 TO 154 156 TO 158 160 TO 162 -
128. 164 TO 166 168 TO 170 172 TO 174 176 TO 178 180 TO 183 PR -11.7
129. ***
130. LOAD 2 LOADTYPE NONE  TITLE NORMAL OPERATING FLUID LOAD ( FL1)
131. JOINT LOAD
```

```
132. 127 FX 0 FY 0 FZ 0
133. 128 FX 0 FY 0 FZ 0
134. LOAD 3 LOADTYPE NONE  TITLE MAX. FLUID LOAD ( FL2)
135. JOINT LOAD
136. 127 FX 0 FY 0 FZ 0
137. 128 FX 0 FY 0 FZ 0
138. LOAD 4 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
139. JOINT LOAD
140. 127 FX 0 FY -34.6 FZ 0
141. 128 FX 0 FY -25.1 FZ 0
142. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD + X ( SLX1 )
143. JOINT LOAD
144. 127 FX 1.4 FY 2.2 FZ -0.1
145. 128 FX 1.3 FY 2.9 FZ -0.1
146. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD - X ( SLX2 )
147. JOINT LOAD
148. 127 FX -1.4 FY -2.2 FZ 0.1
149. 128 FX -1.3 FY -2.9 FZ 0.1
150. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD +Z ( SLZ1)
151. JOINT LOAD
152. 127 FX 0 FY 6 FZ 1.3
153. 128 FX 0 FY -6 FZ 1.3
154. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD -Z ( SLZ2)
155. JOINT LOAD
156. 127 FX 0 FY -6 FZ -1.3
157. 128 FX 0 FY 6 FZ -1.3
158. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
159. JOINT LOAD
160. 127 FX 2.7 FY 3.6 FZ 0
161. 128 FX 2.7 FY 3.4 FZ 0
162. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
163. JOINT LOAD
164. 127 FX -2.7 FY -3.6 FZ 0
165. 128 FX -2.7 FY -3.4 FZ 0
166. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
167. JOINT LOAD
168. 127 FX 0 FY 16.7 FZ 5.3
169. 128 FX 0 FY -16.7 FZ 5.3
170. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
171. JOINT LOAD
172. 127 FX 0 FY -16.7 FZ -5.3
173. 128 FX 0 FY 16.7 FZ -5.3
174. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD ( TL )
175. JOINT LOAD
176. 127 FX 0 FY 0 FZ -0.3
177. 128 FX 0 FY 0 FZ 0.3
178. LOAD 14 LOADTYPE NONE  TITLE LATERAL PIPE LOAD ( PL )
179. JOINT LOAD
180. 127 FX 0 FY -49 FZ 0
181. 128 FX 0 FY -49 FZ 0
182. **********************************************************
183. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
184. LOAD COMB 101 1.4DL+1.4FL1
185. 1 1.4 2 1.4
186. LOAD COMB 102 1.4DL+1.4FL2
187. 1 1.4 3 1.4
```

```
188. LOAD COMB 103 1.2DL+1.2FL1+1.6LL
189. 1 1.2 2 1.2 4 1.6
190. LOAD COMB 104 1.2DL+1.2FL2+1.6LL
191. 1 1.2 3 1.2 4 1.6
192. LOAD COMB 105 1.2DL+1.2FL1+1.2TL+1.6LL
193. 1 1.2 2 1.2 4 1.6 13 1.2
194. LOAD COMB 106 1.2DL+1.2FL2+1.2TL+1.6LL
195. 1 1.2 3 1.2 4 1.6 13 1.2
196. LOAD COMB 107 1.4DL+1.4FL1+1.4PL
197. 1 1.4 2 1.4 14 1.4
198. LOAD COMB 108 1.4DL+1.4FL2+1.4PL
199. 1 1.4 3 1.4 14 1.4
200. LOAD COMB 109 1.2DL+1.2FL1+1.6LL+1.2PL
201. 1 1.2 2 1.2 4 1.6 14 1.2
202. LOAD COMB 110 1.2DL+1.2FL2+1.6LL+1.2PL
203. 1 1.2 3 1.2 4 1.6 14 1.2
204. LOAD COMB 111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
205. 1 1.2 2 1.2 4 1.6 13 1.2 14 1.2
206. LOAD COMB 112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
207. 1 1.2 3 1.2 4 1.6 13 1.2 14 1.2
208. LOAD COMB 113 1.2DL+0.8WLX1
209. 1 1.2 9 0.8
210. LOAD COMB 114 1.2DL+0.8WLX2
211. 1 1.2 10 0.8
212. LOAD COMB 115 1.2DL+0.8WLZ1
213. 1 1.2 11 0.8
214. LOAD COMB 116 1.2DL+0.8WLZ2
215. 1 1.2 12 0.8
216. LOAD COMB 117 1.2DL+1.6WLX1+0.5LL
217. 1 1.2 9 1.6 4 0.5
218. LOAD COMB 118 1.2DL+1.6WLX2+0.5LL
219. 1 1.2 10 1.6 4 0.5
220. LOAD COMB 119 1.2DL+1.6WLZ1+0.5LL
221. 1 1.2 11 1.6 4 0.5
222. LOAD COMB 120 1.2DL+1.6WLZ2+0.5LL
223. 1 1.2 12 1.6 4 0.5
224. LOAD COMB 121 1.2DL+1.0SLX1+0.5LL+0.5FL1
225. 1 1.2 5 1.0 4 0.5 2 0.5
226. LOAD COMB 122 1.2DL+1.0SLX2+0.5LL+0.5FL1
227. 1 1.2 6 1.0 4 0.5 2 0.5
228. LOAD COMB 123 1.2DL+1.0SLZ1+0.5LL+0.5FL1
229. 1 1.2 7 1.0 4 0.5 2 0.5
230. LOAD COMB 124 1.2DL+1.0SLZ2+0.5LL+0.5FL1
231. 1 1.2 8 1.0 4 0.5 2 0.5
232. LOAD COMB 125 0.9DL+1.6WLX1
233. 1 0.9 9 1.6
234. LOAD COMB 126 0.9DL+1.6WLX2
235. 1 0.9 10 1.6
236. LOAD COMB 127 0.9DL+1.6WLZ1
237. 1 0.9 11 1.6
238. LOAD COMB 128 0.9DL+1.6WLZ2
239. 1 0.9 12 1.6
240. LOAD COMB 129 0.9DL+1.0SLX1
241. 1 0.9 5 1.0
242. LOAD COMB 130 0.9DL+1.0SLX2
243. 1 0.9 6 1.0
```

```
244. LOAD COMB 131 0.9DL+1.0SLZ1
245. 1 0.9 7 1.0
246. LOAD COMB 132 0.9DL+1.0SLZ2
247. 1 0.9 8 1.0
248. ********************************************************************************
249. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
250. LOAD COMB 201 1.0DL+1.0FL1
251. 1 1.0 2 1.0
252. LOAD COMB 202 1.0DL+1.0FL2
253. 1 1.0 3 1.0
254. LOAD COMB 203 1.0DL+1.0FL1+1.0LL+1.0TL
255. 1 1.0 2 1.0 4 1.0 13 1.0
256. LOAD COMB 204 1.0DL+1.0FL2+1.0LL+1.0TL
257. 1 1.0 3 1.0 4 1.0 13 1.0
258. LOAD COMB 205 1.0DL+1.0FL1+0.75LL+0.75TL
259. 1 1.0 2 1.0 4 0.75 13 0.75
260. LOAD COMB 206 1.0DL+1.0FL2+0.75LL+0.75TL
261. 1 1.0 3 1.0 4 0.75 13 0.75
262. LOAD COMB 207 1.0DL+1.0FL1+1.0PL
263. 1 1.0 2 1.0 14 1.0
264. LOAD COMB 208 1.0DL+1.0FL2+1.0PL
265. 1 1.0 3 1.0 14 1.0
266. LOAD COMB 209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL
267. 1 1.0 2 1.0 4 1.0 13 1.0 14 1.0
268. LOAD COMB 210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL
269. 1 1.0 3 1.0 4 1.0 13 1.0 14 1.0
270. LOAD COMB 211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL
271. 1 1.0 2 1.0 4 0.75 13 0.75 14 1.0
272. LOAD COMB 212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL
273. 1 1.0 3 1.0 4 0.75 13 0.75 14 1.0
274. LOAD COMB 213 1.0DL+1.0FL1+1.0WLX1
275. 1 1.0 2 1.0 9 1.0
276. LOAD COMB 214 1.0DL+1.0FL1+1.0WLX2
277. 1 1.0 2 1.0 10 1.0
278. LOAD COMB 215 1.0DL+1.0FL1+1.0WLZ1
279. 1 1.0 2 1.0 11 1.0
280. LOAD COMB 216 1.0DL+1.0FL1+1.0WLZ2
281. 1 1.0 2 1.0 12 1.0
282. LOAD COMB 217 1.0DL+1.0FL2+1.0WLX1
283. 1 1.0 3 1.0 9 1.0
284. LOAD COMB 218 1.0DL+1.0FL2+1.0WLX2
285. 1 1.0 3 1.0 10 1.0
286. LOAD COMB 219 1.0DL+1.0FL2+1.0WLZ1
287. 1 1.0 3 1.0 11 1.0
288. LOAD COMB 220 1.0DL+1.0FL2+1.0WLZ2
289. 1 1.0 3 1.0 12 1.0
290. LOAD COMB 221 1.0DL+1.0FL1+0.7SLX1
291. 1 1.0 2 1.0 5 0.7
292. LOAD COMB 222 1.0DL+1.0FL1+0.7SLX2
293. 1 1.0 2 1.0 6 0.7
294. LOAD COMB 223 1.0DL+1.0FL1+0.7SLZ1
295. 1 1.0 2 1.0 7 0.7
296. LOAD COMB 224 1.0DL+1.0FL1+0.7SLZ2
297. 1 1.0 2 1.0 8 0.7
298. LOAD COMB 225 1.0DL+1.0FL2+0.7SLX1
299. 1 1.0 3 1.0 5 0.7
```

```
300. LOAD COMB 226 1.0DL+1.0FL2+0.7SLX2
301. 1 1.0 3 1.0 6 0.7
302. LOAD COMB 227 1.0DL+1.0FL2+0.7SLZ1
303. 1 1.0 3 1.0 7 0.7
304. LOAD COMB 228 1.0DL+1.0FL2+0.7SLZ2
305. 1 1.0 3 1.0 8 0.7
306. LOAD COMB 229 1.0DL+1.0FL1+0.75WLX1+0.75LL
307. 1 1.0 2 1.0 4 0.75 9 0.75
308. LOAD COMB 230 1.0DL+1.0FL1+0.75WLX2+0.75LL
309. 1 1.0 2 1.0 4 0.75 10 0.75
310. LOAD COMB 231 1.0DL+1.0FL1+0.75WLZ1+0.75LL
311. 1 1.0 2 1.0 4 0.75 11 0.75
312. LOAD COMB 232 1.0DL+1.0FL1+0.75WLZ2+0.75LL
313. 1 1.0 2 1.0 4 0.75 12 0.75
314. LOAD COMB 233 1.0DL+1.0FL2+0.75WLX1+0.75LL
315. 1 1.0 3 1.0 4 0.75 9 0.75
316. LOAD COMB 234 1.0DL+1.0FL2+0.75WLX2+0.75LL
317. 1 1.0 3 1.0 4 0.75 10 0.75
318. LOAD COMB 235 1.0DL+1.0FL2+0.75WLZ1+0.75LL
319. 1 1.0 3 1.0 4 0.75 11 0.75
320. LOAD COMB 236 1.0DL+1.0FL2+0.75WLZ2+0.75LL
321. 1 1.0 3 1.0 4 0.75 12 0.75
322. LOAD COMB 237 1.0DL+1.0FL1+0.525SLX1+0.75LL
323. 1 1.0 2 1.0 4 0.75 5 0.525
324. LOAD COMB 238 1.0DL+1.0FL1+0.525SLX2+0.75LL
325. 1 1.0 2 1.0 4 0.75 6 0.525
326. LOAD COMB 239 1.0DL+1.0FL1+0.525SLZ1+0.75LL
327. 1 1.0 2 1.0 4 0.75 7 0.525
328. LOAD COMB 240 1.0DL+1.0FL1+0.525SLZ2+0.75LL
329. 1 1.0 2 1.0 4 0.75 8 0.525
330. LOAD COMB 241 1.0DL+1.0FL2+0.525SLX1+0.75LL
331. 1 1.0 3 1.0 4 0.75 5 0.525
332. LOAD COMB 242 1.0DL+1.0FL2+0.525SLX2+0.75LL
333. 1 1.0 3 1.0 4 0.75 6 0.525
334. LOAD COMB 243 1.0DL+1.0FL2+0.525SLZ1+0.75LL
335. 1 1.0 3 1.0 4 0.75 7 0.525
336. LOAD COMB 244 1.0DL+1.0FL2+0.525SLZ2+0.75LL
337. 1 1.0 3 1.0 4 0.75 8 0.525
338. LOAD COMB 245 0.6DL+1.0WLX1
339. 1 0.6 9 1.0
340. LOAD COMB 246 0.6DL+1.0WLX2
341. 1 0.6 10 1.0
342. LOAD COMB 247 0.6DL+1.0WLZ1
343. 1 0.6 11 1.0
344. LOAD COMB 248 0.6DL+1.0WLZ2
345. 1 0.6 12 1.0
346. LOAD COMB 249 0.6DL+0.6FL1+0.7SLX1
347. 1 0.6 2 0.6 5 0.7
348. LOAD COMB 250 0.6DL+0.6FL1+0.7SLX2
349. 1 0.6 2 0.6 6 0.7
350. LOAD COMB 251 0.6DL+0.6FL1+0.7SLZ1
351. 1 0.6 2 0.6 7 0.7
352. LOAD COMB 252 0.6DL+0.6FL1+0.7SLZ2
353. 1 0.6 2 0.6 8 0.7
354. ********
355. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**130**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**102**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**126**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 128/ 13/ 45 DOF
TOTAL PRIMARY LOAD CASES = 14, TOTAL DEGREES OF FREEDOM = 402
TOTAL LOAD COMBINATION CASES = 84 SO FAR.
SIZE OF STIFFNESS MATRIX = 19 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.8/ 112097.6 MB

356. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 11:55: 2 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                    *
************************************************************
```

ATTACHMENT 10 STAAD INPUT & OUTPUT FILE FOR F14 STABILITY

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    APR  7, 2020                   *
*           Time=     9:51:56                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F14 STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 31-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0 -0.025; 2 0 0 3.175; 127 0 1.4 -0.025; 128 0 1.4 3.175
     9. MEMBER INCIDENCES
    10. 184 1 127; 185 2 128
    11. DEFINE MATERIAL START
    12. ISOTROPIC CONCRETE
    13. E 2.17185E+007
    14. POISSON 0.17
    15. DENSITY 25
    16. ALPHA 1E-005
    17. DAMP 0.05
    18. TYPE CONCRETE
    19. STRENGTH FCU 27579
    20. END DEFINE MATERIAL
    21. MEMBER PROPERTY AMERICAN
    22. 184 185 PRIS YD 0.5 ZD 0.5
    23. CONSTANTS
    24. MATERIAL CONCRETE ALL
    25. ************************************************************
    26. ****************************************************
    27. SUPPORTS
    28. 1 2 FIXED
    29. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
              MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    30. JOINT LOAD
    31. 127 FX 0 FY -21 FZ 0
    32. 128 FX 0 FY -16.2 FZ 0
    33. ***
    34. LOAD 2 LOADTYPE NONE  TITLE NORMAL OPERATING FLUID LOAD ( FL1)
    35. JOINT LOAD
    36. 127 FX 0 FY 0 FZ 0
```

```
37. 128 FX 0 FY 0 FZ 0
38. LOAD 3 LOADTYPE NONE  TITLE MAX. FLUID LOAD ( FL2)
39. JOINT LOAD
40. 127 FX 0 FY 0 FZ 0
41. 128 FX 0 FY 0 FZ 0
42. LOAD 4 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
43. JOINT LOAD
44. 127 FX 0 FY -34.6 FZ 0
45. 128 FX 0 FY -25.1 FZ 0
46. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD + X ( SLX1 )
47. JOINT LOAD
48. 127 FX 1.4 FY 2.2 FZ -0.1
49. 128 FX 1.3 FY 2.9 FZ -0.1
50. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD - X ( SLX2 )
51. JOINT LOAD
52. 127 FX -1.4 FY -2.2 FZ 0.1
53. 128 FX -1.3 FY -2.9 FZ 0.1
54. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD +Z ( SLZ1)
55. JOINT LOAD
56. 127 FX 0 FY 6 FZ 1.3
57. 128 FX 0 FY -6 FZ 1.3
58. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD -Z ( SLZ2)
59. JOINT LOAD
60. 127 FX 0 FY -6 FZ -1.3
61. 128 FX 0 FY 6 FZ -1.3
62. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
63. JOINT LOAD
64. 127 FX 2.7 FY 3.6 FZ 0
65. 128 FX 2.7 FY 3.4 FZ 0
66. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
67. JOINT LOAD
68. 127 FX -2.7 FY -3.6 FZ 0
69. 128 FX -2.7 FY -3.4 FZ 0
70. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
71. JOINT LOAD
72. 127 FX 0 FY 16.7 FZ 5.3
73. 128 FX 0 FY -16.7 FZ 5.3
74. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
75. JOINT LOAD
76. 127 FX 0 FY -16.7 FZ -5.3
77. 128 FX 0 FY 16.7 FZ -5.3
78. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD ( TL )
79. JOINT LOAD
80. 127 FX 0 FY 0 FZ -0.3
81. 128 FX 0 FY 0 FZ 0.3
82. LOAD 14 LOADTYPE NONE  TITLE LATERAL PIPE LOAD ( PL )
83. JOINT LOAD
84. 127 FX 0 FY -49 FZ 0
85. 128 FX 0 FY -49 FZ 0
86. ******************************************************
87. ********************************************************************************
88. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
89. LOAD COMB 201 1.0DL+1.0FL1
90. 1 1.0 2 1.0
91. LOAD COMB 202 1.0DL+1.0FL2
92. 1 1.0 3 1.0
```

```
 93. LOAD COMB 203 1.0DL+1.0FL1+1.0LL+1.0TL
 94. 1 1.0 2 1.0 4 1.0 13 1.0
 95. LOAD COMB 204 1.0DL+1.0FL2+1.0LL+1.0TL
 96. 1 1.0 3 1.0 4 1.0 13 1.0
 97. LOAD COMB 205 1.0DL+1.0FL1+0.75LL+0.75TL
 98. 1 1.0 2 1.0 4 0.75 13 0.75
 99. LOAD COMB 206 1.0DL+1.0FL2+0.75LL+0.75TL
100. 1 1.0 3 1.0 4 0.75 13 0.75
101. LOAD COMB 207 1.0DL+1.0FL1+1.0PL
102. 1 1.0 2 1.0 14 1.0
103. LOAD COMB 208 1.0DL+1.0FL2+1.0PL
104. 1 1.0 3 1.0 14 1.0
105. LOAD COMB 209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL
106. 1 1.0 2 1.0 4 1.0 13 1.0 14 1.0
107. LOAD COMB 210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL
108. 1 1.0 3 1.0 4 1.0 13 1.0 14 1.0
109. LOAD COMB 211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL
110. 1 1.0 2 1.0 4 0.75 13 0.75 14 1.0
111. LOAD COMB 212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL
112. 1 1.0 3 1.0 4 0.75 13 0.75 14 1.0
113. LOAD COMB 213 1.0DL+1.0FL1+1.0WLX1
114. 1 1.0 2 1.0 9 1.0
115. LOAD COMB 214 1.0DL+1.0FL1+1.0WLX2
116. 1 1.0 2 1.0 10 1.0
117. LOAD COMB 215 1.0DL+1.0FL1+1.0WLZ1
118. 1 1.0 2 1.0 11 1.0
119. LOAD COMB 216 1.0DL+1.0FL1+1.0WLZ2
120. 1 1.0 2 1.0 12 1.0
121. LOAD COMB 217 1.0DL+1.0FL2+1.0WLX1
122. 1 1.0 3 1.0 9 1.0
123. LOAD COMB 218 1.0DL+1.0FL2+1.0WLX2
124. 1 1.0 3 1.0 10 1.0
125. LOAD COMB 219 1.0DL+1.0FL2+1.0WLZ1
126. 1 1.0 3 1.0 11 1.0
127. LOAD COMB 220 1.0DL+1.0FL2+1.0WLZ2
128. 1 1.0 3 1.0 12 1.0
129. LOAD COMB 221 1.0DL+1.0FL1+0.7SLX1
130. 1 1.0 2 1.0 5 0.7
131. LOAD COMB 222 1.0DL+1.0FL1+0.7SLX2
132. 1 1.0 2 1.0 6 0.7
133. LOAD COMB 223 1.0DL+1.0FL1+0.7SLZ1
134. 1 1.0 2 1.0 7 0.7
135. LOAD COMB 224 1.0DL+1.0FL1+0.7SLZ2
136. 1 1.0 2 1.0 8 0.7
137. LOAD COMB 225 1.0DL+1.0FL2+0.7SLX1
138. 1 1.0 3 1.0 5 0.7
139. LOAD COMB 226 1.0DL+1.0FL2+0.7SLX2
140. 1 1.0 3 1.0 6 0.7
141. LOAD COMB 227 1.0DL+1.0FL2+0.7SLZ1
142. 1 1.0 3 1.0 7 0.7
143. LOAD COMB 228 1.0DL+1.0FL2+0.7SLZ2
144. 1 1.0 3 1.0 8 0.7
145. LOAD COMB 229 1.0DL+1.0FL1+0.75WLX1+0.75LL
146. 1 1.0 2 1.0 4 0.75 9 0.75
147. LOAD COMB 230 1.0DL+1.0FL1+0.75WLX2+0.75LL
148. 1 1.0 2 1.0 4 0.75 10 0.75
```

```
149. LOAD COMB 231 1.0DL+1.0FL1+0.75WLZ1+0.75LL
150. 1 1.0 2 1.0 4 0.75 11 0.75
151. LOAD COMB 232 1.0DL+1.0FL1+0.75WLZ2+0.75LL
152. 1 1.0 2 1.0 4 0.75 12 0.75
153. LOAD COMB 233 1.0DL+1.0FL2+0.75WLX1+0.75LL
154. 1 1.0 3 1.0 4 0.75 9 0.75
155. LOAD COMB 234 1.0DL+1.0FL2+0.75WLX2+0.75LL
156. 1 1.0 3 1.0 4 0.75 10 0.75
157. LOAD COMB 235 1.0DL+1.0FL2+0.75WLZ1+0.75LL
158. 1 1.0 3 1.0 4 0.75 11 0.75
159. LOAD COMB 236 1.0DL+1.0FL2+0.75WLZ2+0.75LL
160. 1 1.0 3 1.0 4 0.75 12 0.75
161. LOAD COMB 237 1.0DL+1.0FL1+0.525SLX1+0.75LL
162. 1 1.0 2 1.0 4 0.75 5 0.525
163. LOAD COMB 238 1.0DL+1.0FL1+0.525SLX2+0.75LL
164. 1 1.0 2 1.0 4 0.75 6 0.525
165. LOAD COMB 239 1.0DL+1.0FL1+0.525SLZ1+0.75LL
166. 1 1.0 2 1.0 4 0.75 7 0.525
167. LOAD COMB 240 1.0DL+1.0FL1+0.525SLZ2+0.75LL
168. 1 1.0 2 1.0 4 0.75 8 0.525
169. LOAD COMB 241 1.0DL+1.0FL2+0.525SLX1+0.75LL
170. 1 1.0 3 1.0 4 0.75 5 0.525
171. LOAD COMB 242 1.0DL+1.0FL2+0.525SLX2+0.75LL
172. 1 1.0 3 1.0 4 0.75 6 0.525
173. LOAD COMB 243 1.0DL+1.0FL2+0.525SLZ1+0.75LL
174. 1 1.0 3 1.0 4 0.75 7 0.525
175. LOAD COMB 244 1.0DL+1.0FL2+0.525SLZ2+0.75LL
176. 1 1.0 3 1.0 4 0.75 8 0.525
177. LOAD COMB 245 0.6DL+1.0WLX1
178. 1 0.6 9 1.0
179. LOAD COMB 246 0.6DL+1.0WLX2
180. 1 0.6 10 1.0
181. LOAD COMB 247 0.6DL+1.0WLZ1
182. 1 0.6 11 1.0
183. LOAD COMB 248 0.6DL+1.0WLZ2
184. 1 0.6 12 1.0
185. LOAD COMB 249 0.6DL+0.6FL1+0.7SLX1
186. 1 0.6 2 0.6 5 0.7
187. LOAD COMB 250 0.6DL+0.6FL1+0.7SLX2
188. 1 0.6 2 0.6 6 0.7
189. LOAD COMB 251 0.6DL+0.6FL1+0.7SLZ1
190. 1 0.6 2 0.6 7 0.7
191. LOAD COMB 252 0.6DL+0.6FL1+0.7SLZ2
192. 1 0.6 2 0.6 8 0.7
193. ********
194. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 4 NUMBER OF MEMBERS 2
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 2

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 2/ 2/ 6 DOF
TOTAL PRIMARY LOAD CASES = 14, TOTAL DEGREES OF FREEDOM = 12
TOTAL LOAD COMBINATION CASES = 52 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 112121.4 MB

195. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 9:52: 3 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                     *
*************************************************************
```

ATTACHMENT 11 STAAD INPUT & OUTPUT FILE FOR F14A DESIGN

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  7, 2020                *
*           Time=    11:57:33                    *
*                                                *
*      USER ID: LT Construction                 *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F14A DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 31-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 129 4.625 0 -0.185; 130 4.9 0 -0.185; 131 4.9 0 -0.45; 132 4.625 0 -0.45
     9. 133 5.15 0 -0.185; 134 5.15 0 -0.45; 135 5.4 0 -0.185; 136 5.4 0 -0.45
    10. 137 4.9 0 -0.775; 138 4.625 0 -0.775; 139 5.15 0 -0.775; 140 5.4 0 -0.775
    11. 141 5.65 0 -0.185; 142 5.65 0 -0.45; 143 5.9 0 -0.185; 144 5.9 0 -0.45
    12. 145 6.175 0 -0.185; 146 6.175 0 -0.45; 147 5.65 0 -0.775; 148 5.9 0 -0.775
    13. 149 6.175 0 -0.775; 150 4.625 0 3.925; 151 4.9 0 3.925; 152 4.9 0 3.55
    14. 153 4.625 0 3.55; 154 5.15 0 3.925; 155 5.15 0 3.55; 156 5.4 0 3.925
    15. 157 5.4 0 3.55; 158 4.9 0 3.335; 159 4.625 0 3.335; 160 5.15 0 3.335
    16. 161 5.4 0 3.335; 162 5.65 0 3.925; 163 5.65 0 3.55; 164 5.9 0 3.925
    17. 165 5.9 0 3.55; 166 6.175 0 3.925; 167 6.175 0 3.55; 168 5.65 0 3.335
    18. 169 5.9 0 3.335; 170 6.175 0 3.335; 171 4.9 0 2.90769; 172 4.625 0 2.90769
    19. 173 5.15 0 2.90769; 174 5.4 0 2.90769; 175 4.9 0 2.66538; 176 4.625 0 2.66538
    20. 177 5.15 0 2.66538; 178 5.4 0 2.66538; 179 4.9 0 2.42308; 180 4.625 0 2.42308
    21. 181 5.15 0 2.42308; 182 5.4 0 2.42308; 183 4.9 0 2.18077; 184 4.625 0 2.18077
    22. 185 5.15 0 2.18077; 186 5.4 0 2.18077; 187 4.9 0 1.93846; 188 4.625 0 1.93846
    23. 189 5.15 0 1.93846; 190 5.4 0 1.93846; 191 4.9 0 1.69615; 192 4.625 0 1.69615
    24. 193 5.15 0 1.69615; 194 5.4 0 1.69615; 195 4.9 0 1.45385; 196 4.625 0 1.45385
    25. 197 5.15 0 1.45385; 198 5.4 0 1.45385; 199 4.9 0 1.21154; 200 4.625 0 1.21154
    26. 201 5.15 0 1.21154; 202 5.4 0 1.21154; 203 4.9 0 0.969231
    27. 204 4.625 0 0.969231; 205 5.15 0 0.969231; 206 5.4 0 0.969231
    28. 207 4.9 0 0.726923; 208 4.625 0 0.726923; 209 5.15 0 0.726923
    29. 210 5.4 0 0.726923; 211 4.9 0 0.484615; 212 4.625 0 0.484615
    30. 213 5.15 0 0.484615; 214 5.4 0 0.484615; 215 4.9 0 0.242308
    31. 216 4.625 0 0.242308; 217 5.15 0 0.242308; 218 5.4 0 0.242308
    32. 219 5.65 0 2.90769; 220 5.9 0 2.90769; 221 6.175 0 2.90769; 222 5.65 0 2.66538
    33. 223 5.9 0 2.66538; 224 6.175 0 2.66538; 225 5.65 0 2.42308; 226 5.9 0 2.42308
    34. 227 6.175 0 2.42308; 228 5.65 0 2.18077; 229 5.9 0 2.18077
    35. 230 6.175 0 2.18077; 231 5.65 0 1.93846; 232 5.9 0 1.93846
    36. 233 6.175 0 1.93846; 234 5.65 0 1.69615; 235 5.9 0 1.69615
    37. 236 6.175 0 1.69615; 237 5.65 0 1.45385; 238 5.9 0 1.45385
    38. 239 6.175 0 1.45385; 240 5.65 0 1.21154; 241 5.9 0 1.21154
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39. 242 6.175 0 1.21154; 243 5.65 0 0.969231; 244 5.9 0 0.969231
40. 245 6.175 0 0.969231; 246 5.65 0 0.726923; 247 5.9 0 0.726923
41. 248 6.175 0 0.726923; 249 5.65 0 0.484615; 250 5.9 0 0.484615
42. 251 6.175 0 0.484615; 252 5.65 0 0.242308; 253 5.9 0 0.242308
43. 254 6.175 0 0.242308; 255 5.4 1.4 -0.185; 257 5.65 1.4 3.335
44. 260 5.4 0.2 -0.185; 261 5.65 0.2 3.335; 262 4.9 0 -0.5875; 263 4.625 0 -0.5875
45. 264 5.15 0 -0.5875; 265 5.4 0 -0.5875; 266 5.65 0 -0.5875; 267 5.9 0 -0.5875
46. 268 6.175 0 -0.5875; 269 4.9 0 3.7375; 270 4.625 0 3.7375; 271 5.15 0 3.7375
47. 272 5.4 0 3.7375; 273 5.65 0 3.7375; 274 5.9 0 3.7375; 275 6.175 0 3.7375
48. 276 4.9 0 3.12135; 277 4.625 0 3.12135; 278 5.15 0 3.12135; 279 5.4 0 3.12135
49. 280 4.9 0 0.028654; 281 4.625 0 0.028654; 282 5.15 0 0.028654
50. 283 5.4 0 0.028654; 284 5.65 0 3.12135; 285 5.9 0 3.12135; 286 6.175 0 3.12135
51. 287 5.65 0 0.028654; 288 5.9 0 0.028654; 289 6.175 0 0.028654
52. MEMBER INCIDENCES
53. 288 135 260; 290 168 261; 293 260 255; 294 261 257
54. ELEMENT INCIDENCES SHELL
55. 186 129 130 131 132; 187 130 133 134 131; 188 133 135 136 134
56. 192 135 141 142 136; 193 141 143 144 142; 194 143 145 146 144
57. 201 153 152 158 159; 202 152 155 160 158; 203 155 157 161 160
58. 207 157 163 168 161; 208 163 165 169 168; 209 165 167 170 169
59. 213 172 171 175 176; 214 171 173 177 175; 215 173 174 178 177
60. 216 176 175 179 180; 217 175 177 181 179; 218 177 178 182 181
61. 219 180 179 183 184; 220 179 181 185 183; 221 181 182 186 185
62. 222 184 183 187 188; 223 183 185 189 187; 224 185 186 190 189
63. 225 188 187 191 192; 226 187 189 193 191; 227 189 190 194 193
64. 228 192 191 195 196; 229 191 193 197 195; 230 193 194 198 197
65. 231 196 195 199 200; 232 195 197 201 199; 233 197 198 202 201
66. 234 200 199 203 204; 235 199 201 205 203; 236 201 202 206 205
67. 237 204 203 207 208; 238 203 205 209 207; 239 205 206 210 209
68. 240 208 207 211 212; 241 207 209 213 211; 242 209 210 214 213
69. 243 212 211 215 216; 244 211 213 217 215; 245 213 214 218 217
70. 252 174 219 222 178; 253 219 220 223 222; 254 220 221 224 223
71. 255 178 222 225 182; 256 222 223 226 225; 257 223 224 227 226
72. 258 182 225 228 186; 259 225 226 229 228; 260 226 227 230 229
73. 261 186 228 231 190; 262 228 229 232 231; 263 229 230 233 232
74. 264 190 231 234 194; 265 231 232 235 234; 266 232 233 236 235
75. 267 194 234 237 198; 268 234 235 238 237; 269 235 236 239 238
76. 270 198 237 240 202; 271 237 238 241 240; 272 238 239 242 241
77. 273 202 240 243 206; 274 240 241 244 243; 275 241 242 245 244
78. 276 206 243 246 210; 277 243 244 247 246; 278 244 245 248 247
79. 279 210 246 249 214; 280 246 247 250 249; 281 247 248 251 250
80. 282 214 249 252 218; 283 249 250 253 252; 284 250 251 254 253
81. 295 132 131 262 263; 296 263 262 137 138; 297 131 134 264 262
82. 298 262 264 139 137; 299 134 136 265 264; 300 264 265 140 139
83. 301 136 142 266 265; 302 265 266 147 140; 303 142 144 267 266
84. 304 266 267 148 147; 305 144 146 268 267; 306 267 268 149 148
85. 307 150 151 269 270; 308 270 269 152 153; 309 151 154 271 269
86. 310 269 271 155 152; 311 154 156 272 271; 312 271 272 157 155
87. 313 156 162 273 272; 314 272 273 163 157; 315 162 164 274 273
88. 316 273 274 165 163; 317 164 166 275 274; 318 274 275 167 165
89. 319 159 158 276 277; 320 277 276 171 172; 321 158 160 278 276
90. 322 276 278 173 171; 323 160 161 279 278; 324 278 279 174 173
91. 325 216 215 280 281; 326 281 280 130 129; 327 215 217 282 280
92. 328 280 282 133 130; 329 217 218 283 282; 330 282 283 135 133
93. 331 161 168 284 279; 332 279 284 219 174; 333 168 169 285 284
94. 334 284 285 220 219; 335 169 170 286 285; 336 285 286 221 220
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 95. 337 218 252 287 283; 338 283 287 141 135; 339 252 253 288 287
 96. 340 287 288 143 141; 341 253 254 289 288; 342 288 289 145 143
 97. START GROUP DEFINITION
 98. ELEMENT
 99. _RAFT 186 TO 188 192 TO 194 201 TO 203 207 TO 209 213 TO 245 252 TO 284 295 -
100. 296 TO 342
101. _RAFT_MOM 186 187 193 194 201 202 213 TO 245 252 TO 284 295 TO 311 313 315 -
102. 317 319 TO 322 325 TO 329 337 339 TO 342
103. _RAFT_SHEAR 186 194 201 213 216 TO 245 254 TO 284 295 296 305 TO 308 319 320 -
104. 325 326 341 342
105. MEMBER
106. _C10 293
107. _C10A 294
108. END GROUP DEFINITION
109. ELEMENT PROPERTY
110. 186 TO 188 192 TO 194 201 TO 203 207 TO 209 213 TO 245 252 TO 284 295 TO 341 -
111. 342 THICKNESS 0.5
112. DEFINE MATERIAL START
113. ISOTROPIC CONCRETE
114. E 2.17185E+007
115. POISSON 0.17
116. DENSITY 25
117. ALPHA 1E-005
118. DAMP 0.05
119. TYPE CONCRETE
120. STRENGTH FCU 27579
121. END DEFINE MATERIAL
122. MEMBER PROPERTY AMERICAN
123. 288 293 PRIS YD 0.5 ZD 0.5
124. MEMBER PROPERTY AMERICAN
125. 290 294 PRIS YD 0.825 ZD 0.825
126. CONSTANTS
127. MATERIAL CONCRETE ALL
128. ************************************************************
129. SUPPORTS
130. 186 TO 188 192 TO 194 201 TO 203 207 TO 209 213 TO 245 252 TO 284 295 TO 341 -
131. 342 PLATE MAT DIRECT Y SUBGRADE 40500 PRINT
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JOINT	INFLUENCE AREA
129	0.032907
130	0.062823
131	0.052828
132	0.027672
133	0.059832
134	0.050312
135	0.059832
136	0.050313
137	0.024609
138	0.012891
139	0.023437
140	0.023438
141	0.059832
142	0.050313
143	0.062823

144	0.052828
145	0.032907
146	0.027672
147	0.023438
148	0.024609
149	0.012891
150	0.012891
151	0.024609
152	0.052828
153	0.027672
154	0.023437
155	0.050312
156	0.023437
157	0.050313
158	0.056260
159	0.029470
160	0.053581
161	0.053581
162	0.023437
163	0.050313
164	0.024609
165	0.052828
166	0.012891
167	0.027672
168	0.053581
169	0.056260
170	0.029470
171	0.059846
172	0.031348
173	0.056996
174	0.056996
175	0.063605
176	0.033317
177	0.060576
178	0.060576
179	0.063605
180	0.033317
181	0.060576
182	0.060576
183	0.063606
184	0.033318
185	0.060577
186	0.060578
187	0.063606
188	0.033318
189	0.060577
190	0.060578
191	0.063605
192	0.033317
193	0.060576
194	0.060576
195	0.063605
196	0.033317
197	0.060576
198	0.060576
199	0.063606

200	0.033318
201	0.060577
202	0.060577
203	0.063606
204	0.033317
205	0.060577
206	0.060577
207	0.063606
208	0.033317
209	0.060577
210	0.060577
211	0.063606
212	0.033317
213	0.060577
214	0.060577
215	0.059845
216	0.031347
217	0.056995
218	0.056995
219	0.056996
220	0.059846
221	0.031348
222	0.060576
223	0.063605
224	0.033317
225	0.060576
226	0.063605
227	0.033317
228	0.060578
229	0.063606
230	0.033318
231	0.060578
232	0.063606
233	0.033318
234	0.060576
235	0.063605
236	0.033317
237	0.060576
238	0.063605
239	0.033317
240	0.060577
241	0.063606
242	0.033318
243	0.060577
244	0.063606
245	0.033317
246	0.060577
247	0.063606
248	0.033317
249	0.060577
250	0.063606
251	0.033317
252	0.056995
253	0.059845
254	0.031347
262	0.042656

263	0.022344
264	0.040625
265	0.040625
266	0.040625
267	0.042656
268	0.022344
269	0.049219
270	0.025781
271	0.046875
272	0.046875
273	0.046875
274	0.049219
275	0.025781
276	0.056084
277	0.029378
278	0.053414
279	0.053414
280	0.056084
281	0.029377
282	0.053413
283	0.053414
284	0.053414
285	0.056084
286	0.029378
287	0.053414
288	0.056084
289	0.029377
TOTAL	7.284999

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132. **************************************************
133. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
134. JOINT LOAD
135. 257 FX -3.3 FY -7.3 FZ -0.3
136. 255 FX 0 FY -19.4 FZ -1.2
137. 257 FX 0 FY -15.5 FZ 1.4
138. SELFWEIGHT Y -1 LIST ALL
139. ELEMENT LOAD
140. 186 TO 188 192 TO 194 201 TO 203 207 TO 209 213 TO 245 252 TO 284 295 TO 341 -
141. 342 PR -9.9
142. ***
143. LOAD 2 LOADTYPE NONE  TITLE NORMAL OPERATING FLUID LOAD ( FL1)
144. JOINT LOAD
145. 257 FX 0 FY 0 FZ 0
146. 255 FX 0 FY 0 FZ 0
147. 257 FX 0 FY 0 FZ 0
148. LOAD 3 LOADTYPE NONE  TITLE MAX. FLUID LOAD ( FL2)
149. JOINT LOAD
150. 257 FX 0 FY 0 FZ 0
151. 255 FX 0 FY 0 FZ 0
152. 257 FX 0 FY 0 FZ 0
153. LOAD 4 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
154. JOINT LOAD
155. 257 FX -14.1 FY -28.3 FZ -1.
156. 255 FX 0 FY -44.7 FZ -1.9
157. 257 FX 0 FY -25.6 FZ 2.4
158. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD + X ( SLX1 )
159. JOINT LOAD
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160. 257 FX 1.3 FY 0.8 FZ 0
161. 255 FX 1.4 FY -2.4 FZ -0.3
162. 257 FX 1.3 FY -2.7 FZ 0.1
163. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD - X ( SLX2 )
164. JOINT LOAD
165. 257 FX -1.3 FY -0.8 FZ 0
166. 255 FX -1.4 FY 2.4 FZ 0.3
167. 257 FX -1.3 FY 2.7 FZ -0.1
168. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD +Z ( SLZ1)
169. JOINT LOAD
170. 257 FX 0.1 FY 0.1 FZ 0
171. 255 FX 0 FY 4.7 FZ 2
172. 257 FX 0 FY -4.7 FZ 2.2
173. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD -Z ( SLZ2)
174. JOINT LOAD
175. 257 FX -0.1 FY -0.1 FZ 0
176. 255 FX 0 FY -4.7 FZ -2
177. 257 FX 0 FY 4.7 FZ -2.2
178. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
179. JOINT LOAD
180. 257 FX 0.2 FY 0.2 FZ 0
181. 255 FX 2.3 FY -3.7 FZ -0.4
182. 257 FX 2.3 FY -3.3 FZ 0.2
183. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
184. JOINT LOAD
185. 257 FX -0.2 FY -0.2 FZ 0
186. 255 FX -2.3 FY 3.7 FZ 0.4
187. 257 FX -2.3 FY 3.3 FZ -0.2
188. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
189. JOINT LOAD
190. 257 FX 0.1 FY 0.1 FZ 1
191. 255 FX 0 FY 9.7 FZ 4
192. 257 FX 0 FY -9.7 FZ 4.4
193. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
194. JOINT LOAD
195. 257 FX -0.1 FY -0.1 FZ -1.
196. 255 FX 0 FY -9.7 FZ -4
197. 257 FX 0 FY 9.7 FZ -4.4
198. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD ( TL )
199. JOINT LOAD
200. 257 FX -1.7 FY 0 FZ 0
201. 255 FX 0 FY 0.5 FZ -16.9
202. 257 FX 0 FY -0.5 FZ 17.6
203. LOAD 14 LOADTYPE NONE  TITLE LATERAL PIPE LOAD ( PL )
204. JOINT LOAD
205. 257 FX -0.3 FY -49 FZ 0
206. 255 FX 0 FY -49 FZ -4.2
207. 257 FX 0 FY -49 FZ 4.2
208. ************************************************************
209. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
210. LOAD COMB 101 1.4DL+1.4FL1
211. 1 1.4 2 1.4
212. LOAD COMB 102 1.4DL+1.4FL2
213. 1 1.4 3 1.4
214. LOAD COMB 103 1.2DL+1.2FL1+1.6LL
215. 1 1.2 2 1.2 4 1.6
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216. LOAD COMB 104 1.2DL+1.2FL2+1.6LL
217. 1 1.2 3 1.2 4 1.6
218. LOAD COMB 105 1.2DL+1.2FL1+1.2TL+1.6LL
219. 1 1.2 2 1.2 4 1.6 13 1.2
220. LOAD COMB 106 1.2DL+1.2FL2+1.2TL+1.6LL
221. 1 1.2 3 1.2 4 1.6 13 1.2
222. LOAD COMB 107 1.4DL+1.4FL1+1.4PL
223. 1 1.4 2 1.4 14 1.4
224. LOAD COMB 108 1.4DL+1.4FL2+1.4PL
225. 1 1.4 3 1.4 14 1.4
226. LOAD COMB 109 1.2DL+1.2FL1+1.6LL+1.2PL
227. 1 1.2 2 1.2 4 1.6 14 1.2
228. LOAD COMB 110 1.2DL+1.2FL2+1.6LL+1.2PL
229. 1 1.2 3 1.2 4 1.6 14 1.2
230. LOAD COMB 111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
231. 1 1.2 2 1.2 4 1.6 13 1.2 14 1.2
232. LOAD COMB 112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
233. 1 1.2 3 1.2 4 1.6 13 1.2 14 1.2
234. LOAD COMB 113 1.2DL+0.8WLX1
235. 1 1.2 9 0.8
236. LOAD COMB 114 1.2DL+0.8WLX2
237. 1 1.2 10 0.8
238. LOAD COMB 115 1.2DL+0.8WLZ1
239. 1 1.2 11 0.8
240. LOAD COMB 116 1.2DL+0.8WLZ2
241. 1 1.2 12 0.8
242. LOAD COMB 117 1.2DL+1.6WLX1+0.5LL
243. 1 1.2 9 1.6 4 0.5
244. LOAD COMB 118 1.2DL+1.6WLX2+0.5LL
245. 1 1.2 10 1.6 4 0.5
246. LOAD COMB 119 1.2DL+1.6WLZ1+0.5LL
247. 1 1.2 11 1.6 4 0.5
248. LOAD COMB 120 1.2DL+1.6WLZ2+0.5LL
249. 1 1.2 12 1.6 4 0.5
250. LOAD COMB 121 1.2DL+1.0SLX1+0.5LL+0.5FL1
251. 1 1.2 5 1.0 4 0.5 2 0.5
252. LOAD COMB 122 1.2DL+1.0SLX2+0.5LL+0.5FL1
253. 1 1.2 6 1.0 4 0.5 2 0.5
254. LOAD COMB 123 1.2DL+1.0SLZ1+0.5LL+0.5FL1
255. 1 1.2 7 1.0 4 0.5 2 0.5
256. LOAD COMB 124 1.2DL+1.0SLZ2+0.5LL+0.5FL1
257. 1 1.2 8 1.0 4 0.5 2 0.5
258. LOAD COMB 125 0.9DL+1.6WLX1
259. 1 0.9 9 1.6
260. LOAD COMB 126 0.9DL+1.6WLX2
261. 1 0.9 10 1.6
262. LOAD COMB 127 0.9DL+1.6WLZ1
263. 1 0.9 11 1.6
264. LOAD COMB 128 0.9DL+1.6WLZ2
265. 1 0.9 12 1.6
266. LOAD COMB 129 0.9DL+1.0SLX1
267. 1 0.9 5 1.0
268. LOAD COMB 130 0.9DL+1.0SLX2
269. 1 0.9 6 1.0
270. LOAD COMB 131 0.9DL+1.0SLZ1
271. 1 0.9 7 1.0
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272. LOAD COMB 132 0.9DL+1.0SLZ2
273. 1 0.9 8 1.0
274. ********************************************************************************
275. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
276. LOAD COMB 201 1.0DL+1.0FL1
277. 1 1.0 2 1.0
278. LOAD COMB 202 1.0DL+1.0FL2
279. 1 1.0 3 1.0
280. LOAD COMB 203 1.0DL+1.0FL1+1.0LL+1.0TL
281. 1 1.0 2 1.0 4 1.0 13 1.0
282. LOAD COMB 204 1.0DL+1.0FL2+1.0LL+1.0TL
283. 1 1.0 3 1.0 4 1.0 13 1.0
284. LOAD COMB 205 1.0DL+1.0FL1+0.75LL+0.75TL
285. 1 1.0 2 1.0 4 0.75 13 0.75
286. LOAD COMB 206 1.0DL+1.0FL2+0.75LL+0.75TL
287. 1 1.0 3 1.0 4 0.75 13 0.75
288. LOAD COMB 207 1.0DL+1.0FL1+1.0PL
289. 1 1.0 2 1.0 14 1.0
290. LOAD COMB 208 1.0DL+1.0FL2+1.0PL
291. 1 1.0 3 1.0 14 1.0
292. LOAD COMB 209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL
293. 1 1.0 2 1.0 4 1.0 13 1.0 14 1.0
294. LOAD COMB 210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL
295. 1 1.0 3 1.0 4 1.0 13 1.0 14 1.0
296. LOAD COMB 211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL
297. 1 1.0 2 1.0 4 0.75 13 0.75 14 1.0
298. LOAD COMB 212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL
299. 1 1.0 3 1.0 4 0.75 13 0.75 14 1.0
300. LOAD COMB 213 1.0DL+1.0FL1+1.0WLX1
301. 1 1.0 2 1.0 9 1.0
302. LOAD COMB 214 1.0DL+1.0FL1+1.0WLX2
303. 1 1.0 2 1.0 10 1.0
304. LOAD COMB 215 1.0DL+1.0FL1+1.0WLZ1
305. 1 1.0 2 1.0 11 1.0
306. LOAD COMB 216 1.0DL+1.0FL1+1.0WLZ2
307. 1 1.0 2 1.0 12 1.0
308. LOAD COMB 217 1.0DL+1.0FL2+1.0WLX1
309. 1 1.0 3 1.0 9 1.0
310. LOAD COMB 218 1.0DL+1.0FL2+1.0WLX2
311. 1 1.0 3 1.0 10 1.0
312. LOAD COMB 219 1.0DL+1.0FL2+1.0WLZ1
313. 1 1.0 3 1.0 11 1.0
314. LOAD COMB 220 1.0DL+1.0FL2+1.0WLZ2
315. 1 1.0 3 1.0 12 1.0
316. LOAD COMB 221 1.0DL+1.0FL1+0.7SLX1
317. 1 1.0 2 1.0 5 0.7
318. LOAD COMB 222 1.0DL+1.0FL1+0.7SLX2
319. 1 1.0 2 1.0 6 0.7
320. LOAD COMB 223 1.0DL+1.0FL1+0.7SLZ1
321. 1 1.0 2 1.0 7 0.7
322. LOAD COMB 224 1.0DL+1.0FL1+0.7SLZ2
323. 1 1.0 2 1.0 8 0.7
324. LOAD COMB 225 1.0DL+1.0FL2+0.7SLX1
325. 1 1.0 3 1.0 5 0.7
326. LOAD COMB 226 1.0DL+1.0FL2+0.7SLX2
327. 1 1.0 3 1.0 6 0.7
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328. LOAD COMB 227 1.0DL+1.0FL2+0.7SLZ1
329. 1 1.0 3 1.0 7 0.7
330. LOAD COMB 228 1.0DL+1.0FL2+0.7SLZ2
331. 1 1.0 3 1.0 8 0.7
332. LOAD COMB 229 1.0DL+1.0FL1+0.75WLX1+0.75LL
333. 1 1.0 2 1.0 4 0.75 9 0.75
334. LOAD COMB 230 1.0DL+1.0FL1+0.75WLX2+0.75LL
335. 1 1.0 2 1.0 4 0.75 10 0.75
336. LOAD COMB 231 1.0DL+1.0FL1+0.75WLZ1+0.75LL
337. 1 1.0 2 1.0 4 0.75 11 0.75
338. LOAD COMB 232 1.0DL+1.0FL1+0.75WLZ2+0.75LL
339. 1 1.0 2 1.0 4 0.75 12 0.75
340. LOAD COMB 233 1.0DL+1.0FL2+0.75WLX1+0.75LL
341. 1 1.0 3 1.0 4 0.75 9 0.75
342. LOAD COMB 234 1.0DL+1.0FL2+0.75WLX2+0.75LL
343. 1 1.0 3 1.0 4 0.75 10 0.75
344. LOAD COMB 235 1.0DL+1.0FL2+0.75WLZ1+0.75LL
345. 1 1.0 3 1.0 4 0.75 11 0.75
346. LOAD COMB 236 1.0DL+1.0FL2+0.75WLZ2+0.75LL
347. 1 1.0 3 1.0 4 0.75 12 0.75
348. LOAD COMB 237 1.0DL+1.0FL1+0.525SLX1+0.75LL
349. 1 1.0 2 1.0 4 0.75 5 0.525
350. LOAD COMB 238 1.0DL+1.0FL1+0.525SLX2+0.75LL
351. 1 1.0 2 1.0 4 0.75 6 0.525
352. LOAD COMB 239 1.0DL+1.0FL1+0.525SLZ1+0.75LL
353. 1 1.0 2 1.0 4 0.75 7 0.525
354. LOAD COMB 240 1.0DL+1.0FL1+0.525SLZ2+0.75LL
355. 1 1.0 2 1.0 4 0.75 8 0.525
356. LOAD COMB 241 1.0DL+1.0FL2+0.525SLX1+0.75LL
357. 1 1.0 3 1.0 4 0.75 5 0.525
358. LOAD COMB 242 1.0DL+1.0FL2+0.525SLX2+0.75LL
359. 1 1.0 3 1.0 4 0.75 6 0.525
360. LOAD COMB 243 1.0DL+1.0FL2+0.525SLZ1+0.75LL
361. 1 1.0 3 1.0 4 0.75 7 0.525
362. LOAD COMB 244 1.0DL+1.0FL2+0.525SLZ2+0.75LL
363. 1 1.0 3 1.0 4 0.75 8 0.525
364. LOAD COMB 245 0.6DL+1.0WLX1
365. 1 0.6 9 1.0
366. LOAD COMB 246 0.6DL+1.0WLX2
367. 1 0.6 10 1.0
368. LOAD COMB 247 0.6DL+1.0WLZ1
369. 1 0.6 11 1.0
370. LOAD COMB 248 0.6DL+1.0WLZ2
371. 1 0.6 12 1.0
372. LOAD COMB 249 0.6DL+0.6FL1+0.7SLX1
373. 1 0.6 2 0.6 5 0.7
374. LOAD COMB 250 0.6DL+0.6FL1+0.7SLX2
375. 1 0.6 2 0.6 6 0.7
376. LOAD COMB 251 0.6DL+0.6FL1+0.7SLZ1
377. 1 0.6 2 0.6 7 0.7
378. LOAD COMB 252 0.6DL+0.6FL1+0.7SLZ2
379. 1 0.6 2 0.6 8 0.7
380. ********
381. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 158 NUMBER OF MEMBERS 4
NUMBER OF PLATES 126 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 154

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 149/ 13/ 48 DOF
TOTAL PRIMARY LOAD CASES = 14, TOTAL DEGREES OF FREEDOM = 486
TOTAL LOAD COMBINATION CASES = 84 SO FAR.
SIZE OF STIFFNESS MATRIX = 24 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.0/ 112092.9 MB

382. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 11:57:47 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 12 STAAD INPUT & OUTPUT FILE FOR F14A STABILITY

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    APR  7, 2020                   *
*           Time=     9:55:12                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F14A STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 31-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 135 5.4 0 -0.345; 168 5.65 0 3.175; 255 5.4 1.4 -0.345; 257 5.65 1.4 3.175
     9. MEMBER INCIDENCES
    10. 288 135 255; 290 168 257
    11. DEFINE MATERIAL START
    12. ISOTROPIC CONCRETE
    13. E 2.17185E+007
    14. POISSON 0.17
    15. DENSITY 25
    16. ALPHA 1E-005
    17. DAMP 0.05
    18. TYPE CONCRETE
    19. STRENGTH FCU 27579
    20. END DEFINE MATERIAL
    21. MEMBER PROPERTY AMERICAN
    22. 288 PRIS YD 0.5 ZD 0.5
    23. MEMBER PROPERTY
    24. 290 PRIS YD 0.825 ZD 0.825
    25. CONSTANTS
    26. MATERIAL CONCRETE ALL
    27. ************************************************************
    28. ****************************************************
    29. SUPPORTS
    30. 135 168 FIXED
    31. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
              MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    32. JOINT LOAD
    33. 257 FX -3.3 FY -7.3 FZ -0.3
    34. 255 FX 0 FY -19.4 FZ -1.2
    35. 257 FX 0 FY -15.5 FZ 1.4
    36. ***
```

```
37. LOAD 2 LOADTYPE NONE  TITLE NORMAL OPERATING FLUID LOAD ( FL1)
38. JOINT LOAD
39. 257 FX 0 FY 0 FZ 0
40. 255 FX 0 FY 0 FZ 0
41. 257 FX 0 FY 0 FZ 0
42. LOAD 3 LOADTYPE NONE  TITLE MAX. FLUID LOAD ( FL2)
43. JOINT LOAD
44. 257 FX 0 FY 0 FZ 0
45. 255 FX 0 FY 0 FZ 0
46. 257 FX 0 FY 0 FZ 0
47. LOAD 4 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
48. JOINT LOAD
49. 257 FX -14.1 FY -28.3 FZ -1.
50. 255 FX 0 FY -44.7 FZ -1.9
51. 257 FX 0 FY -25.6 FZ 2.4
52. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD + X ( SLX1 )
53. JOINT LOAD
54. 257 FX 1.3 FY 0.8 FZ 0
55. 255 FX 1.4 FY -2.4 FZ -0.3
56. 257 FX 1.3 FY -2.7 FZ 0.1
57. LOAD 6 LOADTYPE NONE  TITLE SEISMIC LOAD - X ( SLX2 )
58. JOINT LOAD
59. 257 FX -1.3 FY -0.8 FZ 0
60. 255 FX -1.4 FY 2.4 FZ 0.3
61. 257 FX -1.3 FY 2.7 FZ -0.1
62. LOAD 7 LOADTYPE NONE  TITLE SEISMIC LOAD +Z ( SLZ1)
63. JOINT LOAD
64. 257 FX 0.1 FY 0.1 FZ 0
65. 255 FX 0 FY 4.7 FZ 2
66. 257 FX 0 FY -4.7 FZ 2.2
67. LOAD 8 LOADTYPE NONE  TITLE SEISMIC LOAD -Z ( SLZ2)
68. JOINT LOAD
69. 257 FX -0.1 FY -0.1 FZ 0
70. 255 FX 0 FY -4.7 FZ -2
71. 257 FX 0 FY 4.7 FZ -2.2
72. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
73. JOINT LOAD
74. 257 FX 0.2 FY 0.2 FZ 0
75. 255 FX 2.3 FY -3.7 FZ -0.4
76. 257 FX 2.3 FY -3.3 FZ 0.2
77. LOAD 10 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
78. JOINT LOAD
79. 257 FX -0.2 FY -0.2 FZ 0
80. 255 FX -2.3 FY 3.7 FZ 0.4
81. 257 FX -2.3 FY 3.3 FZ -0.2
82. LOAD 11 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
83. JOINT LOAD
84. 257 FX 0.1 FY 0.1 FZ 1
85. 255 FX 0 FY 9.7 FZ 4
86. 257 FX 0 FY -9.7 FZ 4.4
87. LOAD 12 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
88. JOINT LOAD
89. 257 FX -0.1 FY -0.1 FZ -1.
90. 255 FX 0 FY -9.7 FZ -4
91. 257 FX 0 FY 9.7 FZ -4.4
92. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD ( TL )
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 93. JOINT LOAD
 94. 257 FX -1.7 FY 0 FZ 0
 95. 255 FX 0 FY 0.5 FZ -16.9
 96. 257 FX 0 FY -0.5 FZ 17.6
 97. LOAD 14 LOADTYPE NONE  TITLE LATERAL PIPE LOAD ( PL )
 98. JOINT LOAD
 99. 257 FX -0.3 FY -49 FZ 0
100. 255 FX 0 FY -49 FZ -4.2
101. 257 FX 0 FY -49 FZ 4.2
102. ******************************************************
103. ********************************************************************************
104. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
105. LOAD COMB 201 1.0DL+1.0FL1
106. 1 1.0 2 1.0
107. LOAD COMB 202 1.0DL+1.0FL2
108. 1 1.0 3 1.0
109. LOAD COMB 203 1.0DL+1.0FL1+1.0LL+1.0TL
110. 1 1.0 2 1.0 4 1.0 13 1.0
111. LOAD COMB 204 1.0DL+1.0FL2+1.0LL+1.0TL
112. 1 1.0 3 1.0 4 1.0 13 1.0
113. LOAD COMB 205 1.0DL+1.0FL1+0.75LL+0.75TL
114. 1 1.0 2 1.0 4 0.75 13 0.75
115. LOAD COMB 206 1.0DL+1.0FL2+0.75LL+0.75TL
116. 1 1.0 3 1.0 4 0.75 13 0.75
117. LOAD COMB 207 1.0DL+1.0FL1+1.0PL
118. 1 1.0 2 1.0 14 1.0
119. LOAD COMB 208 1.0DL+1.0FL2+1.0PL
120. 1 1.0 3 1.0 14 1.0
121. LOAD COMB 209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL
122. 1 1.0 2 1.0 4 1.0 13 1.0 14 1.0
123. LOAD COMB 210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL
124. 1 1.0 3 1.0 4 1.0 13 1.0 14 1.0
125. LOAD COMB 211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL
126. 1 1.0 2 1.0 4 0.75 13 0.75 14 1.0
127. LOAD COMB 212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL
128. 1 1.0 3 1.0 4 0.75 13 0.75 14 1.0
129. LOAD COMB 213 1.0DL+1.0FL1+1.0WLX1
130. 1 1.0 2 1.0 9 1.0
131. LOAD COMB 214 1.0DL+1.0FL1+1.0WLX2
132. 1 1.0 2 1.0 10 1.0
133. LOAD COMB 215 1.0DL+1.0FL1+1.0WLZ1
134. 1 1.0 2 1.0 11 1.0
135. LOAD COMB 216 1.0DL+1.0FL1+1.0WLZ2
136. 1 1.0 2 1.0 12 1.0
137. LOAD COMB 217 1.0DL+1.0FL2+1.0WLX1
138. 1 1.0 3 1.0 9 1.0
139. LOAD COMB 218 1.0DL+1.0FL2+1.0WLX2
140. 1 1.0 3 1.0 10 1.0
141. LOAD COMB 219 1.0DL+1.0FL2+1.0WLZ1
142. 1 1.0 3 1.0 11 1.0
143. LOAD COMB 220 1.0DL+1.0FL2+1.0WLZ2
144. 1 1.0 3 1.0 12 1.0
145. LOAD COMB 221 1.0DL+1.0FL1+0.7SLX1
146. 1 1.0 2 1.0 5 0.7
147. LOAD COMB 222 1.0DL+1.0FL1+0.7SLX2
148. 1 1.0 2 1.0 6 0.7
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```
149. LOAD COMB 223 1.0DL+1.0FL1+0.7SLZ1
150. 1 1.0 2 1.0 7 0.7
151. LOAD COMB 224 1.0DL+1.0FL1+0.7SLZ2
152. 1 1.0 2 1.0 8 0.7
153. LOAD COMB 225 1.0DL+1.0FL2+0.7SLX1
154. 1 1.0 3 1.0 5 0.7
155. LOAD COMB 226 1.0DL+1.0FL2+0.7SLX2
156. 1 1.0 3 1.0 6 0.7
157. LOAD COMB 227 1.0DL+1.0FL2+0.7SLZ1
158. 1 1.0 3 1.0 7 0.7
159. LOAD COMB 228 1.0DL+1.0FL2+0.7SLZ2
160. 1 1.0 3 1.0 8 0.7
161. LOAD COMB 229 1.0DL+1.0FL1+0.75WLX1+0.75LL
162. 1 1.0 2 1.0 4 0.75 9 0.75
163. LOAD COMB 230 1.0DL+1.0FL1+0.75WLX2+0.75LL
164. 1 1.0 2 1.0 4 0.75 10 0.75
165. LOAD COMB 231 1.0DL+1.0FL1+0.75WLZ1+0.75LL
166. 1 1.0 2 1.0 4 0.75 11 0.75
167. LOAD COMB 232 1.0DL+1.0FL1+0.75WLZ2+0.75LL
168. 1 1.0 2 1.0 4 0.75 12 0.75
169. LOAD COMB 233 1.0DL+1.0FL2+0.75WLX1+0.75LL
170. 1 1.0 3 1.0 4 0.75 9 0.75
171. LOAD COMB 234 1.0DL+1.0FL2+0.75WLX2+0.75LL
172. 1 1.0 3 1.0 4 0.75 10 0.75
173. LOAD COMB 235 1.0DL+1.0FL2+0.75WLZ1+0.75LL
174. 1 1.0 3 1.0 4 0.75 11 0.75
175. LOAD COMB 236 1.0DL+1.0FL2+0.75WLZ2+0.75LL
176. 1 1.0 3 1.0 4 0.75 12 0.75
177. LOAD COMB 237 1.0DL+1.0FL1+0.525SLX1+0.75LL
178. 1 1.0 2 1.0 4 0.75 5 0.525
179. LOAD COMB 238 1.0DL+1.0FL1+0.525SLX2+0.75LL
180. 1 1.0 2 1.0 4 0.75 6 0.525
181. LOAD COMB 239 1.0DL+1.0FL1+0.525SLZ1+0.75LL
182. 1 1.0 2 1.0 4 0.75 7 0.525
183. LOAD COMB 240 1.0DL+1.0FL1+0.525SLZ2+0.75LL
184. 1 1.0 2 1.0 4 0.75 8 0.525
185. LOAD COMB 241 1.0DL+1.0FL2+0.525SLX1+0.75LL
186. 1 1.0 3 1.0 4 0.75 5 0.525
187. LOAD COMB 242 1.0DL+1.0FL2+0.525SLX2+0.75LL
188. 1 1.0 3 1.0 4 0.75 6 0.525
189. LOAD COMB 243 1.0DL+1.0FL2+0.525SLZ1+0.75LL
190. 1 1.0 3 1.0 4 0.75 7 0.525
191. LOAD COMB 244 1.0DL+1.0FL2+0.525SLZ2+0.75LL
192. 1 1.0 3 1.0 4 0.75 8 0.525
193. LOAD COMB 245 0.6DL+1.0WLX1
194. 1 0.6 9 1.0
195. LOAD COMB 246 0.6DL+1.0WLX2
196. 1 0.6 10 1.0
197. LOAD COMB 247 0.6DL+1.0WLZ1
198. 1 0.6 11 1.0
199. LOAD COMB 248 0.6DL+1.0WLZ2
200. 1 0.6 12 1.0
201. LOAD COMB 249 0.6DL+0.6FL1+0.7SLX1
202. 1 0.6 2 0.6 5 0.7
203. LOAD COMB 250 0.6DL+0.6FL1+0.7SLX2
204. 1 0.6 2 0.6 6 0.7
```

```
205. LOAD COMB 251 0.6DL+0.6FL1+0.7SLZ1
206. 1 0.6 2 0.6 7 0.7
207. LOAD COMB 252 0.6DL+0.6FL1+0.7SLZ2
208. 1 0.6 2 0.6 8 0.7
209. ********
210. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**4**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 2/ 2/ 6 DOF
TOTAL PRIMARY LOAD CASES = 14, TOTAL DEGREES OF FREEDOM = 12
TOTAL LOAD COMBINATION CASES = 52 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 112121.3 MB

```
211. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR  7,2020   TIME=  9:55:21 ****
```

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
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```